GRAND CANYON NATIONAL PARK
WORLD HERITAGE SITE
REHABILITATE NORTH RIM INFRASTRUCTURE
VOLUME 1 – PIPELINES – NORTH RIM

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 3	PIPELINES - NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES - ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 6	NORTH RIM WATER TREATMENT PLANT
VOLUME 7	NOT USED
VOLUME 8	NORTH RIM WASTEWATER COLLECTION SYSTEM
VOLUME 9	POWER DISTRIBUTION - NORTH RIM
VOLUME 10	POWER DISTRIBUTION - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	010G001	GENERAL	SHEET INDEX
2	010C101	GENERAL	SITE PLAN
3	010D101	GENERAL	RAW WATER HYDRAULIC PROFILE
4	010D102	GENERAL	FINISHED WATER HYDRAULIC PROFILE
5	011X101	DEMOLITION	TREE REMOVAL AND PROTECTION SHEET MAP
6	011X102	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN BALLFIELD SITE
7	011X103	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN BALLFIELD ROAD
8	011X104	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN BALLFIELD ROAD
9	011X105	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN BALLFIELD ROAD
10	011X106	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN BALLFIELD ROAD
11	011X107	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN BALLFIELD ROAD
12	011X108	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN BALLFIELD ROAD AT NORTON CT
13	011X109	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN BRIDLE PATH TO WTP
14	011X110	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN BRIDLE PATH TO WTP
15	011X111	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN BRIDLE PATH TO WTP
16	011X112	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN WTP TO PDC
17	011X113	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN WTP TO PDC
18	011X114	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN WTP TO PDC
19	011X115	DEMOLITION	TREE REMOVAL AND PROTECTION PLAN PDC TO WWTP
20	011C101	SITEWORK	GESC SHEET MAP
21	011C102	SITEWORK	GESC PLAN BALLFIELD SITE
22	011C103	SITEWORK	GESC PLAN BALLFIELD ROAD
23	011C104	SITEWORK	GESC PLAN BALLFIELD ROAD
24	011C105	SITEWORK	GESC PLAN BALLFIELD ROAD
25	011C106	SITEWORK	GESC PLAN BALLFIELD ROAD
26	011C107	SITEWORK	GESC PLAN BALLFIELD ROAD
27	011C108	SITEWORK	GESC PLAN BALLFIELD ROAD AT NORTON CT
28	011C109	SITEWORK	GESC PLAN BRIDLE PATH TO WTP
29	011C110	SITEWORK	GESC PLAN BRIDLE PATH TO WTP
30	011C111	SITEWORK	GESC PLAN BRIDLE PATH TO WTP
31	011C112	SITEWORK	GESC PLAN WTP TO PDC AT SR 67
32	011C113	SITEWORK	GESC PLAN WTP TO PDC
33	011C114	SITEWORK	GESC PLAN WTP TO PDC
34	011C115	SITEWORK	GESC PLAN PDC TO WWTP
35	011C116	SITEWORK	GESC PLAN PDC TO WWTP
36	011C117	SITEWORK	GESC PLAN LODGE TO NORTON CT
37	011C118	SITEWORK	GESC PLAN LODGE TO NORTON CT
38	011C119	SITEWORK	GESC PLAN LODGE TO NORTON CT
39	011C120	SITEWORK	GESC PLAN LODGE TO NORTON CT
40	011C121	SITEWORK	GESC PLAN LODGE TO NORTON CT
41	011C122	SITEWORK	GESC PLAN LODGE TO NORTON CT
42	011C123	SITEWORK	GESC PLAN LODGE TO NORTON CT
43	011C124	SITEWORK	GESC PLAN LODGE TO NORTON CT
44	012C130	UTILITIES	SHEET MAP
45	012C131	UTILITIES	BORE TO WTP UTILITIES PLAN AND PROFILE W1 AND RW1
46	012C132	UTILITIES	BORE TO WTP UTILITIES PLAN AND PROFILE STA 101+00 TO 106+00
47	012C133	UTILITIES	BORE TO WTP UTILITIES PLAN AND PROFILE STA 106+00 TO 111+00
48	012C134	UTILITIES	BORE TO WTP UTILITIES PLAN AND PROFILE STA 111+00 TO 116+00
49	012C135	UTILITIES	BORE TO WTP UTILITIES PLAN AND PROFILE STA 116+00 TO 121+00
50	012C136	UTILITIES	BORE TO WTP UTILITIES PLAN AND PROFILE STA 121+00 TO 126+00
51	012C137	UTILITIES	BORE TO WTP UTILITIES PLAN AND PROFILE STA 126+00 TO 131+00
52	012C138	UTILITIES	BORE TO WTP UTILITIES PLAN AND PROFILE STA 131+00 TO 136+00
53	012C139	UTILITIES	BORE TO WTP UTILITIES PLAN AND PROFILE STA 136+00 TO 141+00
54	012C140	UTILITIES	BORE TO WTP UTILITIES PLAN AND PROFILE STA 141+00 TO 146+00
55	012C141	UTILITIES	BORE TO WTP UTILITIES PLAN AND PROFILE STA 146+00 TO 151+00
56	012C142	UTILITIES	LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 159+50 TO 164+00
57	012C143	UTILITIES	LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 164+00 TO 169+00
58	012C144	UTILITIES	LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 169+00 TO 174+00
59	012C145	UTILITIES	LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 174+00 TO 179+00
60	012C146	UTILITIES	LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 179+00 TO 184+00
61	012C147	UTILITIES	LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 184+00 TO 189+00
62	012C148	UTILITIES	LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 189+00 TO 194+00
63	012C149	UTILITIES	LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 194+00 TO 199+00
64	012C150	UTILITIES	LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 199+00 TO 202+83
65	012C151	UTILITIES	WWTP TO PDC UTILITIES PLAN AND PROFILE STA 160+00 TO 164+25
66	012C152	UTILITIES	WWTP TO PDC UTILITIES PLAN AND PROFILE STA 164+25 TO 167+85
67	012C153	UTILITIES	PDC TO WTP UTILITIES PLAN AND PROFILE STA 167+85 TO 173+00
68	012C154	UTILITIES	PDC TO WTP UTILITIES PLAN AND PROFILE STA 173+00 TO 177+50
69	012C155	UTILITIES	PDC TO WTP UTILITIES PLAN AND PROFILE STA 177+50 TO 182+00
70	012C156	UTILITIES	MH D-103 TO WTP UTILITIES PLAN AND PROFILE STA 70+00 TO 73+50
71	012C157	UTILITIES	MH D-103 TO WTP UTILITIES PLAN AND PROFILE STA 73+50 TO 77+64
72	012C501	UTILITIES	CIVIL DETAILS CARV VALVES
73	012C502	UTILITIES	CIVIL DETAILS TRENCHING
74	012C503	UTILITIES	CIVIL DETAILS
75	013L101	SITEWORK	LANDSCAPE PLAN
76	013C101	SITEWORK	OVERALL BALLFIELD GRADING PLAN EXISTING SITE CONDITION
77	013C102	SITEWORK	OVERALL BALLFIELD GRADING PLAN POST-CONSTRUCTION
78	013C103	BALLFIELD ROAD	PLAN AND PROFILE STA 9+50 TO 14+50
79	013C104	BALLFIELD ROAD	PLAN AND PROFILE STA 14+50 TO 19+50
80	013C105	BALLFIELD ROAD	PLAN AND PROFILE STA 19+50 TO 24+50
81	013C106	BALLFIELD ROAD	PLAN AND PROFILE STA 24+50 TO 29+50
82	013C107	BALLFIELD ROAD	PLAN AND PROFILE STA 29+50 TO 34+50
83	013C108	BALLFIELD ROAD	PLAN AND PROFILE STA 34+50 TO 38+50
84	013C109	BALLFIELD ROAD	PLAN AND PROFILE STA 38+50 TO 40+50
85	013C110	ADMIN LOOP	ASPHALT PAVING PLAN
86	NOT USED	NOT USED	NOT USED
87	013C301	BALLFIELD ROAD	SECTIONS

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY Date Signed 5-1-2026 ARIZONA, U.S.A.

Nicholas Humphrey

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 41809 RYAN RHOADES Date Signed 5-1-26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: N. LYNCH TECH. REVIEW: M. GOUGH DATE: AUG 2026	SUB SHEET NO. 010 G001	TITLE OF SHEET GENERAL SHEET INDEX NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 1 OF 87

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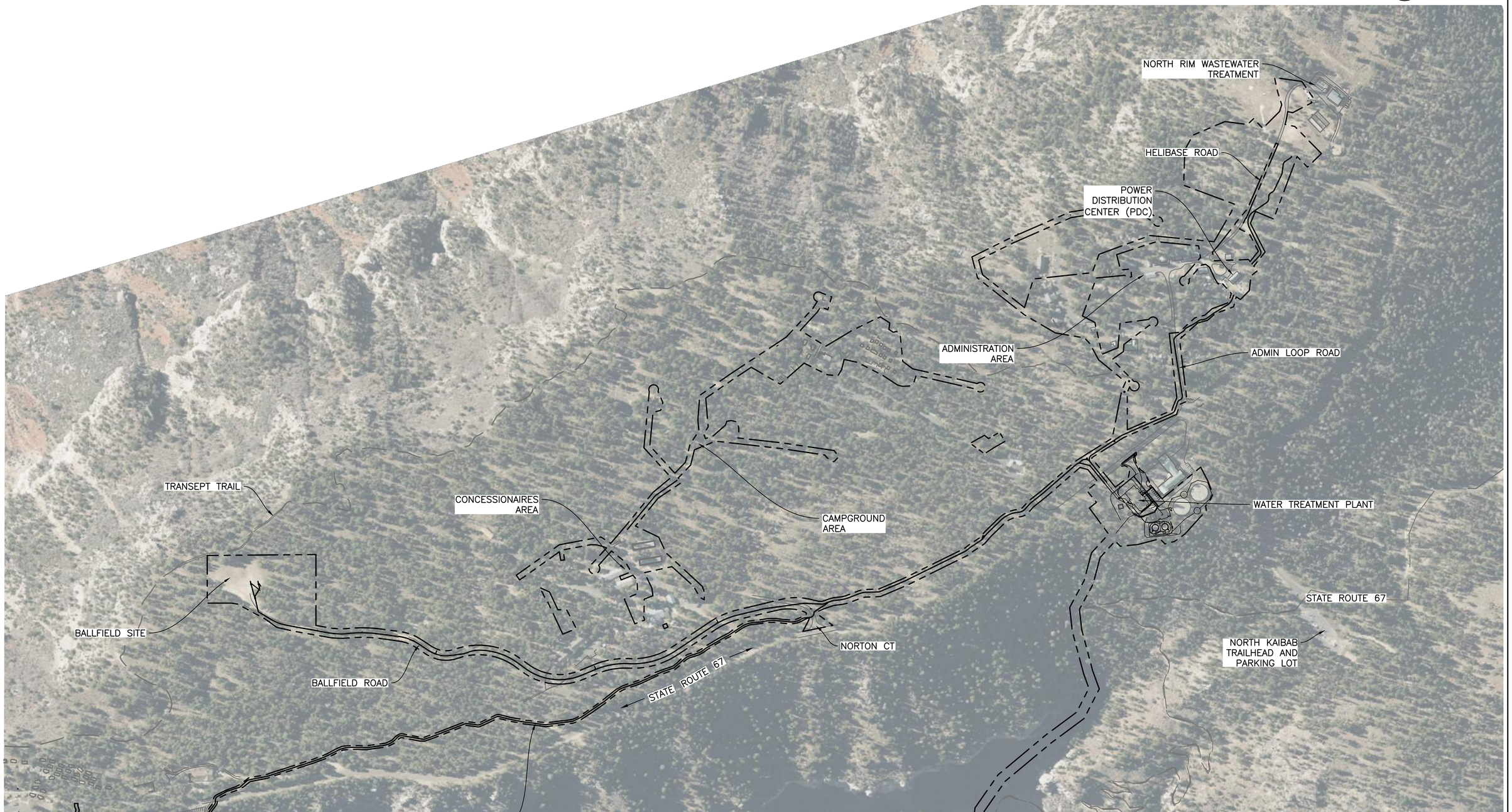


250 0 250 500 1" = 250'

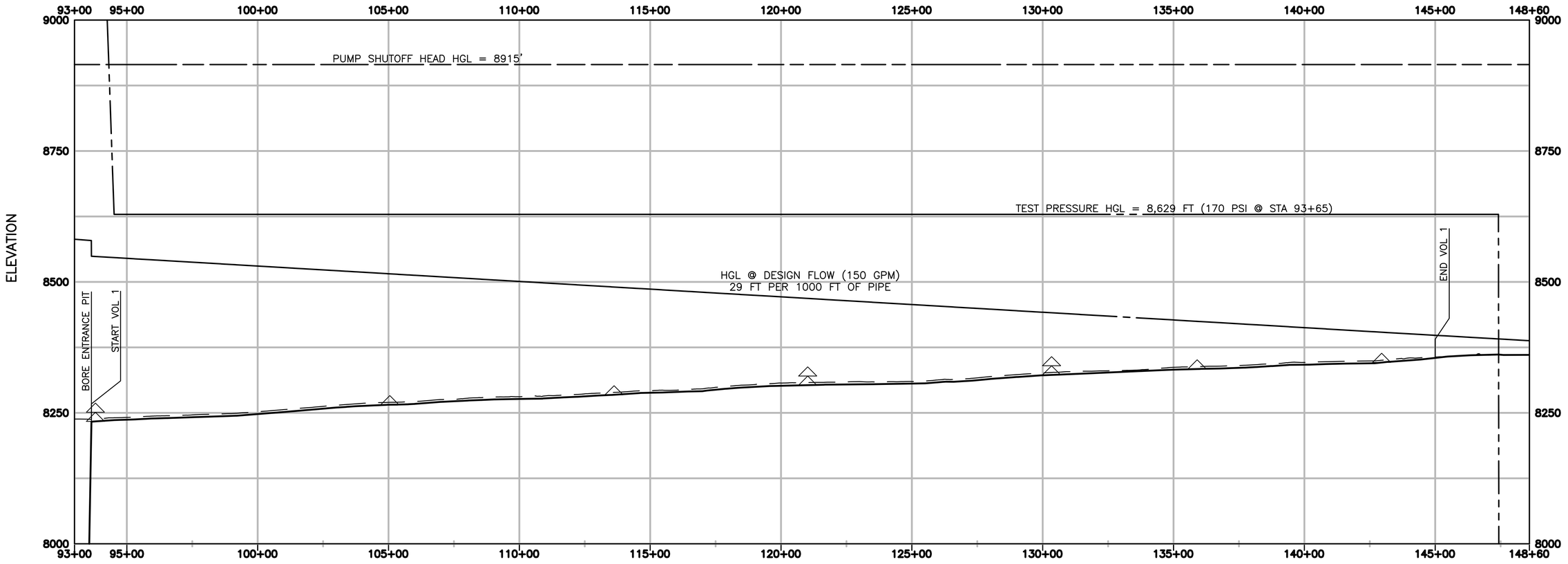
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 8-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	SUB SHEET NO. 010 C101	TITLE OF SHEET GENERAL SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 2 OF 87

8/18/2026 Green, Jake C:\Users\jgreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\010-C101.dwg



NORTH RIM WTP TO BORE ENTRY HYDRAULIC PROFILE

LEGEND:

- △ COMBINATION AIR RELEASE AND VACUUM VALVE
- —— —— —— EXISTING GROUND AT SURFACE
- ——— PIPE CENTERLINE
- ——— – ——— WATERLINE HGL
- ——— – – ——— TEST PRESSURE
- – — — — — – PUMP SHUTOFF HEAD

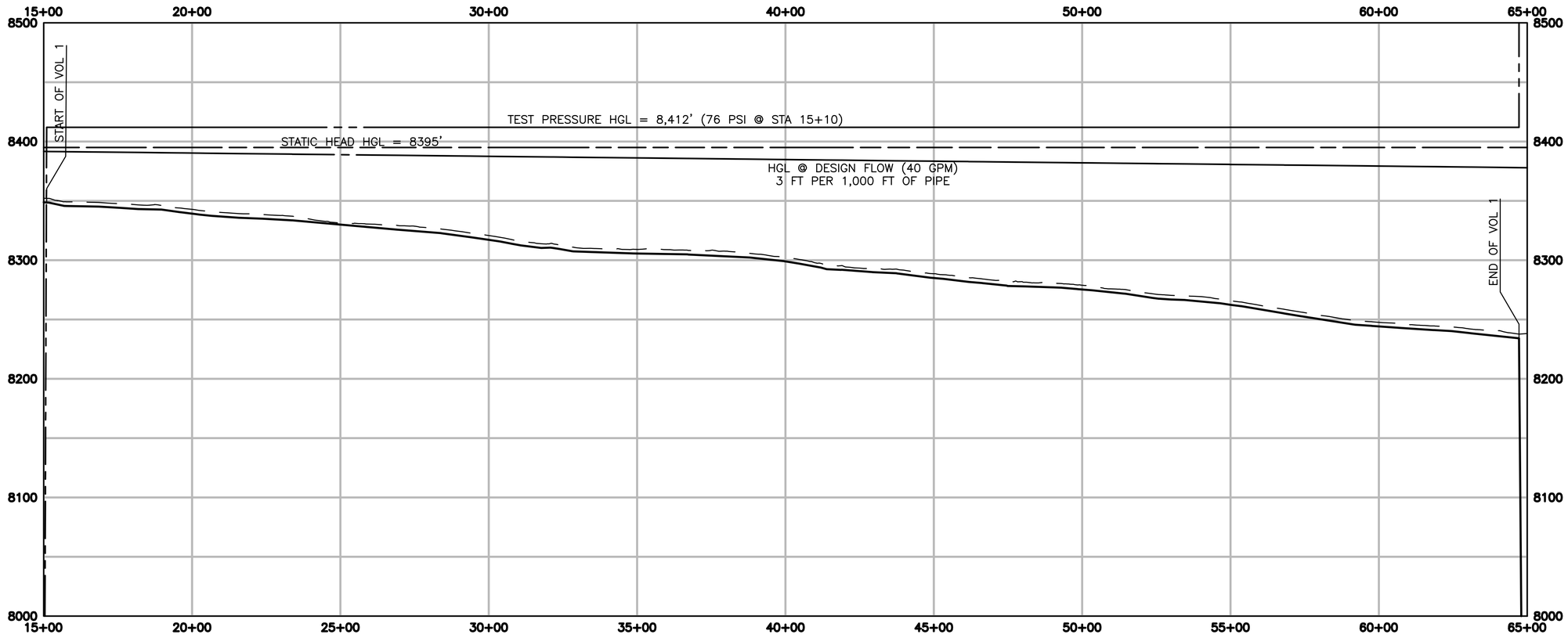
200 0 200 400 HORIZ.
100 0 100 200 VERT.
SCALE OF FEET
H: 1"=400' V: 1"=100'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	N. HUMPHREY ADD C. WITTENMEIER TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	010 D101	GENERAL RAW WATER HYDRAULIC PROFILE NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 3 OF 87

4/17/2026 Dall, Abby C:\Users\adall\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\PROCESS\010-D101.dwg



NORTH RIM WATER TREATMENT PLANT TO ROARING SPRINGS PUMPHOUSE HYDRAULIC PROFILE

LEGEND:

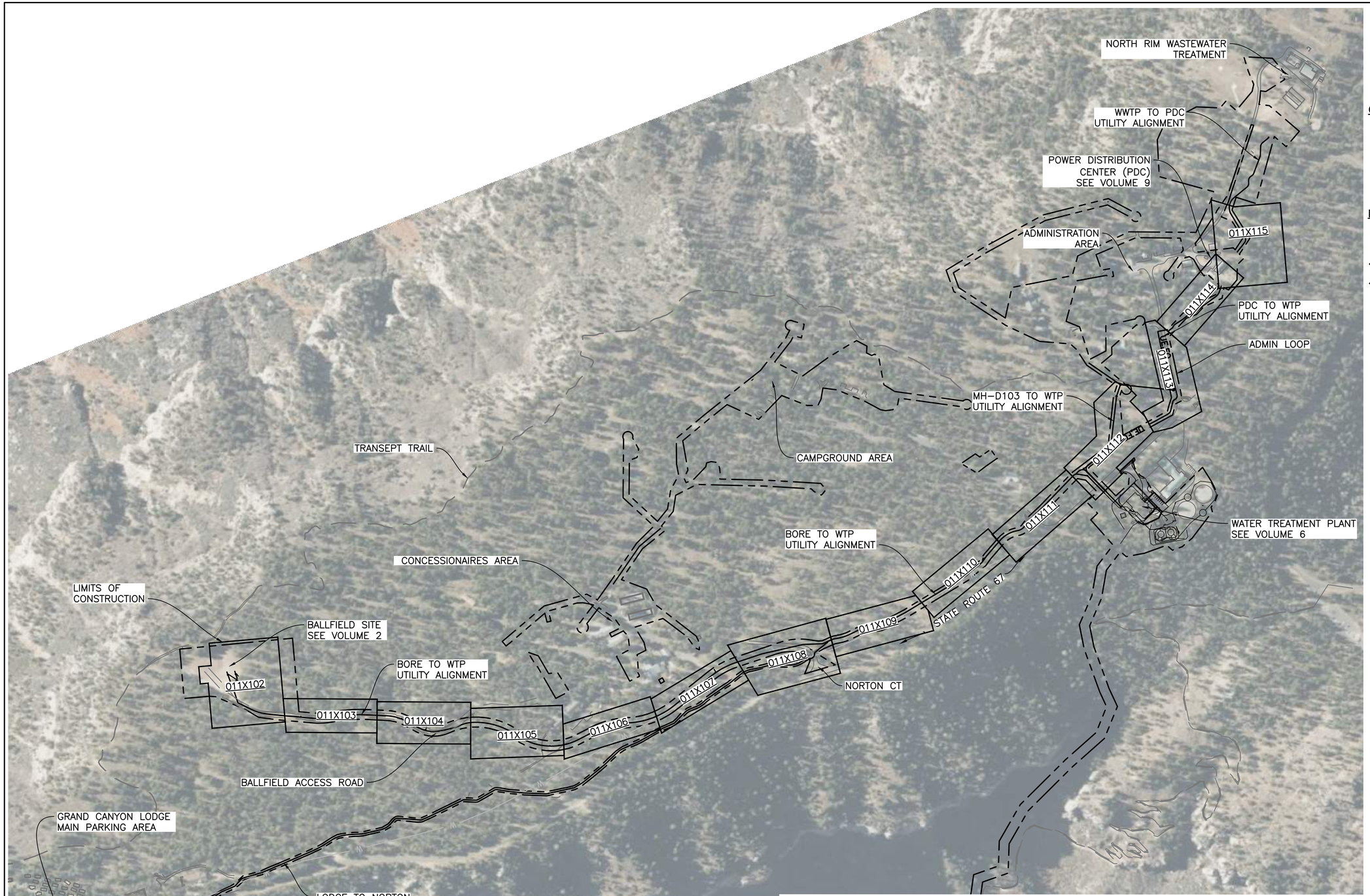
- EXISTING GROUND AT SURFACE
- PIPE CENTERLINE
- WATERLINE HGL
- TEST PRESSURE
- STATIC HEAD

200 0 200 400 HORIZ.
50 0 50 100 VERT.
SCALE OF FEET
H: 1"=200' V: 1"=50'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: M. GOUGH/R. RHODES DATE: MAY 2026	SUB SHEET NO. 010 D102	TITLE OF SHEET GENERAL FINISHED WATER HYDRAULIC PROFILE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 4 OF 87

4/17/2026 Dall, Abby C:\Users\adall\DC\ACCDocs\HDR\103505445_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\PROCESS\010-D102.dwg



GENERAL NOTES:.

A. TOTAL NUMBER OF TREES TO BE DEMOLISHED FOR VOLUME 1 ARE APPROXIMATELY 355 TREES AS NOTED ON PLANS. PROTECT EXISTING TREES FROM DAMAGE TO THE EXTENT POSSIBLE.

LEGEND:

XXX (SEE TREE REMOVAL TABLE)

INDICATES TREE TO BE DEMOLISHED

250 0 250 500 1" = 250'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

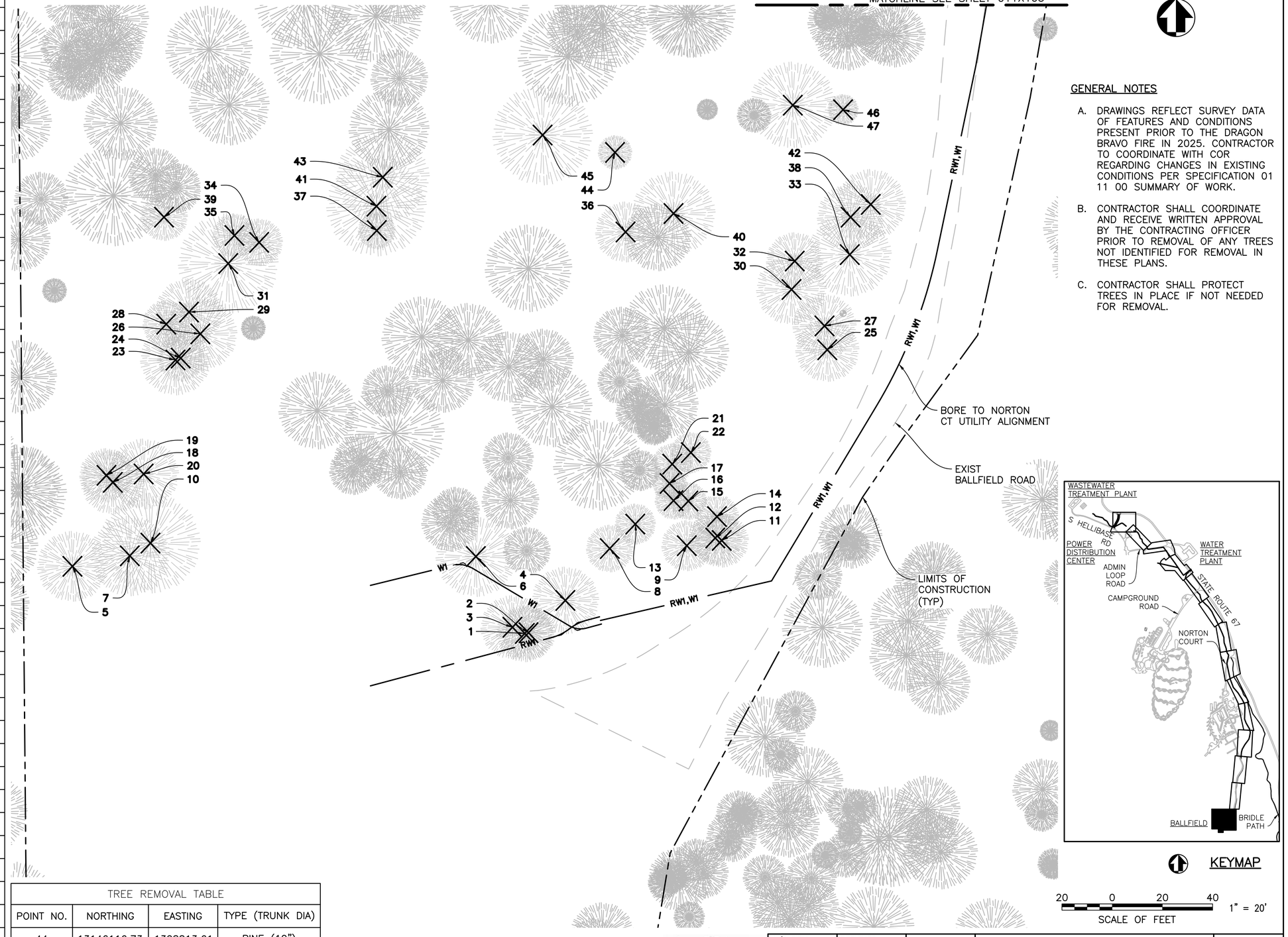
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	SUB SHEET NO. 011 X101	TITLE OF SHEET SITEWORK TREE REMOVAL AND PROTECTION SHEET MAP NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 5 OF 87

8/18/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X101.dwg

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
1	13145925.14	1328777.68	PINE (17")
2	13145925.42	1328779.16	PINE (16")
3	13145927.61	1328772.77	PINE (19")
4	13145938.28	1328793.82	PINE (20")
5	13145951.87	1328597.54	PINE (21")
6	13145955.78	1328758.20	PINE (24")
7	13145956.06	1328620.38	PINE (24")
8	13145958.95	1328811.51	PINE (14")
9	13145960.09	1328842.18	PINE (15")
10	13145961.11	1328628.61	PINE (30")
11	13145962.17	1328856.09	PINE (16")
12	13145963.16	1328853.52	PINE (16")
13	13145968.69	1328821.67	PINE (12")
14	13145971.67	1328854.21	PINE (13")
15	13145977.87	1328842.81	PINE (11")
16	13145978.05	1328836.90	PINE (11")
17	13145984.88	1328835.31	PINE (10")
18	13145985.31	1328613.63	PINE (19")
19	13145988.12	1328611.09	PINE (16")
20	13145988.65	1328625.91	PINE (21")
21	13145992.59	1328836.39	PINE (12")
22	13145997.18	1328843.81	PINE (12")
23	13146034.05	1328638.58	PINE (X")
24	13146034.83	1328640.66	PINE (14")
25	13146038.13	1328898.12	PINE (X")
26	13146044.46	1328648.48	PINE (X")
27	13146047.63	1328897.02	PINE (19")
28	13146048.33	1328634.89	PINE (14")
29	13146053.15	1328643.95	PINE (X")
30	13146062.14	1328883.88	PINE (X")
31	13146072.47	1328659.58	PINE (34")
32	13146073.46	1328885.17	PINE (17")
33	13146076.03	1328907.07	PINE (25")
34	13146080.91	1328671.89	PINE (X")
35	13146083.64	1328662.05	PINE (X")
36	13146085.04	1328817.80	PINE (X")
37	13146085.61	1328718.74	PINE (11")
38	13146090.95	1328907.43	PINE (X")
39	13146090.96	1328634.00	PINE (X")
40	13146092.40	1328836.91	PINE (X")
41	13146095.33	1328718.61	PINE (X")
42	13146096.03	1328915.44	PINE (21")
43	13146106.85	1328721.12	PINE (28")

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
44	13146116.73	1328813.61	PINE (10")
45	13146123.65	1328784.76	PINE (30")
46	13146133.74	1328904.41	PINE (9")
47	13146135.54	1328884.39	PINE (26")



GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.

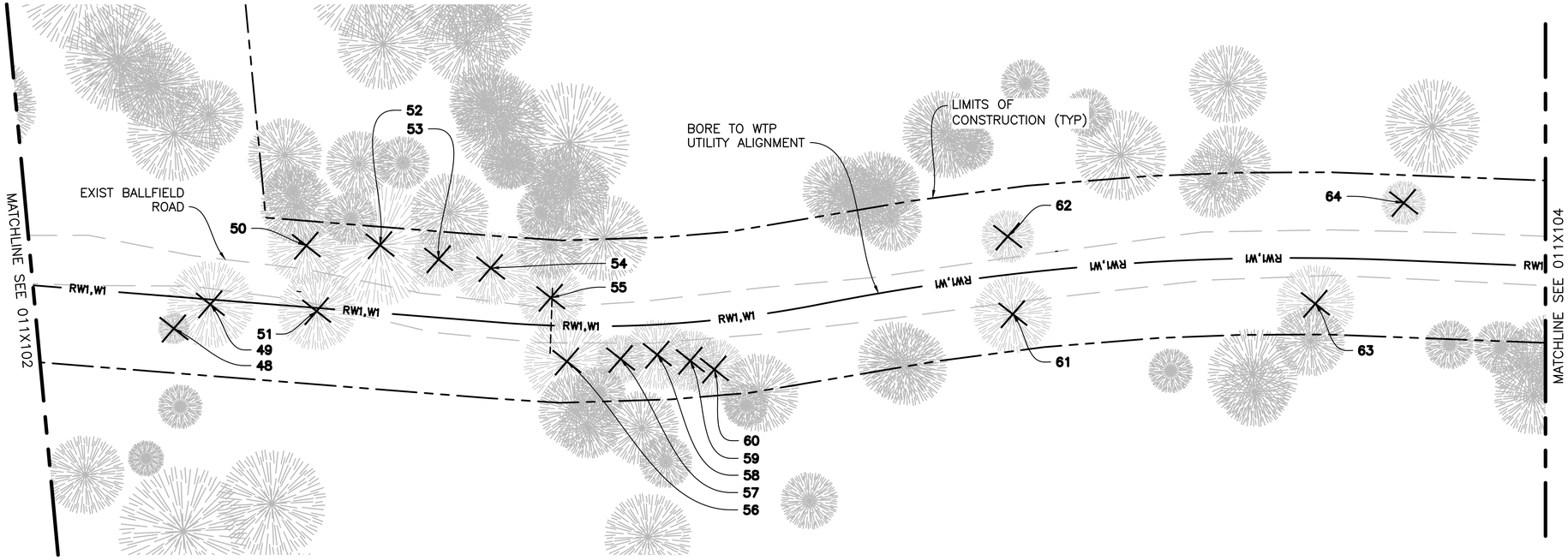
C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5/1/2026 Not Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL (CADD) J. GREEN	011 X102	DEMOLITION TREE REMOVAL AND PROTECTION PLAN BALLFIELD SITE NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
	TECH. REVIEW: GOUGH/LUSTER			PMIS/PKG NO. 318719
	DATE: MAY 2026			SHEET 6 OF 87

4/17/2026 Race, Sara C:\Users\sracex\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



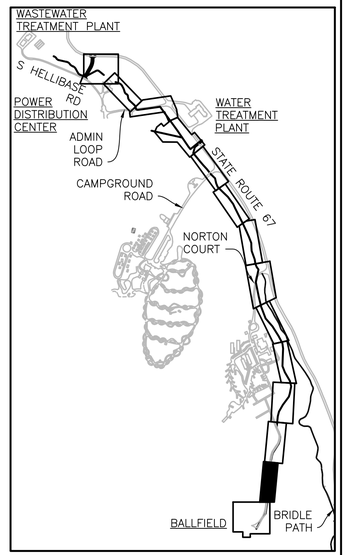
GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.

C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
48	13146217.53	1328978.91	PINE (7")
49	13146229.39	1328972.55	PINE (20")
50	13146260.53	1328957.33	PINE (18")
51	13146261.82	1328977.73	PINE (18")
52	13146283.18	1328959.31	PINE (28")
53	13146300.73	1328965.35	PINE (13")
54	13146316.47	1328969.53	PINE (17")
55	13146334.00	1328980.40	PINE (15")
56	13146337.12	1329000.44	PINE (20")
57	13146353.67	1329001.22	PINE (11")
58	13146365.04	1329000.98	PINE (16")
59	13146374.99	1329003.80	PINE (14")
60	13146382.12	1329007.15	PINE (14")
61	13146475.26	1328998.86	PINE (18")
62	13146475.94	1328974.86	PINE (12")
63	13146568.35	1329004.44	PINE (18")
64	13146598.38	1328976.04	PINE (10")



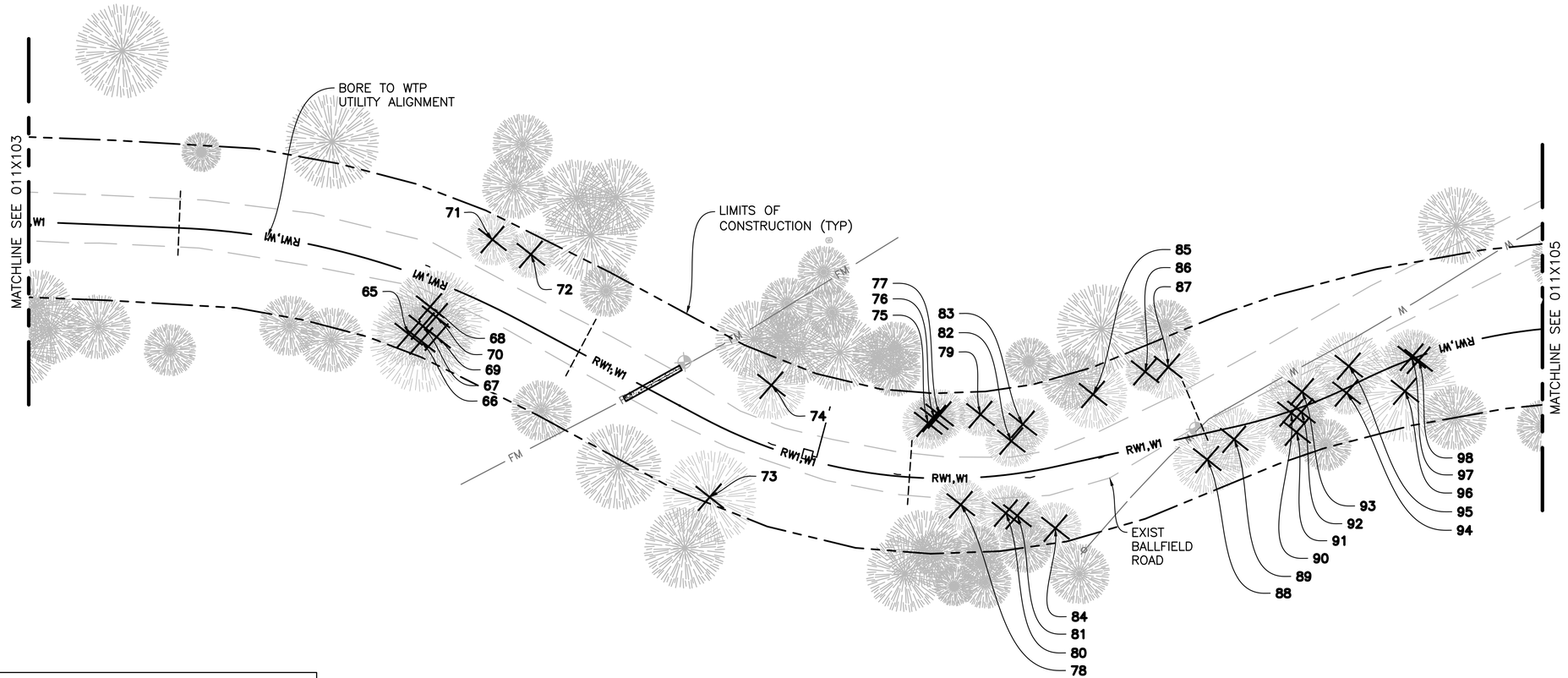
KEYMAP

20 0 20 40 1" = 20'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 X103	TITLE OF SHEET **DEMOLITION** TREE REMOVAL AND PROTECTION PLAN BALLFIELD ROAD NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 7 OF 87

4/17/2026 Race, Sara C:\Users\sgrace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.

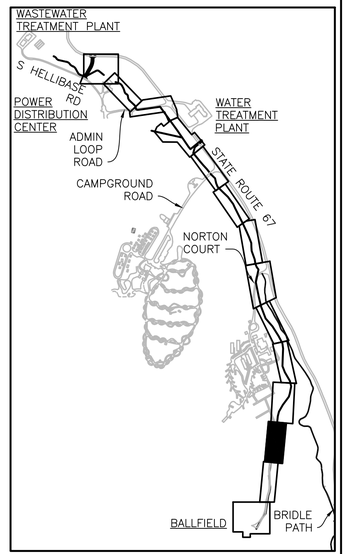
C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.

TREE REMOVAL TABLE

POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
65	13146754.42	1329045.68	PINE (15")
66	13146758.09	1329048.19	PINE (24")
67	13146759.03	1329043.58	PINE (9")
68	13146761.89	1329037.83	PINE (17")
69	13146762.99	1329047.41	PINE (16")
70	13146763.39	1329041.09	PINE (22")
71	13146783.22	1329018.75	PINE (12")
72	13146794.74	1329024.49	PINE (11")
73	13146843.21	1329105.21	PINE (22")
74	13146865.66	1329072.22	PINE (16")
75	13146913.30	1329088.64	PINE (12")
76	13146915.32	1329087.38	PINE (10")
77	13146917.08	1329086.43	PINE (12")
78	13146921.03	1329114.93	PINE (12")
79	13146929.80	1329087.49	PINE (10")
80	13146934.81	1329118.86	PINE (13")
81	13146938.29	1329120.01	PINE (11")
82	13146938.60	1329096.64	PINE (12")
83	13146942.69	1329091.75	PINE (12")
84	13146949.77	1329125.01	PINE (12")

TREE REMOVAL TABLE

POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
85	13146965.39	1329084.53	PINE (16")
86	13146982.24	1329079.01	PINE (9")
87	13146989.55	1329078.30	PINE (15")
88	13146998.97	1329108.15	PINE (16")
89	13147007.96	1329102.65	PINE (15")
90	13147026.50	1329096.14	PINE (16")
91	13147027.57	1329102.27	PINE (12")
92	13147029.65	1329096.14	PINE (10")
93	13147030.44	1329089.88	PINE (15")
94	13147044.26	1329091.55	PINE (10")
95	13147045.74	1329083.29	PINE (16")
96	13147062.47	1329092.71	PINE (24")
97	13147065.55	1329082.35	PINE (13")
98	13147067.63	1329083.52	PINE (13")



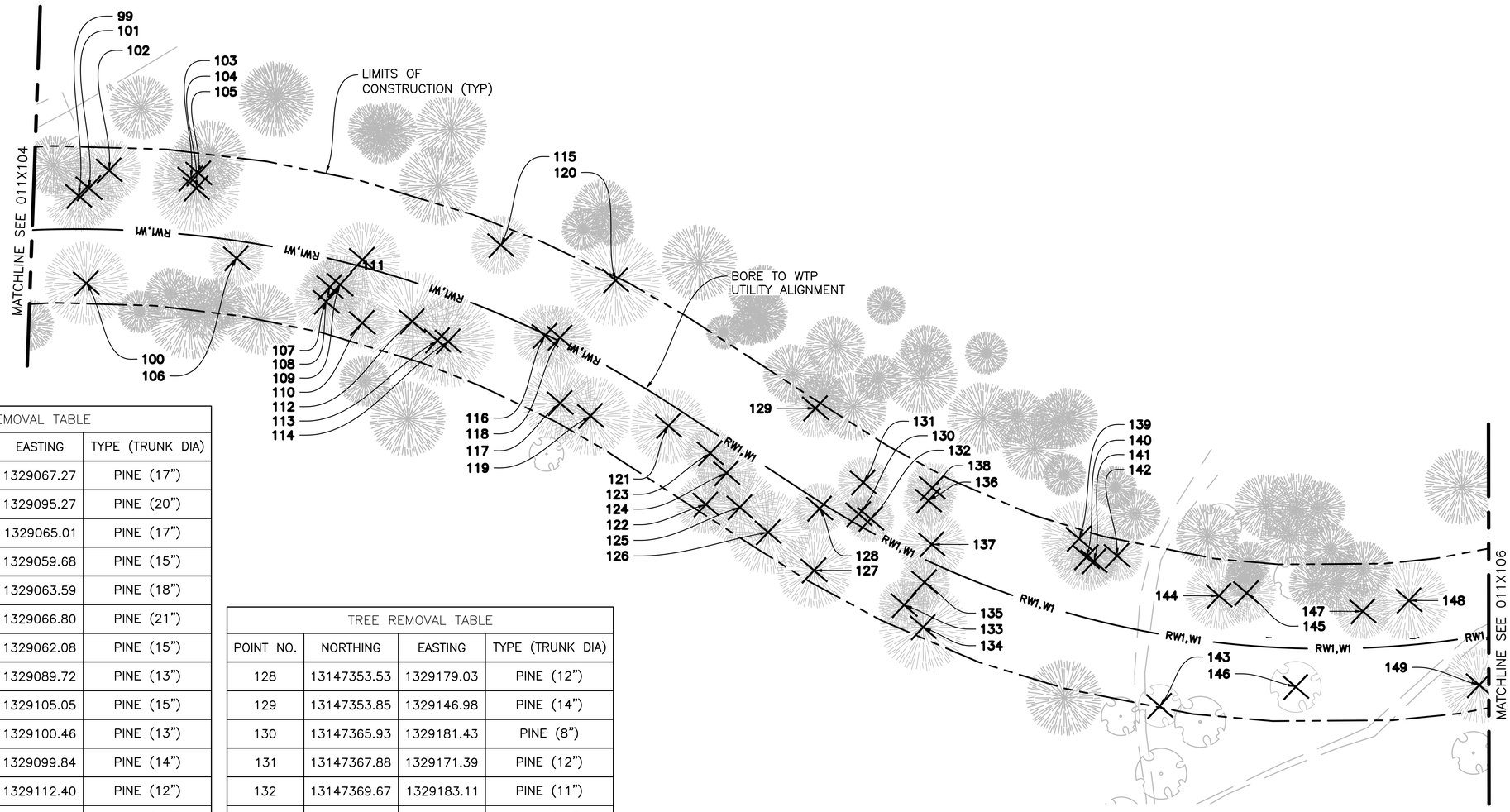
KEYMAP

20 0 20 40 1" = 20'
SCALE OF FEETS

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 X104	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN BALLFIELD ROAD NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 8 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



KEYMAP

20 0 20 40
SCALE OF FEET
1" = 20'

GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.

C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.

TREE REMOVAL TABLE

POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
99	13147122.37	1329067.27	PINE (17")
100	13147123.44	1329095.27	PINE (20")
101	13147125.80	1329065.01	PINE (17")
102	13147132.52	1329059.68	PINE (15")
103	13147158.44	1329063.59	PINE (18")
104	13147159.99	1329066.80	PINE (21")
105	13147161.05	1329062.08	PINE (15")
106	13147171.73	1329089.72	PINE (13")
107	13147199.58	1329105.05	PINE (15")
108	13147201.02	1329100.46	PINE (13")
109	13147204.27	1329099.84	PINE (14")
110	13147210.75	1329112.40	PINE (12")
111	13147211.84	1329092.28	PINE (18")
112	13147226.71	1329112.72	PINE (24")
113	13147234.43	1329119.20	PINE (20")
114	13147237.86	1329119.74	PINE (22")
115	13147256.18	1329089.83	PINE (14")
116	13147268.77	1329119.47	PINE (15")
117	13147272.53	1329141.19	PINE (18")
118	13147273.57	1329120.30	PINE (15")
119	13147282.00	1329145.75	PINE (19")
120	13147292.18	1329102.97	PINE (22")
121	13147306.71	1329150.22	PINE (18")
122	13147317.34	1329175.85	PINE (14")
123	13147319.55	1329159.62	PINE (12")
124	13147324.46	1329166.19	PINE (14")
125	13147328.16	1329177.33	PINE (18")
126	13147336.74	1329185.74	PINE (20")
127	13147350.75	1329198.64	PINE (18")

TREE REMOVAL TABLE

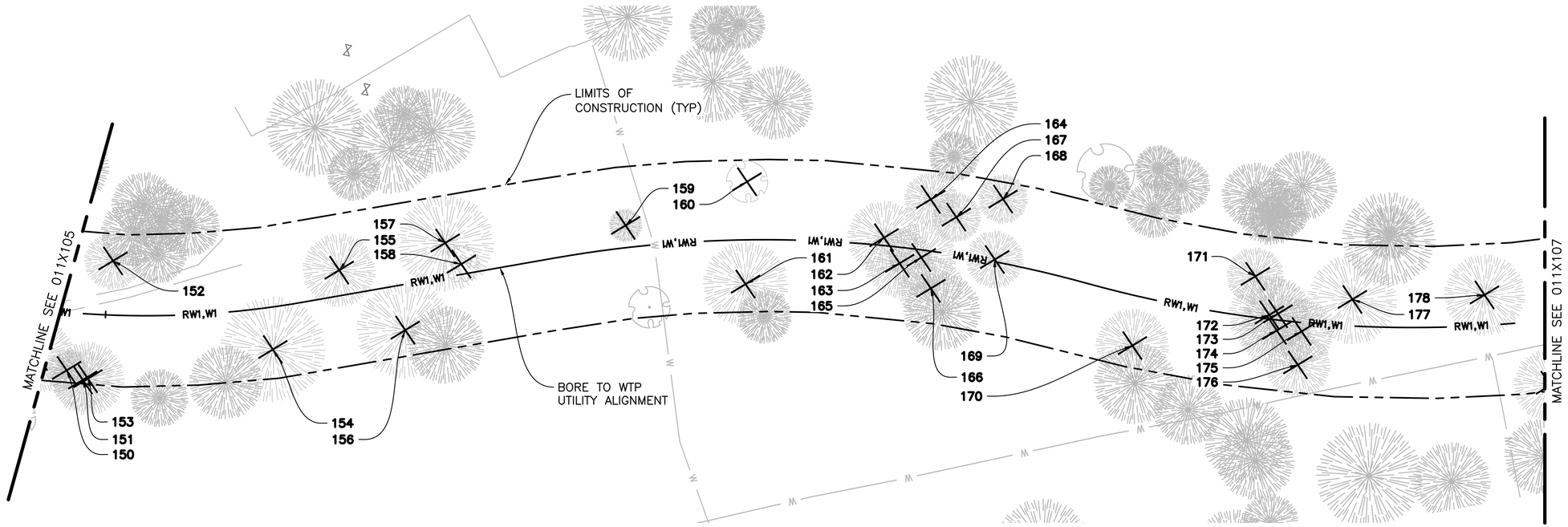
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
128	13147353.53	1329179.03	PINE (12")
129	13147353.85	1329146.98	PINE (14")
130	13147365.93	1329181.43	PINE (8")
131	13147367.88	1329171.39	PINE (12")
132	13147369.67	1329183.11	PINE (11")
133	13147378.90	1329211.11	PINE (12")
134	13147384.63	1329218.51	PINE (21")
135	13147385.77	1329204.22	PINE (12")
136	13147388.35	1329178.25	PINE (12")
137	13147388.70	1329192.26	PINE (16")
138	13147389.80	1329174.21	PINE (12")
139	13147435.74	1329192.84	PINE (18")
140	13147437.80	1329198.87	PINE (16")
141	13147440.15	1329200.26	PINE (11")
142	13147447.61	1329198.92	PINE (12")
143	13147458.72	1329247.45	DEC (10")
144	13147479.46	1329213.29	PINE (16")
145	13147488.31	1329212.86	PINE (14")
146	13147502.40	1329243.61	DEC (12")
147	13147525.06	1329220.59	PINE (13")
148	13147539.92	1329217.92	PINE (19")
149	13147560.95	1329246.01	PINE (18")

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 X105	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN BALLFIELD ROAD NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 9 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



GENERAL NOTES

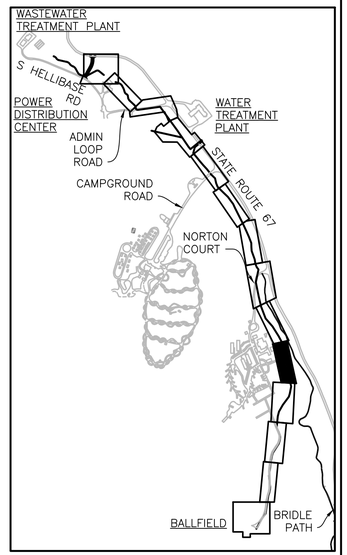
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.

C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
150	13147571.16	1329248.40	PINE (10")
151	13147577.03	1329250.34	PINE (15")
152	13147578.32	1329209.96	PINE (14")
153	13147578.58	1329249.57	PINE (18")
154	13147635.93	1329227.12	PINE (26")
155	13147651.64	1329196.94	PINE (18")
156	13147677.10	1329211.53	PINE (24")
157	13147683.77	1329180.69	PINE (24")
158	13147690.47	1329186.32	DEC (9")
159	13147740.48	1329162.30	PINE (8")
160	13147776.40	1329139.66	DEC (10")
161	13147783.16	1329172.67	PINE (21")
162	13147824.38	1329147.94	PINE (18")
163	13147831.19	1329155.08	PINE (17")
164	13147836.67	1329132.33	PINE (13")

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
165	13147837.74	1329151.68	PINE (17")
166	13147842.99	1329160.92	PINE (16")
167	13147846.13	1329136.25	PINE (12")
168	13147859.88	1329127.11	PINE (12")
169	13147861.49	1329147.12	PINE (14")
170	13147911.78	1329165.05	PINE (18")
171	13147946.38	1329134.11	PINE (14")
172	13147953.59	1329145.52	PINE (20")
173	13147956.32	1329144.50	PINE (17")
174	13147957.13	1329149.98	PINE (19")
175	13147965.15	1329148.52	PINE (16")
176	13147966.29	1329159.55	PINE (16")
177	13147979.21	1329134.65	PINE (22")
178	13148021.33	1329123.85	PINE (20")



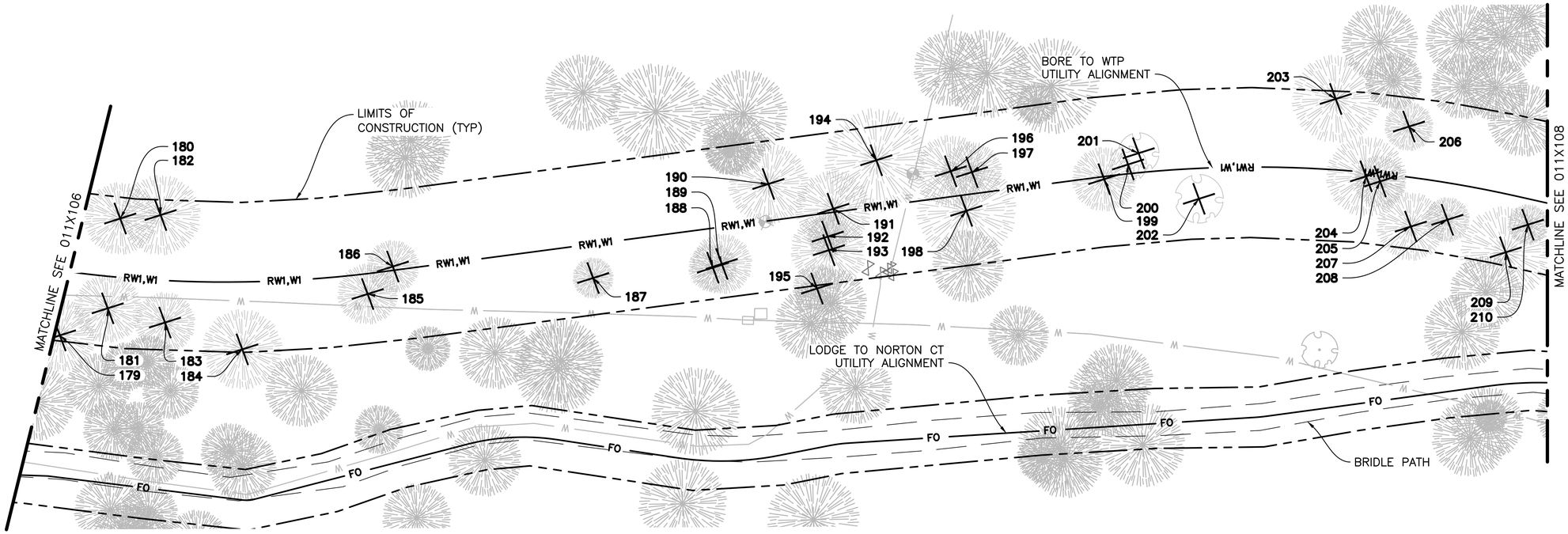
KEYMAP

20 0 20 40 1" = 20'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 X106	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN BALLFIELD ROAD NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 10 OF 87

4/17/2026 Race, Sara C:\Users\sracе\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



GENERAL NOTES

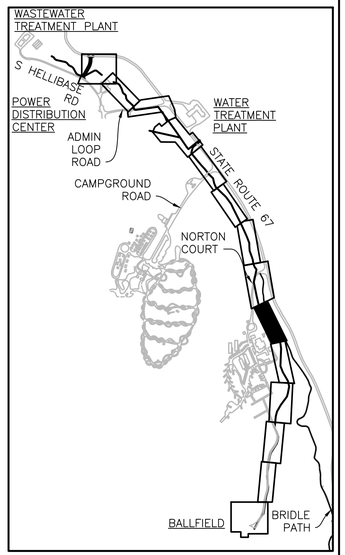
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.

C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
179	13148048.90	1329148.21	PINE (18")
180	13148049.84	1329104.34	PINE (18")
181	13148059.26	1329133.07	PINE (17")
182	13148061.43	1329097.37	PINE (20")
183	13148078.51	1329128.94	PINE (15")
184	13148105.62	1329125.63	PINE (20")
185	13148135.07	1329090.66	PINE (15")
186	13148138.58	1329078.72	PINE (13")
187	13148200.29	1329052.78	PINE (10")
188	13148234.67	1329032.10	PINE (12")
189	13148236.80	1329029.69	PINE (15")
190	13148239.03	1328998.88	PINE (20")
191	13148262.11	1328996.80	PINE (19")
192	13148264.54	1329005.95	PINE (10")
193	13148266.87	1329009.84	PINE (19")
194	13148267.88	1328975.94	PINE (26")

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
195	13148268.93	1329023.08	PINE (12")
196	13148291.79	1328968.34	PINE (18")
197	13148298.29	1328965.47	PINE (15")
198	13148302.29	1328977.88	PINE (25")
199	13148338.95	1328948.18	PINE (16")
200	13148343.96	1328940.28	DEC (9")
201	13148345.47	1328935.33	DEC (10")
202	13148370.24	1328940.16	DEC (12")
203	13148396.58	1328890.17	PINE (22")
204	13148417.04	1328908.90	PINE (20")
205	13148421.97	1328908.38	PINE (13")
206	13148422.57	1328887.71	PINE (12")
207	13148437.80	1328917.40	PINE (17")
208	13148447.76	1328910.00	PINE (11")
209	13148470.02	1328911.19	PINE (23")
210	13148472.63	1328899.64	PINE (10")



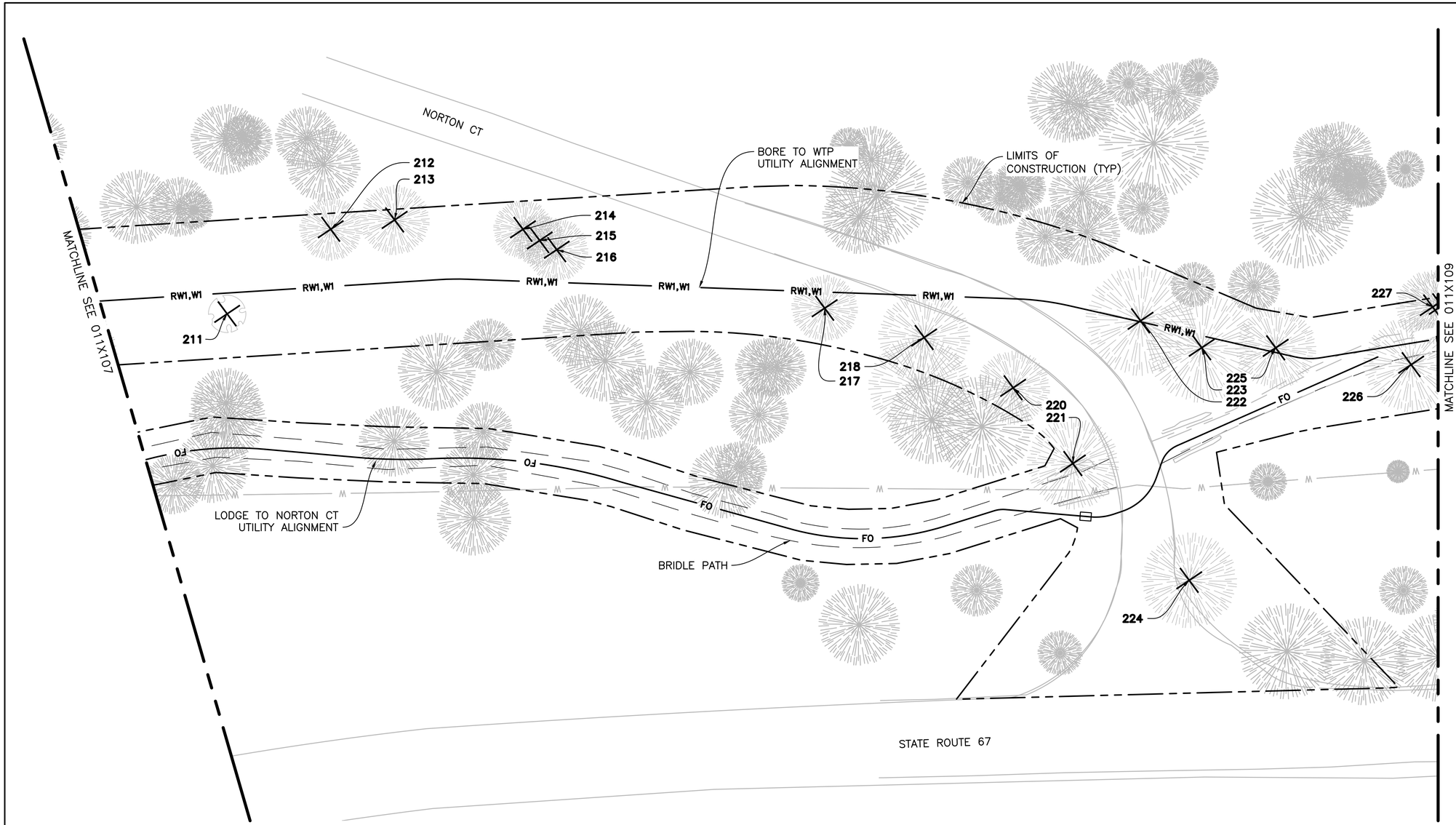
KEYMAP

20 0 20 40 1" = 20'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 X107	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN BALLFIELD ROAD NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 11 OF 87

4/17/2026 Race, Sara C:\Users\sracе\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



GENERAL NOTES

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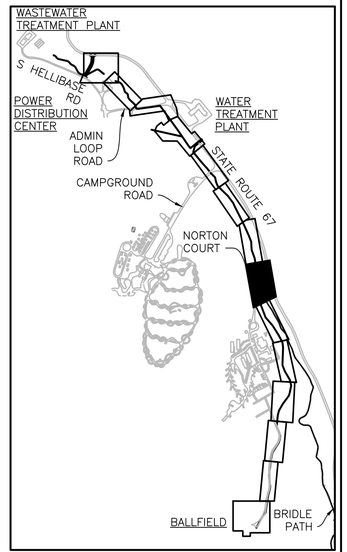


KEYMAP

20 0 20 40 1" = 20'
SCALE OF FEET

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
211	13148521.66	1328886.94	DEC (10")
212	13148553.42	1328850.24	PINE (17")
213	13148575.70	1328842.93	PINE (19")
214	13148622.17	1328838.29	PINE (16")
215	13148628.85	1328841.25	PINE (14")
216	13148635.52	1328843.48	PINE (17")
217	13148734.96	1328847.68	PINE (18")
218	13148772.36	1328851.96	PINE (25")

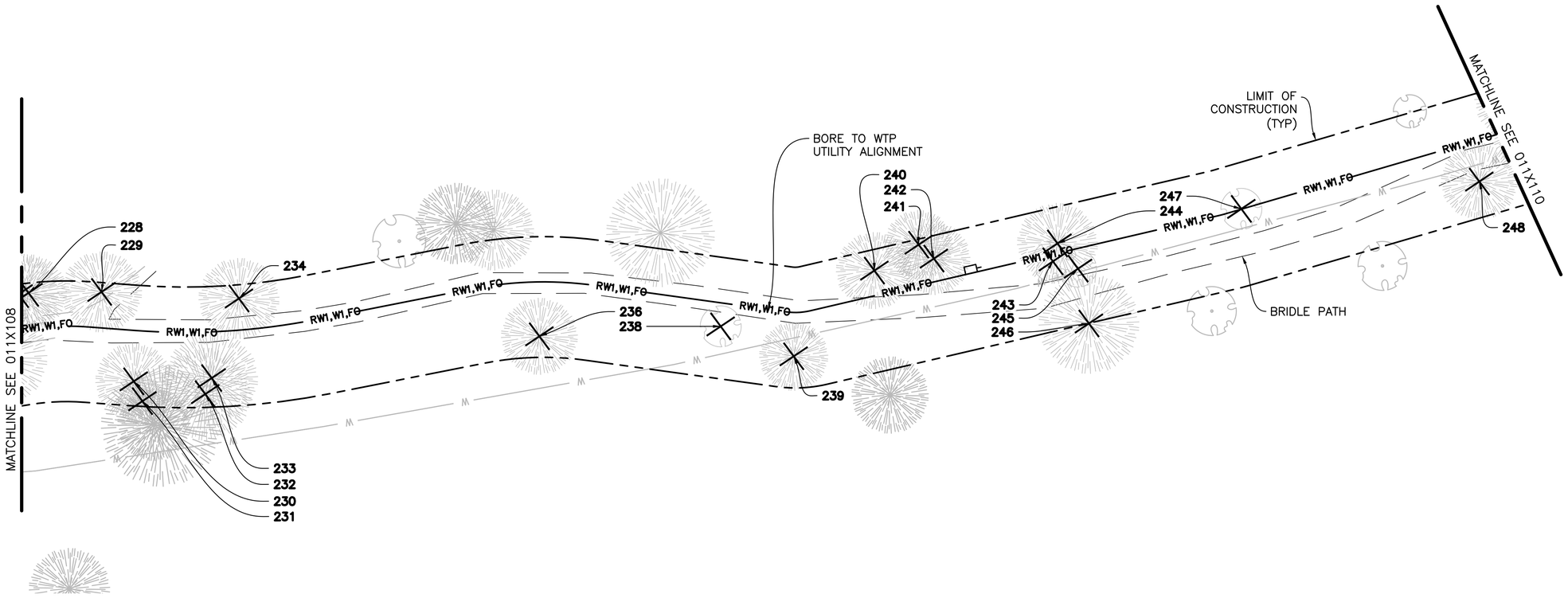
TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
220	13148807.31	1328864.40	PINE (20")
221	13148833.38	1328887.74	PINE (25")
222	13148848.70	1328832.50	PINE (36")
223	13148872.24	1328838.31	PINE (44IN")
224	13148882.27	1328922.23	PINE (26")
225	13148898.70	1328833.74	PINE (22")
226	13148948.20	1328831.32	PINE (26")
227	13148952.30	1328809.38	PINE (20")

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 X108	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN BALLFIELD RD AT NORTON CT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 12 OF 87

4/17/2026 Race, Sara C:\Users\sracE\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

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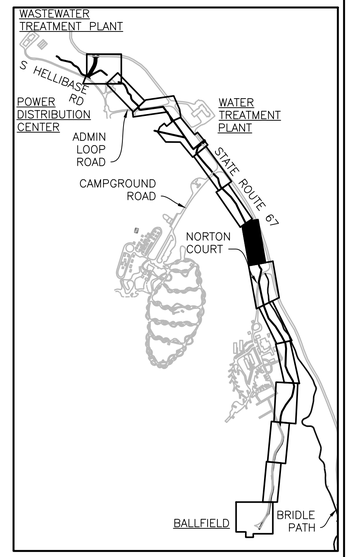
TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
228	13148955.46	1328808.37	PINE (19")
229	13148979.26	1328803.91	PINE (20")
230	13148994.87	1328831.27	PINE (19")
231	13148998.82	1328837.26	PINE (12")
232	13149018.95	1328831.38	PINE (23")
233	13149020.13	1328825.78	PINE (23")
234	13149024.50	1328798.23	PINE (20")
236	13149124.77	1328793.48	PINE (19")
238	13149183.41	1328779.93	DEC (10")
239	13149208.96	1328785.54	PINE (17")

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
240	13149230.43	1328752.95	PINE (19")
241	13149243.41	1328741.97	PINE (17")
242	13149249.18	1328745.64	PINE (18")
243	13149288.21	1328739.78	DEC (10")
244	13149288.68	1328733.67	PINE (20")
245	13149296.83	1328740.74	DEC (9")
246	13149303.50	1328758.01	PINE (25")
247	13149346.75	1328711.92	DEC (10")
248	13149422.91	1328689.45	PINE (19")



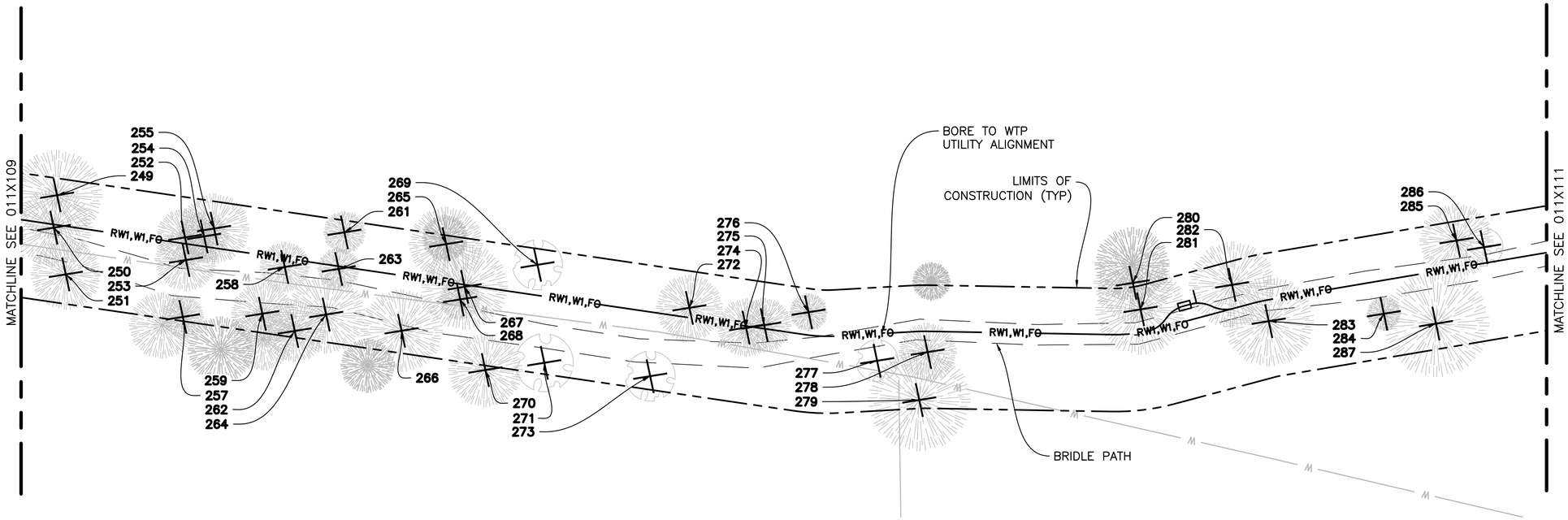
KEYMAP

20 0 20 40 1" = 20'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 X109	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN BRIDLE PATH TO WTP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 13 OF 87

4/17/2026 Race, Sara C:\Users\sracе\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



GENERAL NOTES

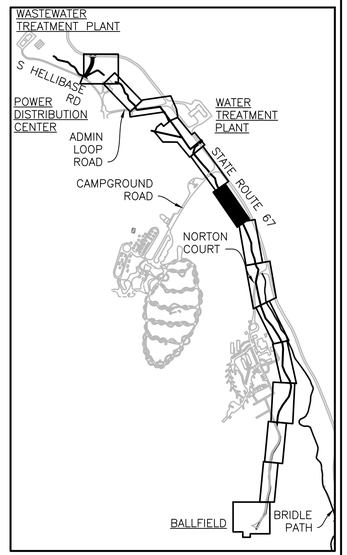
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.

C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
249	13149431.04	1328659.94	PINE (22")
250	13149436.14	1328669.21	PINE (19")
251	13149447.95	1328679.51	PINE (17")
252	13149473.05	1328647.66	PINE (14")
253	13149477.47	1328653.54	PINE (15")
254	13149477.97	1328643.89	PINE (12")
255	13149479.20	1328639.94	PINE (17")
257	13149487.12	1328669.34	PINE (10")
258	13149505.20	1328637.21	PINE (12")
259	13149507.80	1328653.84	PINE (15")
261	13149514.78	1328616.95	PINE (10")
262	13149519.65	1328652.48	PINE (15")
263	13149520.00	1328627.78	PINE (10")
264	13149525.03	1328642.31	PINE (19")
265	13149544.19	1328601.37	PINE (18")
266	13149548.20	1328632.77	PINE (20")
267	13149556.99	1328609.38	PINE (21")
268	13149558.12	1328613.58	PINE (10")
269	13149572.45	1328590.09	DEC (12")

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
270	13149577.88	1328627.83	PINE (18")
271	13149592.31	1328615.05	DEC (14")
272	13149621.04	1328573.62	PINE (16")
273	13149623.12	1328599.14	DEC (12")
274	13149639.84	1328568.48	PINE (14")
275	13149644.24	1328564.46	PINE (13")
276	13149653.73	1328552.85	PINE (9")
277	13149681.02	1328553.22	DEC (10")
278	13149692.95	1328541.59	PINE (14")
279	13149699.53	1328555.96	PINE (25")
280	13149735.14	1328485.28	PINE (19")
281	13149742.08	1328490.95	PINE (22")
282	13149762.07	1328467.52	PINE (21")
283	13149778.56	1328470.50	PINE (22")
284	13149808.01	1328447.31	PINE (8")
285	13149813.99	1328414.35	PINE (18")
286	13149822.58	1328411.04	DEC (9")
287	13149823.78	1328440.36	PINE (26")



KEYMAP

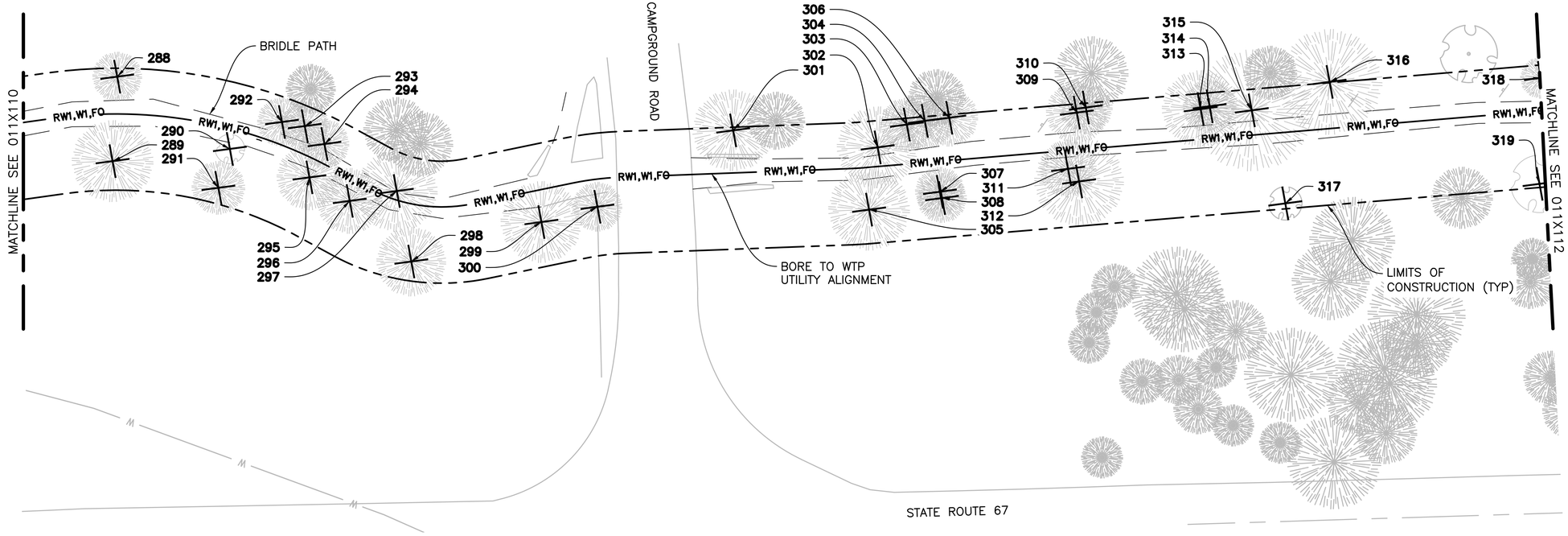
20 0 20 40 1" = 20'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 X110	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN BRIDLE PATH TO WTP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 14 OF 87

4/17/2026 Race, Sara C:\Users\sgrace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



GENERAL NOTES

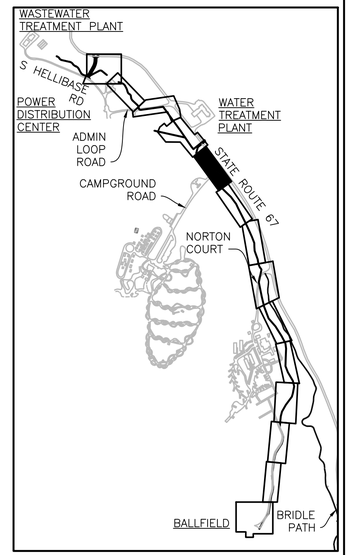
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.

C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
288	13149855.14	1328371.85	PINE (12")
289	13149870.25	1328395.11	PINE (20")
290	13149898.98	1328369.17	DEC (7")
291	13149903.46	1328381.68	PINE (13")
292	13149907.59	1328351.95	PINE (12")
293	13149914.42	1328348.61	PINE (12")
294	13149923.11	1328349.55	PINE (12")
295	13149925.58	1328361.29	PINE (16")
296	13149940.81	1328359.73	PINE (16")
297	13149951.37	1328347.99	PINE (20")
298	13149968.86	1328363.85	PINE (17")
299	13149995.71	1328328.20	PINE (20")
300	13150007.61	1328313.31	PINE (12")
301	13150028.81	1328266.80	PINE (20")
302	13150070.41	1328243.32	PINE (20")
303	13150073.73	1328231.63	PINE (13")

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
304	13150077.78	1328227.16	PINE (14")
305	13150080.18	1328261.47	PINE (20")
306	13150083.67	1328221.60	PINE (14")
307	13150095.49	1328243.02	PINE (12")
308	13150097.27	1328244.28	PINE (12")
309	13150116.48	1328195.54	PINE (15")
310	13150117.89	1328192.46	PINE (13")
311	13150125.03	1328212.14	PINE (14")
312	13150130.40	1328213.34	PINE (22")
313	13150148.64	1328170.64	PINE (19")
314	13150150.46	1328168.15	PINE (13")
315	13150162.42	1328161.07	PINE (30")
316	13150177.56	1328138.46	PINE (26")
317	13150189.31	1328179.22	DEC (8")
318	13150232.80	1328096.46	PINE (10")
319	13150252.92	1328124.82	DEC (14")



KEYMAP

20 0 20 40 1" = 20'
SCALE OF FEET

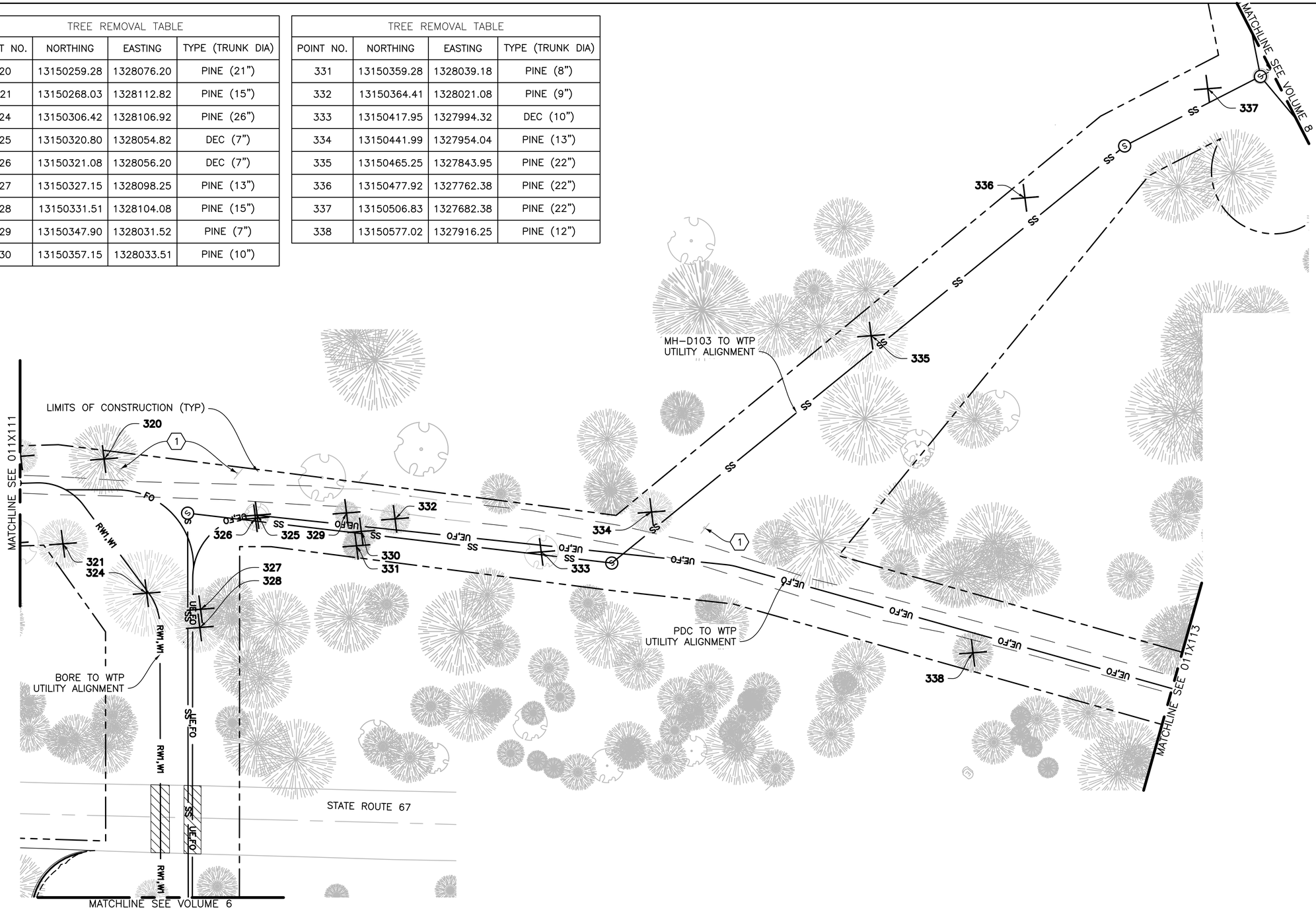
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 X111	TITLE OF SHEET DEMOLITION TREE REMOVAL AND PROTECTION PLAN BRIDLE PATH TO WTP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 15 OF 87

4/17/2026 Race, Sara C:\Users\sgrace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
320	13150259.28	1328076.20	PINE (21")
321	13150268.03	1328112.82	PINE (15")
324	13150306.42	1328106.92	PINE (26")
325	13150320.80	1328054.82	DEC (7")
326	13150321.08	1328056.20	DEC (7")
327	13150327.15	1328098.25	PINE (13")
328	13150331.51	1328104.08	PINE (15")
329	13150347.90	1328031.52	PINE (7")
330	13150357.15	1328033.51	PINE (10")

TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
331	13150359.28	1328039.18	PINE (8")
332	13150364.41	1328021.08	PINE (9")
333	13150417.95	1327994.32	DEC (10")
334	13150441.99	1327954.04	PINE (13")
335	13150465.25	1327843.95	PINE (22")
336	13150477.92	1327762.38	PINE (22")
337	13150506.83	1327682.38	PINE (22")
338	13150577.02	1327916.25	PINE (12")



GENERAL NOTES

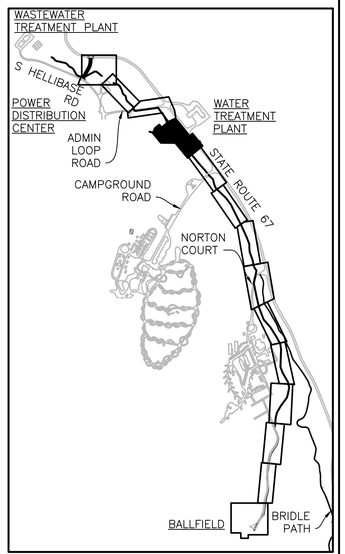
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. CONTRACTOR SHALL COORDINATE AND RECEIVE WRITTEN APPROVAL BY THE CONTRACTING OFFICER PRIOR TO REMOVAL OF ANY TREES NOT IDENTIFIED FOR REMOVAL IN THESE PLANS.

C. CONTRACTOR SHALL PROTECT TREES IN PLACE IF NOT NEEDED FOR REMOVAL.

KEY NOTES: (X)

1. EXISTING SNOW MARKERS NEED TO BE PROTECTED OR IF REMOVED, PUT BACK IN EXACT SAME LOCATION.



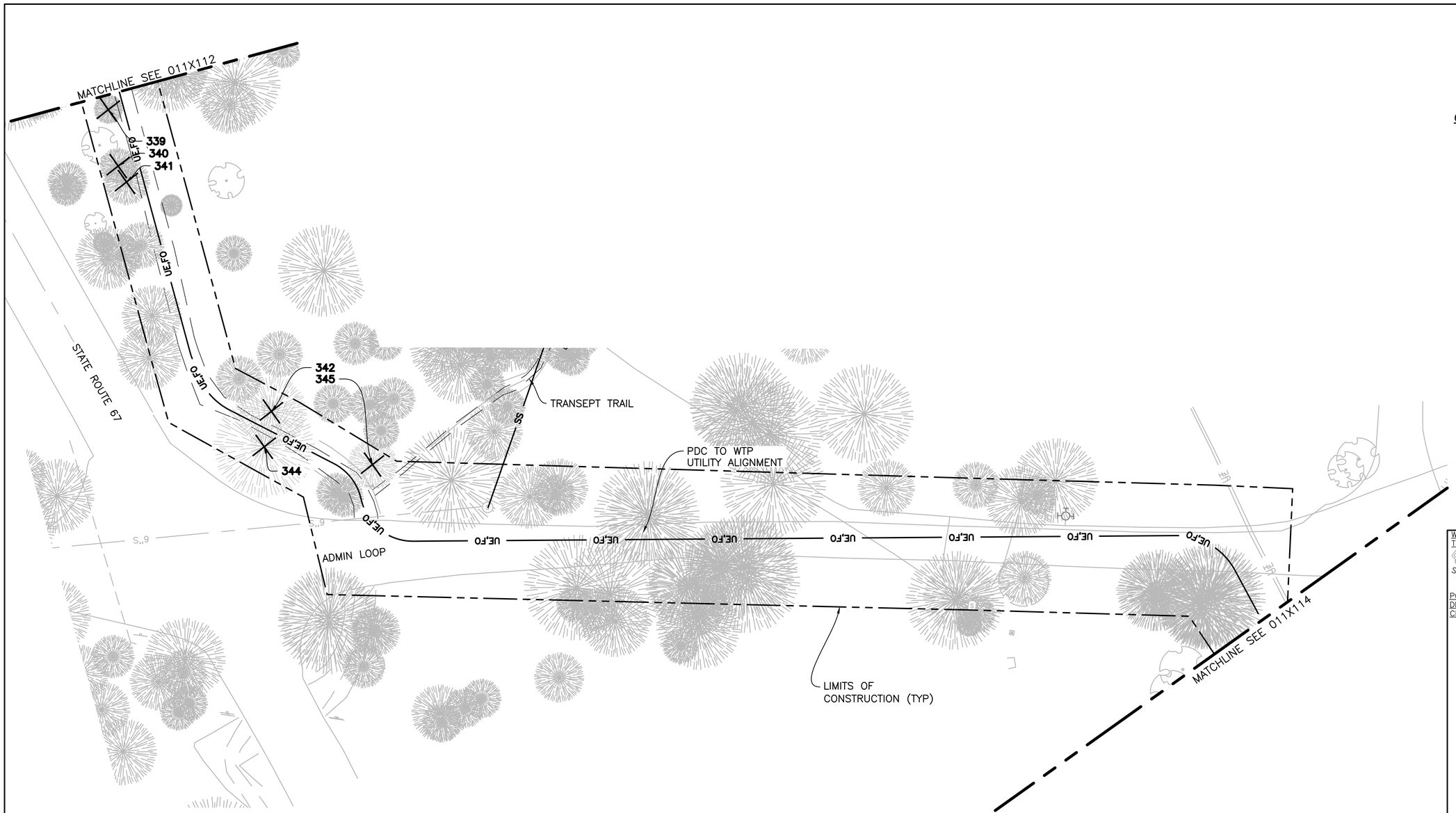
KEYMAP

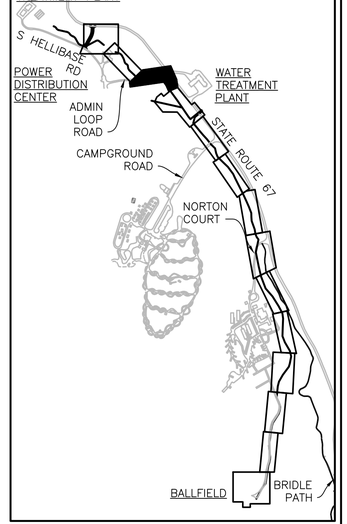
20 0 20 40
SCALE OF FEET 1" = 20'

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	011 X112	**DEMOLITION** TREE REMOVAL AND PROTECTION PLAN WTP TO PDC NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 16 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg





KEYMAP

20 0 20 40 1" = 20'
SCALE OF FEET

GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

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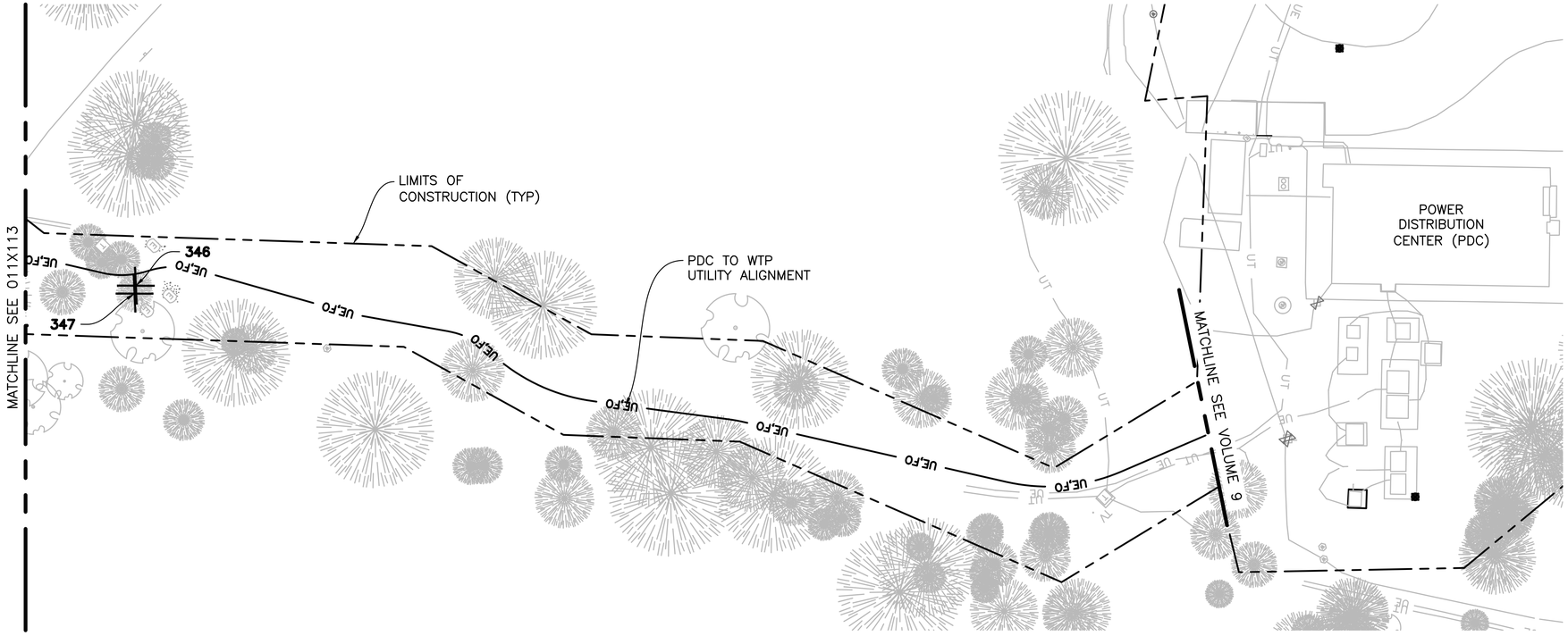
TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
339	13150655.02	1327880.05	PINE (8")
340	13150675.64	1327873.20	PINE (10")
341	13150681.17	1327869.19	PINE (13")
342	13150759.21	1327802.01	PINE (X")
344	13150772.92	1327802.74	PINE (29")
345	13150773.56	1327761.09	PINE (14")

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Registered ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL, GADD, J. GREEN	011 X113	DEMOLITION TREE REMOVAL AND PROTECTION PLAN WTP TO PDC NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
	TECH. REVIEW: GOUGH/LUSTER			PMIS/PKG NO. 318719
	DATE: MAY 2026			SHEET 17 OF 87

4/17/2026 Race, Sara C:\Users\grace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
346	13150811.46	1327401.49	PINE (7")
347	13150812.82	1327403.44	PINE (8")

GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

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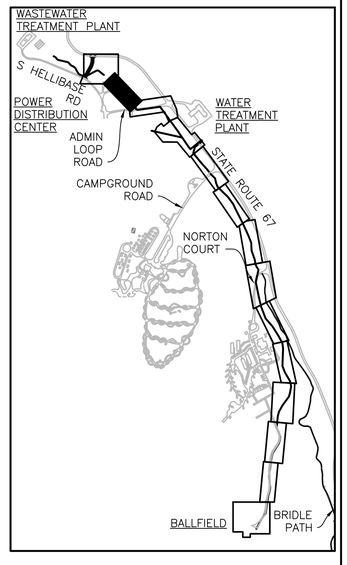


KEYMAP

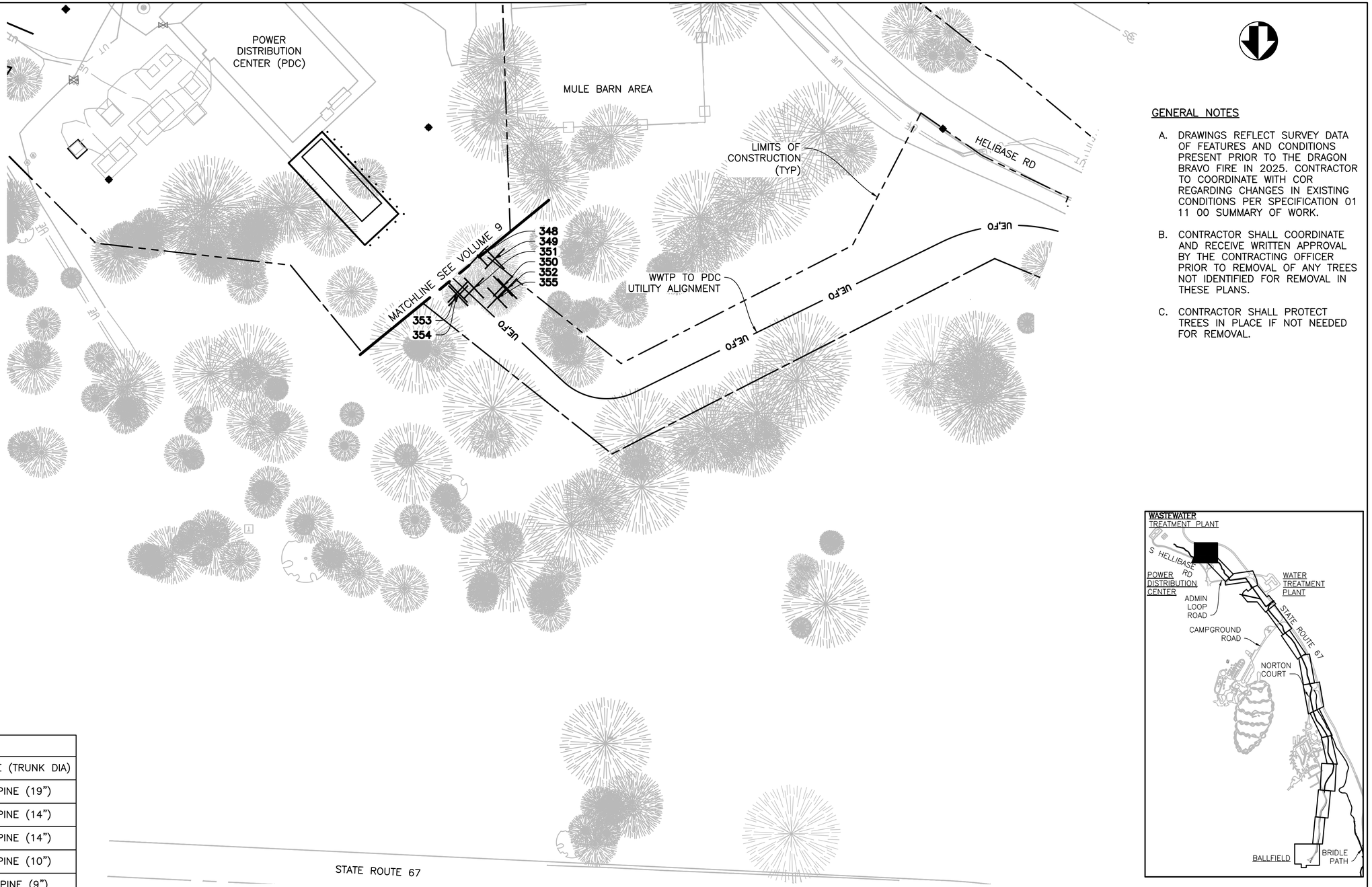
20 0 20 40
SCALE OF FEET
1" = 20'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 X114	TITLE OF SHEET **DEMOLITION** TREE REMOVAL AND PROTECTION PLAN WTP TO PDC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 18 OF 87

4/17/2026 Race, Sara C:\Users\grace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



GENERAL NOTES

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

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TREE REMOVAL TABLE			
POINT NO.	NORTHING	EASTING	TYPE (TRUNK DIA)
348	13151156.71	1327039.40	PINE (19")
349	13151157.82	1327032.88	PINE (14")
350	13151168.59	1327029.64	PINE (14")
351	13151169.33	1327041.04	PINE (10")
352	13151169.59	1327027.98	PINE (9")
353	13151170.97	1327047.44	DEC (9")
354	13151172.37	1327047.23	PINE (8")
355	13151173.16	1327031.94	PINE (11")

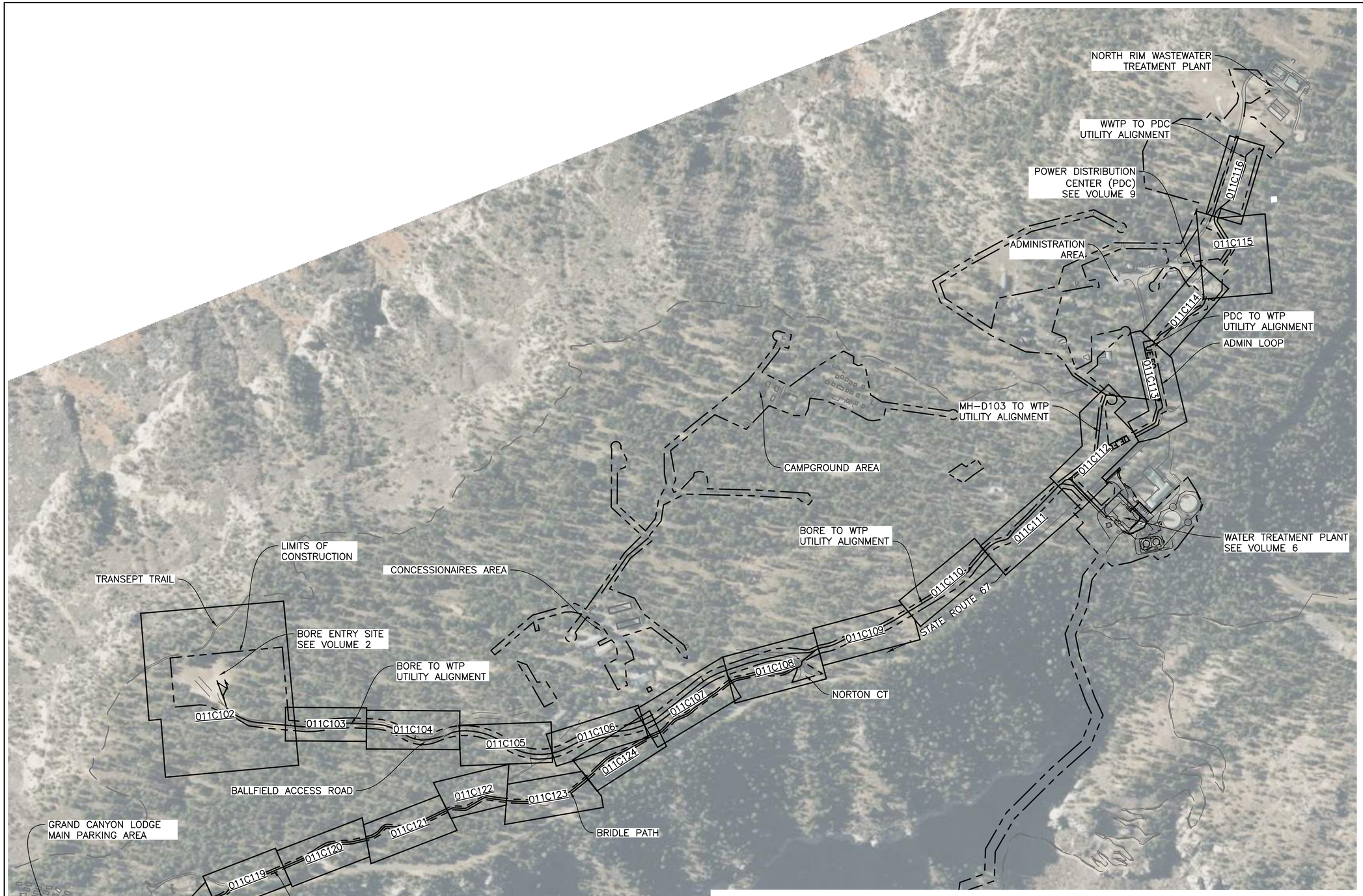
KEYMAP

20 0 20 40 1" = 20'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	011 X115	**DEMOLITION** TREE REMOVAL AND PROTECTION PLAN PDC TO WWTP NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 19 OF 87

8/18/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-X102.dwg



GENERAL NOTES:

A. FOR EROSION CONTROL NOTES AND DETAILS SEE VOLUME 0 SHEETS 002C507 – 002C511

B. SILT FENCE
QUANTITY = 13788 LF (VOLUME 1 TOTAL)

C. CONSTRUCTION FENCE
QUANTITY = 22909 LF (VOLUME 1 TOTAL)

LEGEND:

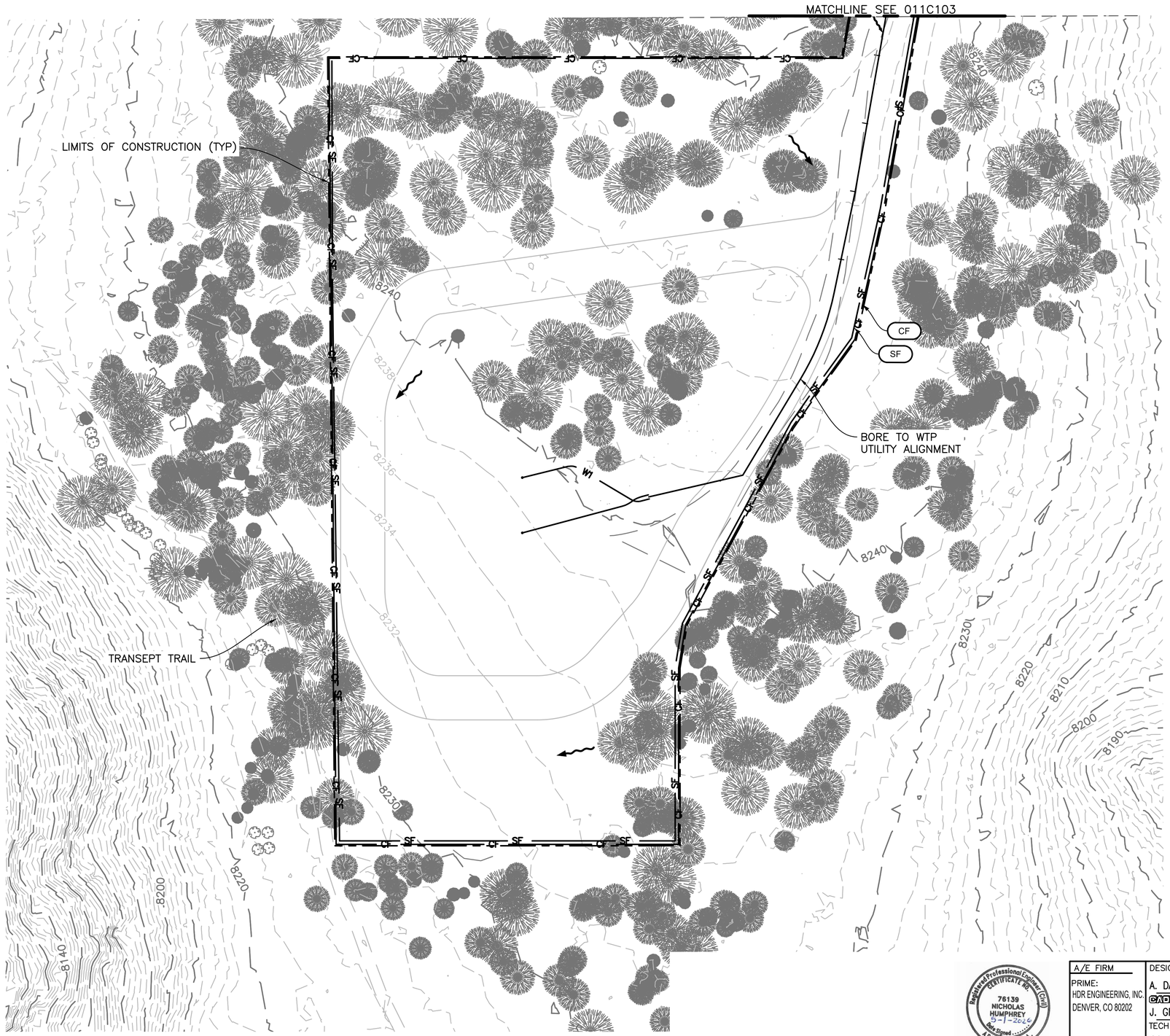
- SF — SILT FENCE, SEE DETAIL 01/002C510
- CF — CONSTRUCTION FENCING, SEE DETAIL 01/002C509
- VTC — VEHICLE TRACKING CONTROL, SEE DETAIL 01/002C511
- SCL — SEDIMENT CONTROL LOG, SEE DETAIL 03/002C510
- FLOW ARROW

250 0 250 500 1" = 250'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 9-1-2026 Date Signed ARIZONA, U.S.A.

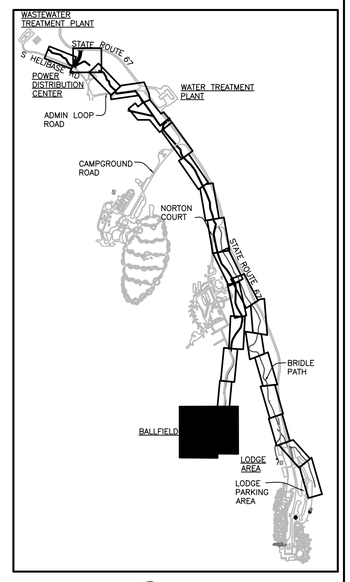
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	SUB SHEET NO. 011 C101	TITLE OF SHEET SITEWORK GESC SHEET MAP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 20 OF 87

8/18/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C101.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

40 0 40 80

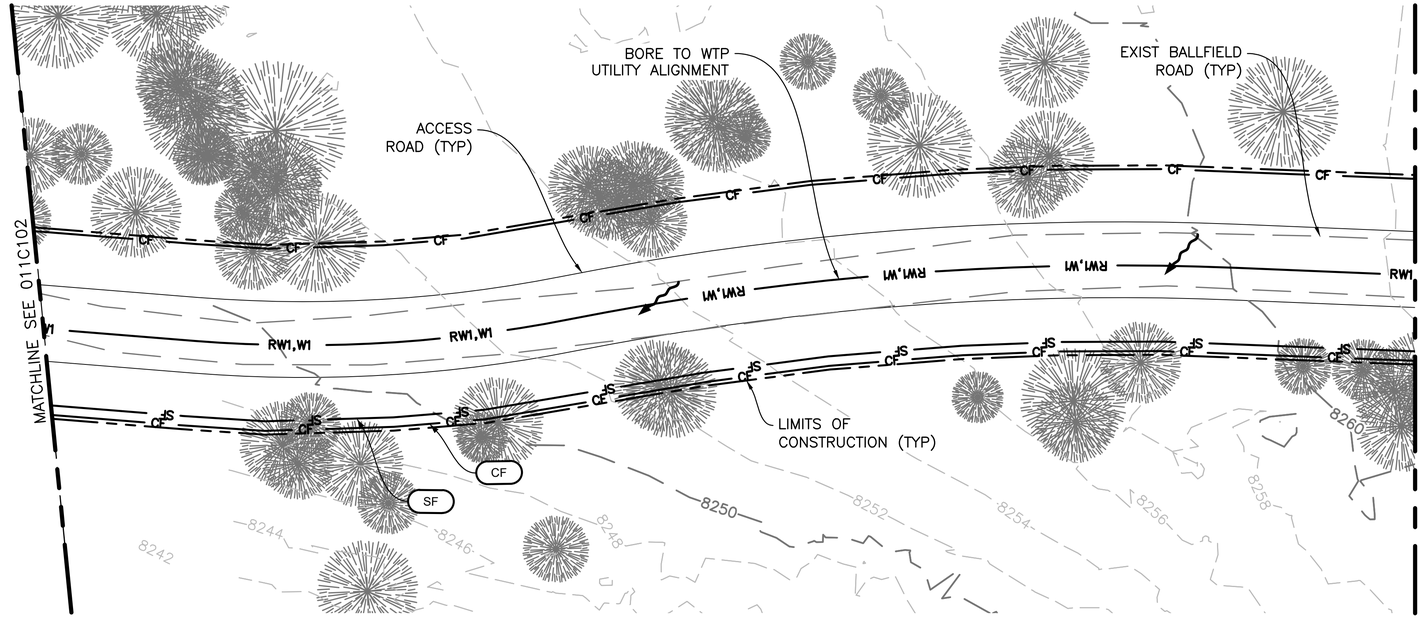
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

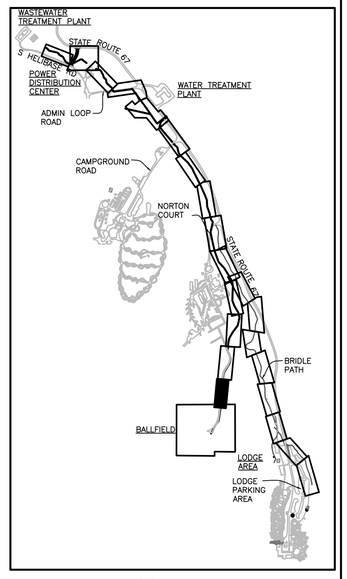
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C102	TITLE OF SHEET SITEWORK GESC PLAN BALLFIELD SITE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 21 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5/1/2026 Date Signed ARIZONA, U.S.A.

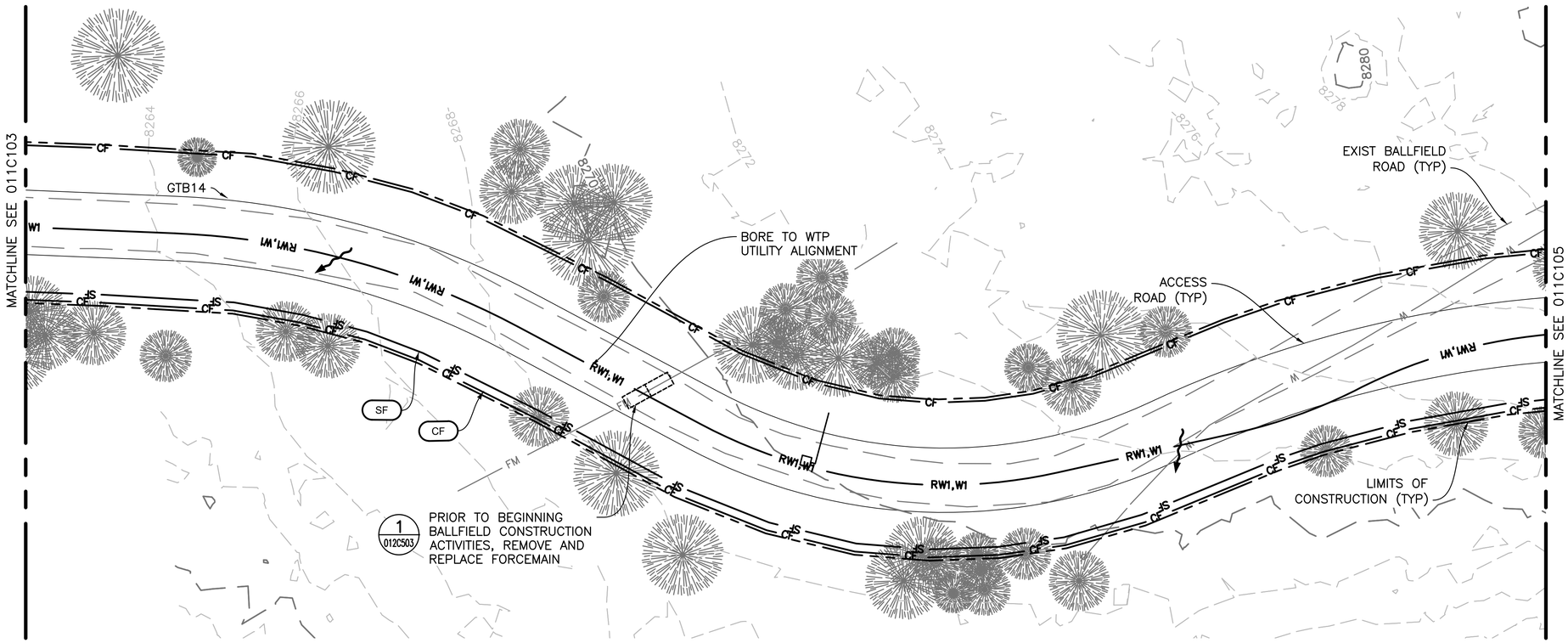
Nicholas Humphrey

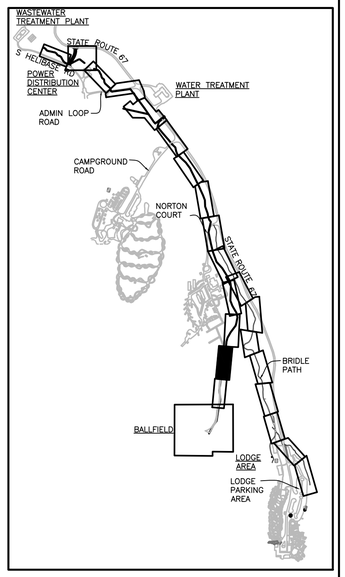
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C103	TITLE OF SHEET SITEWORK GESC PLAN BALLFIELD ROAD NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 22 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg

GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.





KEYMAP

20 0 20 40 1" = 20'

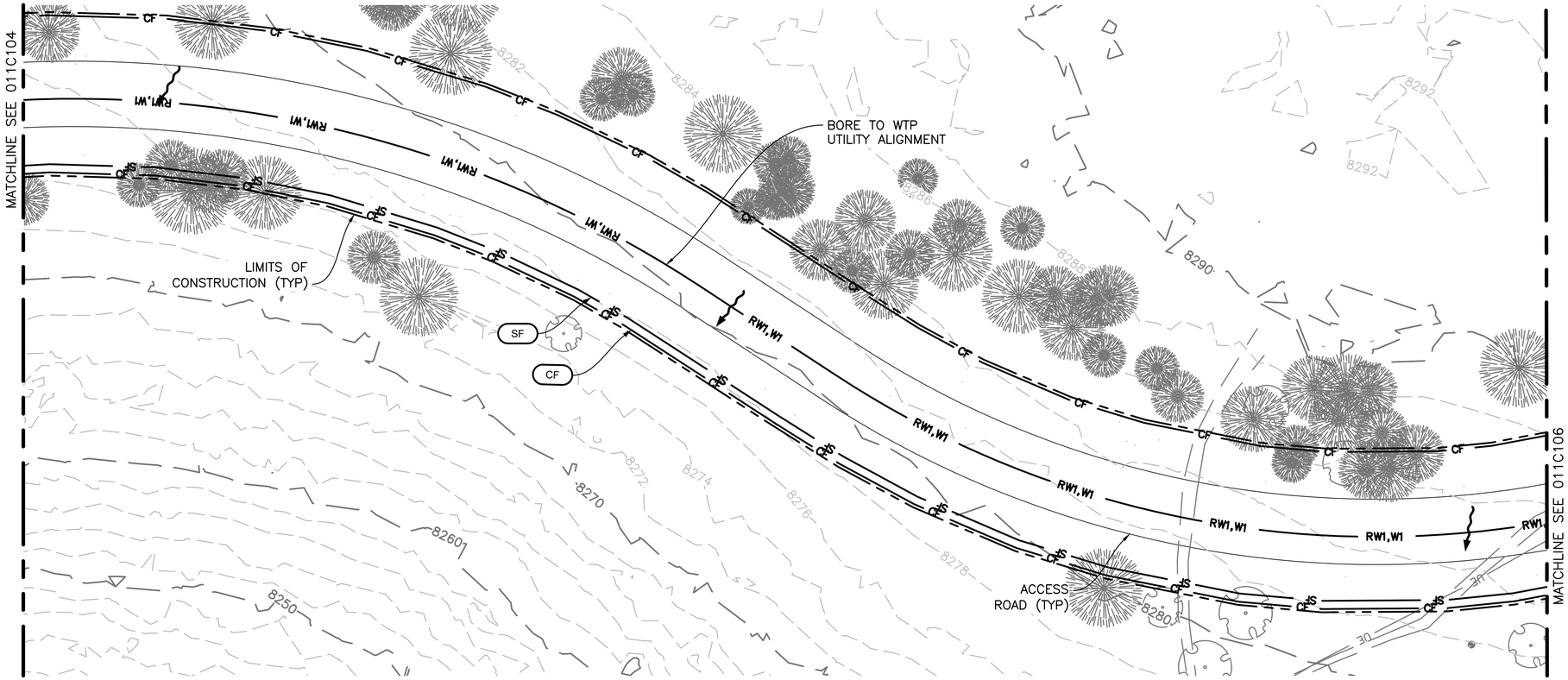
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY Date Signed 5-1-2026 ARIZONA, U.S.A.

Nicholas Humphrey

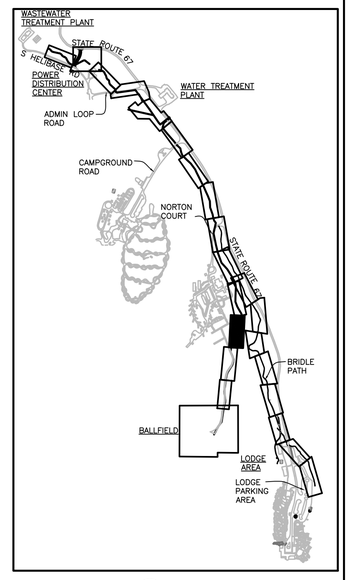
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C104	TITLE OF SHEET SITEWORK GESC PLAN BALLFIELD ROAD NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 23 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

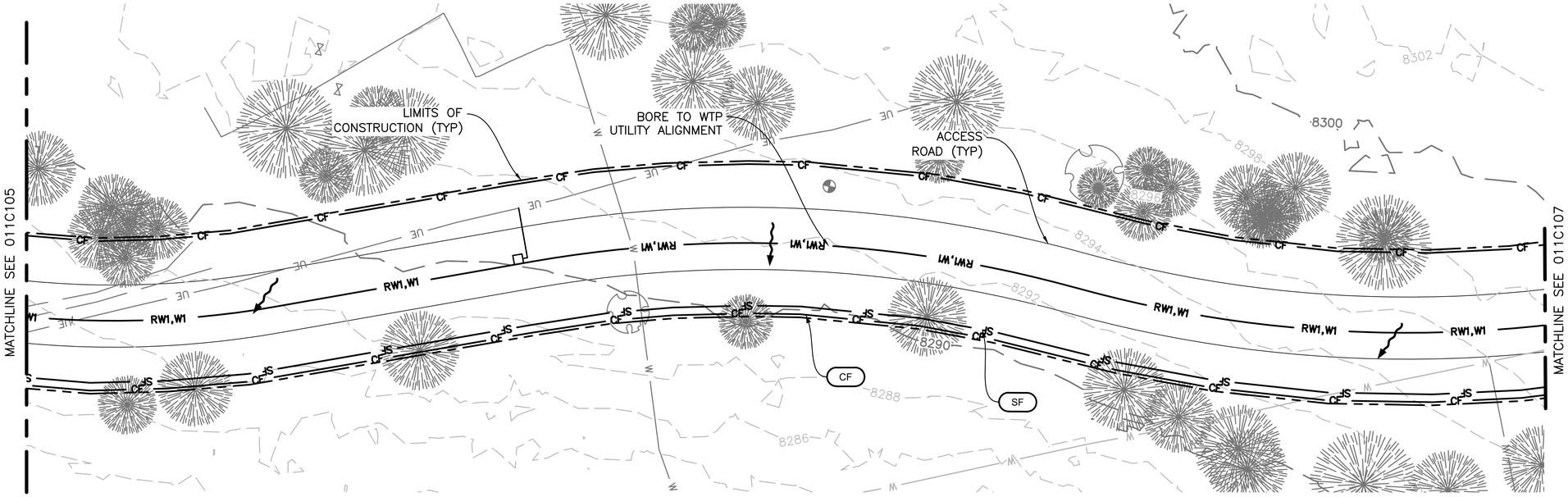
20 0 20 40 1" = 20'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

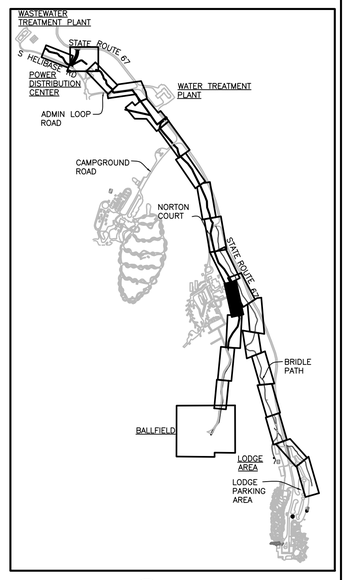
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C105	TITLE OF SHEET SITEWORK GESC PLAN BALLFIELD ROAD NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 24 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

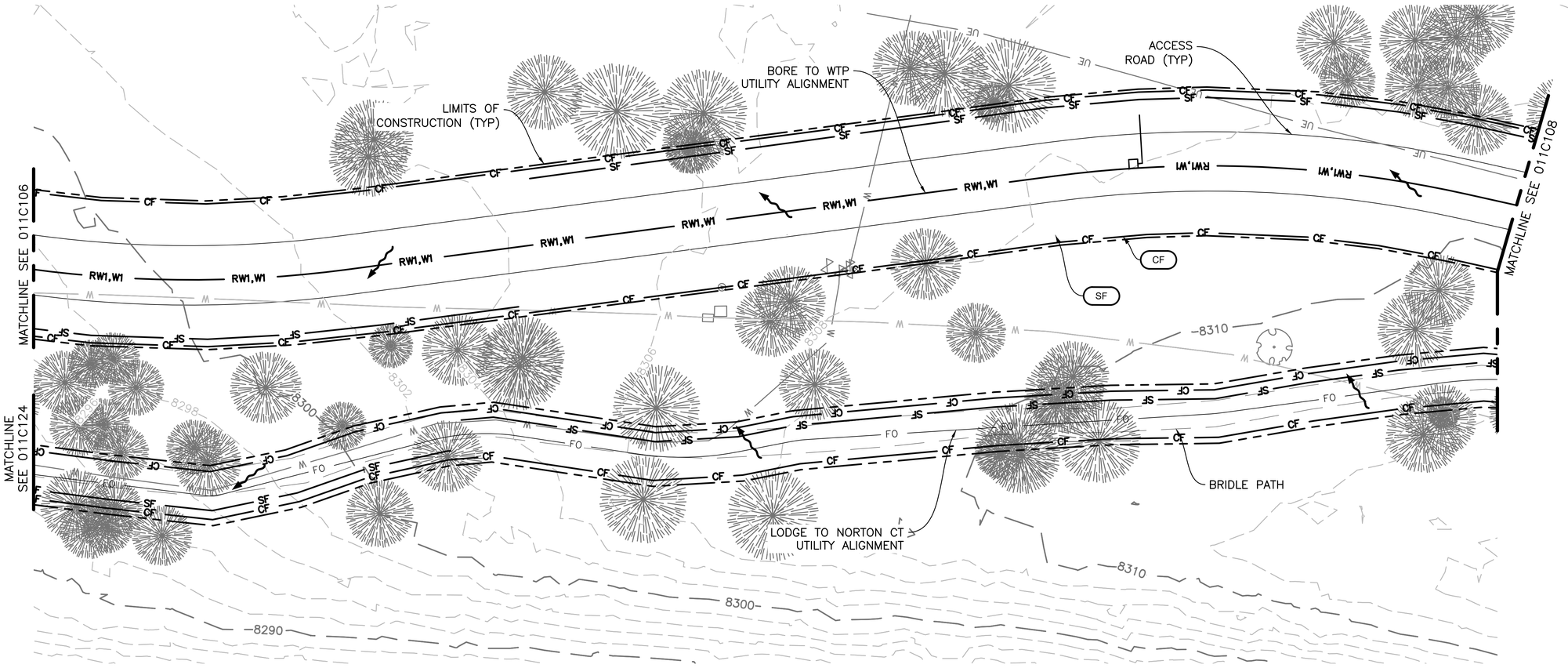
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

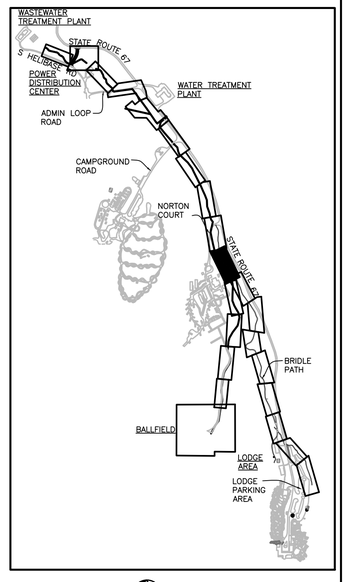
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C106	TITLE OF SHEET SITEWORK GESC PLAN BALLFIELD ROAD NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 25 OF 87

4/17/2026 Race, Sara C:\Users\sracе\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



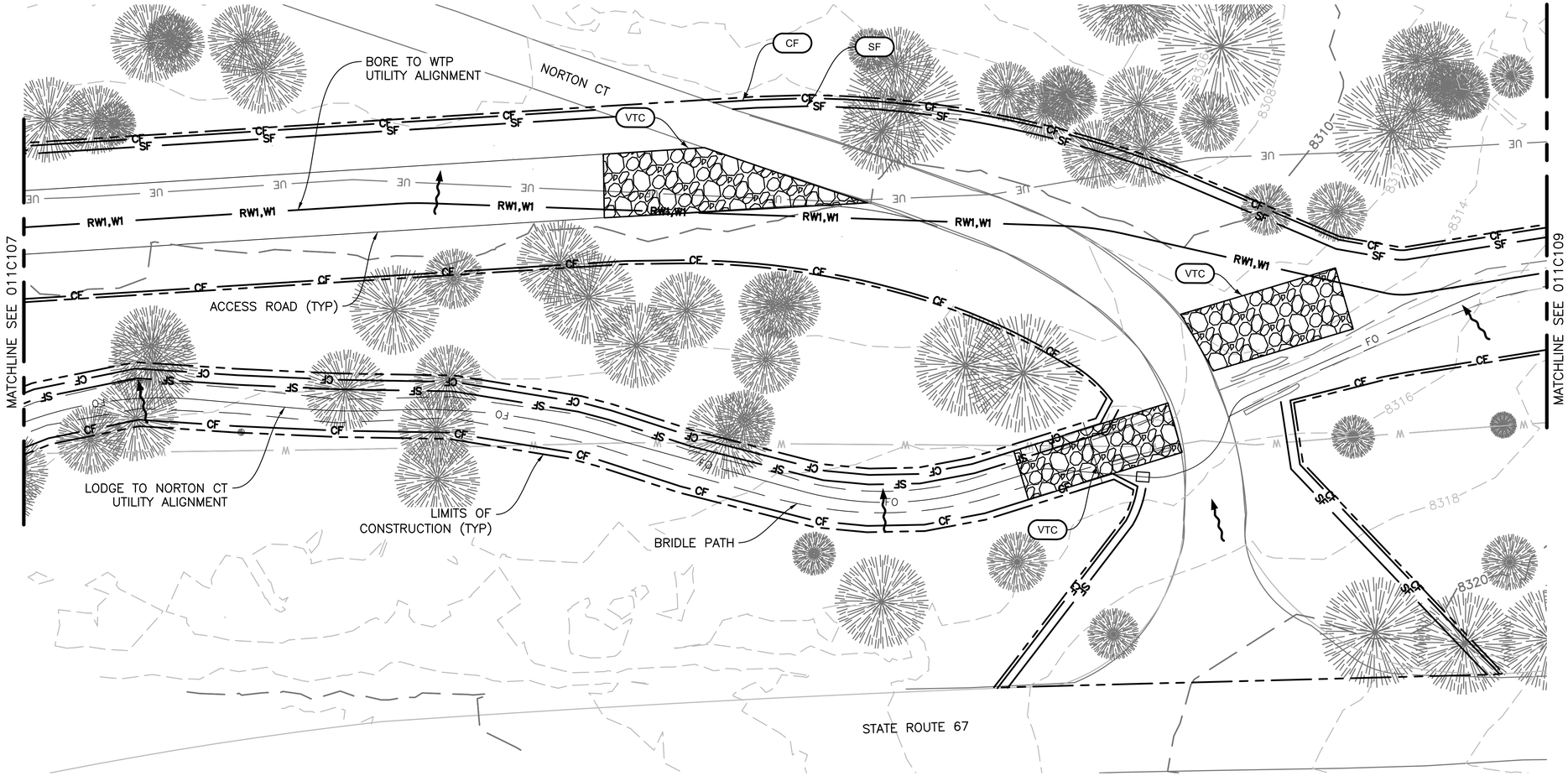
KEYMAP

20 0 20 40 1" = 20'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

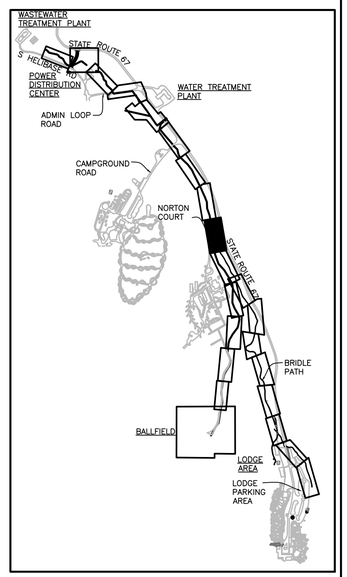
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	011 C107	SITEWORK GESC PLAN BALLFIELD ROAD NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 26 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

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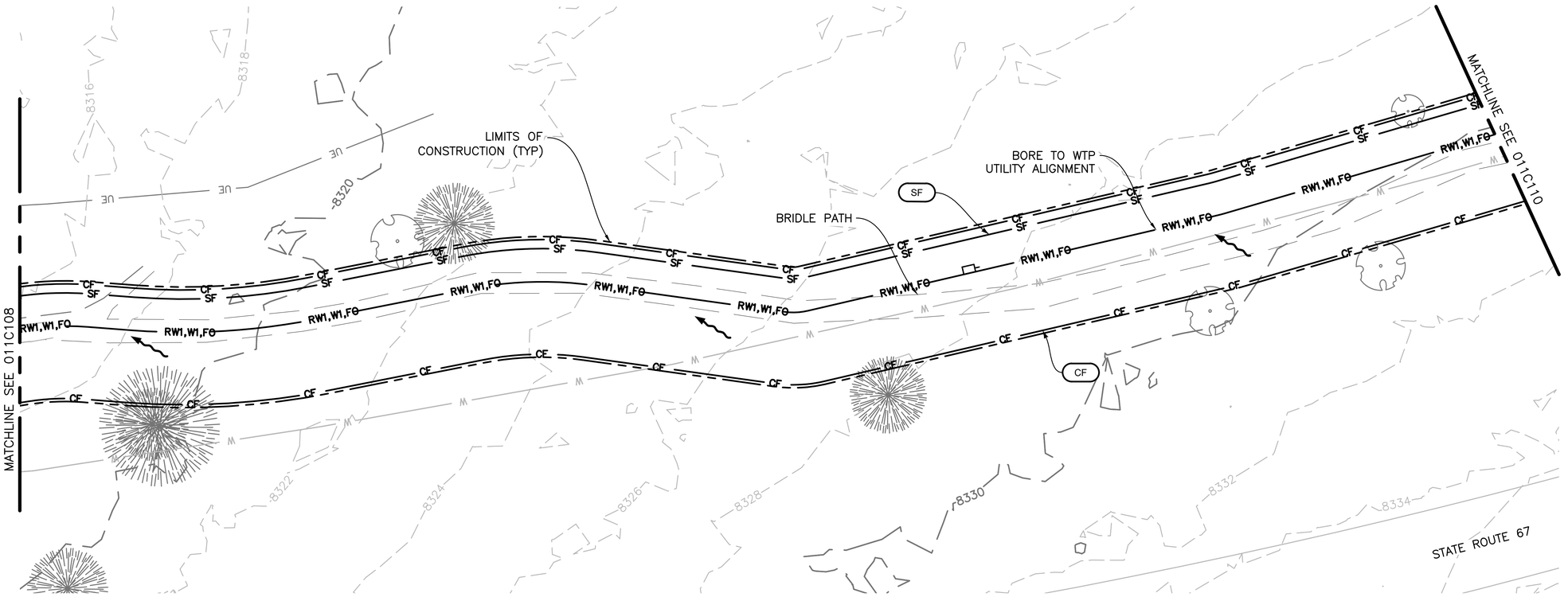
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

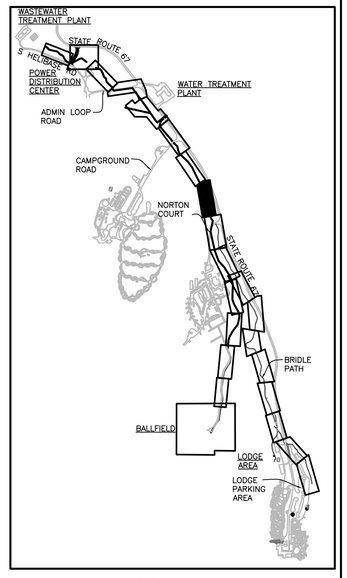
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL	011 C108	SITEWORK GESC PLAN BALLFIELD RD AT NORTON CT NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
	CADD J. GREEN			PMIS/PKG NO. 318719
	TECH. REVIEW: GOUGH/LUSTER			SHEET 27 OF 87
	DATE: MAY 2026			

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

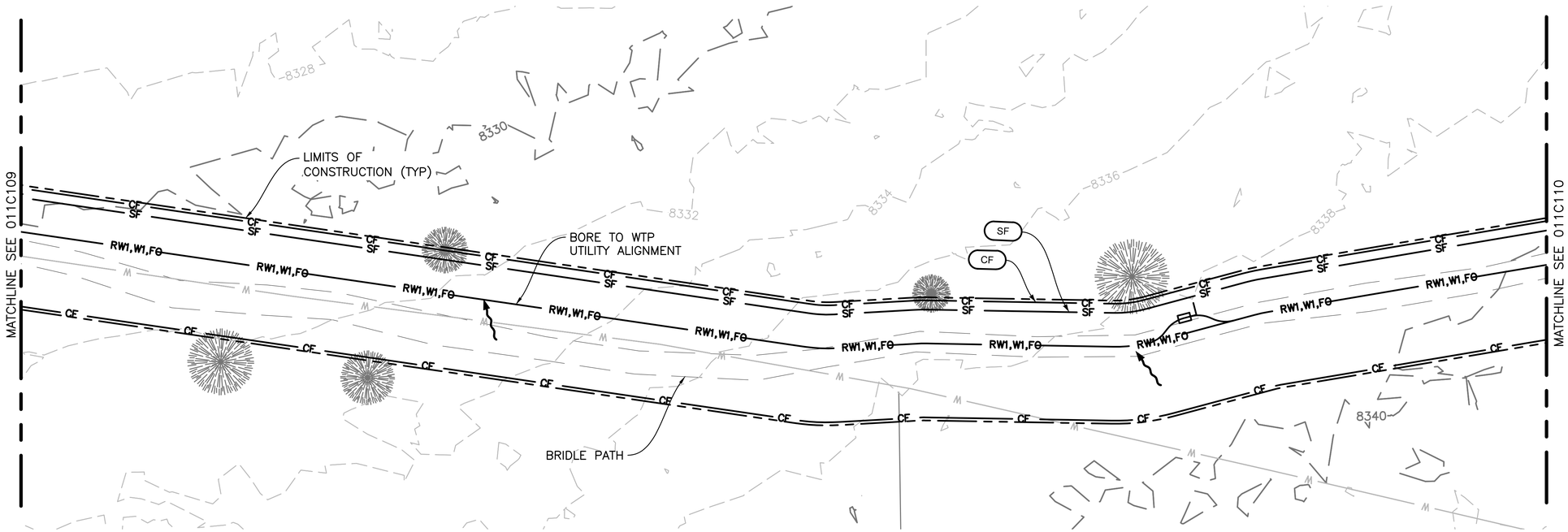
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Signed ARIZONA, U.S.A.

Nicholas Humphrey

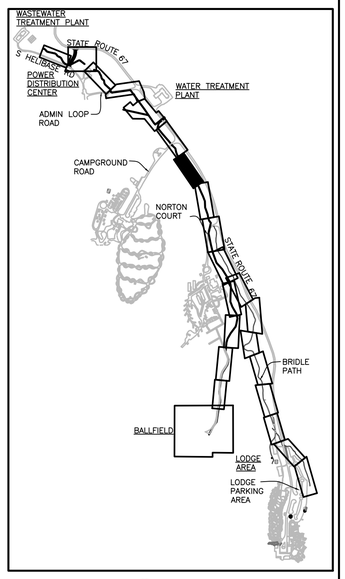
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	011 C109	SITEWORK GESC PLAN BRIDLE PATH TO WTP NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 28 OF 87

4/17/2026 Race, Sara C:\Users\sgrace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

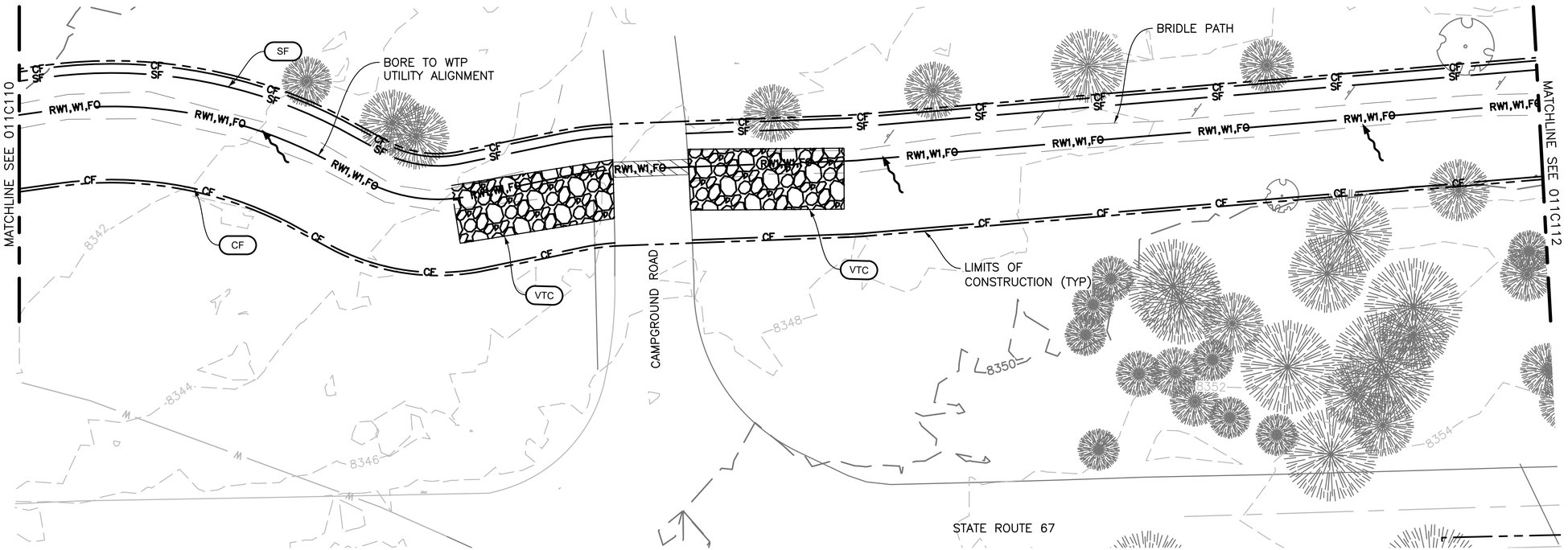
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

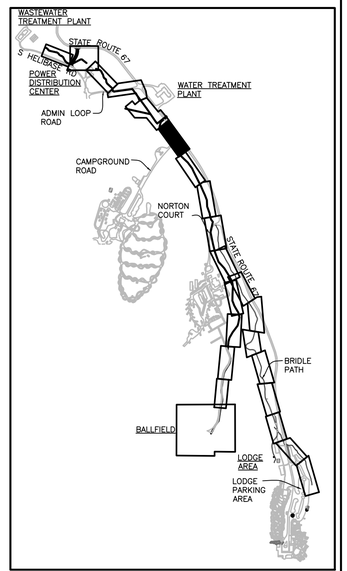
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C110	TITLE OF SHEET SITEWORK GESC PLAN BRIDLE PATH TO WTP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 29 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

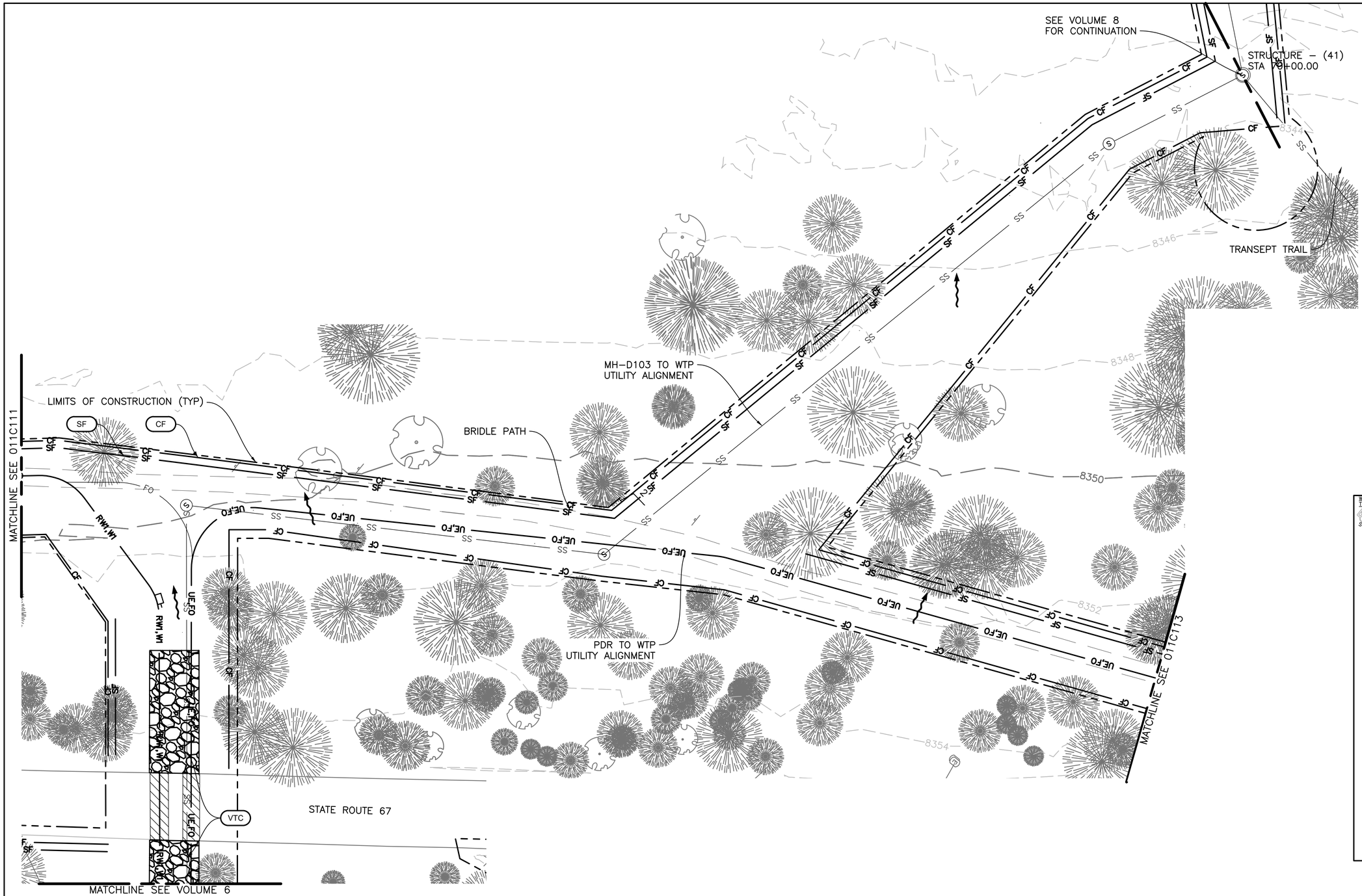
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

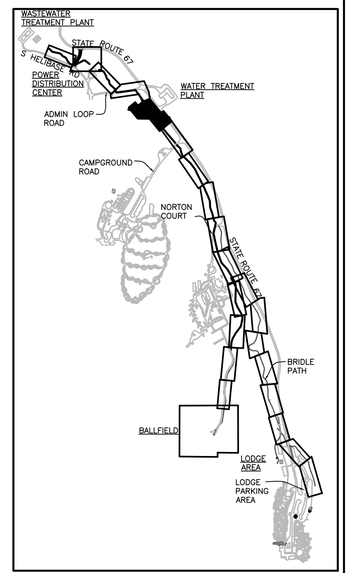
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	011 C111	SITEWORK GESC PLAN BRIDLE PATH TO WTP NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 30 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

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SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	011 C112	SITEWORK GESC PLAN WTP TO PDC AT SR67 NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 31 OF 87

4/17/2026 Race, Sara C:\Users\sracе\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg

GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.

MATCHLINE SEE 011C112

BRIDLE PATH

TRANSEPT TRAIL

PDC TO WTP UTILITY ALIGNMENT

ADMIN LOOP ROAD

LIMITS OF CONSTRUCTION (TYP)

STATE ROUTE 67

MATCHLINE SEE 011C114

CF

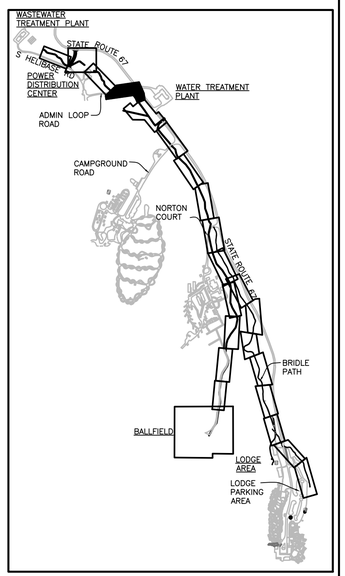
SF

UE,FO

8348

8350

8352



KEYMAP

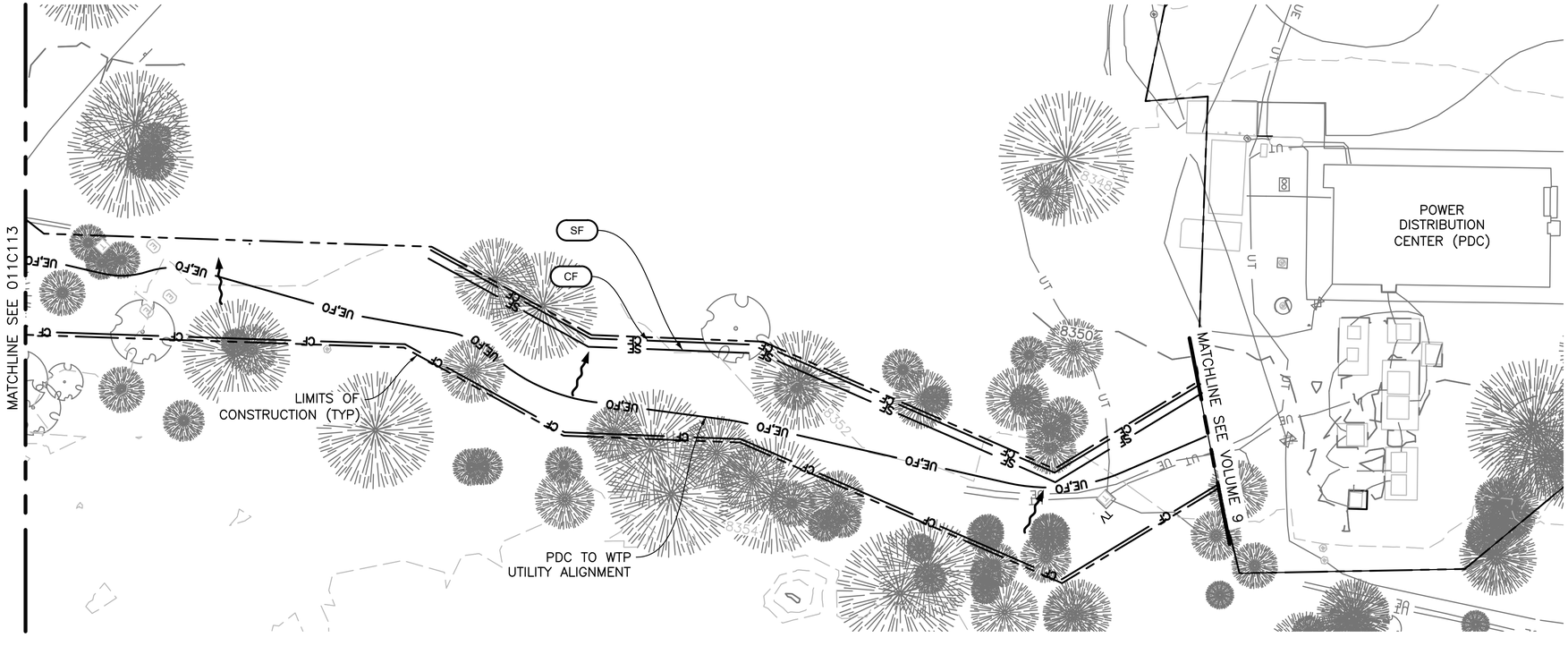
20 0 20 40 1" = 20'

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

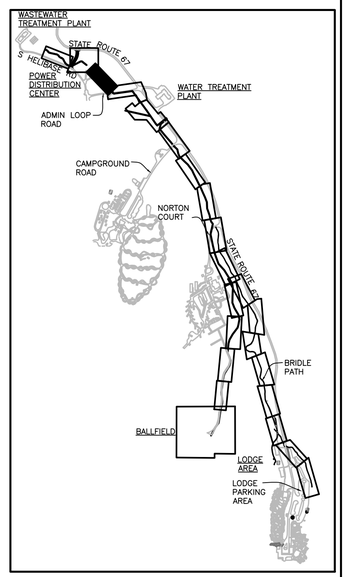
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	011 C113	SITEWORK GESC PLAN WTP TO PDC NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 32 OF 87



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL	011 C114	SITEWORK GESC PLAN WTP TO PDC	113 182752
	CADD J. GREEN			PMIS/PKG NO. 318719
	TECH. REVIEW: GOUGH/LUSTER			SHEET 33 OF 87
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg

POWER DISTRIBUTION CENTER (PDC)

MATCHLINE SEE VOLUME 9

LIMITS OF CONSTRUCTION (TYP)

HELIPAD RD

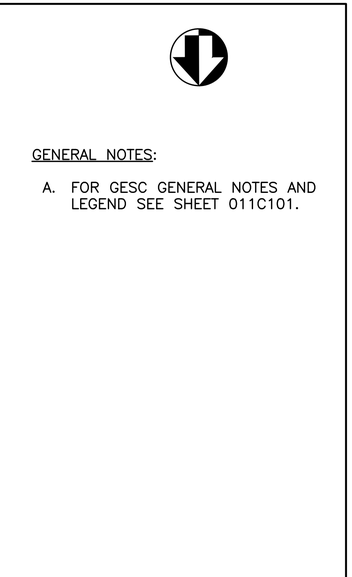
MATCHLINE SEE 011C116

WWTP TO PDC UTILITY ALIGNMENT

STATE ROUTE 67

GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

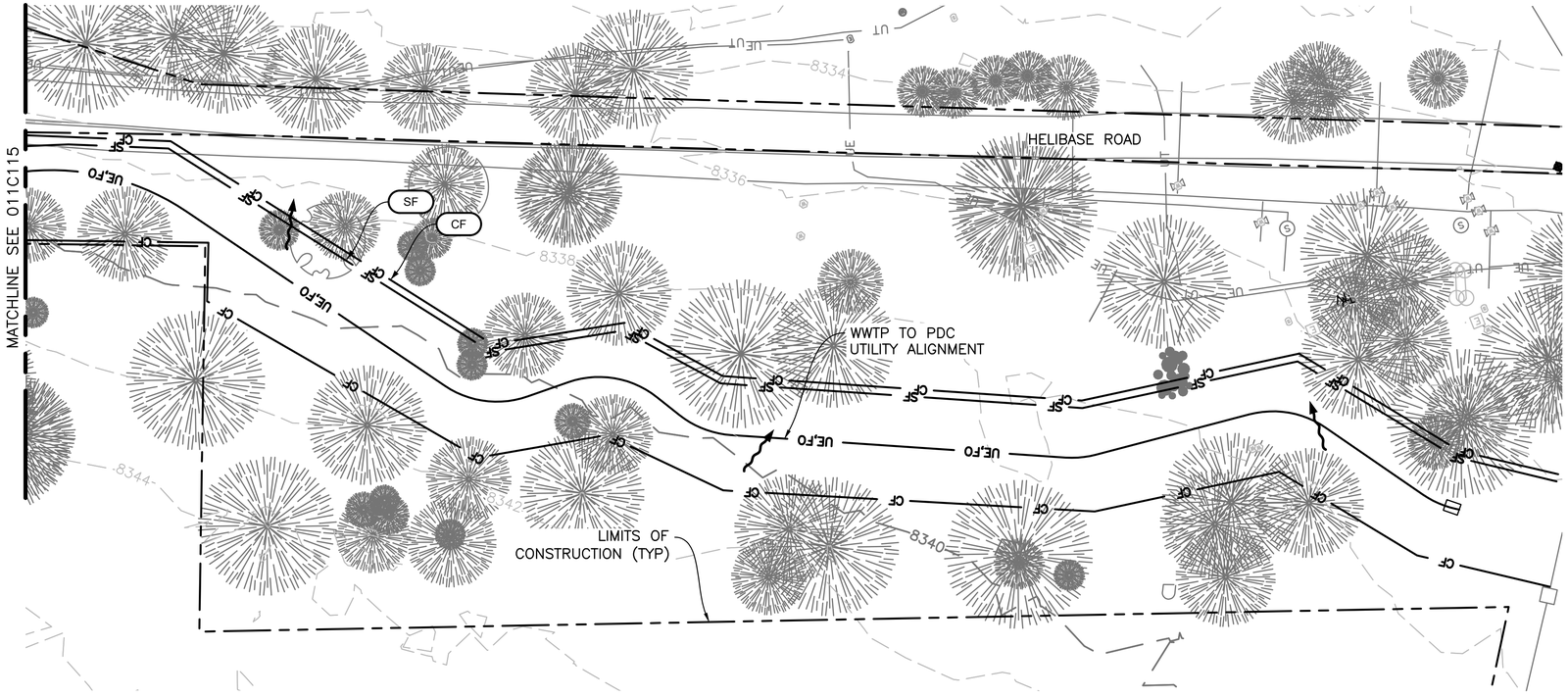
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 8-7-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

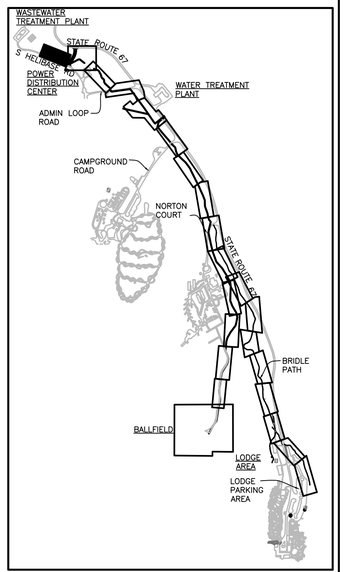
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	011 C115	SITEWORK GESC PLAN PDC TO WWTP NORTH RIM – GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 34 OF 93

8/18/2026 Green, Jake C:\Users\jgreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

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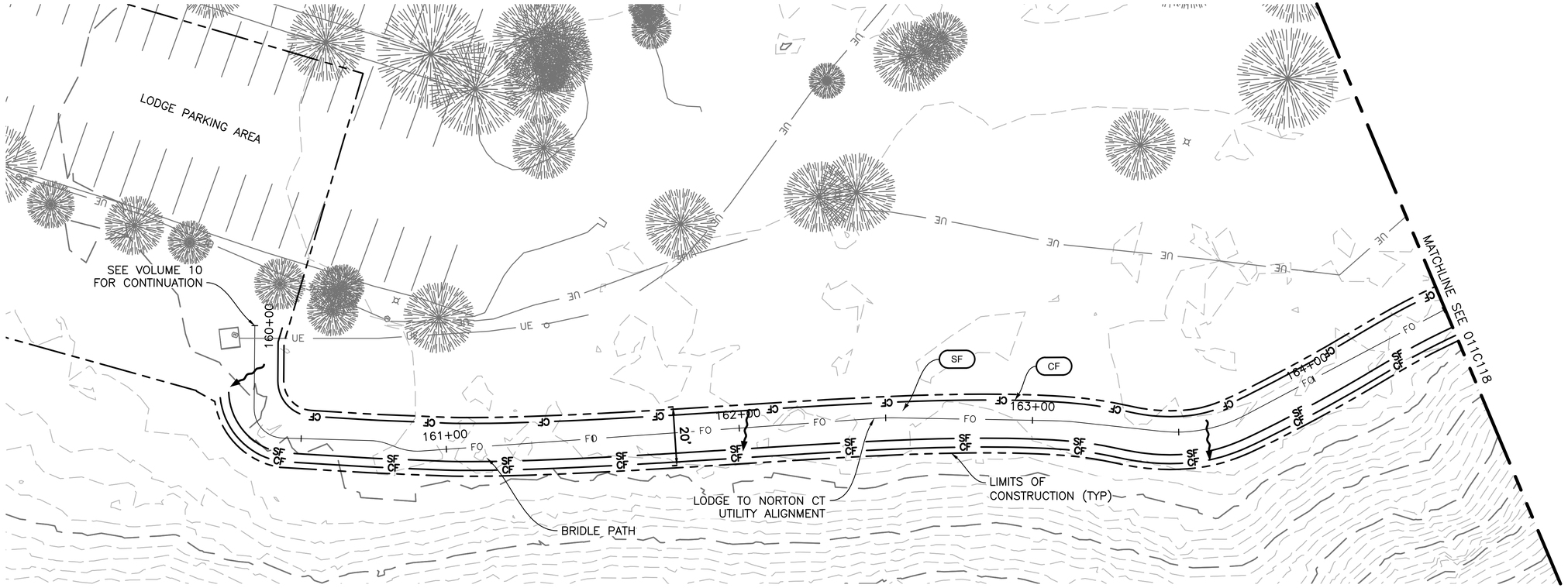
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

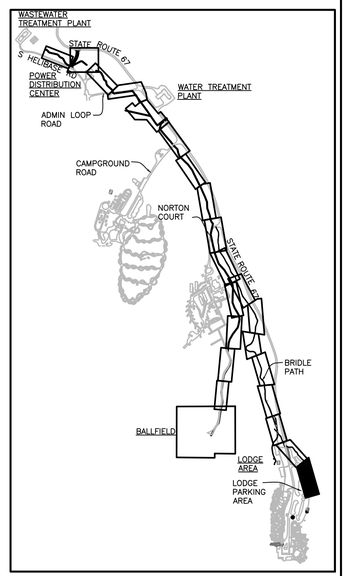
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C116	TITLE OF SHEET SITEWORK GESC PLAN PDC TO WWTP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 35 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C102.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

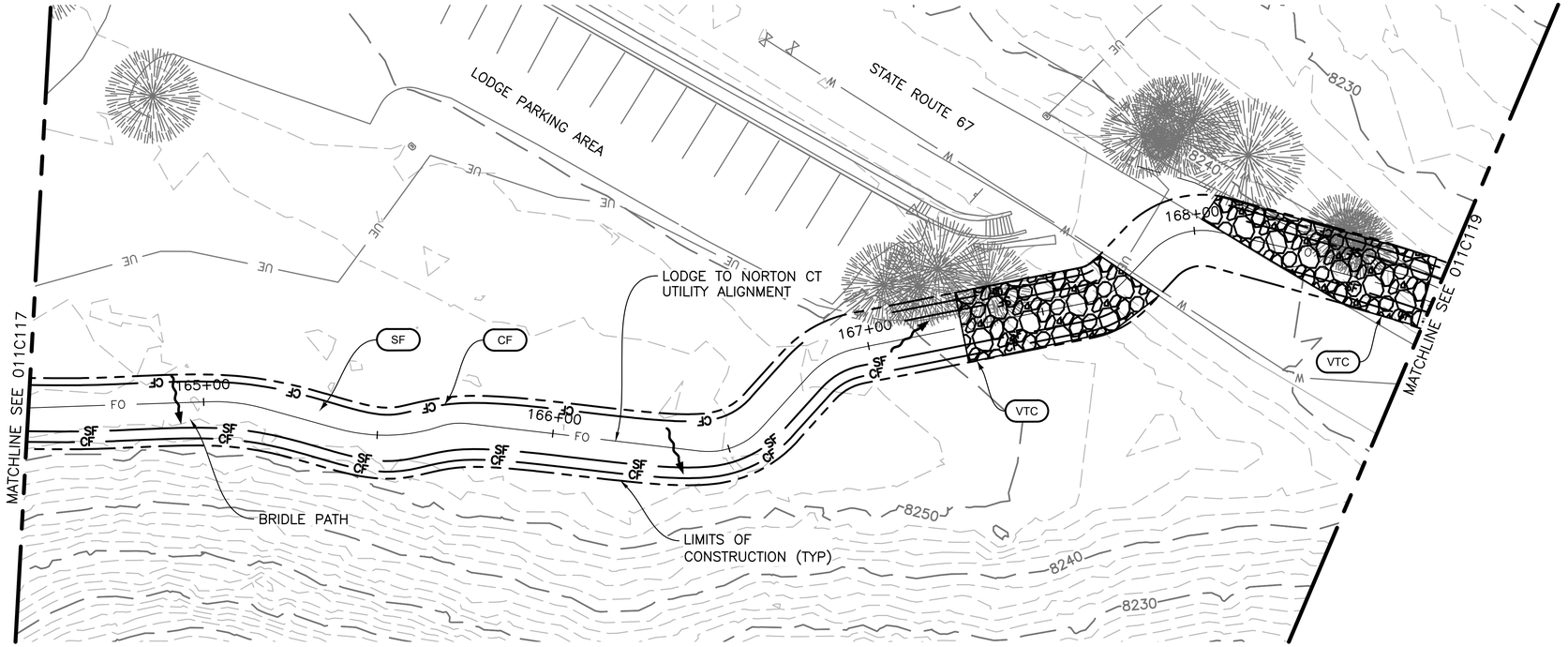
SCALE OF FEET

Registered Professional Engineer (Civil)
CERTIFICATE NO.
76139
NICHOLAS
HUMPHREY
5-1-2026
Date Signed
ARIZONA, U.S.A.

Nicholas Humphrey

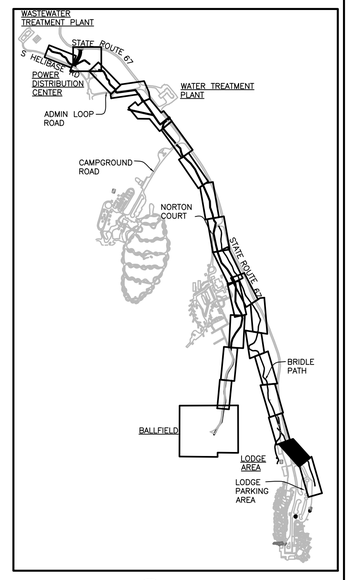
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C117	TITLE OF SHEET SITEWORK GESC PLAN LODGE TO NORTON CT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 36 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C117.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

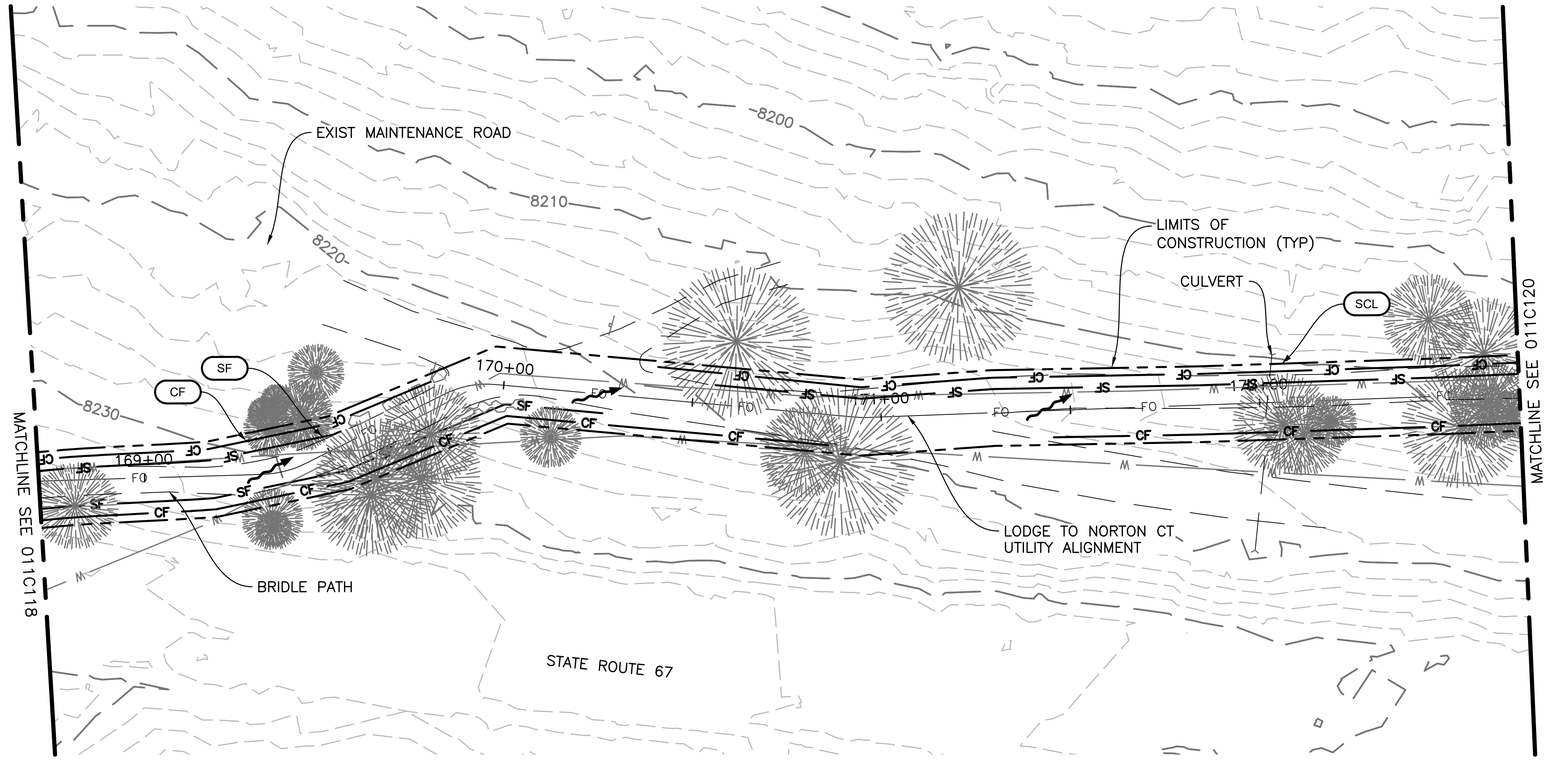
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

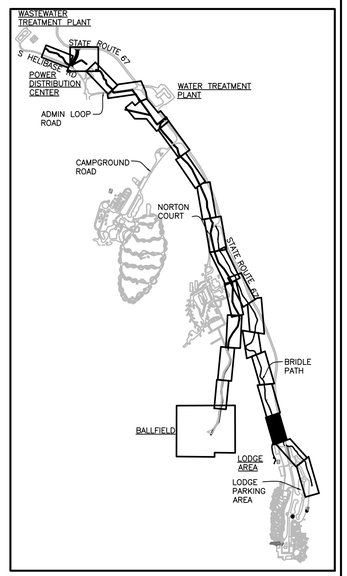
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	011 C118	SITEWORK GESC PLAN LODGE TO NORTON CT NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 37 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C117.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

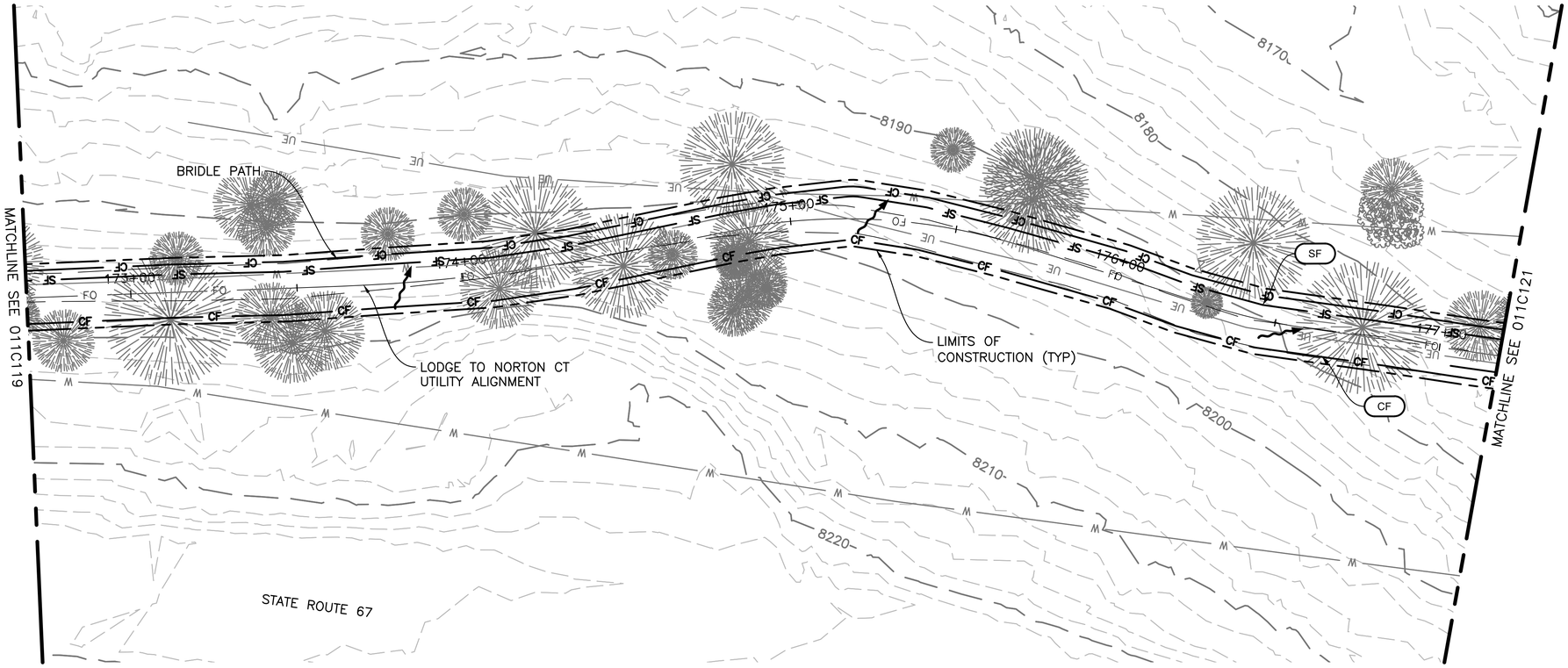
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

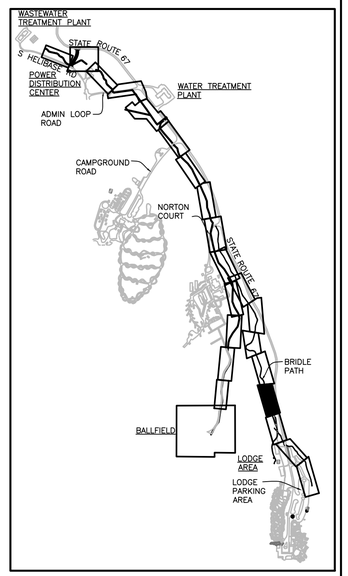
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 0119	TITLE OF SHEET SITEWORK GESC PLAN LODGE TO NORTON CT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 38 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C117.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

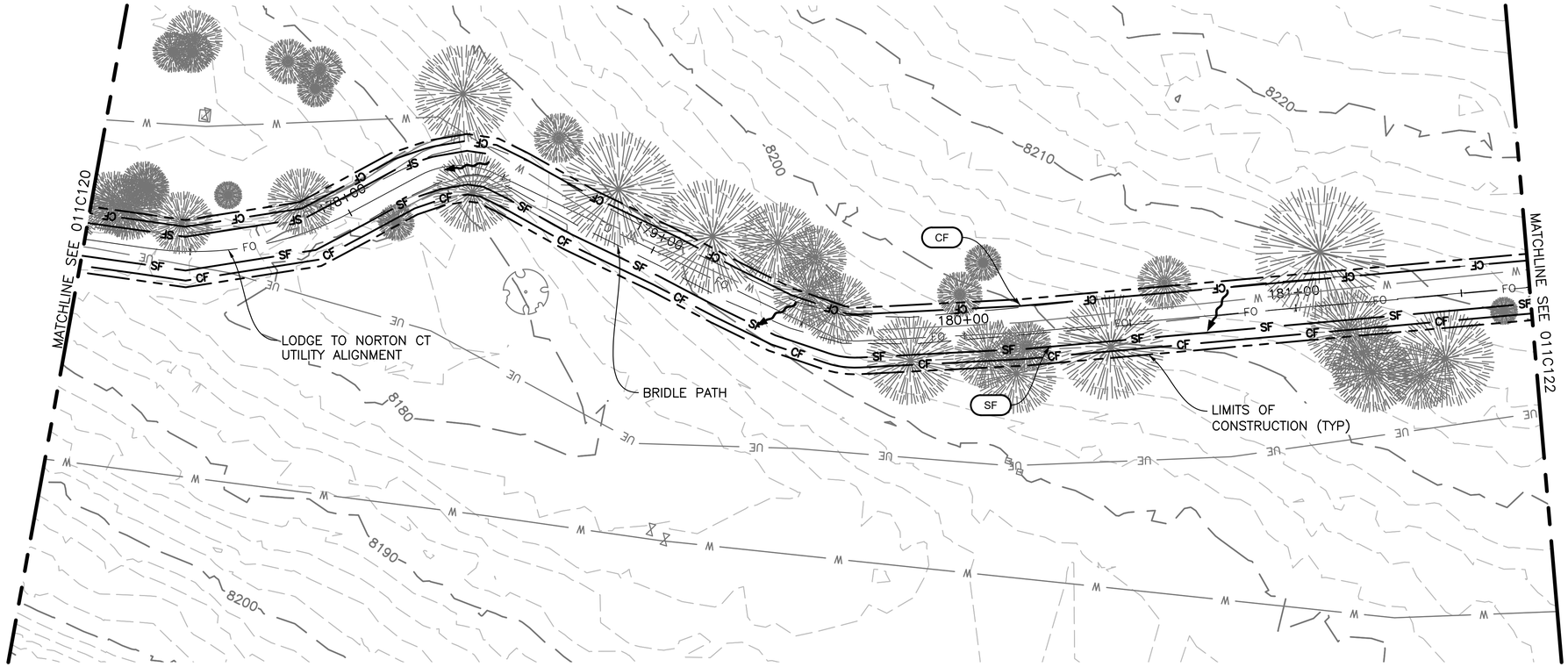
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

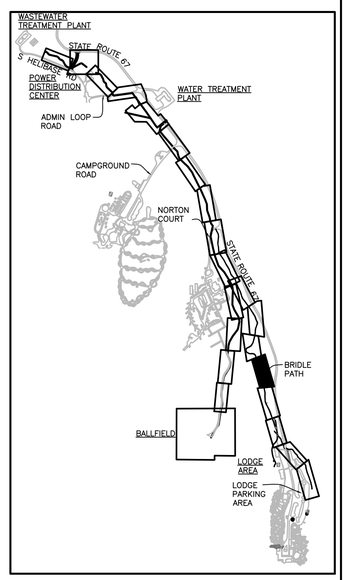
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C120	TITLE OF SHEET SITEWORK GESC PLAN LODGE TO NORTON CT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 39 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C117.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

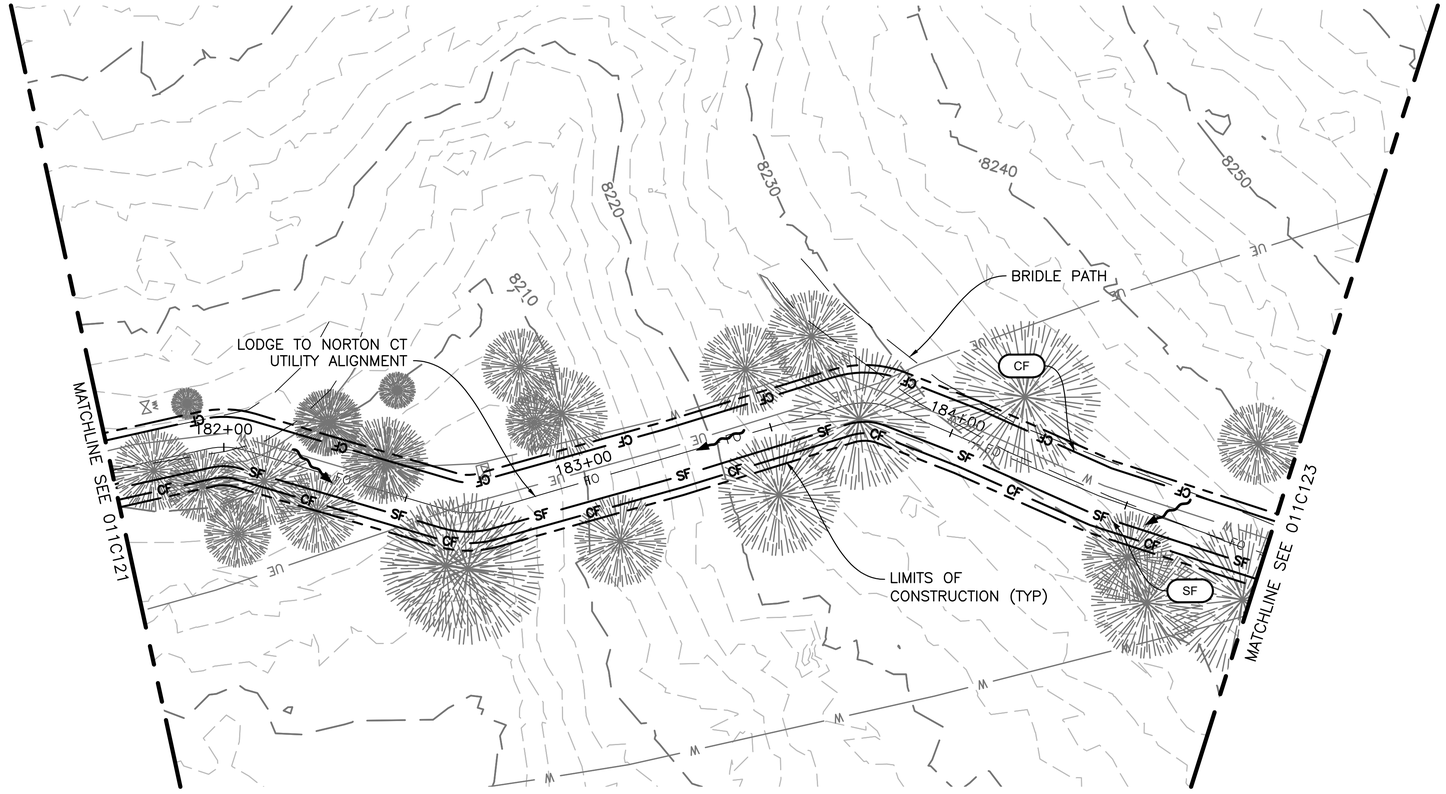
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

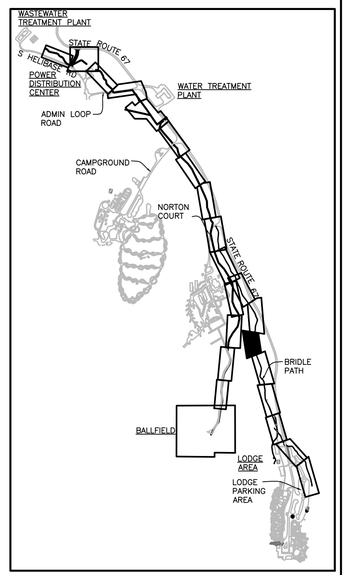
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C121	TITLE OF SHEET SITEWORK GESC PLAN LODGE TO NORTON CT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 40 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C117.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

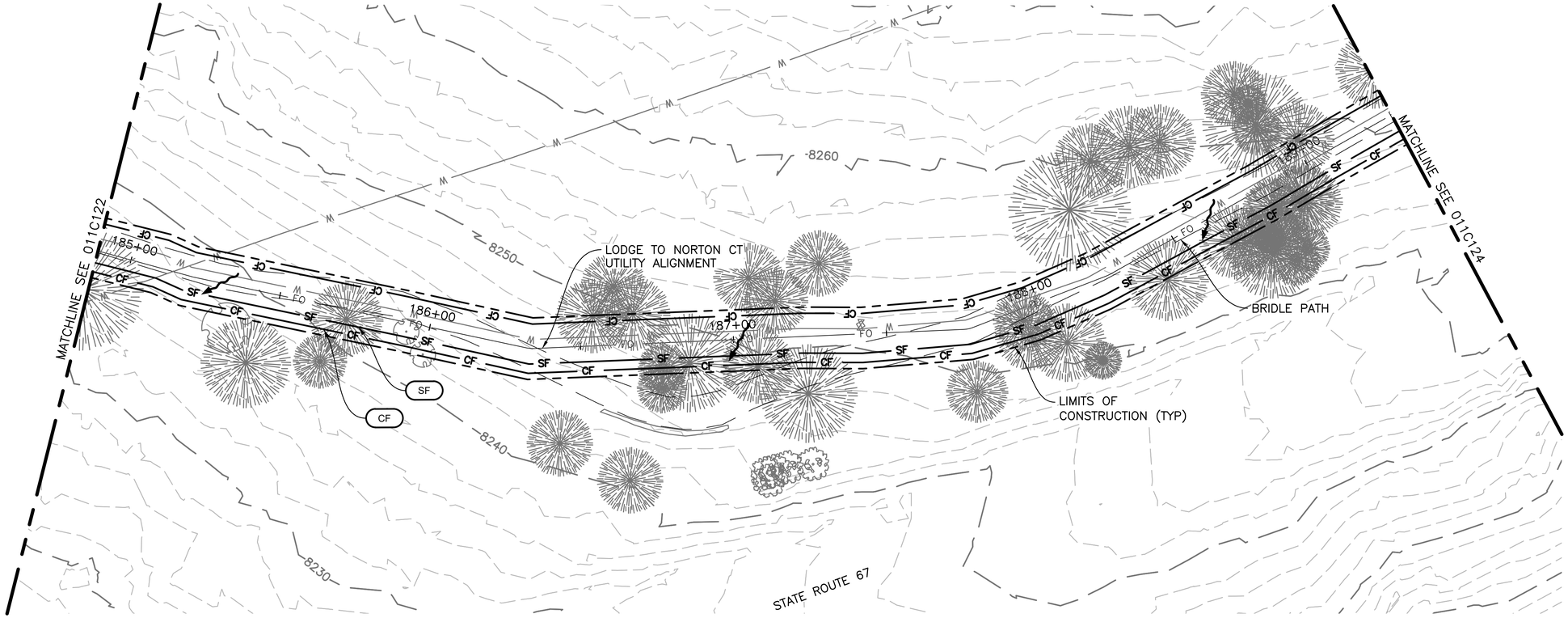
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

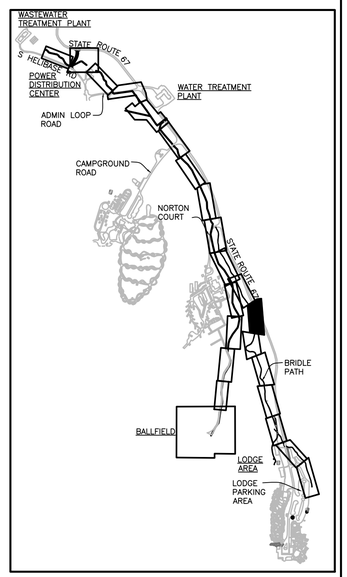
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C122	TITLE OF SHEET SITEWORK GESC PLAN LODGE TO NORTON CT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 41 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C117.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

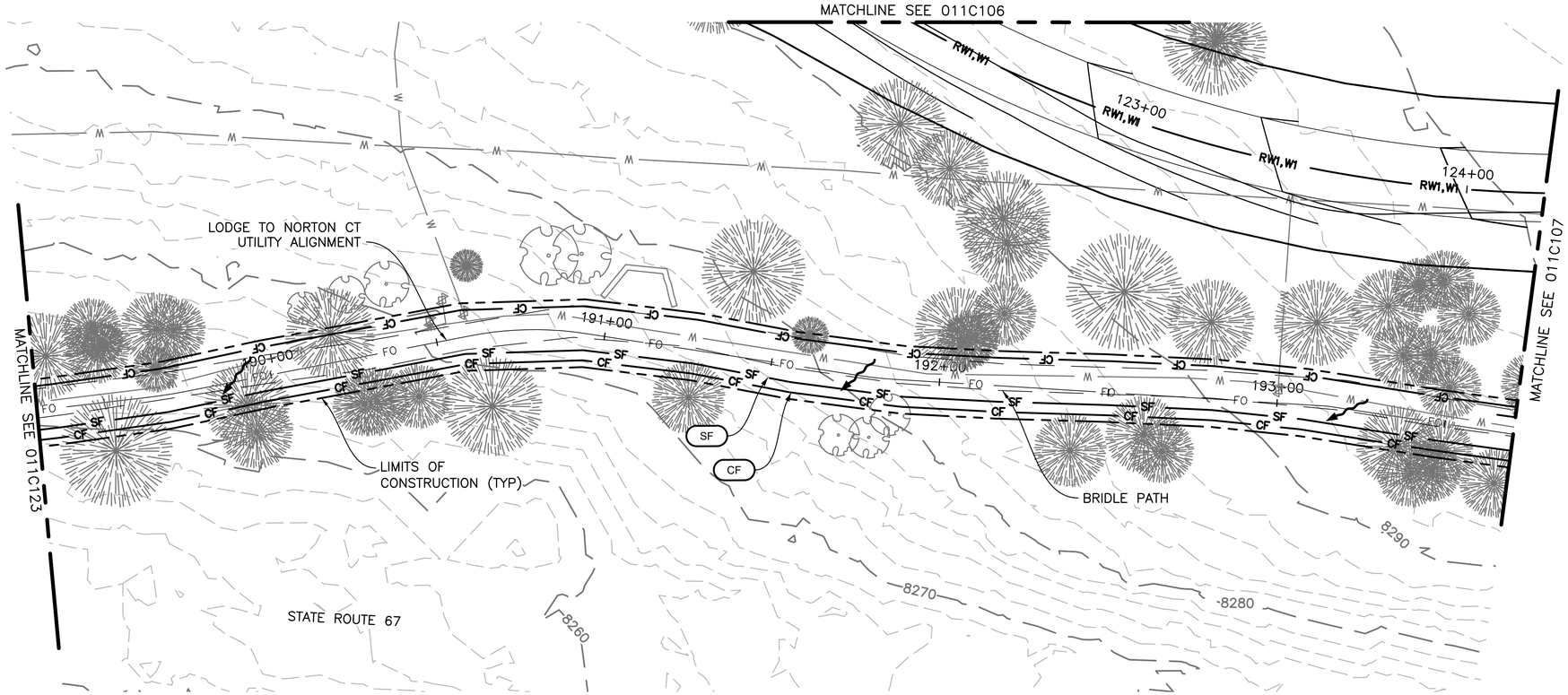
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

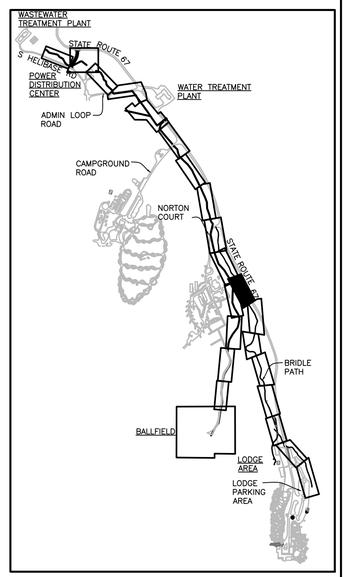
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C123	TITLE OF SHEET SITEWORK GESC PLAN LODGE TO NORTON CT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 42 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C117.dwg



GENERAL NOTES:

A. FOR GESC GENERAL NOTES AND LEGEND SEE SHEET 011C101.



KEYMAP

20 0 20 40 1" = 20'

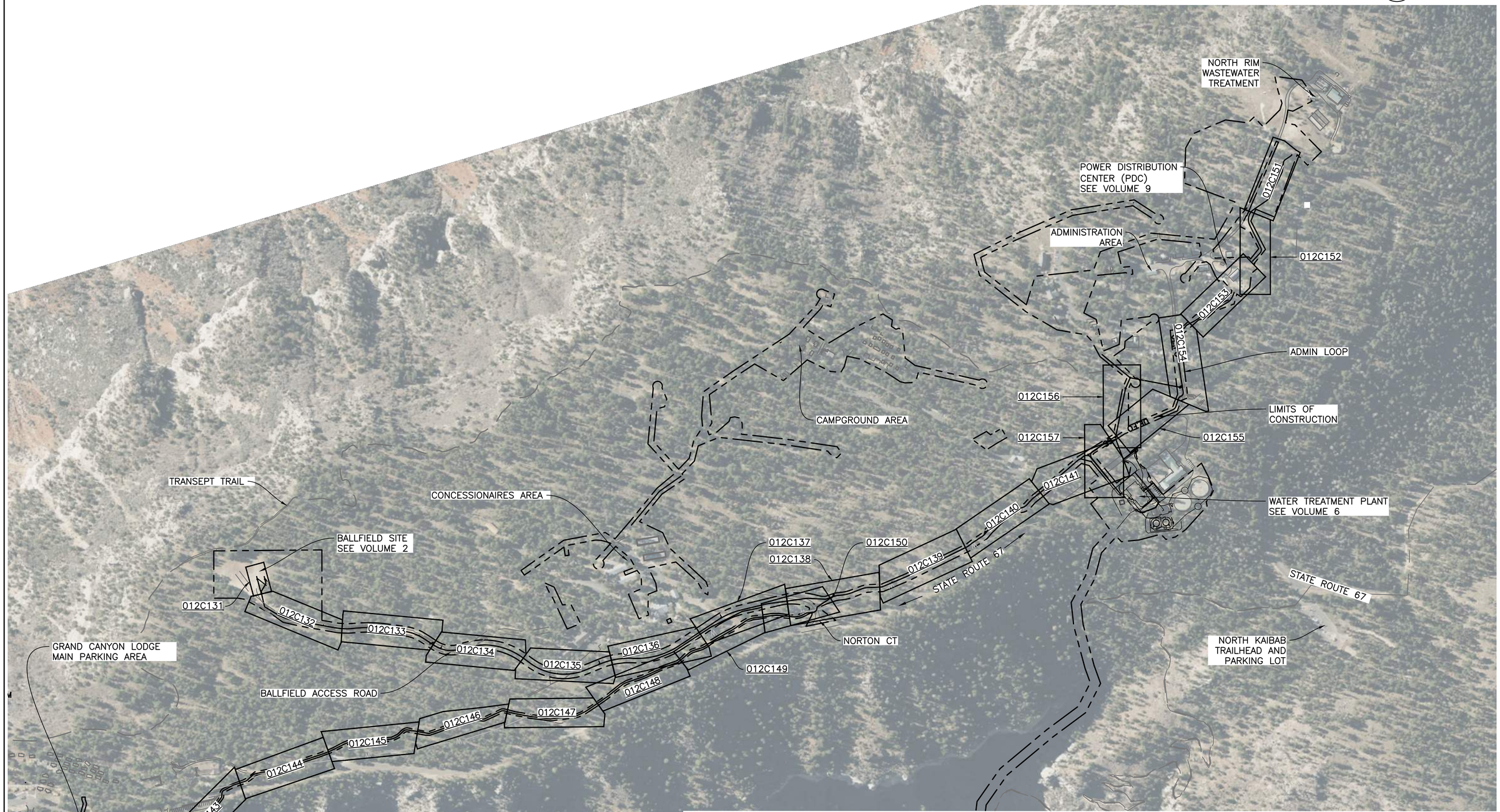
SCALE OF FEET

Registered Professional Engineer (Civil)
CERTIFICATE NO.
76139
NICHOLAS
HUMPHREY
5-1-2026
Date Signed
ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 011 C124	TITLE OF SHEET SITEWORK GESC PLAN LODGE TO NORTON CT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 43 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\011-C117.dwg



250 0 250 500 1" = 250'

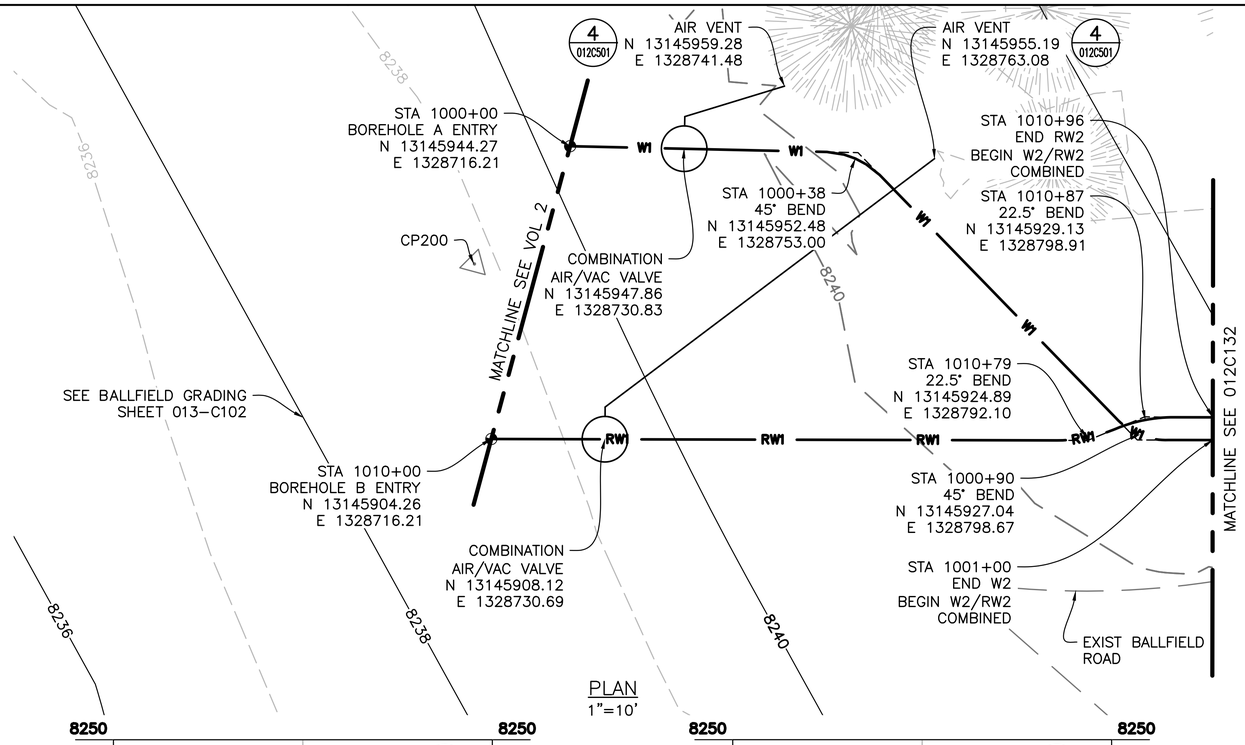
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

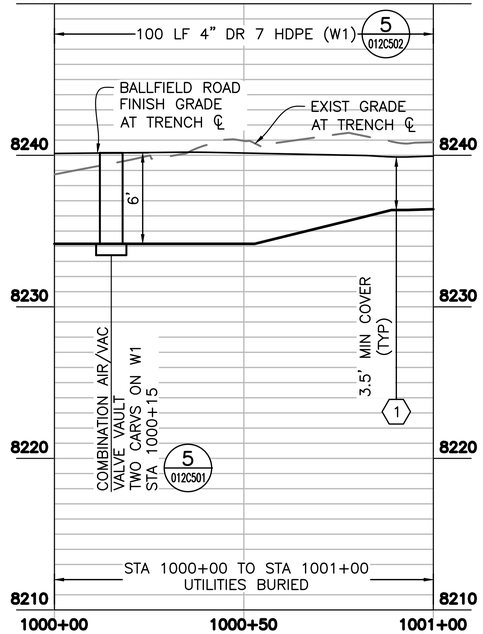
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	SUB SHEET NO. 012 C130	TITLE OF SHEET GENERAL SHEET MAP NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 44 OF 87

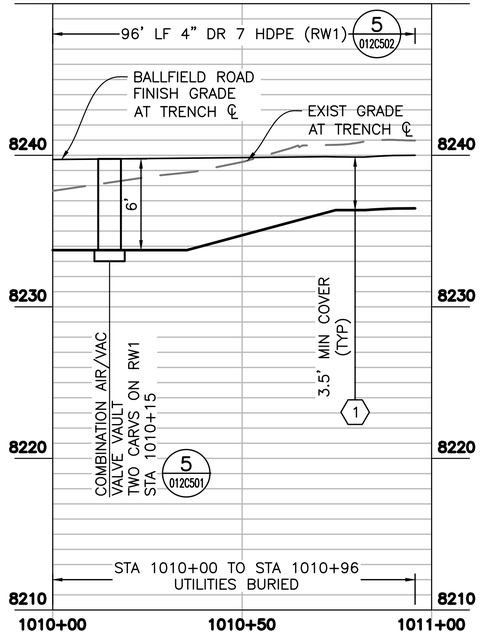
8/18/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C130.dwg



PLAN
1"=10'



W1 PROFILE
HORIZ 1"=20'
VERT 1'=5'



RW1 PROFILE
HORIZ 1"=20'
VERT 1'=5'

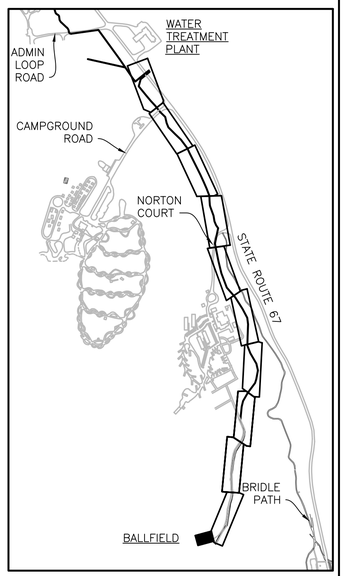
GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND HDPE DR 7 AS SHOWN ON PLANS. BEND RADIUS TO BE A MINIMUM OF 10FT.

KEYNOTES: ⟨X⟩

1. MIN COVER BASED ON FINAL GRADE. CONTRACTOR TO PROVIDE MIN COVER ON UTILITIES IF INSTALLED PRIOR TO CONSTRUCTION OF BALLFIELD ROAD. REFER TO SERIES 013 IN THIS VOLUME FOR BALLFIELD ROAD DETAILS.



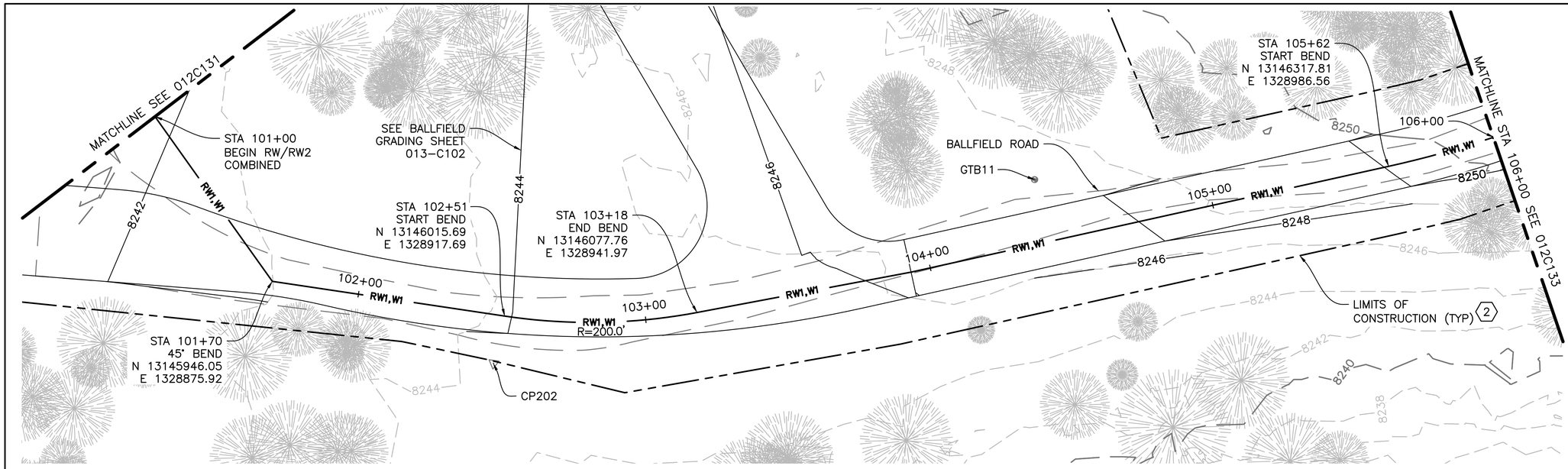
KEYMAP

10 0 10 20 HORIZ. 1" = 10'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

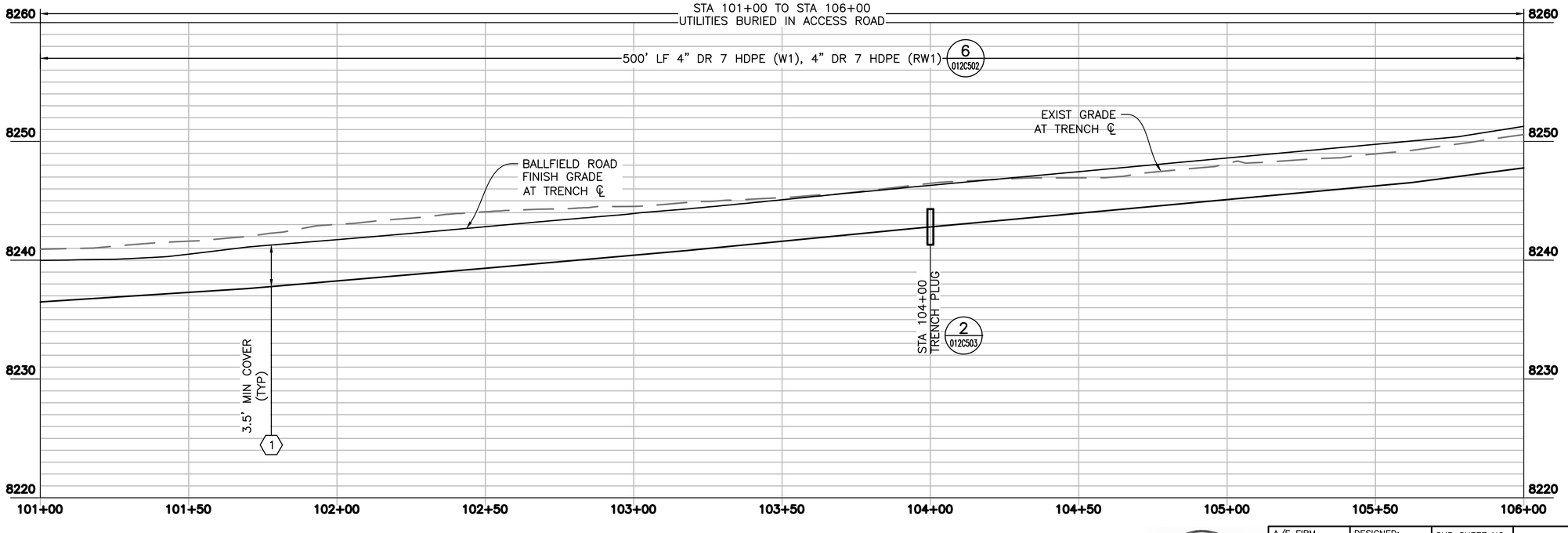
Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 04-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C131	TITLE OF SHEET UTILITIES BORE TO WTP UTILITIES PLAN AND PROFILE W2 AND RW2 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 45 OF 87

4/17/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C131.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
105+00	106+00	26 FT	23 FT
101+00	105+00	SEE SHEET 013C101	



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND HDPE DR 7 AS SHOWN ON PLANS. BEND RADIUS TO BE A MINIMUM OF 10FT.

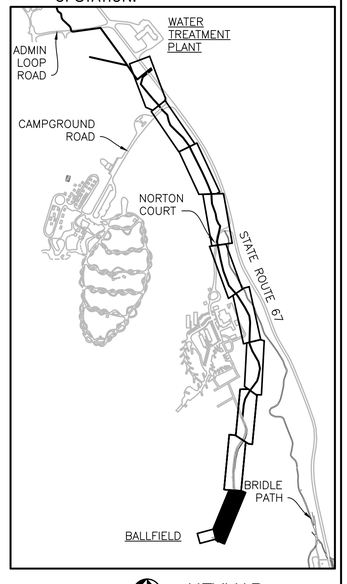
C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION.

D. NORTHINGS AND EASTINGS SHOWN FOR SHARED UTILITY ALIGNMENTS REFLECT TRENCH CENTERLINE.

KEYNOTES: (X)

1. MIN COVER BASED ON FINAL GRADE. CONTRACTOR TO PROVIDE MIN COVER ON UTILITIES IF INSTALLED PRIOR TO CONSTRUCTION OF BALLFIELD ROAD. REFER TO SERIES 013 IN THIS VOLUME FOR BALLFIELD ROAD DETAILS.

2. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.



KEYMAP

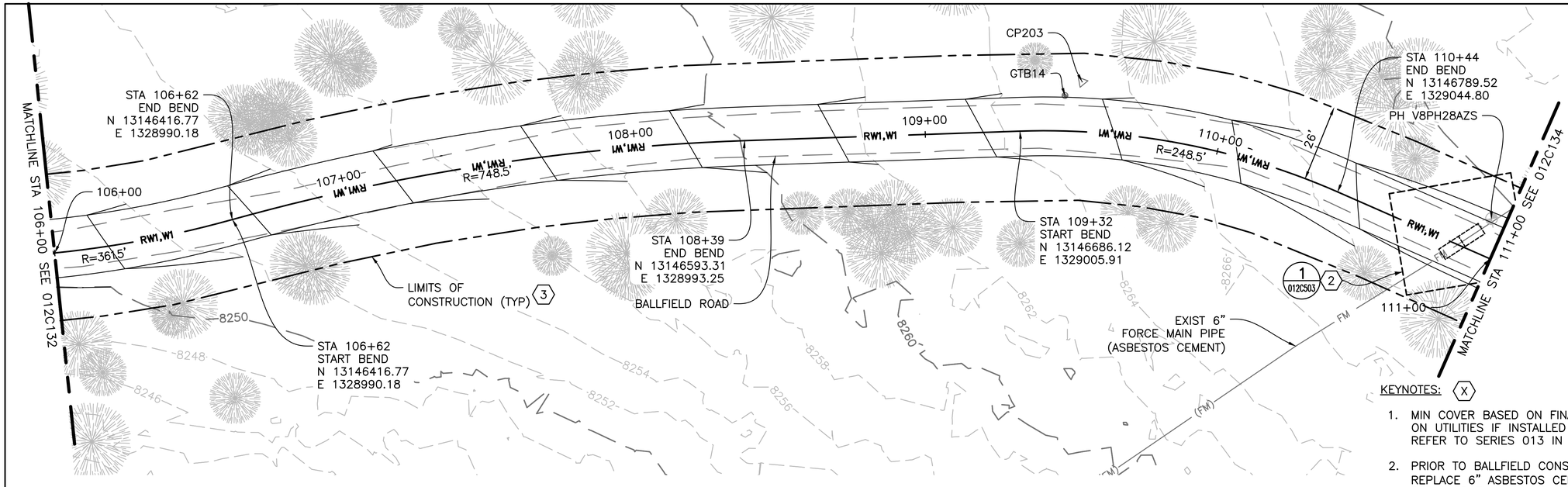
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE	012 C132	UTILITIES BORE TO WTP UTILITIES PLAN AND PROFILE STA 101+00 TO 106+00
	CADD J. GREEN		NORTH RIM - GRAND CANYON NATIONAL PARK
	TECH. REVIEW: GOUGH/LUSTER		
	DATE: MAY 2026		

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	46 OF 87

4/17/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C131.dwg



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND HDPE DR 7 AS SHOWN ON PLANS. BEND RADIUS TO BE A MINIMUM OF 10FT.

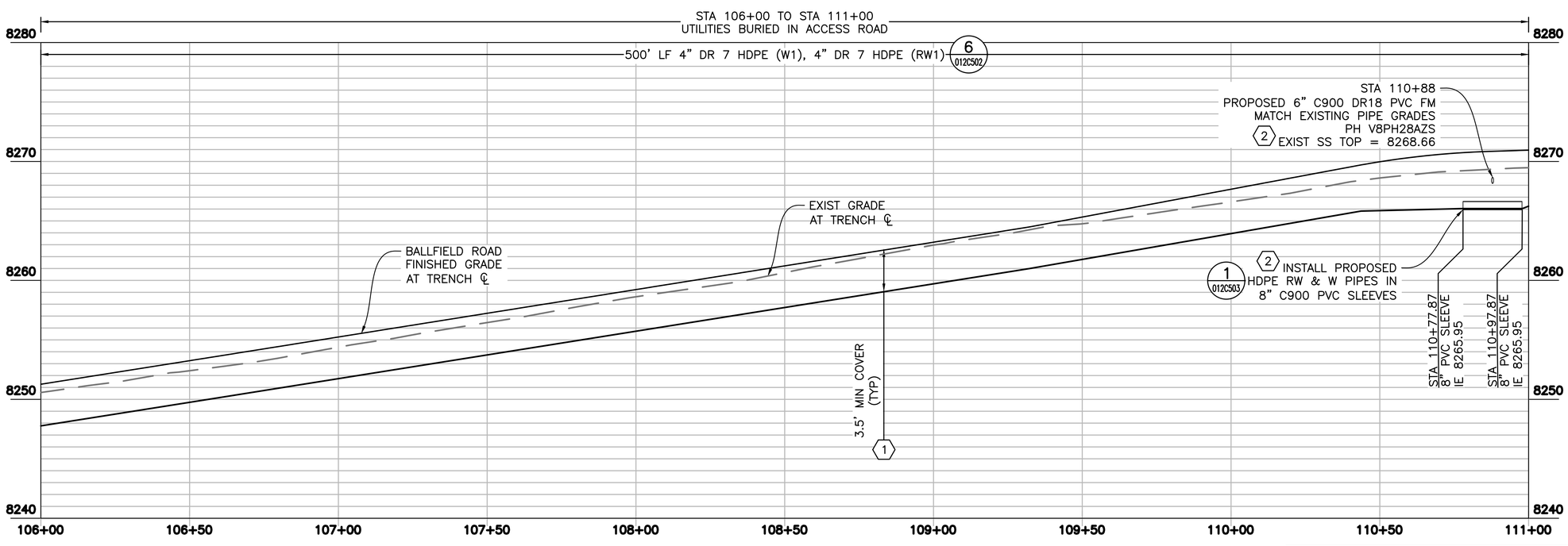
C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION.

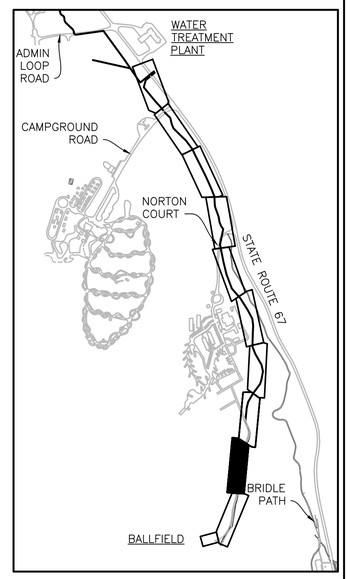
D. NORTHINGS AND EASTINGS SHOWN FOR SHARED UTILITY ALIGNMENTS REFLECT TRENCH CENTERLINE.

KEYNOTES: (X)

1. MIN COVER BASED ON FINAL GRADE. CONTRACTOR TO PROVIDE MIN COVER ON UTILITIES IF INSTALLED PRIOR TO CONSTRUCTION OF BALLFIELD ROAD. REFER TO SERIES 013 IN THIS VOLUME FOR BALLFIELD ROAD DETAILS.
2. PRIOR TO BALLFIELD CONSTRUCTION ACTIVITIES, INSTALL PVC SLEEVES AND REPLACE 6" ASBESTOS CEMENT PIPE WITH DR18 C900 PVC. PROTECT REPLACED FM WITH HS-20 RATED STEEL PLATES DURING CONSTRUCTION AND REMOVE PLATES TO BRING ROAD TO FINAL GRADE. SEE DETAIL 1/012C503.
3. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
106+00	111+00	26 FT	24 FT





KEYMAP

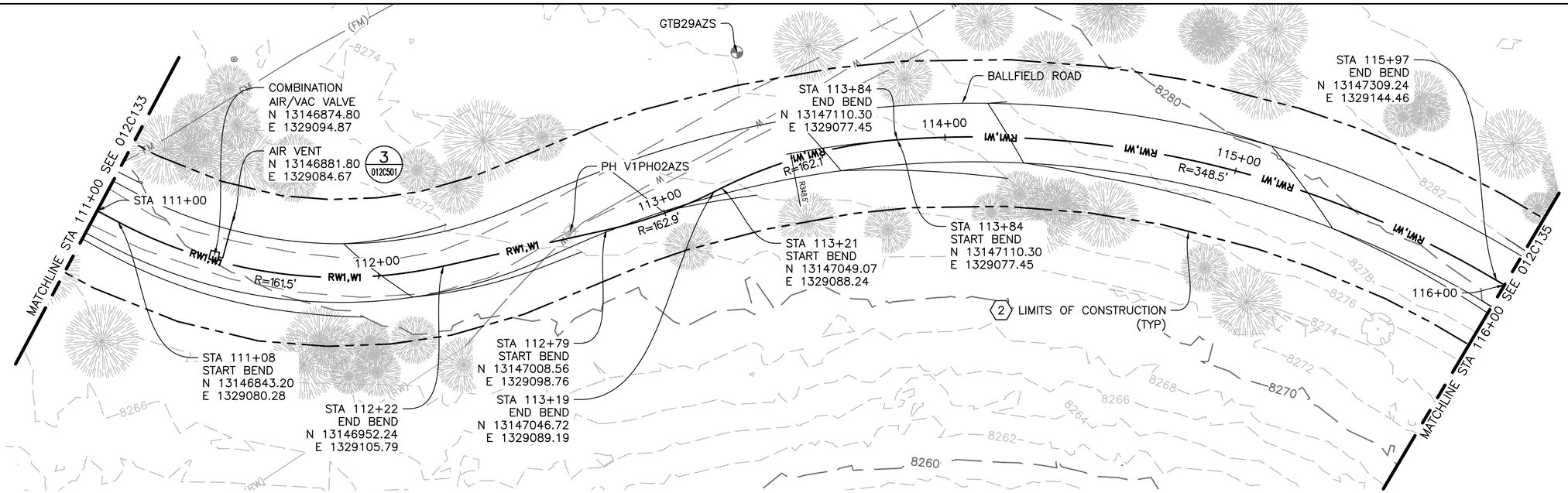
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

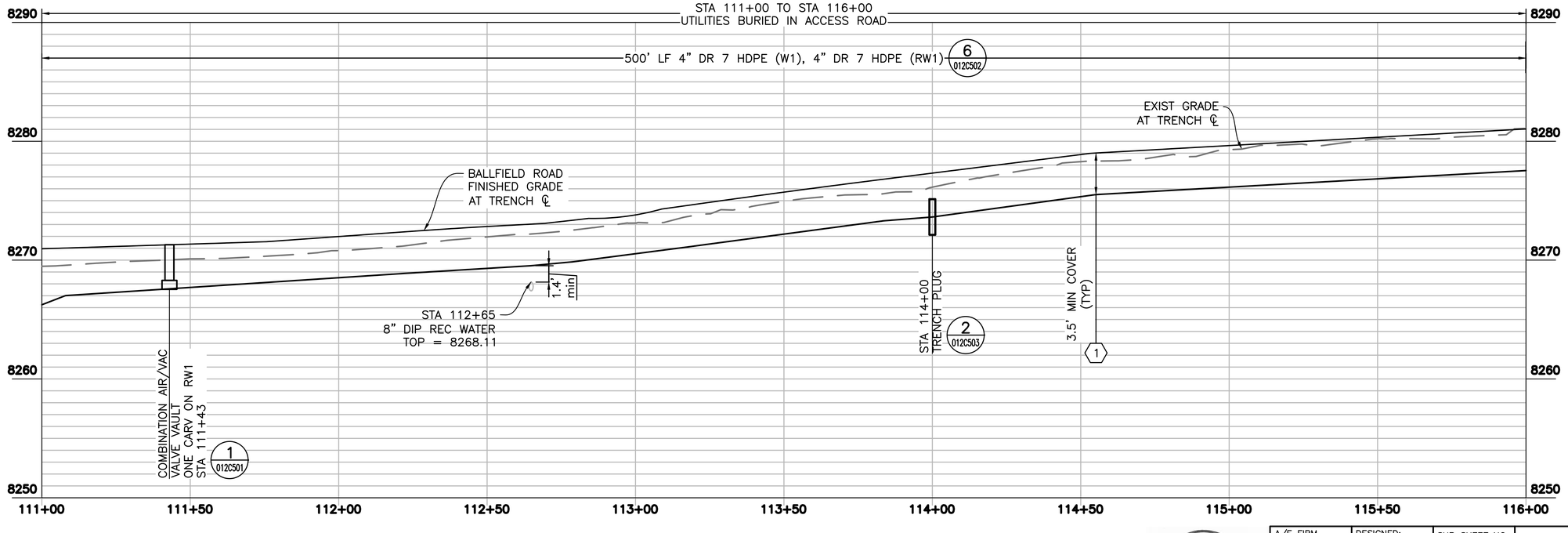
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	012 C133	UTILITIES BORE TO WTP UTILITIES PLAN AND PROFILE STA 106+00 TO 111+00 NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 47 OF 87

4/17/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C131.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
11+00	116+00	34 FT	15 FT



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND HDPE DR 7 AS SHOWN ON PLANS. BEND RADIUS TO BE MINIMUM OF 10 FT.

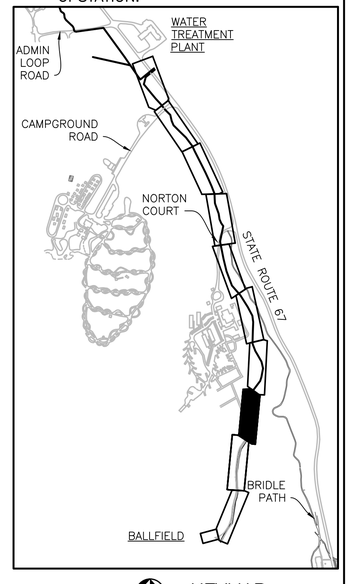
C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

D. NORTHINGS AND EASTINGS SHOWN FOR SHARED UTILITY ALIGNMENTS REFLECT TRENCH CENTERLINE.

KEYNOTES: (X)

1. MIN COVER BASED ON FINAL GRADE. CONTRACTOR TO PROVIDE MIN COVER ON UTILITIES IF INSTALLED PRIOR TO CONSTRUCTION OF BALLFIELD ROAD. REFER TO SERIES 013 IN THIS VOLUME FOR BALLFIELD ROAD DETAILS.

2. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.



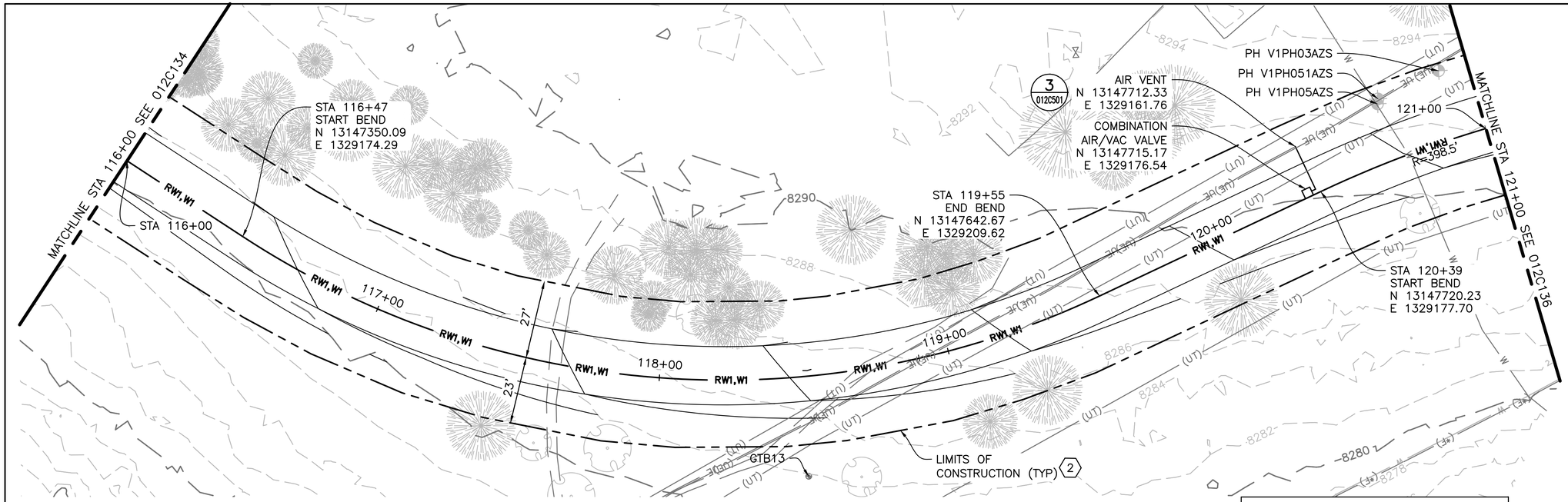
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE	012 C134	UTILITIES BORE TO WTP UTILITIES PLAN AND PROFILE STA 111+00 TO 116+00 NORTH RIM - GRAND CANYON NATIONAL PARK
	DRAFTED: J. GREEN		DRAWING NO. 113 182752
	TECH. REVIEW: GOUGH/LUSTER		PMIS/PKG NO. 318719
	DATE: MAY 2026		SHEET 48 OF 87

4/17/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C131.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
116+00	121+00	27 FT	23 FT

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND HDPE DR 7 AS SHOWN ON PLANS. BEND RADIUS TO BE A MINIMUM OF 10 FT.

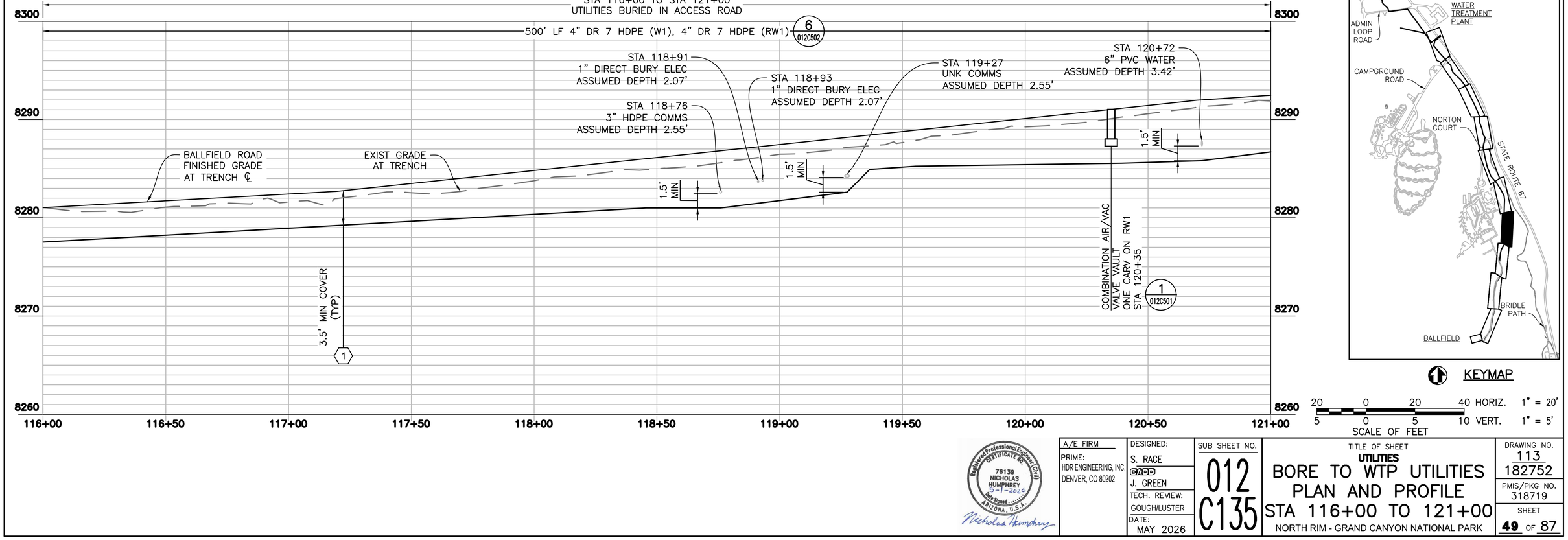
C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

D. NORTHINGS AND EASTINGS SHOWN FOR SHARED UTILITY ALIGNMENTS REFLECT TRENCH CENTERLINE.

KEYNOTES: (X)

1. MIN COVER BASED ON FINAL GRADE. CONTRACTOR TO PROVIDE MIN COVER ON UTILITIES IF INSTALLED PRIOR TO CONSTRUCTION OF BALLFIELD ROAD. REFER TO SERIES 013 IN THIS VOLUME FOR BALLFIELD ROAD DETAILS.

2. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.



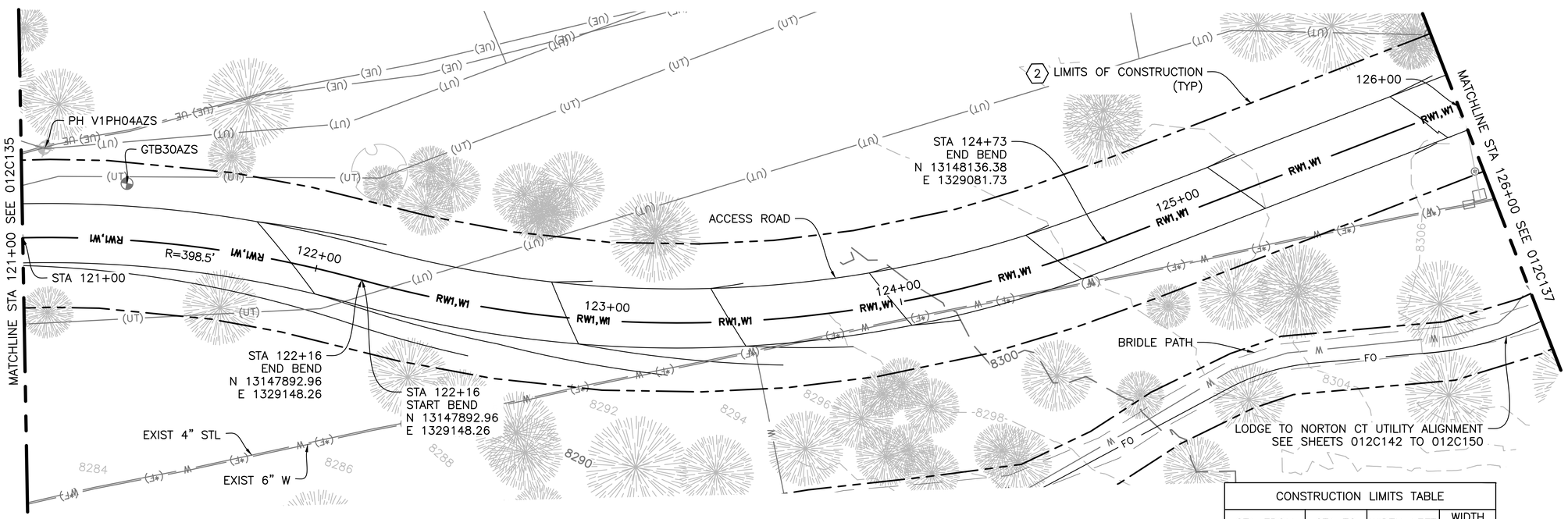
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

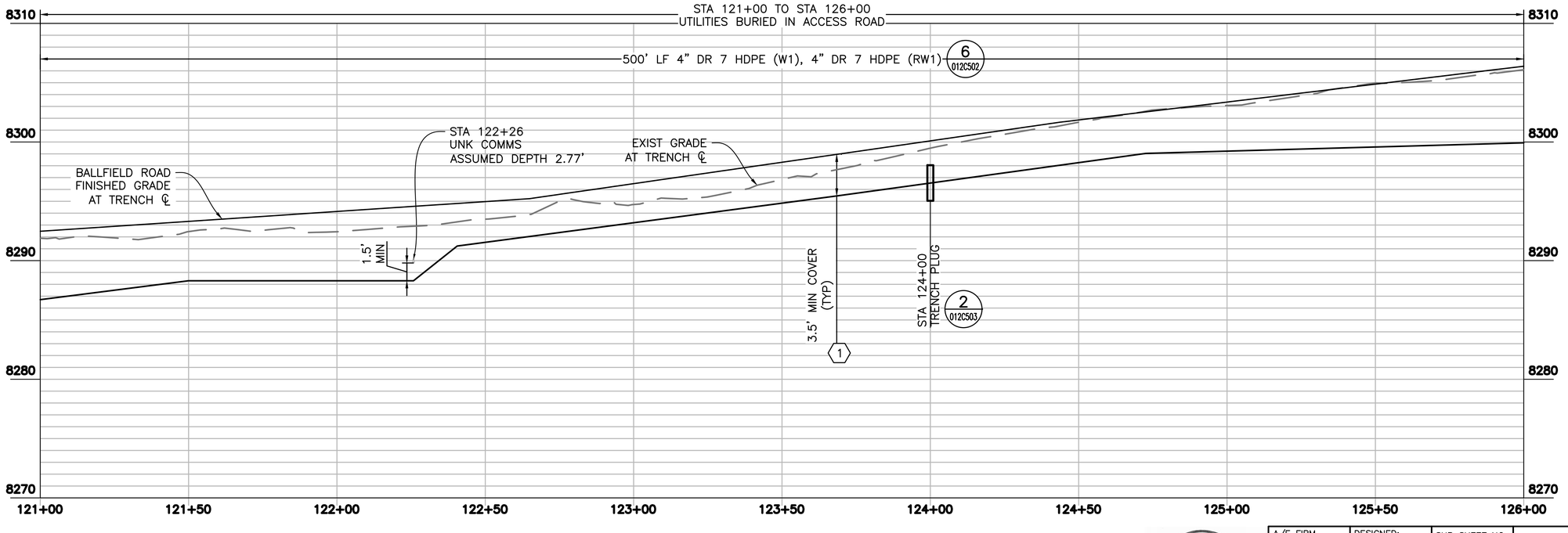
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C135	TITLE OF SHEET UTILITIES BORE TO WTP UTILITIES PLAN AND PROFILE STA 116+00 TO 121+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 49 OF 87

4/17/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C131.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
121+00	126+00	26 FT	23 FT



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND HDPE DR 7 AS SHOWN ON PLANS. BEND RADIUS TO BE A MINIMUM OF 10 FT.

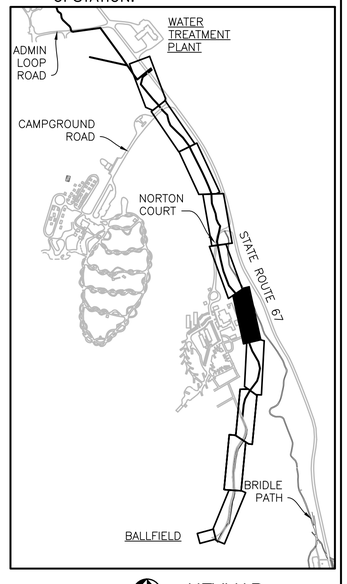
C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

D. NORTHINGS AND EASTINGS SHOWN FOR SHARED UTILITY ALIGNMENTS REFLECT TRENCH CENTERLINE.

KEYNOTES: (X)

1. MIN COVER BASED ON FINAL GRADE. CONTRACTOR TO PROVIDE MIN COVER ON UTILITIES IF INSTALLED PRIOR TO CONSTRUCTION OF BALLFIELD ROAD. REFER TO SERIES 013 IN THIS VOLUME FOR BALLFIELD ROAD DETAILS.

2. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.



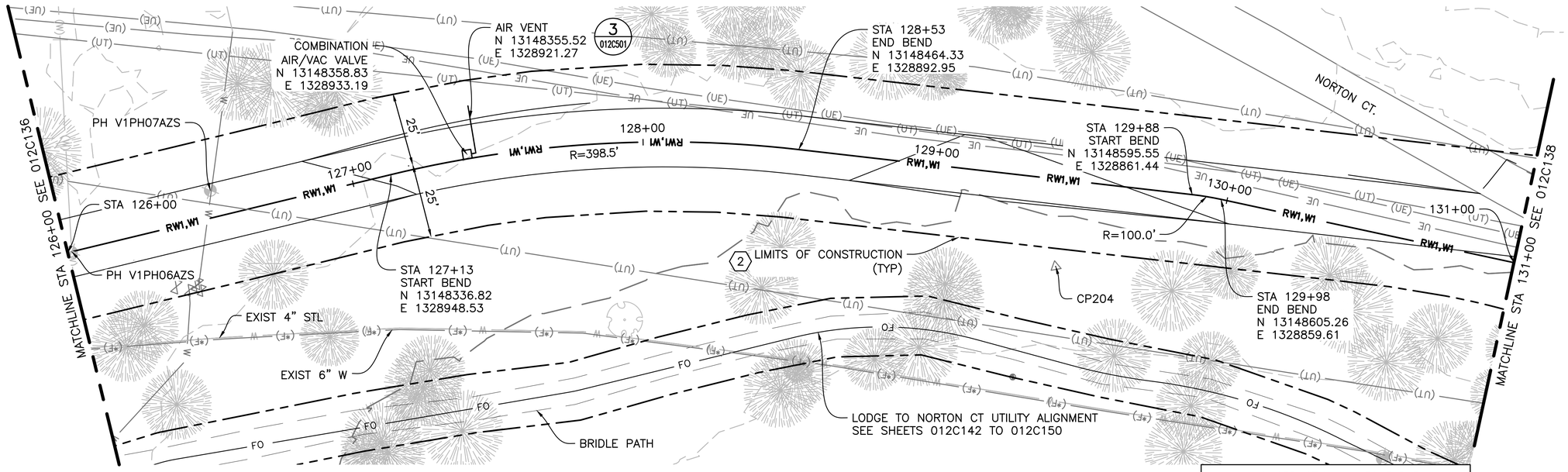
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

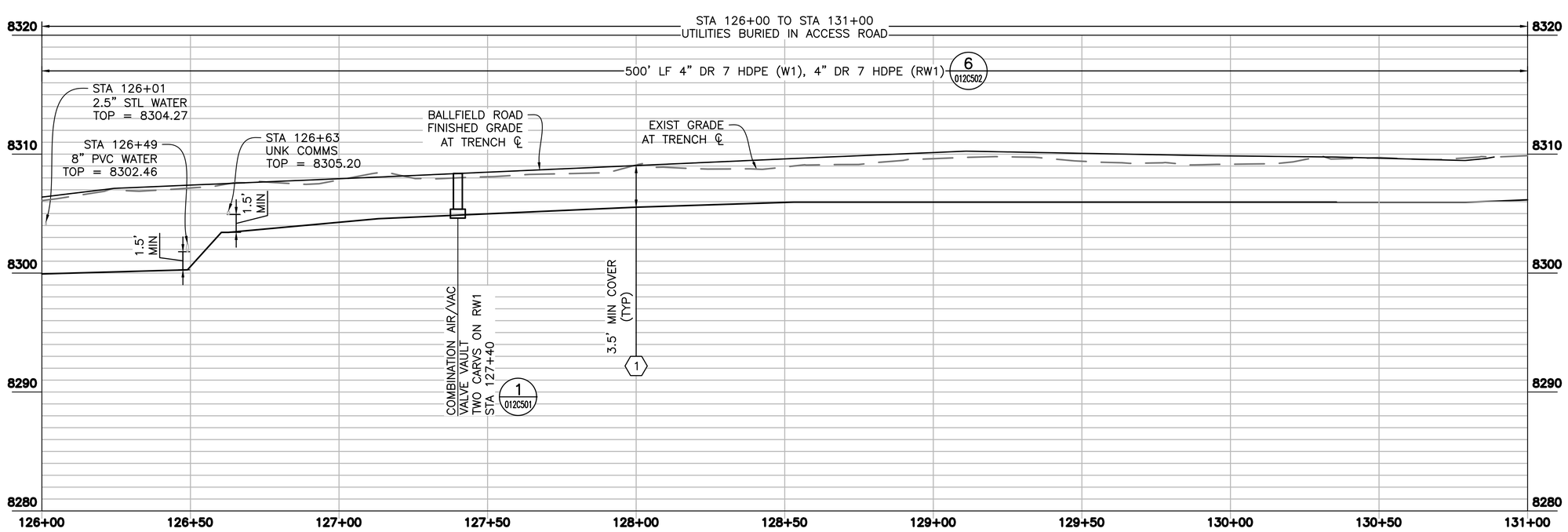
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	
DESIGNED:	S. RACE
CADD	J. GREEN
TECH. REVIEW:	GOUGH/LUSTER
DATE:	MAY 2026
SUB SHEET NO.	012 C136
TITLE OF SHEET	UTILITIES BORE TO WTP UTILITIES PLAN AND PROFILE STA 121+00 TO 126+00 NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	50 OF 87

4/17/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C131.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
116+00	121+00	27 FT	23 FT



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND HDPE DR 7 AS SHOWN ON PLANS. BEND RADIUS TO BE A MINIMUM OF 10 FT.

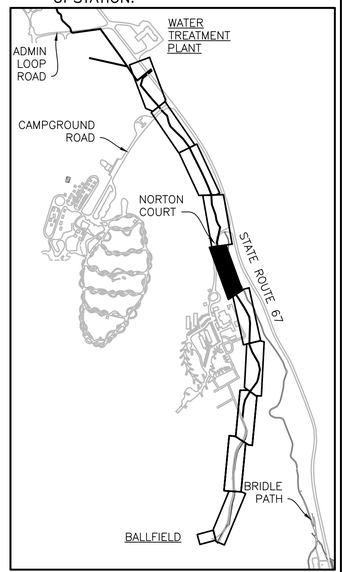
C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

D. NORTHINGS AND EASTINGS SHOWN FOR SHARED UTILITY ALIGNMENTS REFLECT TRENCH CENTERLINE.

KEYNOTES: (X)

1. MIN COVER BASED ON FINAL GRADE. CONTRACTOR TO PROVIDE MIN COVER ON UTILITIES IF INSTALLED PRIOR TO CONSTRUCTION OF BALLFIELD ROAD. REFER TO SERIES 013 IN THIS VOLUME FOR BALLFIELD ROAD DETAILS.

2. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.



KEYMAP

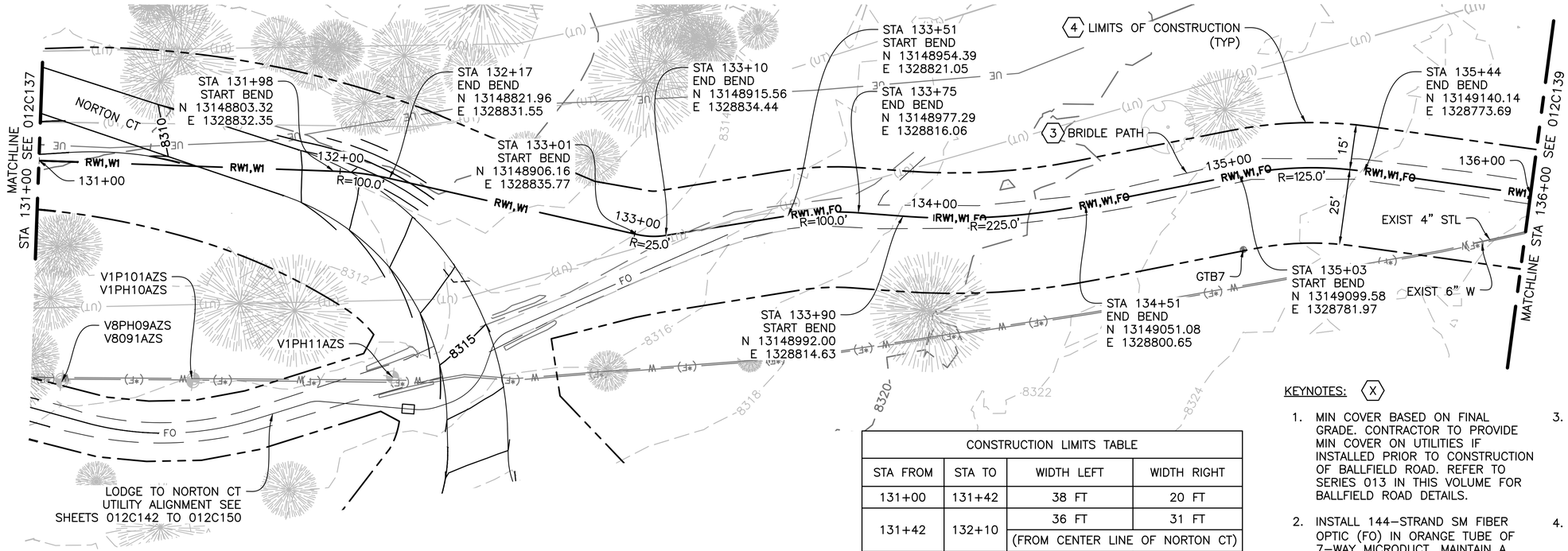
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.	
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE	012 C137	UTILITIES BORE TO WTP UTILITIES PLAN AND PROFILE STA 126+00 TO 131+00 NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752	
	DRAWN: J. GREEN			PMIS/PKG NO. 318719	
	TECH. REVIEW: GOUGH/LUSTER			SHEET 51 OF 87	
	DATE: MAY 2026				

4/17/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\103505445_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C131.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
131+00	131+42	38 FT	20 FT
131+42	132+10	36 FT	31 FT
		(FROM CENTER LINE OF NORTON CT)	
133+10	136+00	15 FT	25 FT

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND HDPE DR 7 AS SHOWN ON PLANS. BEND RADIUS TO BE A MINIMUM OF 10 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

D. NORTHINGS AND EASTINGS SHOWN FOR SHARED UTILITY ALIGNMENTS REFLECT TRENCH CENTERLINE.

E. CONTRACTORS SHALL MAINTAIN A MINIMUM OF ONE LANE OF ACCESS PER SPECIFICATION OF 01 50 00

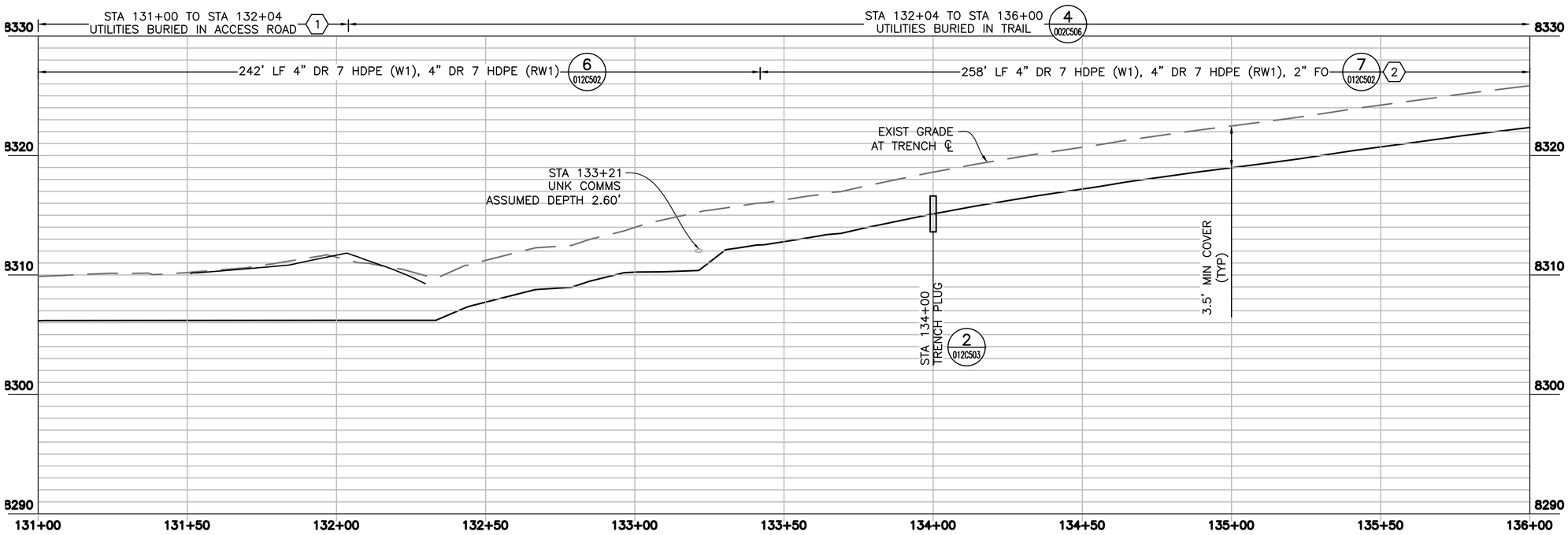
KEYNOTES: (X)

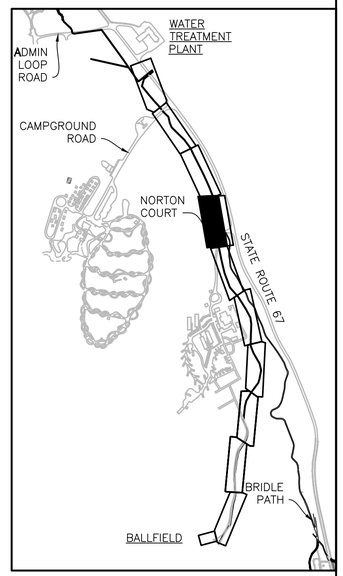
1. MIN COVER BASED ON FINAL GRADE. CONTRACTOR TO PROVIDE MIN COVER ON UTILITIES IF INSTALLED PRIOR TO CONSTRUCTION OF BALLFIELD ROAD. REFER TO SERIES 013 IN THIS VOLUME FOR BALLFIELD ROAD DETAILS.

2. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT. MAINTAIN A MINIMUM 24" COVER.

3. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.

4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.





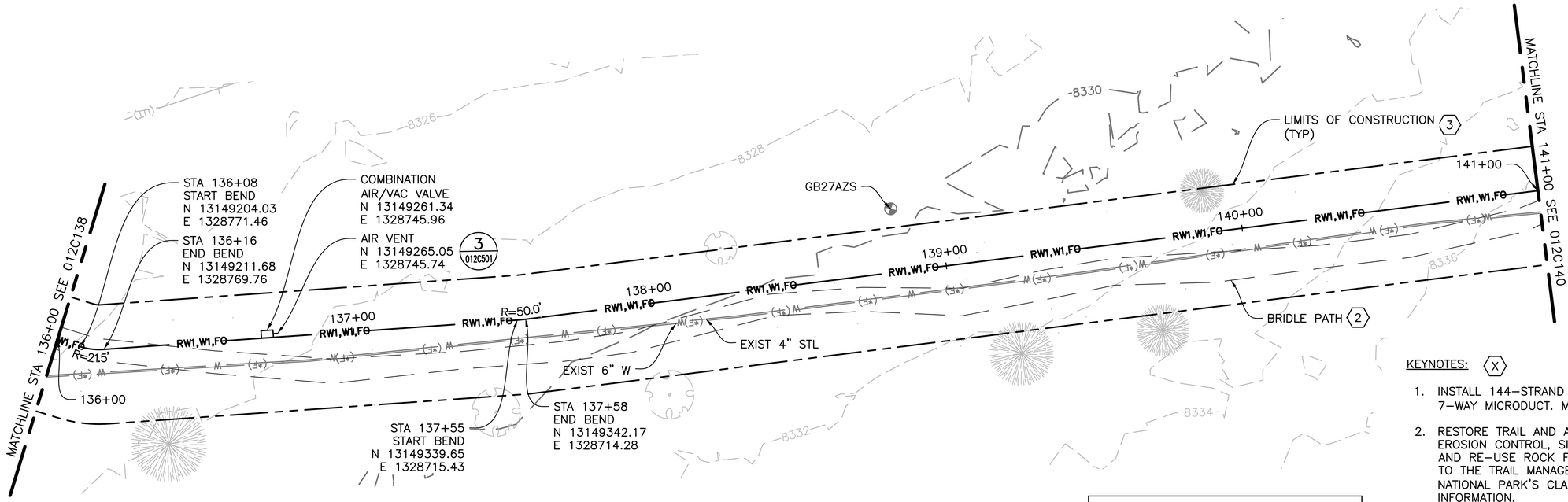
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	SUB SHEET NO. 012 C138	TITLE OF SHEET UTILITIES BORE TO WTP UTILITIES PLAN AND PROFILE STA 131+00 TO 136+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 52 OF 87

8/18/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C138.dwg



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND HDPE DR 7 AS SHOWN ON PLANS. BEND RADIUS TO BE A MINIMUM OF 10 FT.

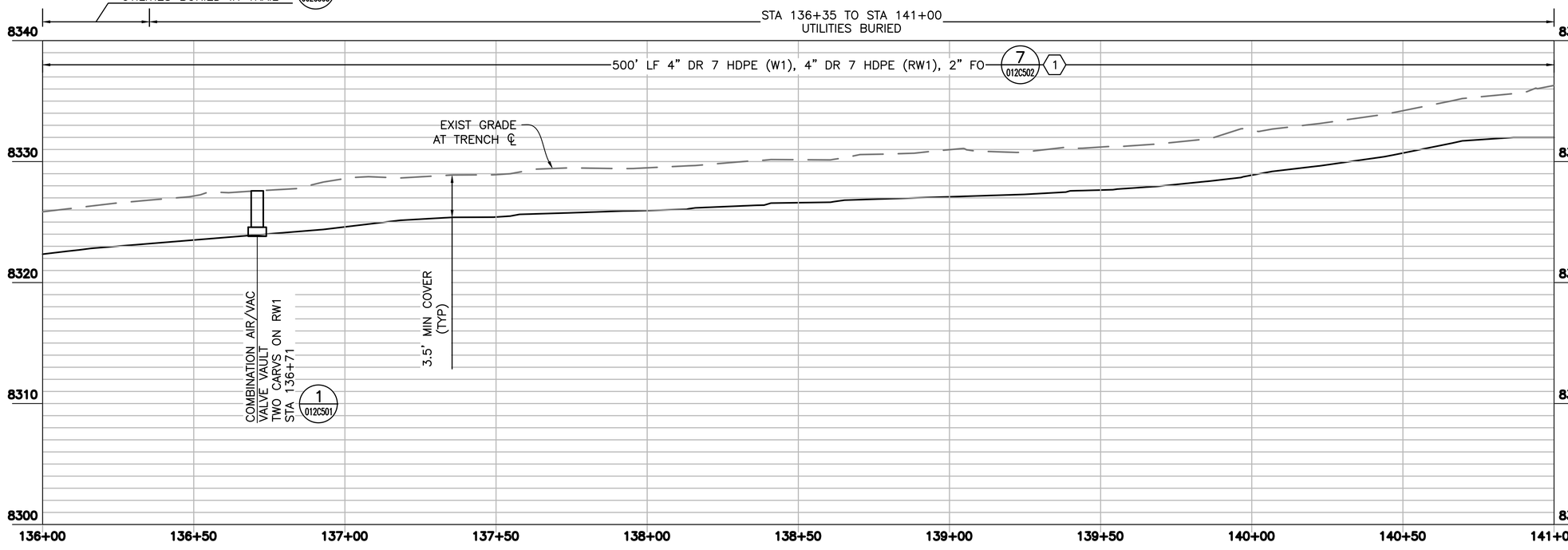
C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

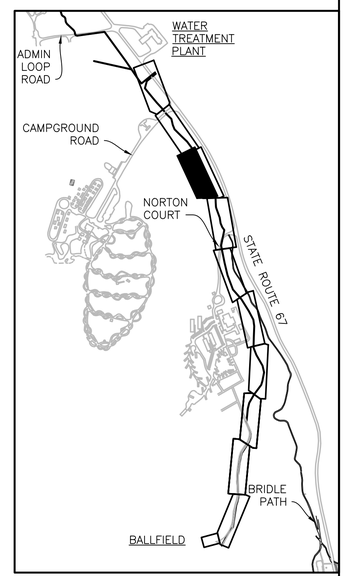
D. NORTHINGS AND EASTINGS SHOWN FOR SHARED UTILITY ALIGNMENTS REFLECT TRENCH CENTERLINE.

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT. MAINTAIN A MINIMUM 24" COVER.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
3. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
136+00	141+00	15 FT	25 FT





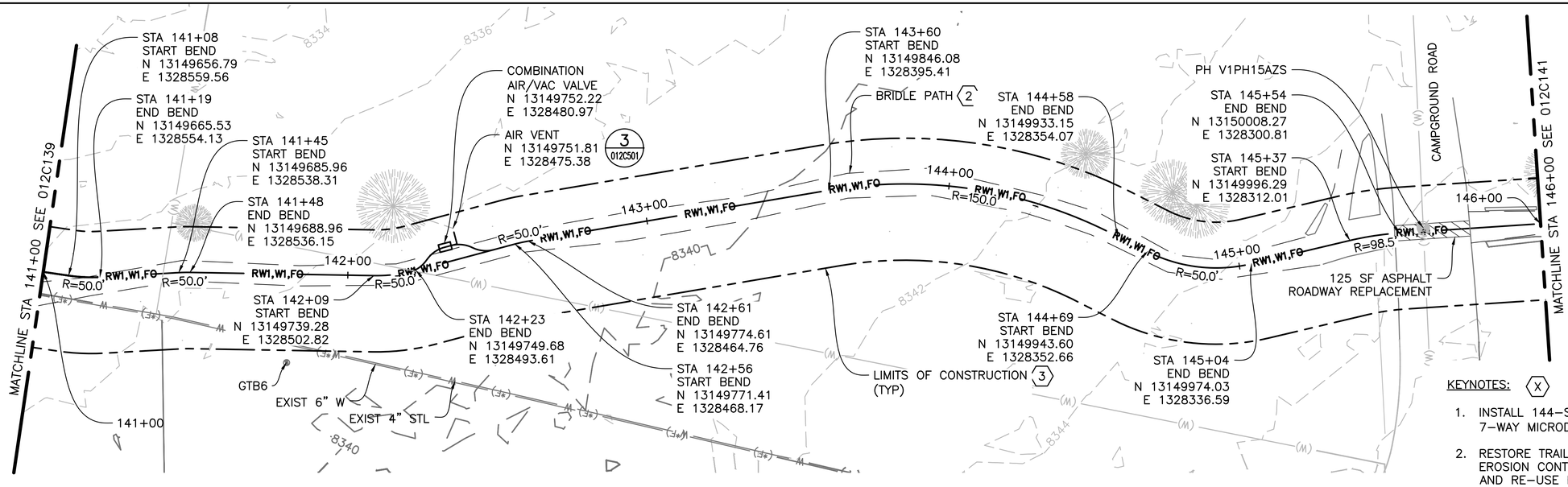
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C139	TITLE OF SHEET UTILITIES BORE TO WTP UTILITIES PLAN AND PROFILE STA 136+00 TO 141+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 53 OF 87

4/17/2026 Green, Jake C:\Users\jogreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C138.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
141+00	146+00	15 FT	25 FT

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND HDPE DR 7 AS SHOWN ON PLANS. BEND RADIUS TO BE A MINIMUM OF 10 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

D. NORTHINGS AND EASTINGS SHOWN FOR SHARED UTILITY ALIGNMENTS REFLECT TRENCH CENTERLINE.

E. CONTRACTORS SHALL MAINTAIN A MINIMUM OF ONE LANE OF ACCESS PER SPECIFICATION OF 01 50 00

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT MAINTAIN. A MINIMUM 24" COVER.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
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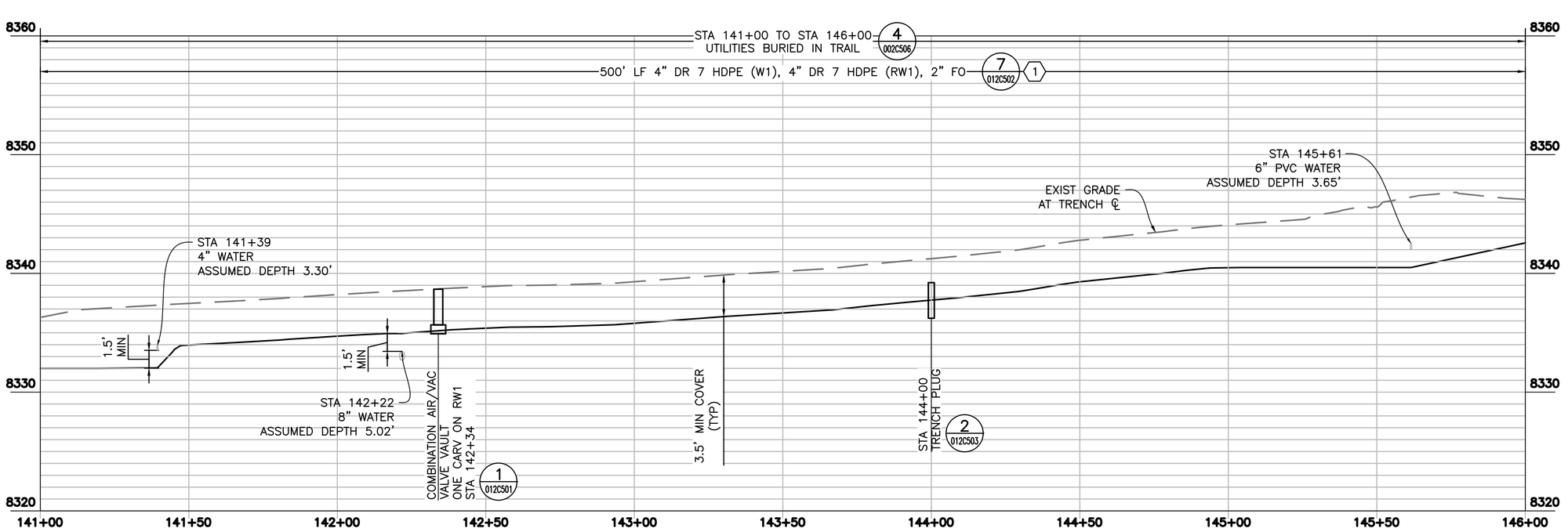
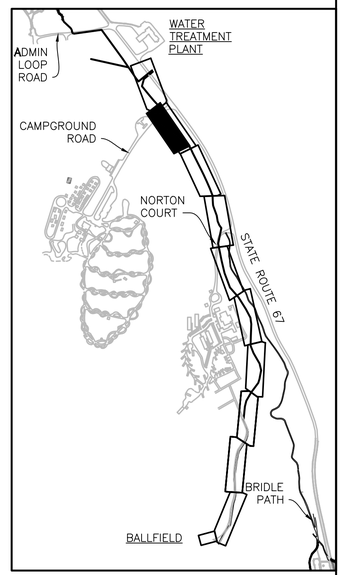




KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	012 C140	UTILITIES BORE TO WTP UTILITIES PLAN AND PROFILE STA 141+00 TO 146+00 NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
				PMIS/PKG NO. 318719
				SHEET 54 OF 87

8/18/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C138.dwg

STA 148+63
START BEND
N 13150244.55
E 1328100.55

STA 149+29
START BEND
N 13150307.08
E 1328106.38

COMBINATION
AIR/VAC VALVE
N 13150315.06
E 1328107.15

AIR VENT
N 13150317.28
E 1328110.13

3
012C501

V1PH16AZS

2 BRIDLE PATH

STA 148+85
END BEND
N 13150264.79
E 1328095.73

STA 146+29
START BEND
N 13150067.24
E 1328254.24

STA 146+36
END BEND
N 13150072.52
E 1328249.77

STA 149+43
END BEND
N 13150318.96
E 1328113.90

187 SF ASPHALT
REMOVAL AND
REPLACEMENT

201 SF ASPHALT
REMOVAL AND REPLACEMENT

END STA 150+59
SEE VOLUME 6
FOR CONTINUATION

SEE VOLUME 6 FOR
W1 CONTINUATION
N 13150391.777
E 1328203.668

SEE VOLUME 6 FOR
RW1 CONTINUATION
N 13150392.938
E 1328202.719

MATCHLINE STA 146+00 SEE 012C139

146+00

LIMITS OF CONSTRUCTION (TYP) 4

15'

25'

R=22.5'

R=22.5'

RW1,W1,FO

RW1,W1

UE,FO

SS

FO

8348

8346

8352

8350

8354 (UE)

KEYNOTES: (X)

1. INSTALL 144–STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7–WAY MICRODUCT. MAINTAIN A MINIMUM 24" COVER.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE–USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION
3. EXISTING SNOW MARKERS NEED TO BE PROTECTED OR IF REMOVED, PUT BACK IN EXACT SAME LOCATION.
4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON WATERLINE ALIGNMENT LOOKING UPSTATION.

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND HDPE DR 7 AS SHOWN ON PLANS. BEND RADIUS TO BE A MINIMUM OF 10 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

D. NORTHINGS AND EASTINGS SHOWN FOR SHARED UTILITY ALIGNMENTS REFLECT TRENCH CENTERLINE.

E. CONTRACTORS SHALL MAINTAIN A MINIMUM OF ONE LANE OF ACCESS PER SPECIFICATION OF 01 50 00

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
146+00	148+65	15 FT	25 FT
148+65	150+30	SEE SHEET 012C157	22 FT

1 002C506 STA 150+06 TO STA 150+33 UTILITIES BURIED IN ASPHALT ROAD

STA 150+33 TO STA 150+70 UTILITIES BURIED

STA 146+00 TO STA 151+00 UTILITIES BURIED IN TRAIL 4 002C506

275' LF 4" DR 7 HDPE (W1), 4" DR 7 HDPE (RW1), 2" FO 7 012C502 1

183' LF 4" DR 7 HDPE (W1), 4" DR 7 HDPE (RW1) 6 012C502

STA 148+15
3" HDPE ELEC
TOP = 8344.72

EXIST GRADE
AT TRENCH ℄

1.5' MIN

SEE VOL 6 FOR
TIE IN ELEVATION

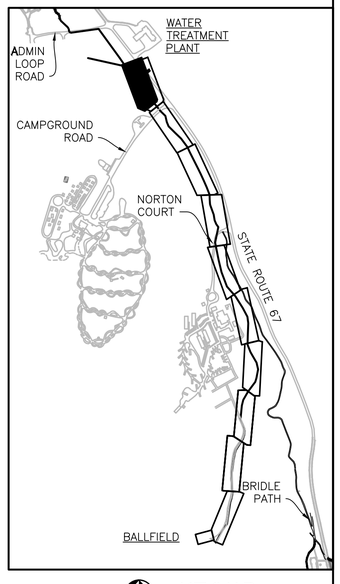
COMBINATION AIR/VAC VALVE VAULT ONE CARV ON RW1 STA 149+32

1 012C501

3.5' MIN COVER (TYP)

8370
8360
8350
8340
8330

146+00 146+50 147+00 147+50 148+00 148+50 149+00 149+50 150+00 150+50 151+00



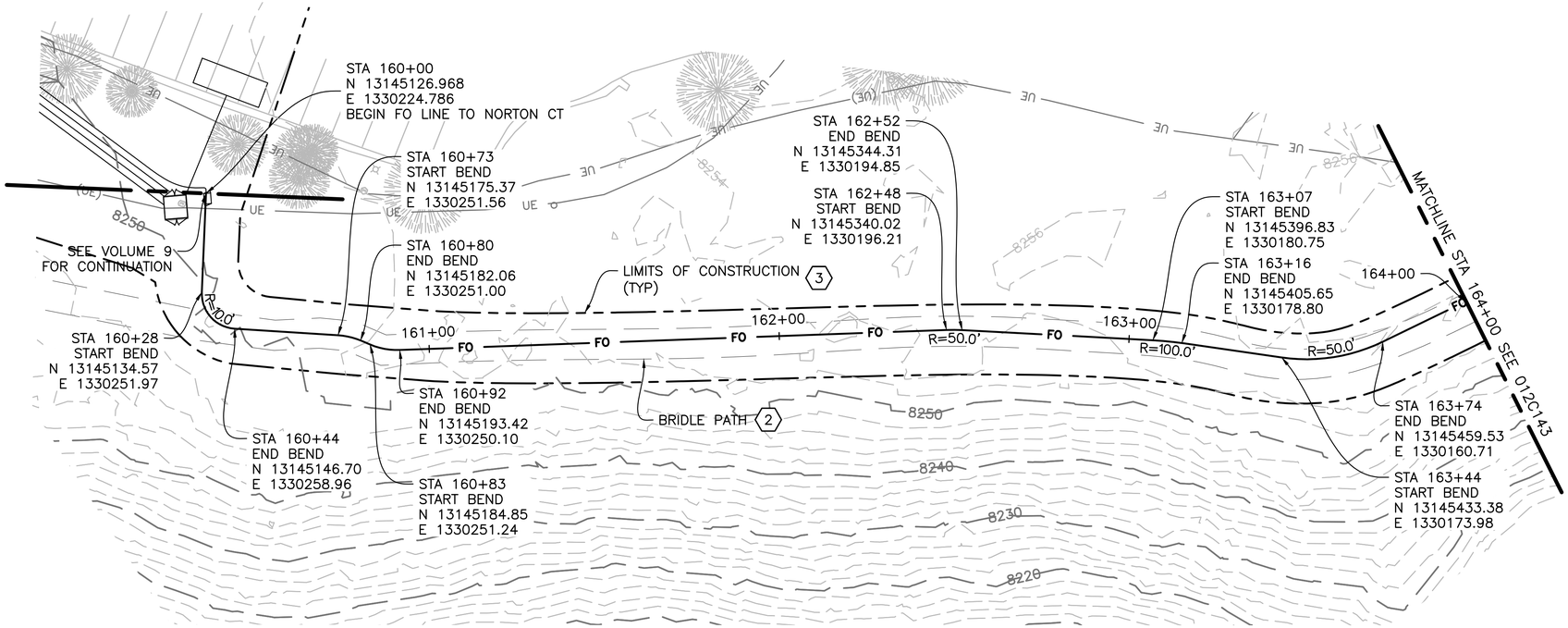
KEYMAP

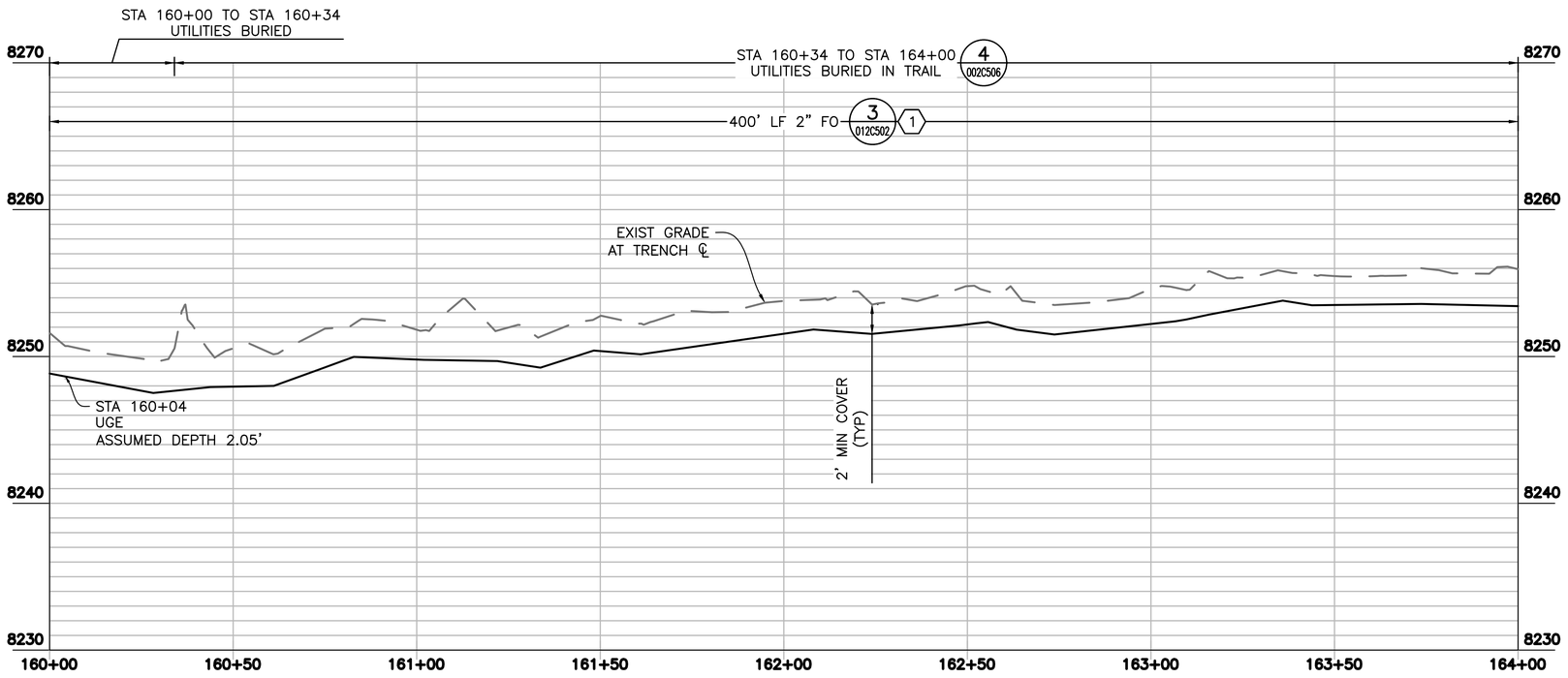
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	SUB SHEET NO. 012 C141	TITLE OF SHEET UTILITIES BORE TO WTP UTILITIES PLAN AND PROFILE STA 146+00 TO 151+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 55 OF 87

8/18/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C138.dwg





GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

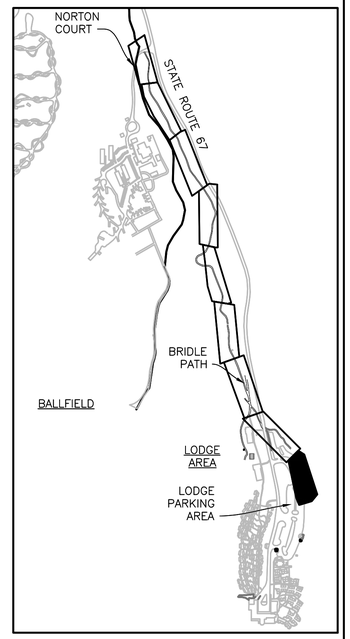
B. BEND FO WITH A MINIMUM RADIUS OF 10 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
3. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTHS ARE LEFT AND RIGHT BASED ON UTILITY ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
160+00	162+00	9 FT	11 FT
162+00	164+00	8 FT	12 FT



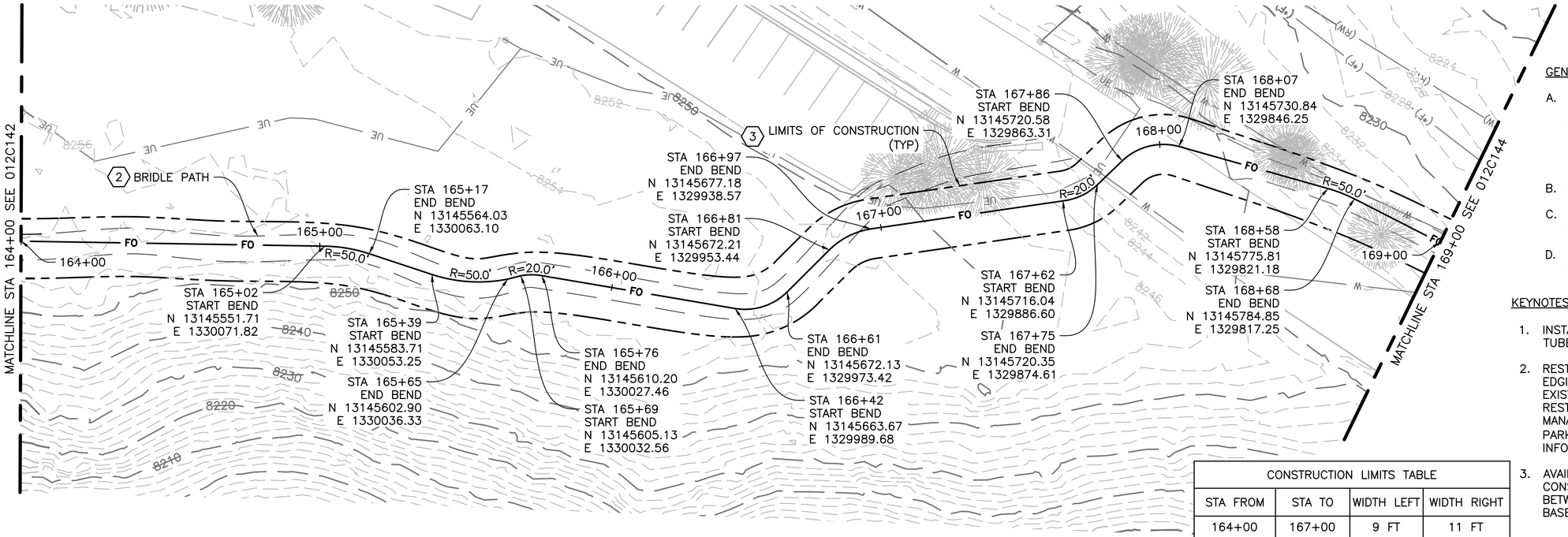
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C142	TITLE OF SHEET UTILITIES LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 159+50 TO 164+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 56 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C142.dwg



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND FO WITH A MINIMUM RADIUS OF 10 FT.

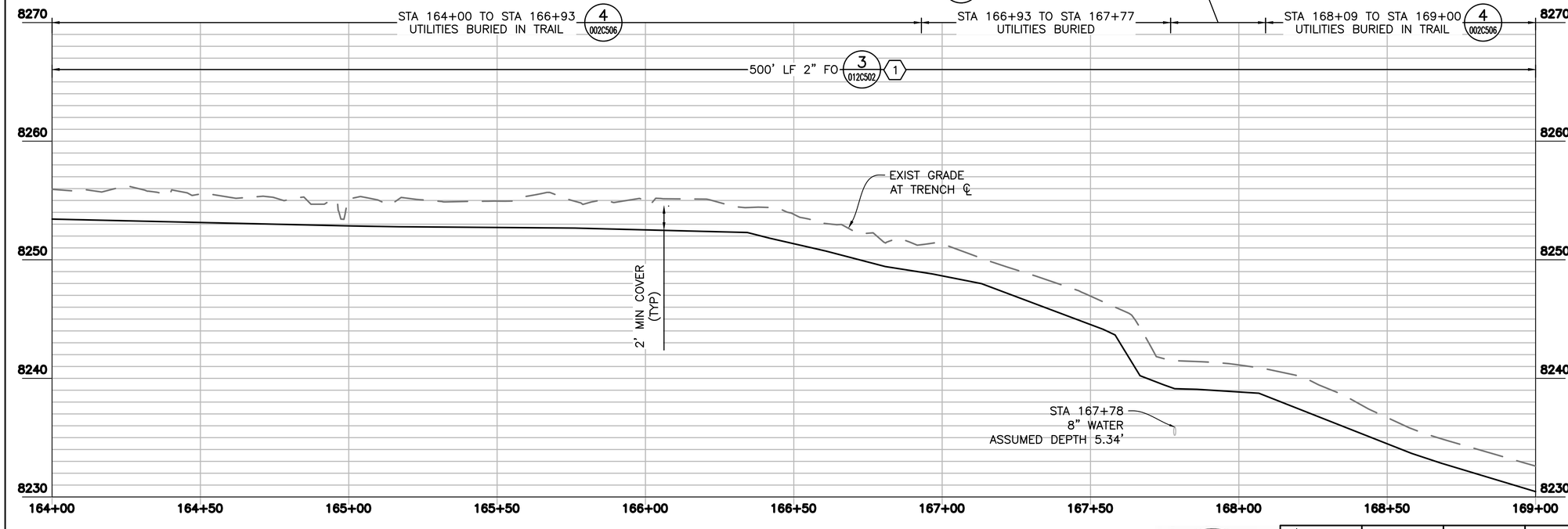
C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

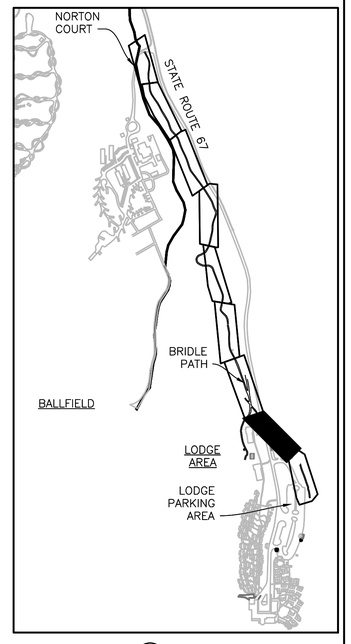
D. CONTRACTORS SHALL MAINTAIN A MINIMUM OF ONE LANE OF ACCESS PER SPECIFICATION OF 01 50 00

KEYNOTES: (X)

1. INSTALL 144–STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7–WAY MICRODUCT.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE–USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
3. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTHS ARE LEFT AND RIGHT BASED ON UTILITY ALIGNMENT LOOKING UPSTATION.

CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
164+00	167+00	9 FT	11 FT
167+00	169+00	10 FT	10 FT





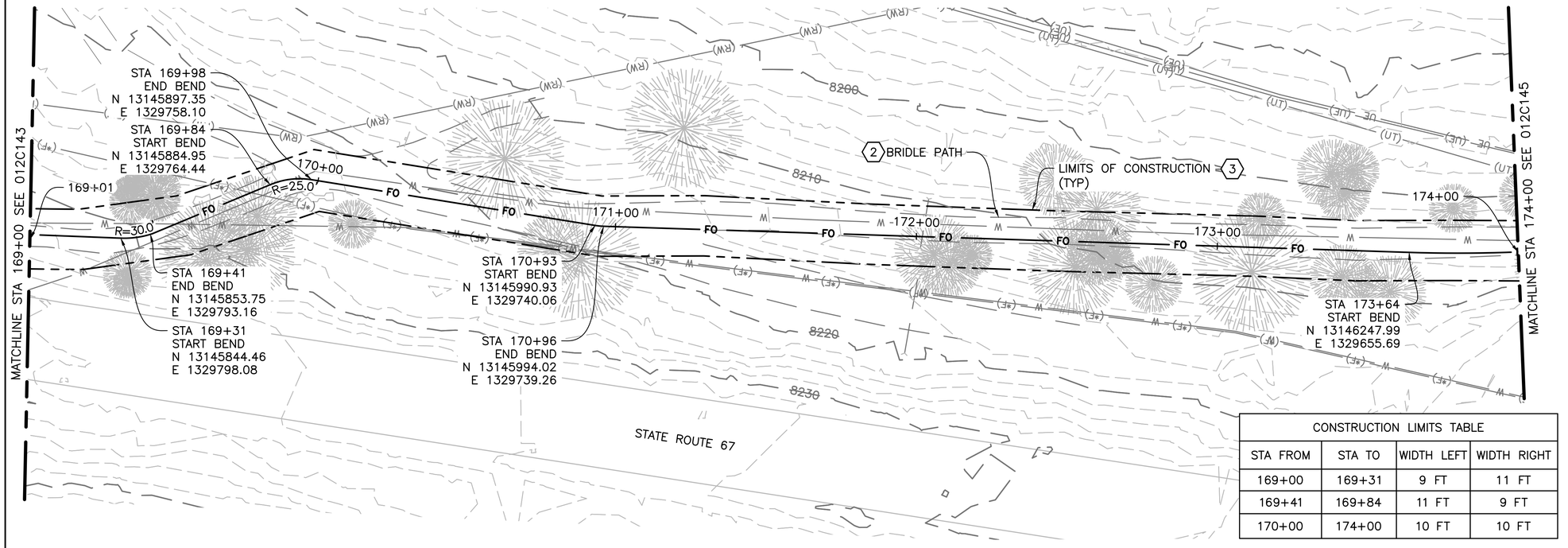
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

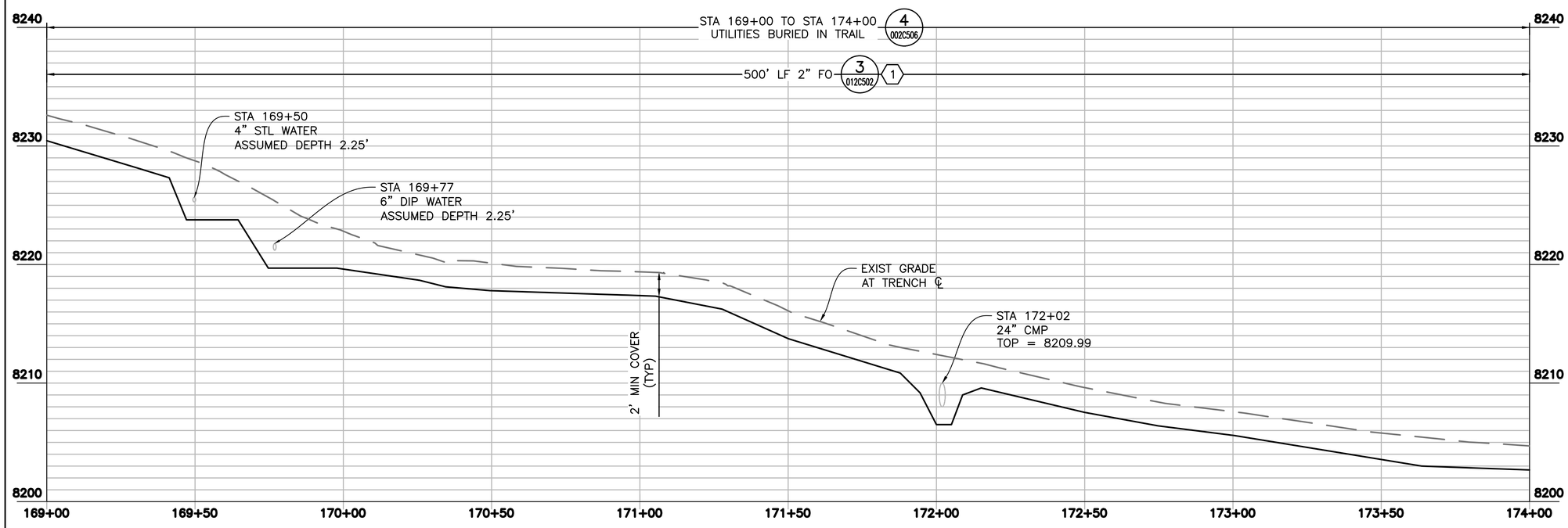
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	SUB SHEET NO. 012 C143	TITLE OF SHEET UTILITIES LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 164+00 TO 169+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 57 OF 87

8/18/2026 Green, Jake C:\Users\jgreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWING\CIVIL\012-C142.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
169+00	169+31	9 FT	11 FT
169+41	169+84	11 FT	9 FT
170+00	174+00	10 FT	10 FT



GENERAL NOTES:

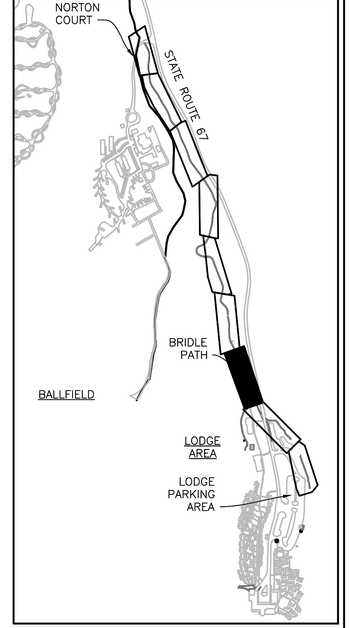
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND FO WITH A MINIMUM RADIUS OF 10 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
3. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTHS ARE LEFT AND RIGHT BASED ON UTILITY ALIGNMENT LOOKING UPSTATION.



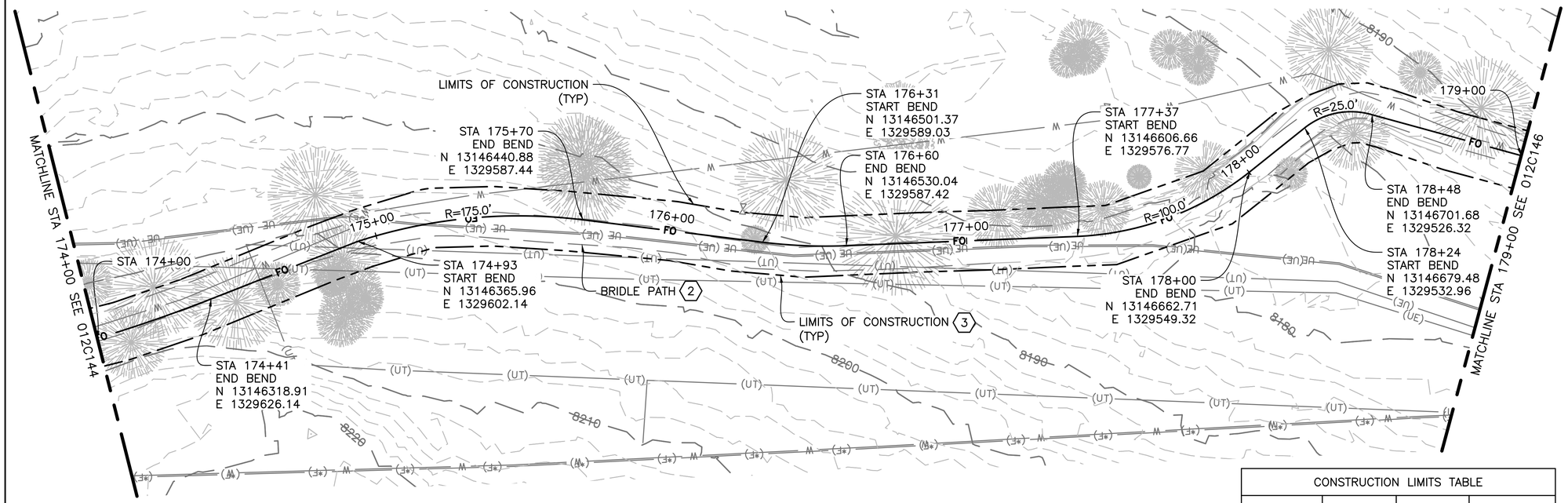
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

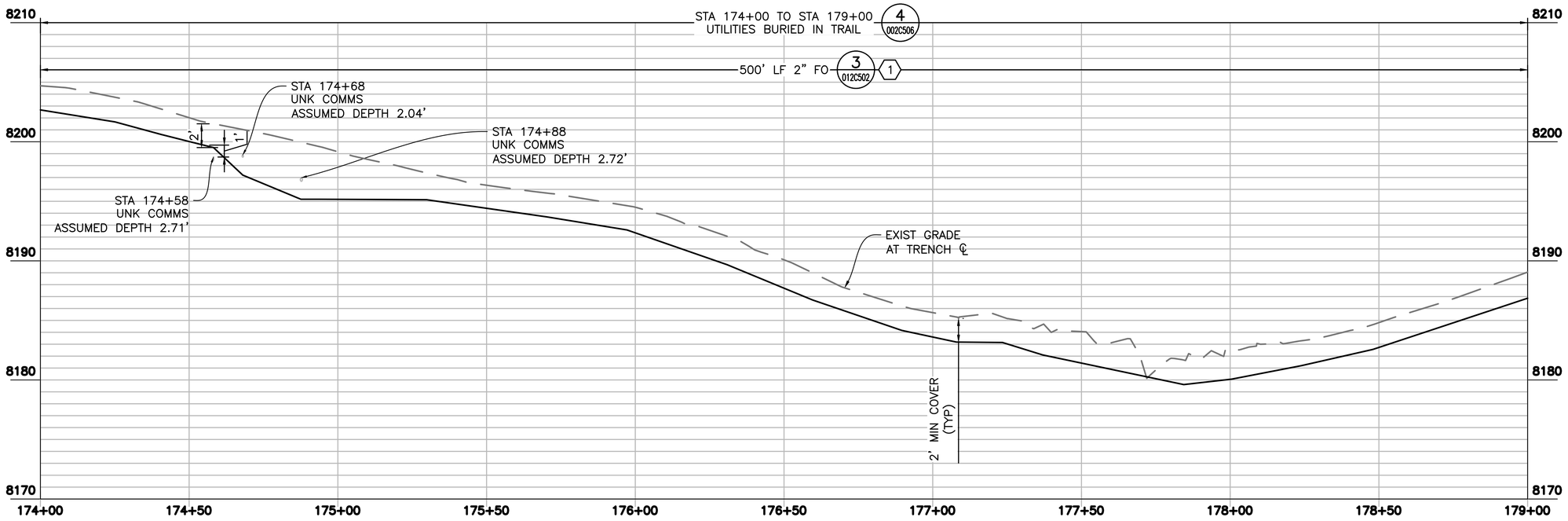
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C144	TITLE OF SHEET UTILITIES LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 169+00 TO 174+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 58 OF 87

4/17/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C142.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
174+00	179+00	10 FT	10 FT



GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND FO WITH A MINIMUM RADIUS OF 10 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
3. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTHS ARE LEFT AND RIGHT BASED ON UTILITY ALIGNMENT LOOKING UPSTATION.

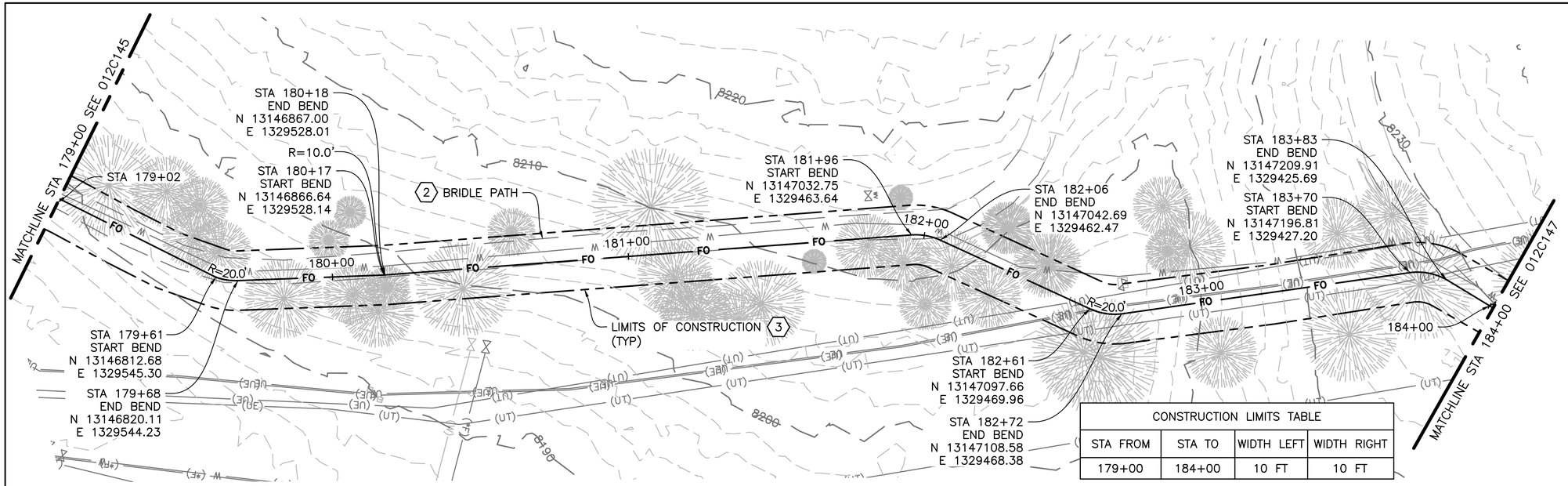
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

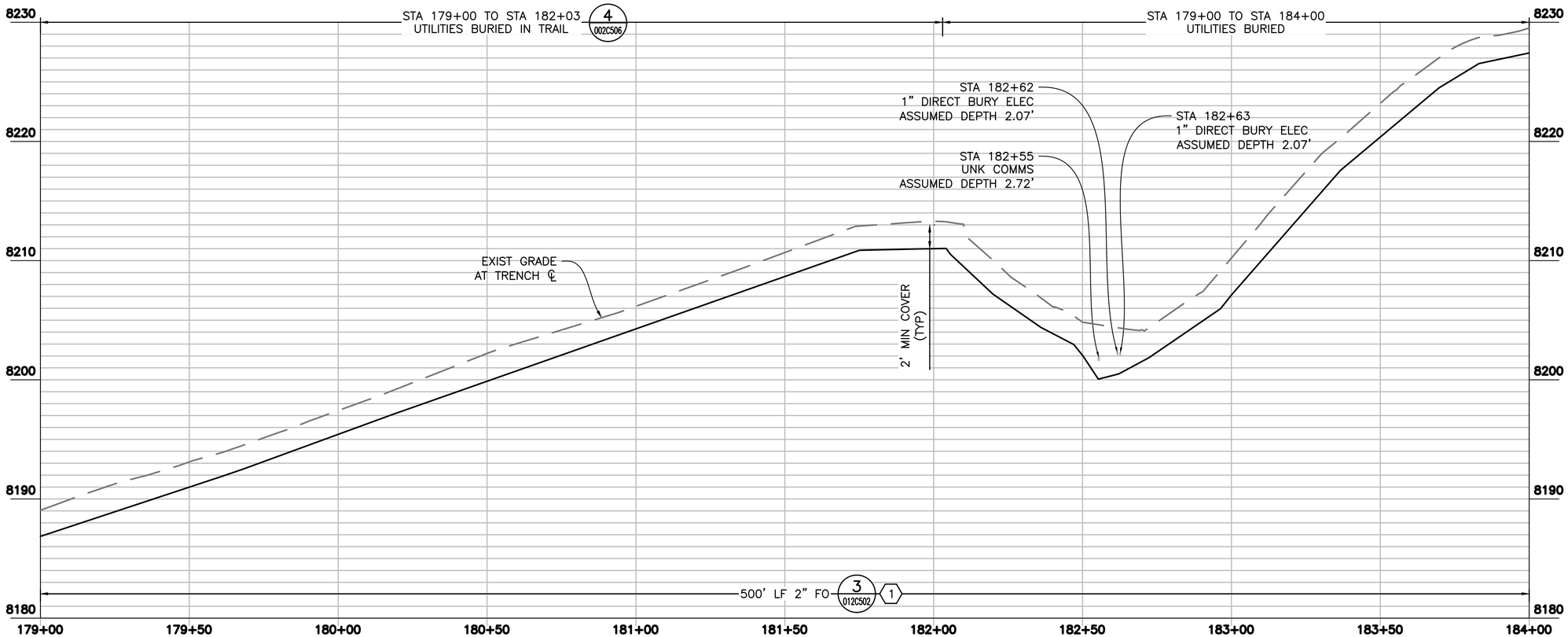
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C145	TITLE OF SHEET UTILITIES LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 174+00 TO 179+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 59 OF 87

4/17/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C142.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
179+00	184+00	10 FT	10 FT



GENERAL NOTES:

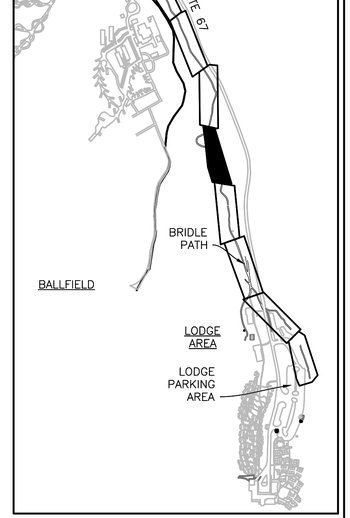
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND FO WITH A MINIMUM RADIUS OF 10 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
3. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTHS ARE LEFT AND RIGHT BASED ON UTILITY ALIGNMENT LOOKING UPSTATION.



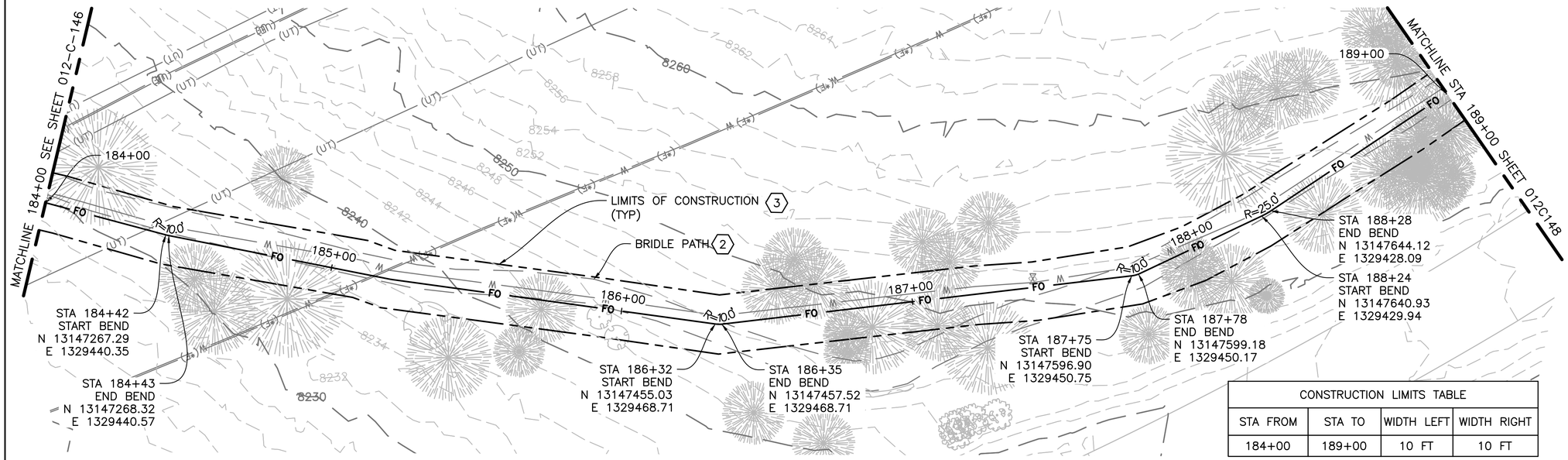
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	012 C146	UTILITIES LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 179+00 TO 184+00 NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
				PMIS/PKG NO. 318719
				SHEET 60 OF 87

4/17/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C142.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
184+00	189+00	10 FT	10 FT

GENERAL NOTES:

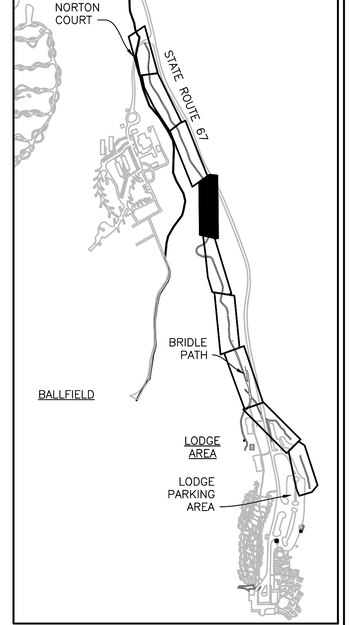
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND FO WITH A MINIMUM RADIUS OF 10 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
3. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTHS ARE LEFT AND RIGHT BASED ON UTILITY ALIGNMENT LOOKING UPSTATION.



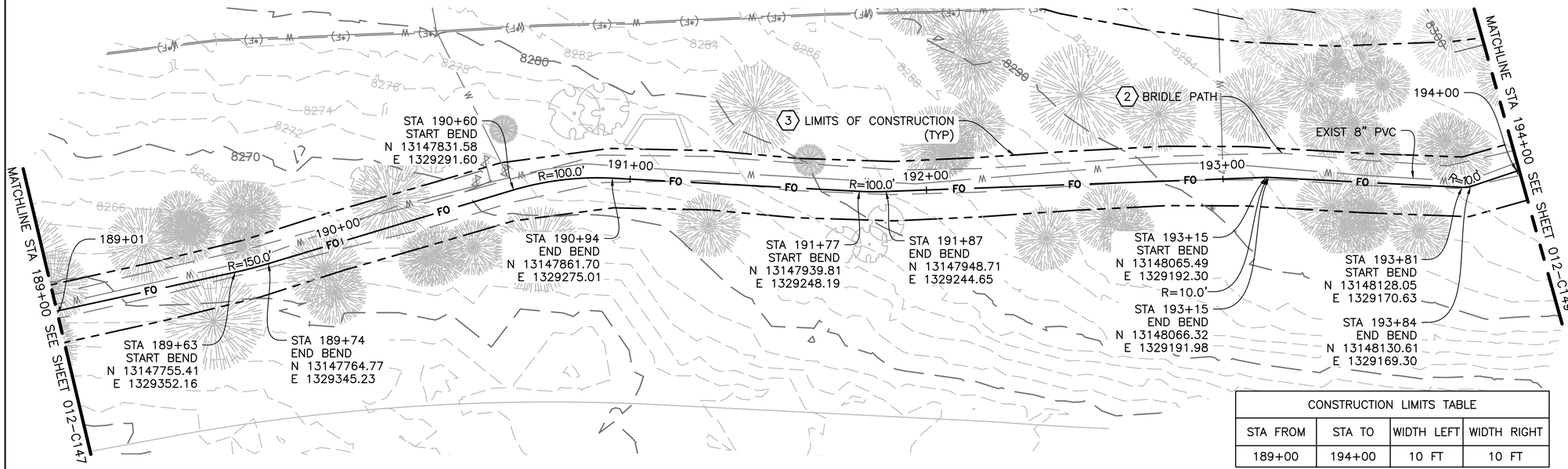
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	012 C147	UTILITIES LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 184+00 TO 189+00 NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 61 OF 87

4/17/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C142.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
189+00	194+00	10 FT	10 FT

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

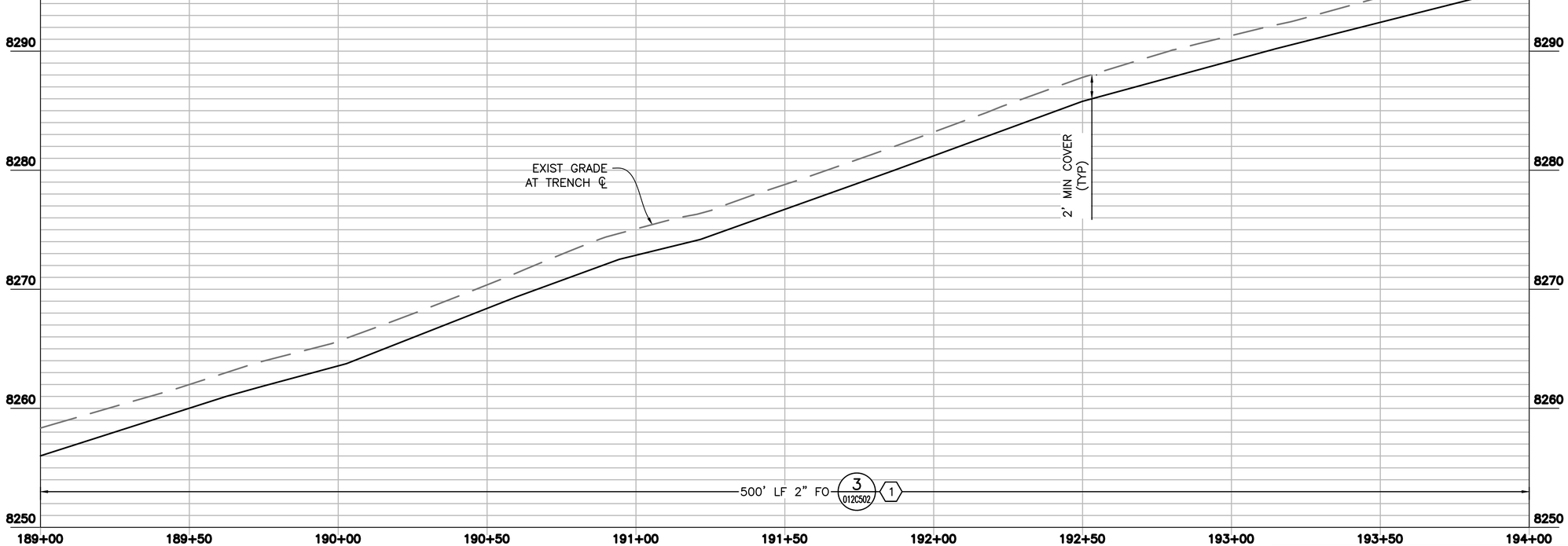
B. BEND FO WITH A MINIMUM RADIUS OF 10 FT.

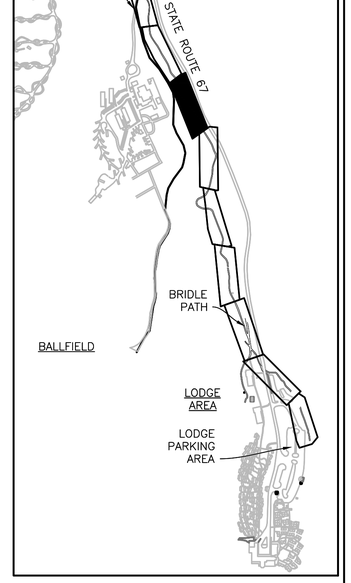
C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
3. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTHS ARE LEFT AND RIGHT BASED ON UTILITY ALIGNMENT LOOKING UPSTATION.

STA 189+00 TO STA 194+00 UTILITIES BURIED IN TRAIL (4 / 002C506)





KEYMAP

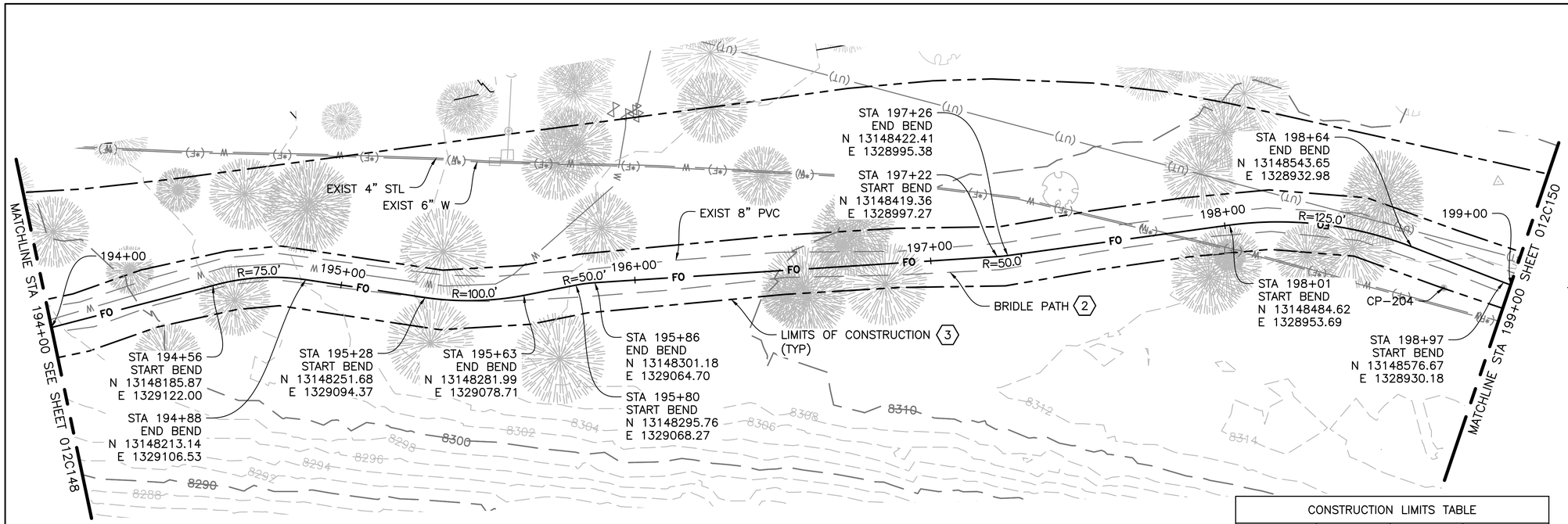
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

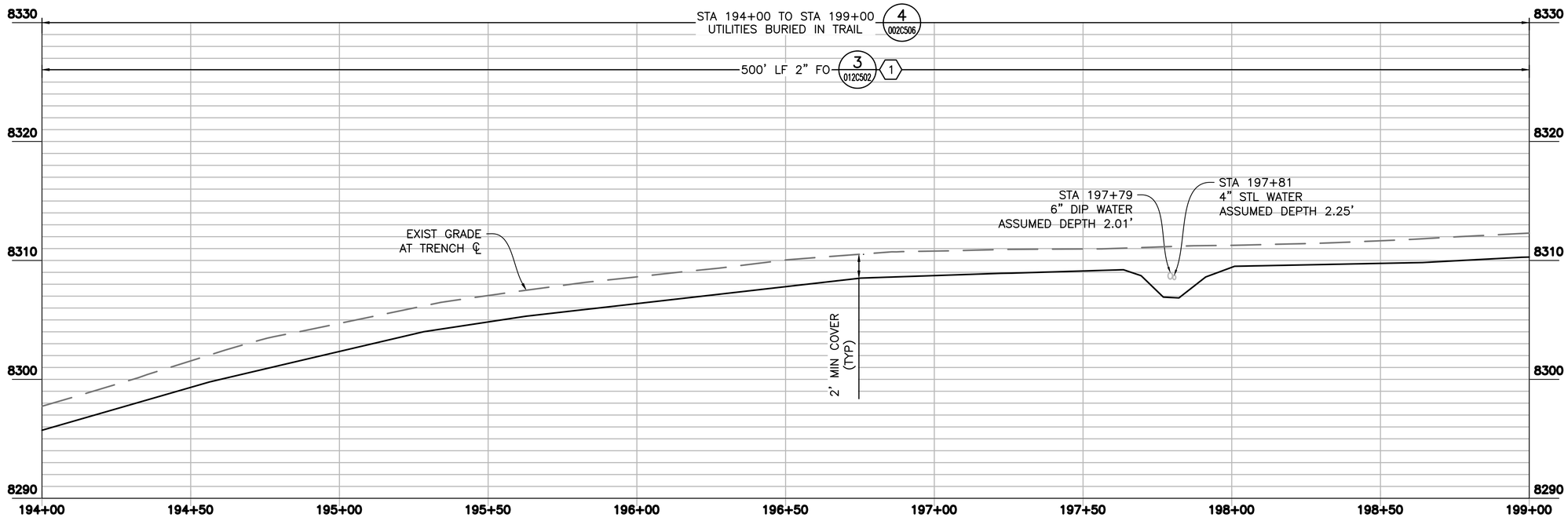
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE	012 C148	UTILITIES LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 189+00 TO 194+00 NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
	ADD J. GREEN			PMIS/PKG NO. 318719
	TECH. REVIEW: GOUGH/LUSTER			SHEET 62 OF 87
	DATE: MAY 2026			

4/17/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C142.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
194+00	199+00	10 FT	10 FT



GENERAL NOTES:

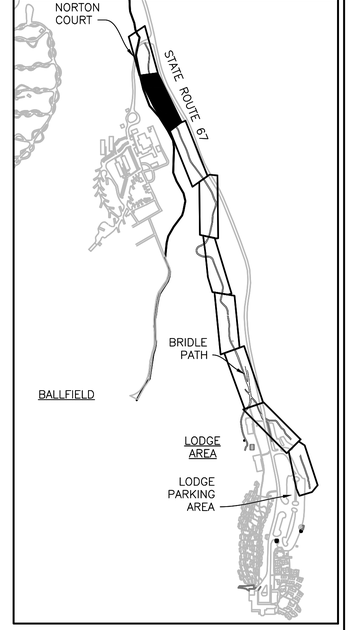
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND FO WITH A MINIMUM RADIUS OF 10 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
3. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTHS ARE LEFT AND RIGHT BASED ON UTILITY ALIGNMENT LOOKING UPSTATION.



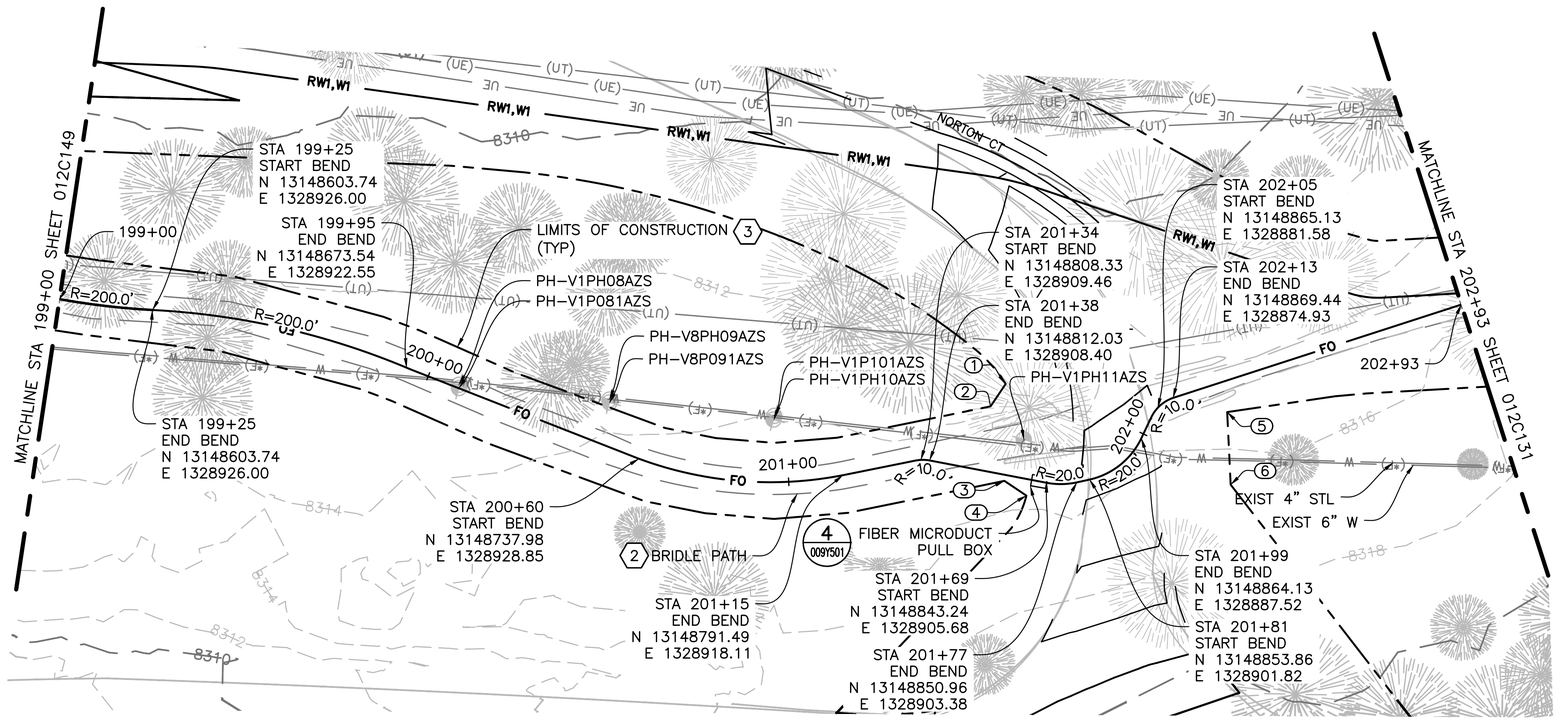
KEYMAP

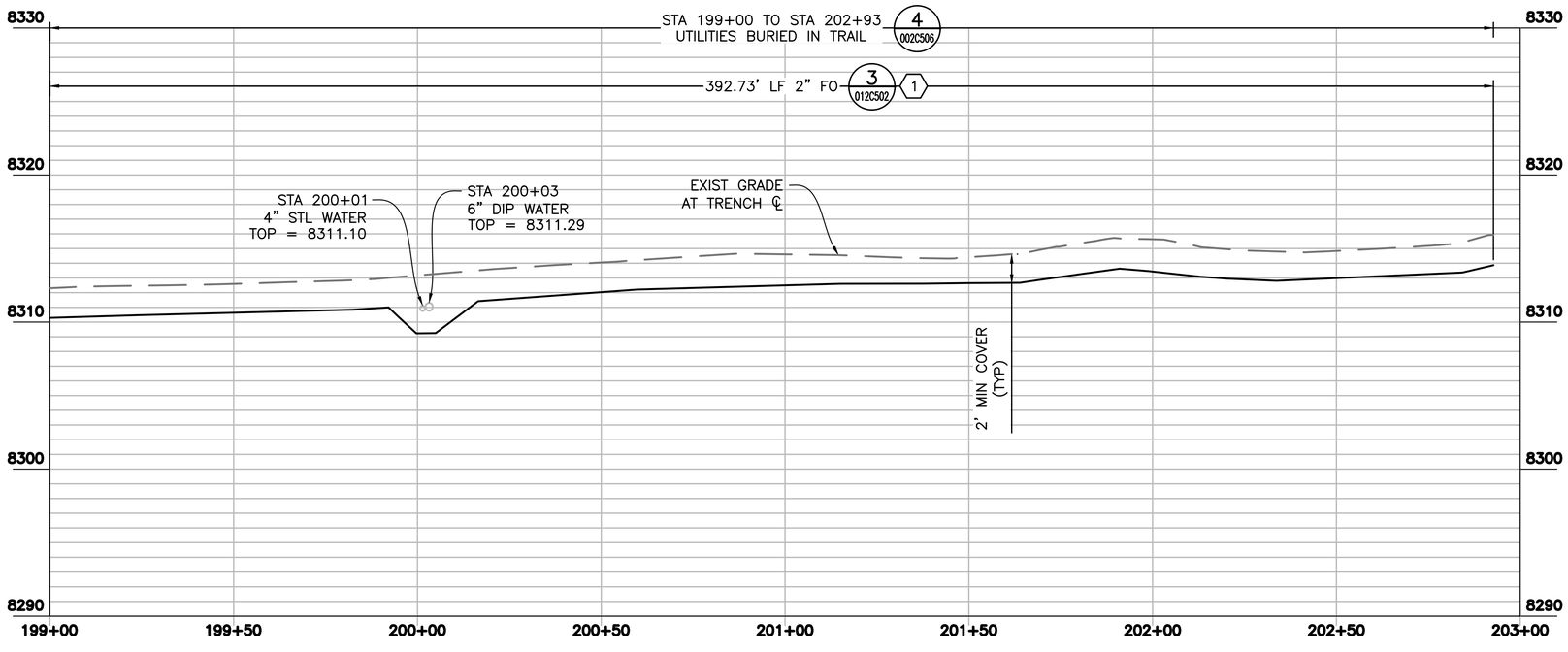
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C149	TITLE OF SHEET UTILITIES LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 184+00 TO 199+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 63 OF 87

4/17/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C142.dwg





GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND FO WITH A MINIMUM RADIUS OF 10 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

D. CONTRACTORS SHALL MAINTAIN A MINIMUM OF ONE LANE OF ACCESS PER SPECIFICATION OF 01 50 00

KEYNOTES: (X)

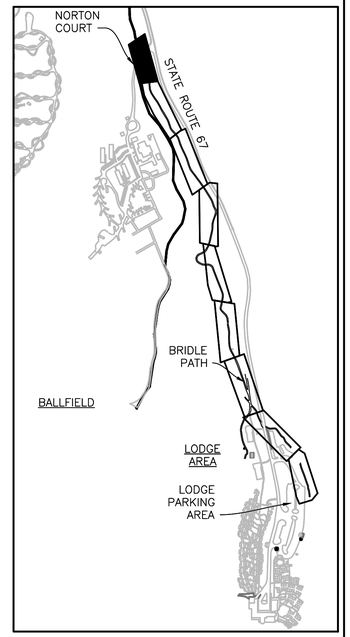
1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT.
2. RESTORE TRAIL AND ALL FEATURES INCLUDING, ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO THE TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
3. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTHS ARE LEFT AND RIGHT BASED ON UTILITY ALIGNMENT LOOKING UPSTATION.

NORTON CT.
CONSTRUCTION LIMITS POINTS

POINT NO.	NORTHING	EASTING
1	13148825.76	1328883.50
2	13148823.54	1328890.28
3	13148832.40	1328908.22
4	13148839.05	1328910.40
5	13148884.05	1328874.75
6	13148890.15	1328892.93

CONSTRUCTION LIMITS TABLE

STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
199+00	201+34	10 FT	10 FT



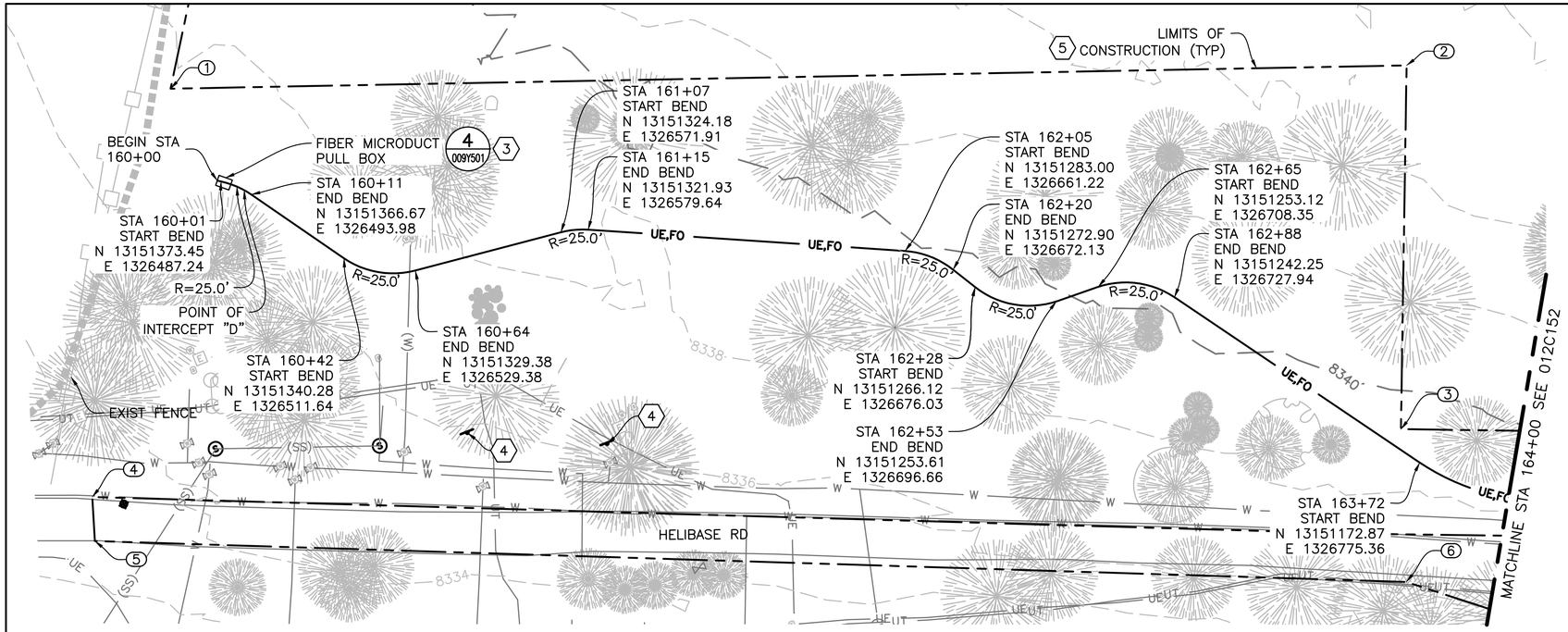
KEYMAP

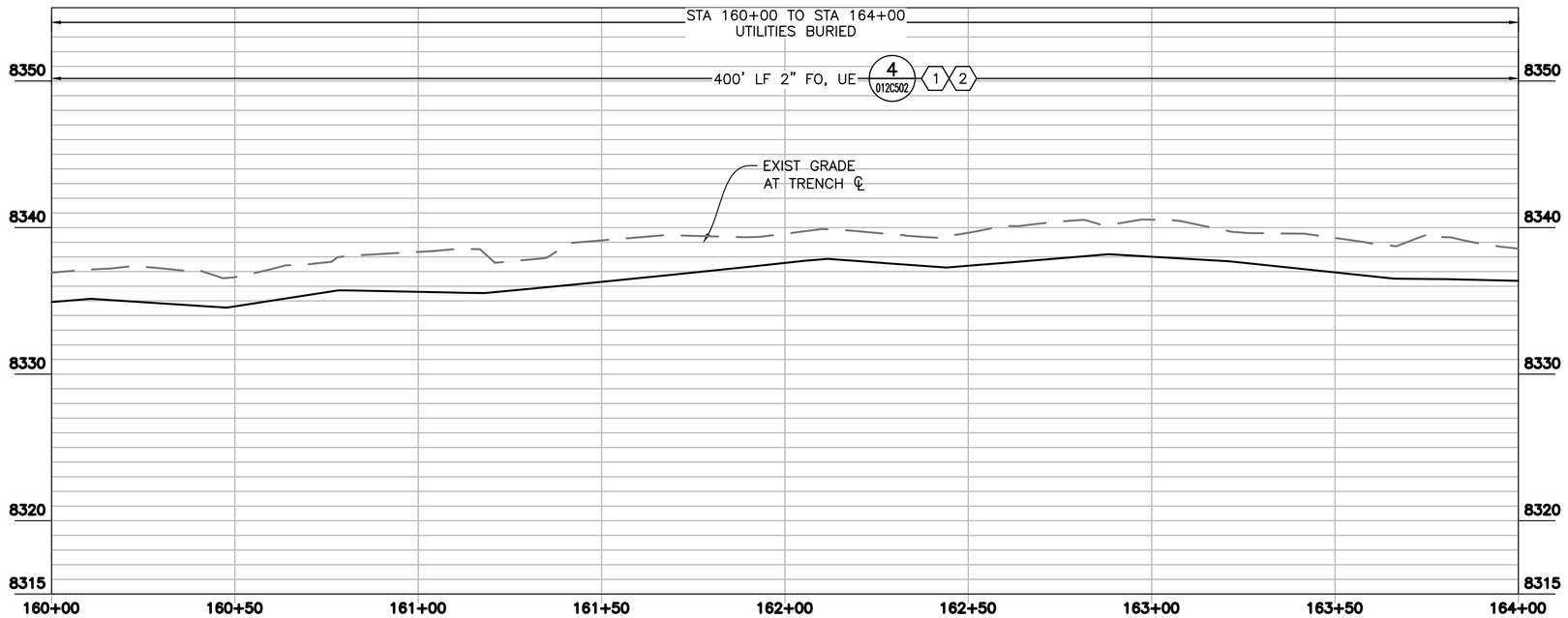
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	SUB SHEET NO. 012 C150	TITLE OF SHEET UTILITIES LODGE TO NORTON CT FIBER PLAN AND PROFILE STA 199+00 TO 202+83 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 64 OF 87

8/18/2026 Green, Jake C:\Users\jgreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C142.dwg





CONSTRUCTION LIMITS POINT TABLE		
POINT NO.	NORTHING	EASTING
1	13151403.33	1326483.64
2	13151279.31	1326813.87
3	13151183.49	1326774.08
4	13151303.36	1326419.89
5	13151291.40	1326415.94
6	13151142.42	1326759.41

GENERAL NOTES:

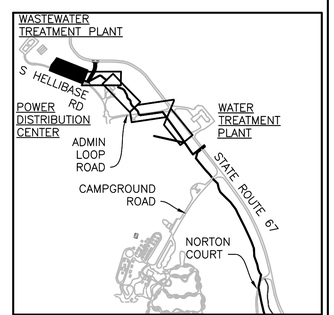
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND UE AND FO WITH A MINIMUM RADIUS OF 20 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 7-WAY MICRODUCT (FO).
2. INSTALL 25kV (3/C) 4/0 AWG ALUMINUM ARMORED POWER CABLE FROM SWITCHGEAR ESB-12 AT THE PDC TO POINT OF INTERCEPT "D" LOCATED OUTSIDE OF WWTP AREA FENCE ADJACENT TO MICRODUCT PULL BOX. STUB AND PROTECT POWER CABLE ENDS IN PLACE FOR FUTURE SPLICE INTO WWTP.
3. TERMINATE/CAP 7-WAY MICRODUCT INNER TUBES.
4. EXISTING SNOW MARKERS NEED TO BE PROTECTED OR IF REMOVED, PUT BACK IN EXACT SAME LOCATION.
5. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS POINT TABLE.



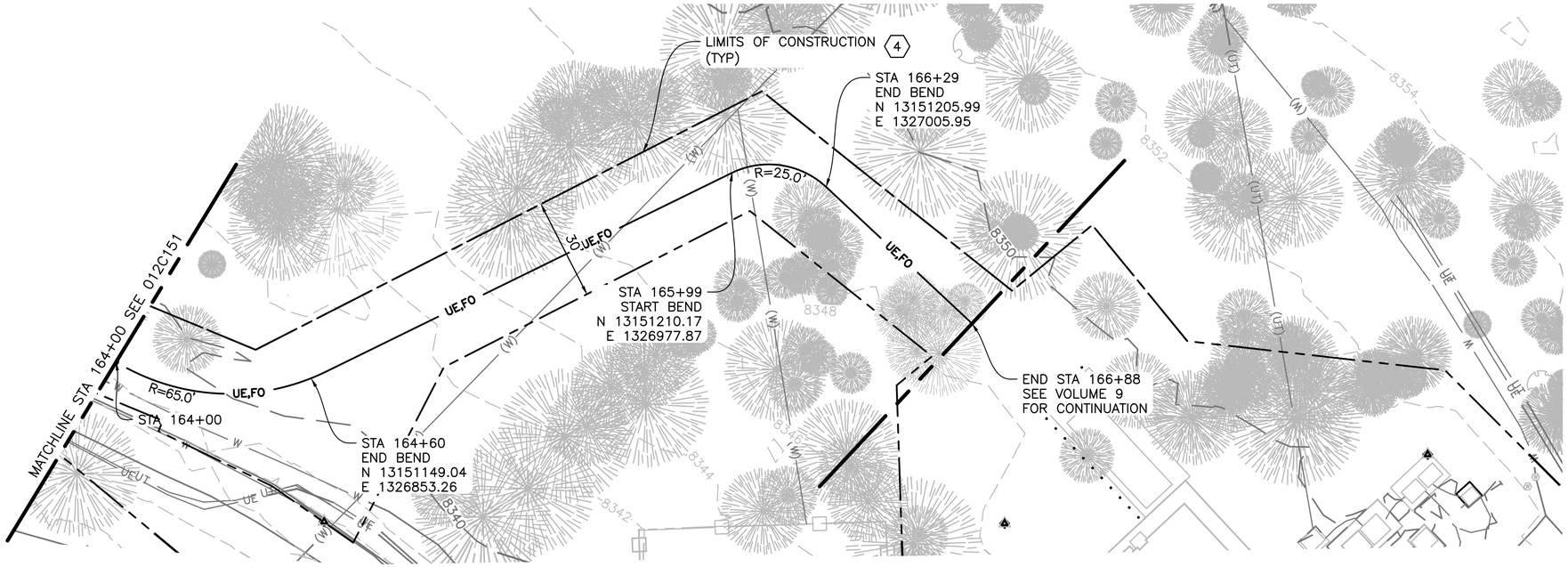
KEYMAP

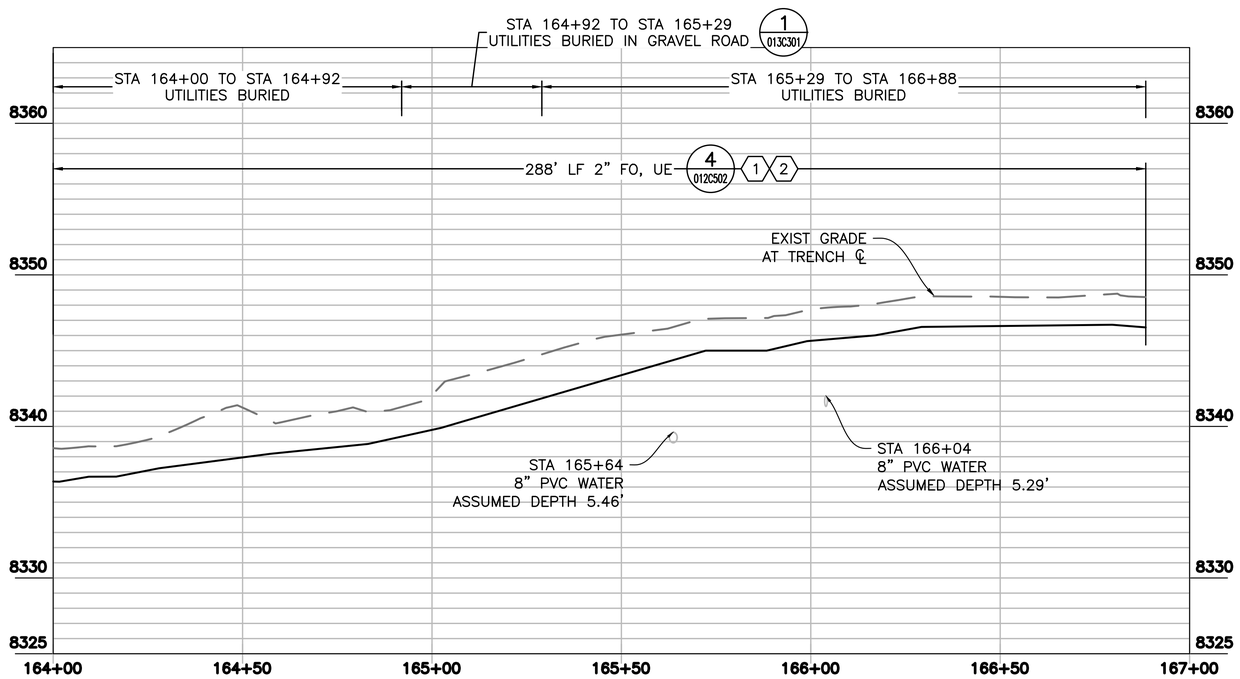
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C151	TITLE OF SHEET UTILITIES WWTP TO PDC UTILITIES PLAN AND PROFILE STA 160+00 TO 164+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 65 OF 87

4/17/2026 Doll, Abby C:\Users\adoll\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C151.dwg





CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
CENTERLINE OF TEMP. CONSTRUCTION ACCESS ROAD		15 FT	15 FT
165+13	165+99	17 FT	13 FT
166+29	166+88	12 FT	18 FT

GENERAL NOTES:

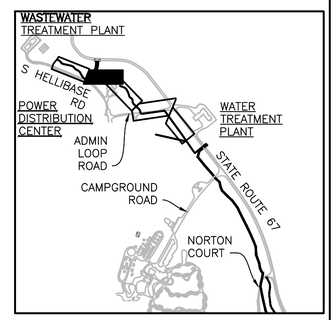
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND UE AND FO WITH A MINIMUM RADIUS OF 20 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 7-WAY MICRODUCT (FO).
2. INSTALL 25kV (3/C) 4/0 AWG ALUMINUM ARMORED POWER CABLE FROM SWITCHGEAR ESB-12 AT THE PDC TO POINT OF INTERCEPT "D" LOCATED OUTSIDE OF WWTP AREA FENCE ADJACENT TO MICRODUCT PULL BOX. STUB AND PROTECT POWER CABLE ENDS IN PLACE FOR FUTURE SPLICE INTO WWTP.
3. TERMINATE/CAP 7-WAY MICRODUCT INNER TUBES.
4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON ALIGNMENT LOOKING UPSTATION.



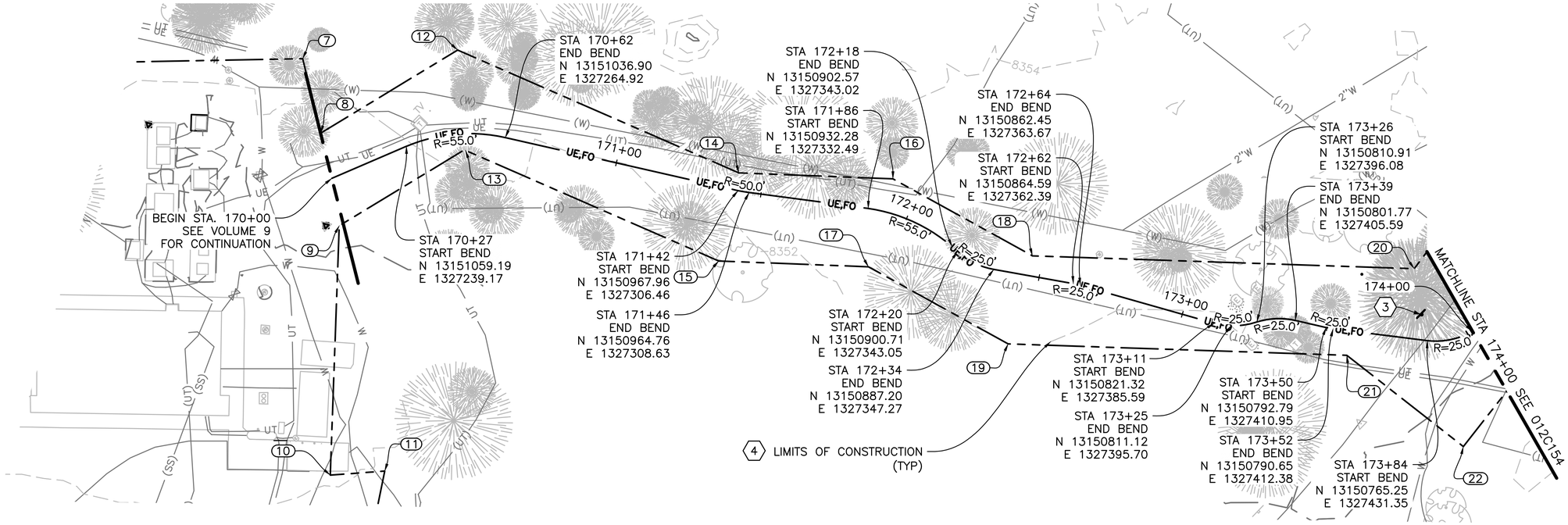
KEYMAP

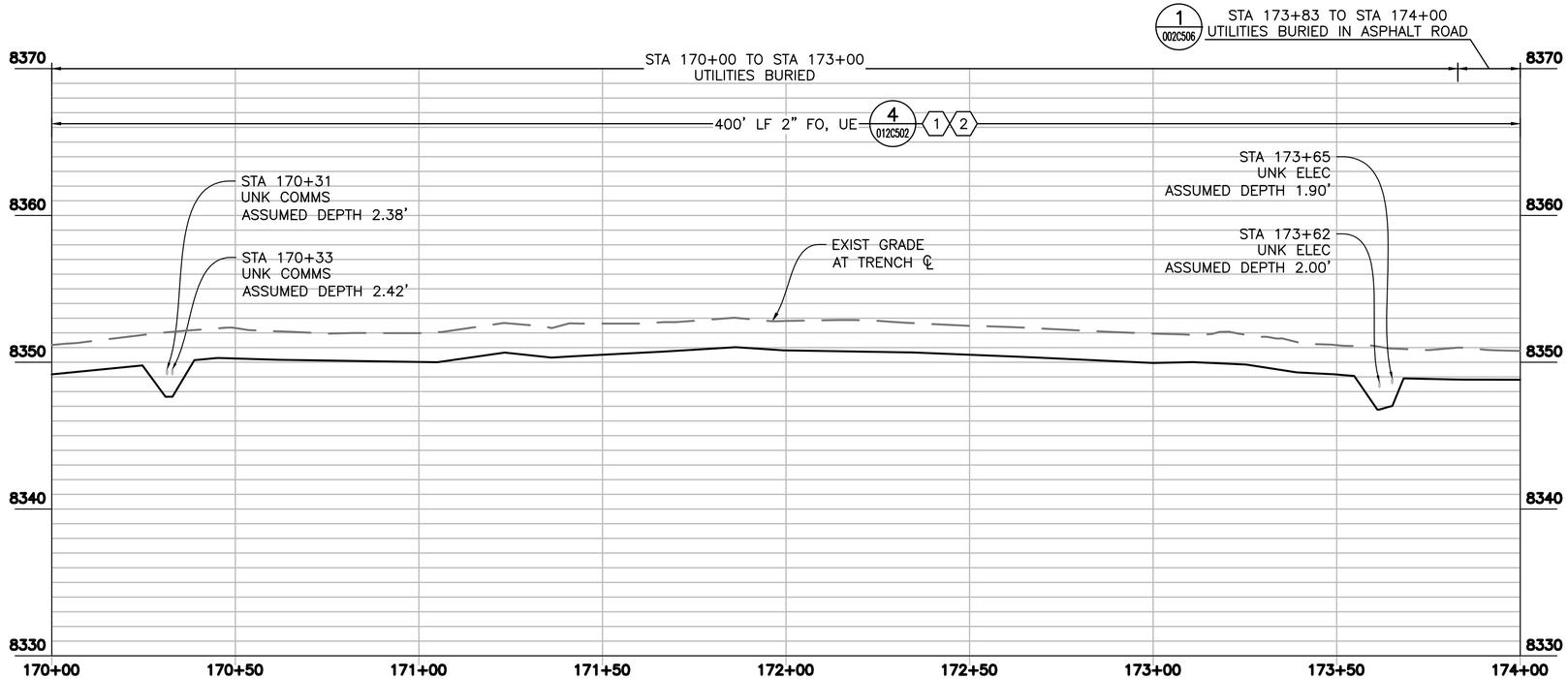
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	SUB SHEET NO. 012 C152	TITLE OF SHEET UTILITIES WWTP TO PDC UTILITIES PLAN AND PROFILE STA 164+00 TO 167+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 66 OF 87

8/18/2026 Green, Jake C:\Users\jgreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C151.dwg





GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND UE AND FO WITH A MINIMUM RADIUS OF 20 FT.

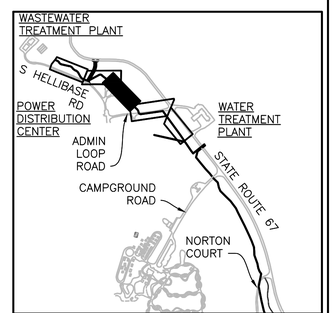
C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 144–STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7–WAY MICRODUCT.
2. INSTALL 25kV (3/C) 4/0 AWG ALUMINUM ARMORED POWER CABLE FROM SWITCHGEAR ESB–12 AT THE PDC TO TRANSFORMER T–49 AT THE WTP.
3. EXISTING SNOW MARKERS NEED TO BE PROTECTED OR IF REMOVED, PUT BACK IN EXACT SAME LOCATION.
4. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS POINT TABLE.

CONSTRUCTION LIMITS POINT TABLE

POINT NO.	NORTHING	EASTING
7	13151104.82	1327236.54
8	13151082.97	1327222.70
9	13151056.76	1327204.48
10	13151000.51	1327141.68
11	13150988.21	1327155.43
12	13151069.12	1327274.98
13	13151043.67	1327252.39
14	13150971.89	1327310.99
15	13150956.16	1327284.82
16	13150932.99	1327345.66
17	13150918.34	1327318.53
18	13150880.99	1327359.61
19	13150866.27	1327332.51
20	13150785.07	1327445.95
21	13150781.16	1327409.11
22	13150731.42	1327414.20



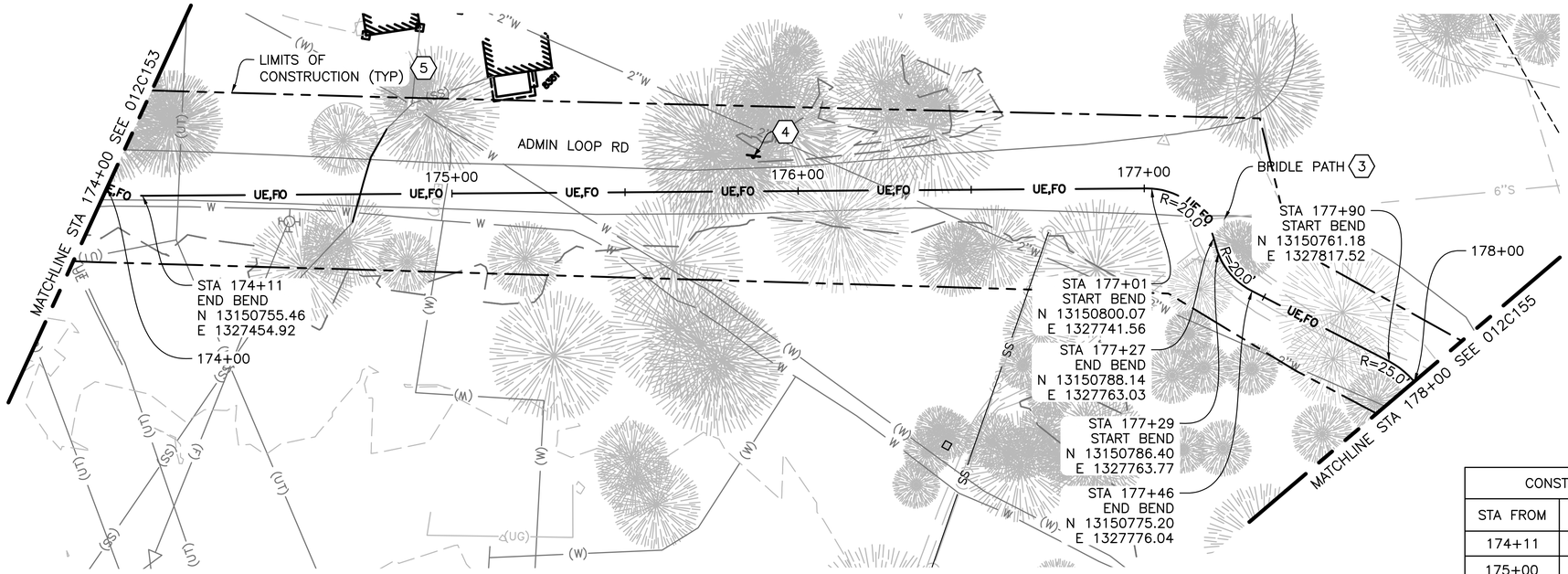
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	012 C153	UTILITIES PDC TO WTP UTILITIES PLAN AND PROFILE STA 170+00 TO 174+00 NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 67 OF 87

4/17/2026 Dall, Abby C:\Users\adall\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C151.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
174+11	175+00	30 FT	20 FT
175+00	176+00	26 FT	24 FT
176+00	177+00	24 FT	26 FT
177+46	178+00	17 FT	13 FT

GENERAL NOTES:

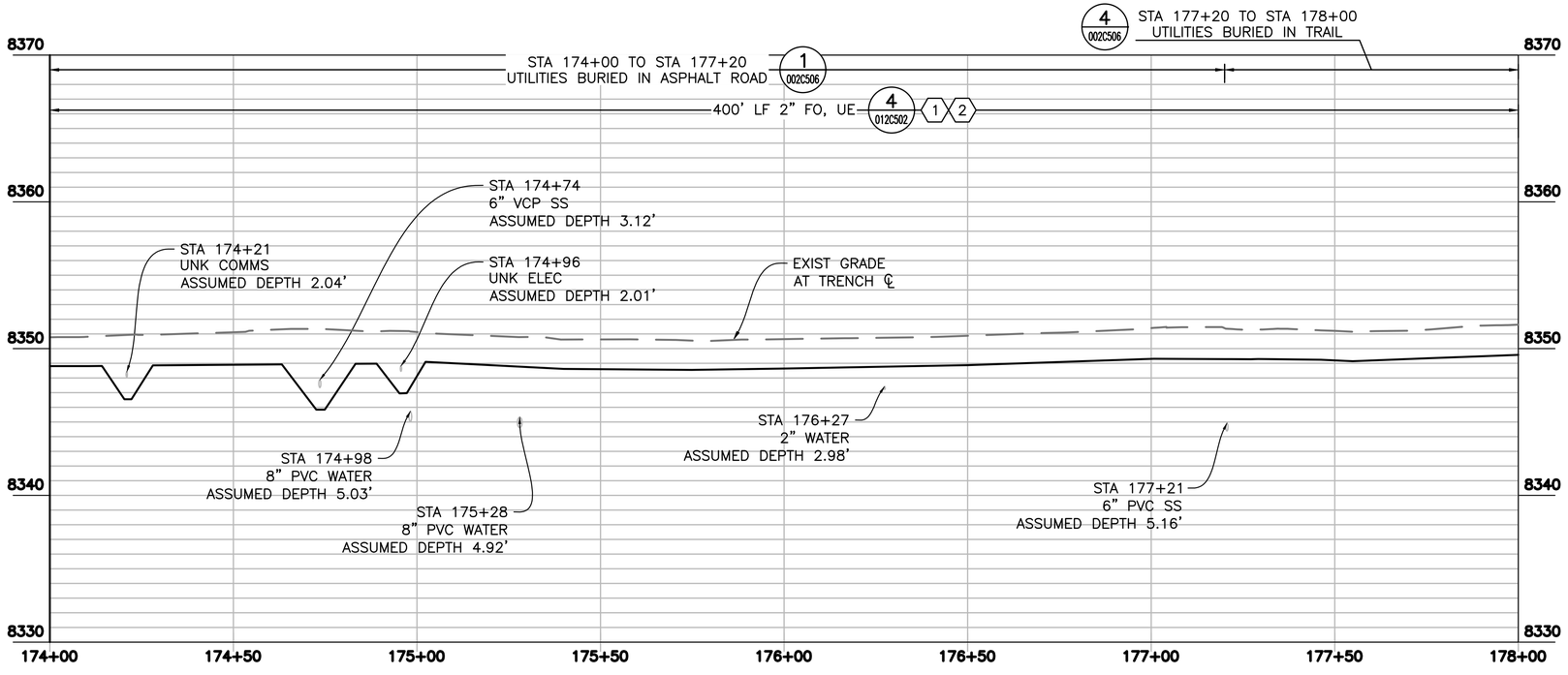
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

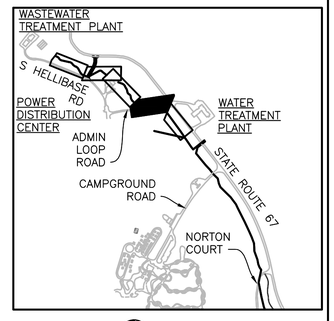
B. BEND UE AND FO WITH A MINIMUM RADIUS OF 20 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OR 7-WAY MICRODUCT.
2. INSTALL 25kV (3/C) 4/0 AWG ALUMINUM ARMORED POWER CABLE FROM SWITCHGEAR ESB-12 AT THE PDC TO TRANSFORMER T-49 AT THE WTP.
3. RESTORE TRAIL AND ALL FEATURES INCLUDING ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
4. EXISTING SNOW MARKERS NEED TO BE PROTECTED OR IF REMOVED, PUT BACK IN EXACT SAME LOCATION.
5. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON ALIGNMENT LOOKING UPSTATION.





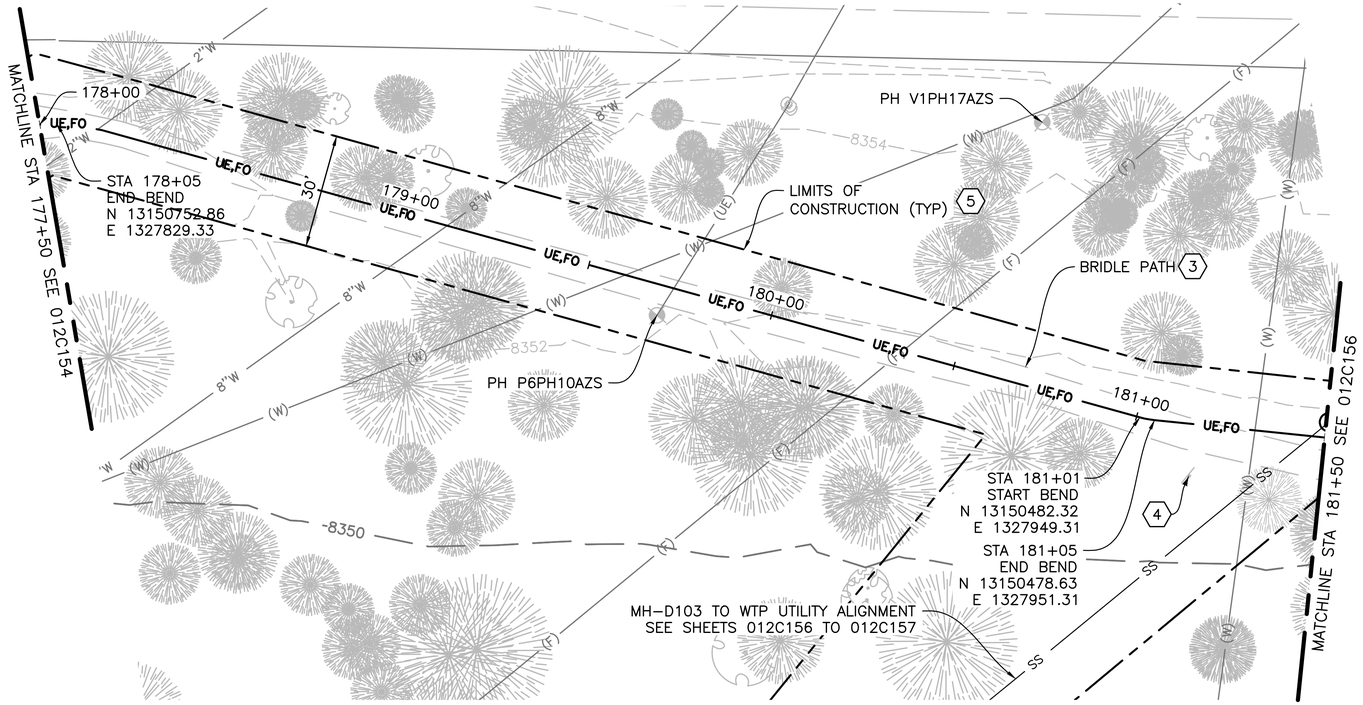
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Not Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	012 C154	UTILITIES PDC TO WTP UTILITIES PLAN AND PROFILE STA 174+00 TO 178+00 NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 68 OF 87

4/17/2026 Dall, Abby C:\Users\adall\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C151.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
178+05	180+71	15 FT	15 FT
180+71	181+50	15 FT	SEE SHEET 012-C156

GENERAL NOTES:

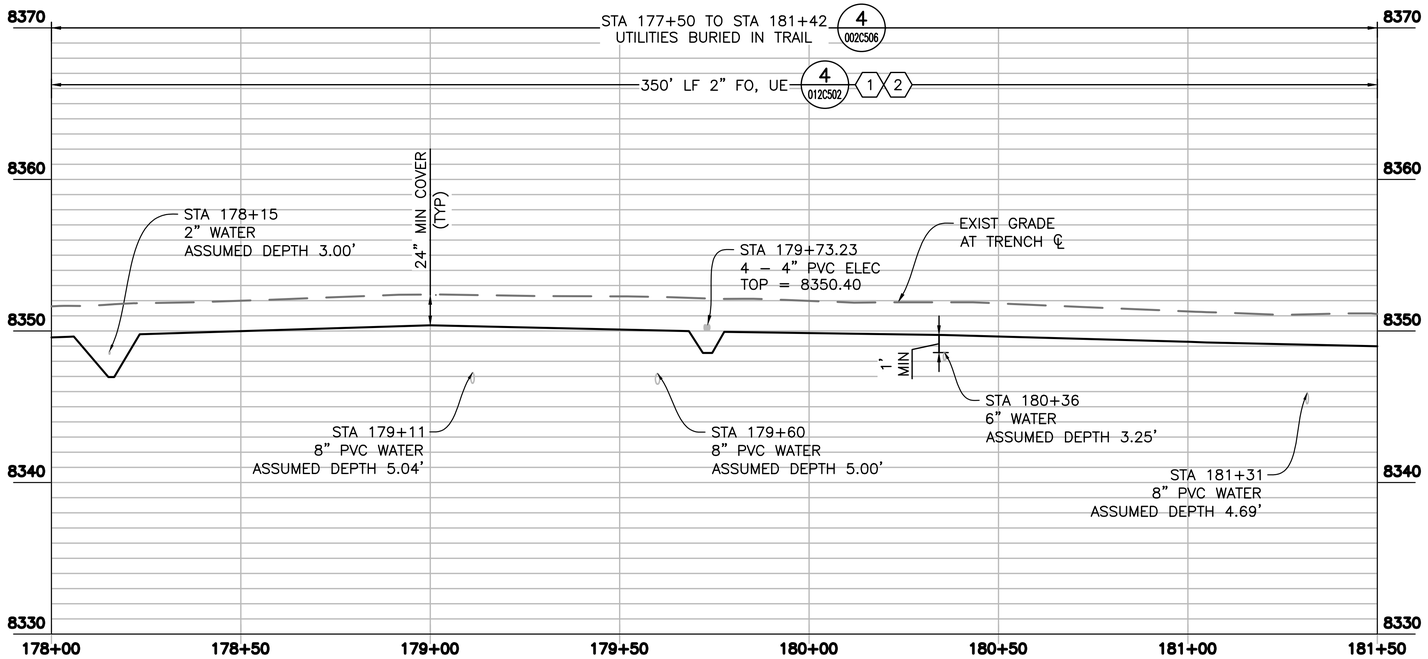
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

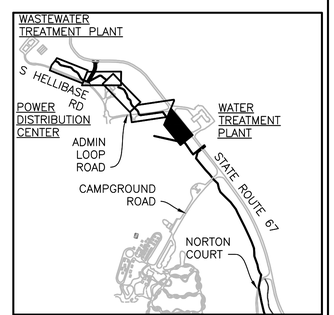
B. BEND UE AND FO WITH A MINIMUM RADIUS OF 20 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OR 7-WAY MICRODUCT.
2. INSTALL 25kV (3/C) 4/0 AWG ALUMINUM ARMORED POWER CABLE FROM SWITCHGEAR ESB-12 AT THE PDC TO TRANSFORMER T-49 AT THE WTP.
3. RESTORE TRAIL AND ALL FEATURES INCLUDING ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
4. EXISTING SNOW MARKERS NEED TO BE PROTECTED OR IF REMOVED, PUT BACK IN EXACT SAME LOCATION.
5. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN STATIONS. WIDTH LEFT AND RIGHT ARE BASED ON ALIGNMENT LOOKING UPSTATION.





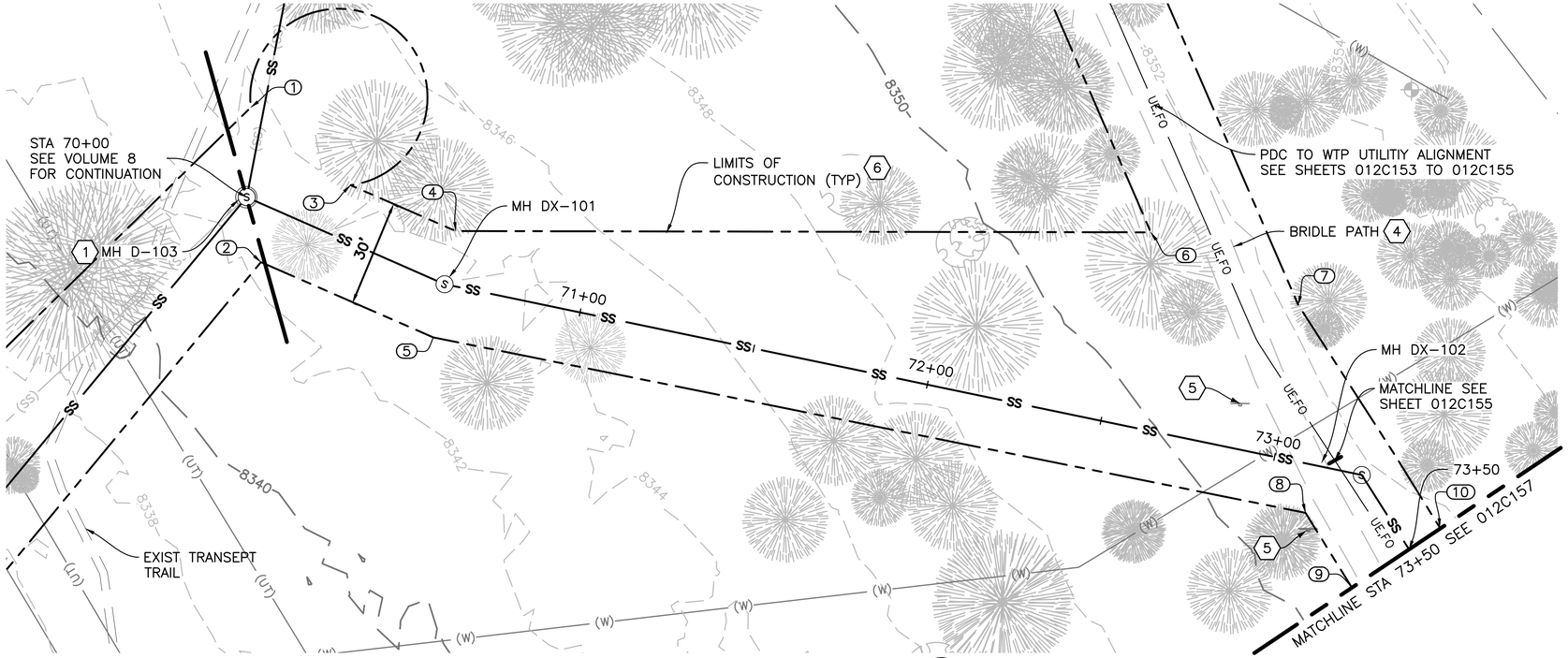
KEYMAP

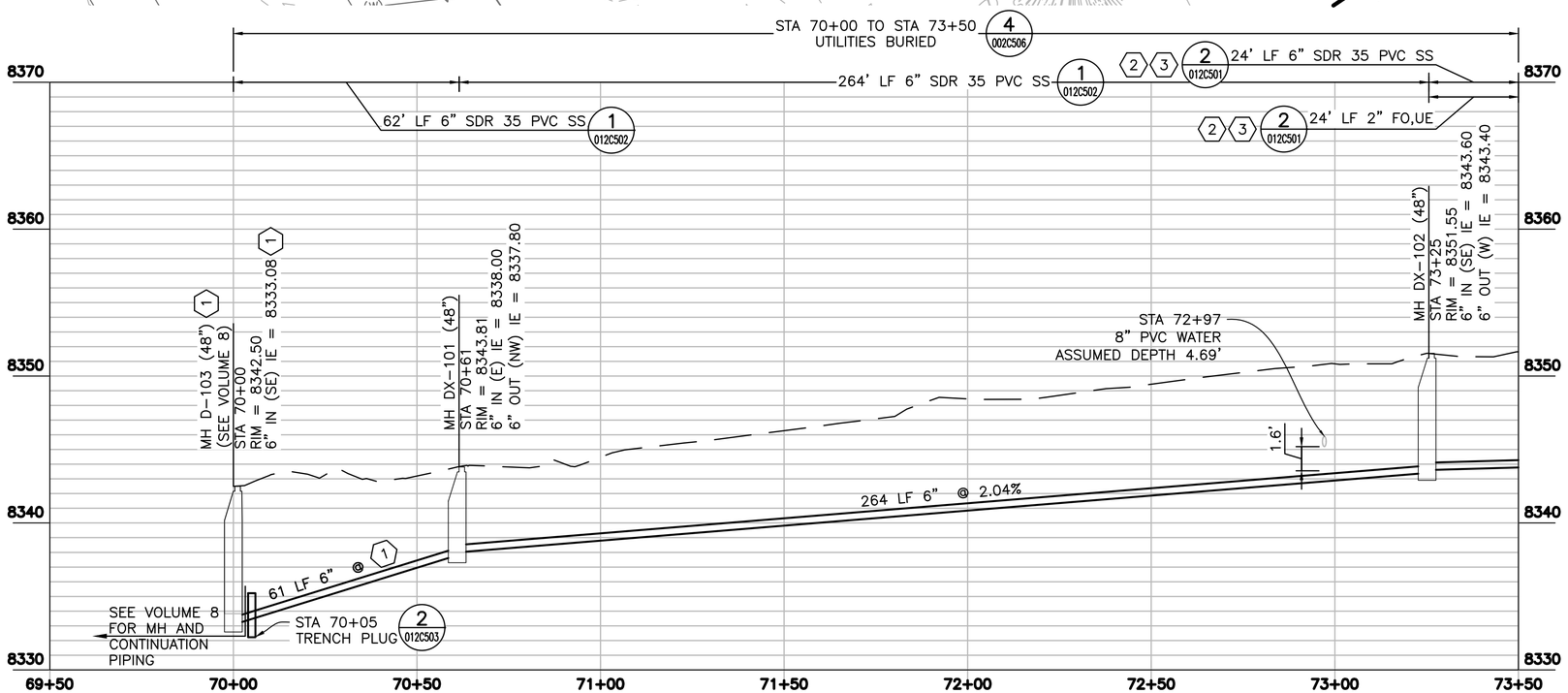
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Not Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C155	TITLE OF SHEET UTILITIES PDC TO WTP UTILITIES PLAN AND PROFILE STA 178+00 TO 181+50 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 69 OF 87

4/17/2026 Doll, Abby C:\Users\adoll\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C151.dwg





KEYNOTES: (X)

1. IF VOLUME 8 (CLIN 8, SPECIFICATION 01 27 00) WORK IS NOT INCLUDED IN BID DOCUMENTS, PIPE SLOPE TO BE ADJUSTED TO TIE INTO EXISTING MH D–103 AT EXISTING MH INVERT ELEVATION. CONTRACTOR TO CORE DRILL EXISTING MH D–103 AND PROVIDE PIPE BOOT FOR NEW 6" SS.
2. INSTALL 144–STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7–WAY MICRODUCT.
3. INSTALL 25kV (3/C) 4/0 AWG ALUMINUM ARMORED POWER CABLE FROM SWITCHGEAR ESB–12 AT THE PDC TO TRANSFORMER T–49 AT THE WTP.
4. RESTORE TRAIL AND ALL FEATURES INCLUDING ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE–USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
5. EXISTING SNOW MARKERS NEED TO BE PROTECTED OR IF REMOVED, PUT BACK IN EXACT SAME LOCATION.
6. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE.

GENERAL NOTES:

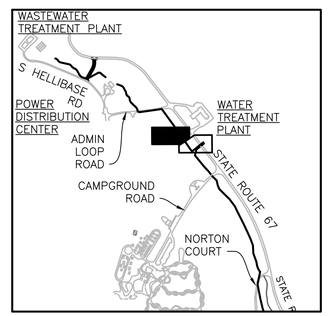
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND UE AND FO WITH A MINIMUM RADIUS OF 20 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

MH–D103 TO WTP
CONSTRUCTION LIMITS POINTS TABLES

POINT NO.	NORTHING	EASTING
1	13150545.30	1327666.31
2	13150501.69	1327669.29
3	13150523.45	1327694.54
4	13150510.63	1327724.20
5	13150480.62	1327718.01
6	13150510.86	1327920.21
7	13150490.55	1327961.81
8	13150431.88	1327964.07
9	13150411.43	1327976.83
10	13150427.89	1328001.91



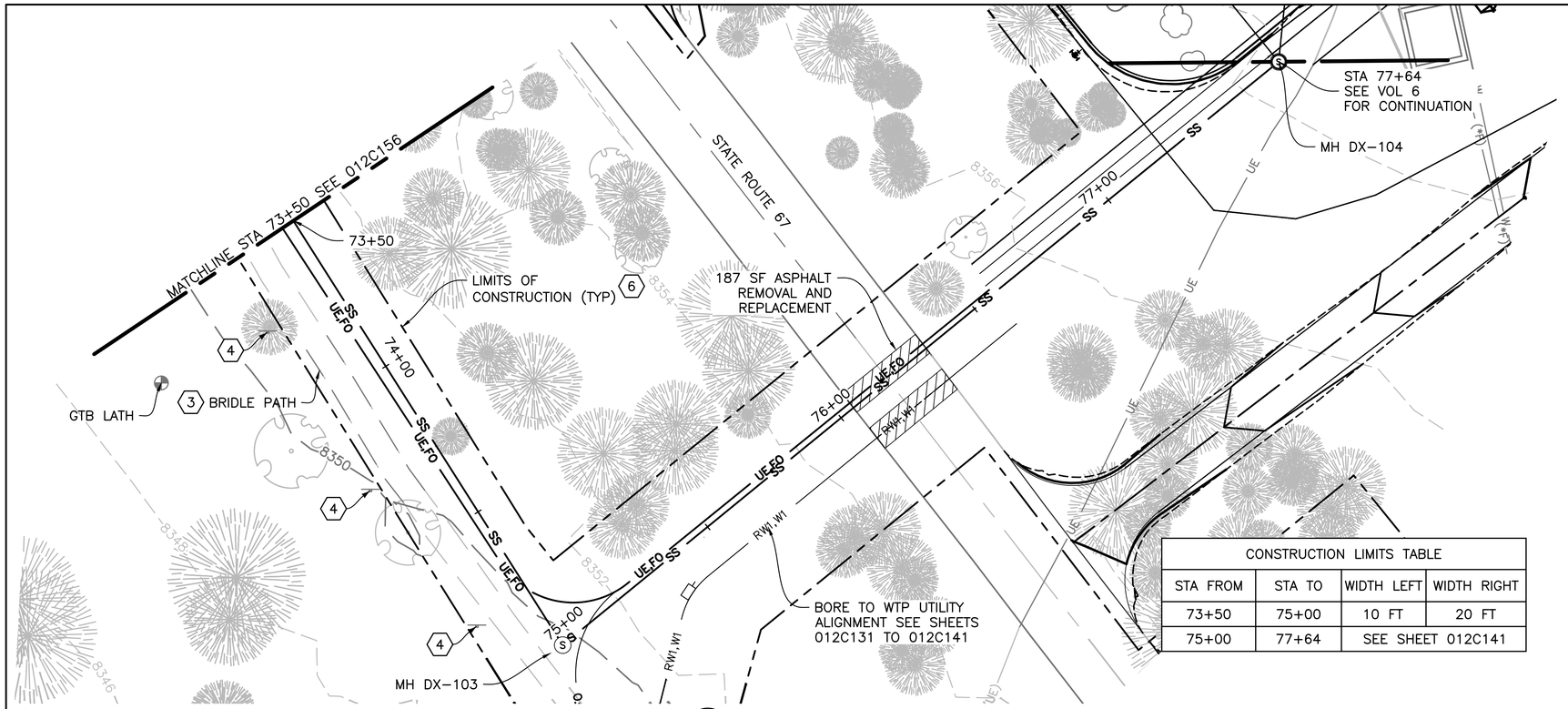
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C156	TITLE OF SHEET UTILITIES MH D–103 TO WTP UTILITIES PLAN AND PROFILE STA 70+00 TO 73+25 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 70 OF 87

4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012–C156.dwg



CONSTRUCTION LIMITS TABLE			
STA FROM	STA TO	WIDTH LEFT	WIDTH RIGHT
73+50	75+00	10 FT	20 FT
75+00	77+64	SEE SHEET 012C141	

KEYNOTES: (X)

1. INSTALL 144-STRAND SM FIBER OPTIC (FO) IN ORANGE TUBE OF 7-WAY MICRODUCT. MAINTAIN A MINIMUM 24" COVER.
2. INSTALL 25kV (3/C) 4/0 AWG ALUMINUM ARMORED POWER CABLE FROM SWITCHGEAR ESB-12 AT THE PDC TO TRANSFORMER T-49 AT THE WTP.
3. RESTORE TRAIL AND ALL FEATURES INCLUDING ROCK EDGING, ROCK EROSION CONTROL, SIGNS, ETC TO EXISTING CONDITION. SALVAGE AND RE-USE ROCK FOR RESTORATION OF TRAIL FEATURES. REFER TO TRAIL MANAGEMENT STANDARDS FOR GRAND CANYON NATIONAL PARK'S CLASS IV STOCK TRAILS FOR ADDITIONAL INFORMATION.
4. EXISTING SNOW MARKERS NEED TO BE PROTECTED OR IF REMOVED, PUT BACK IN EXACT SAME LOCATION.
5. APPROXIMATE 187 SF ASPHALT ROADWAY REPLACEMENT.
6. AVAILABLE CONSTRUCTION WIDTHS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE. TRANSITION IN WIDTH BETWEEN LISTED STATIONS. WIDTHS ARE LEFT AND RIGHT BASED ON UTILITY ALIGNMENT LOOKING UPSTATION.

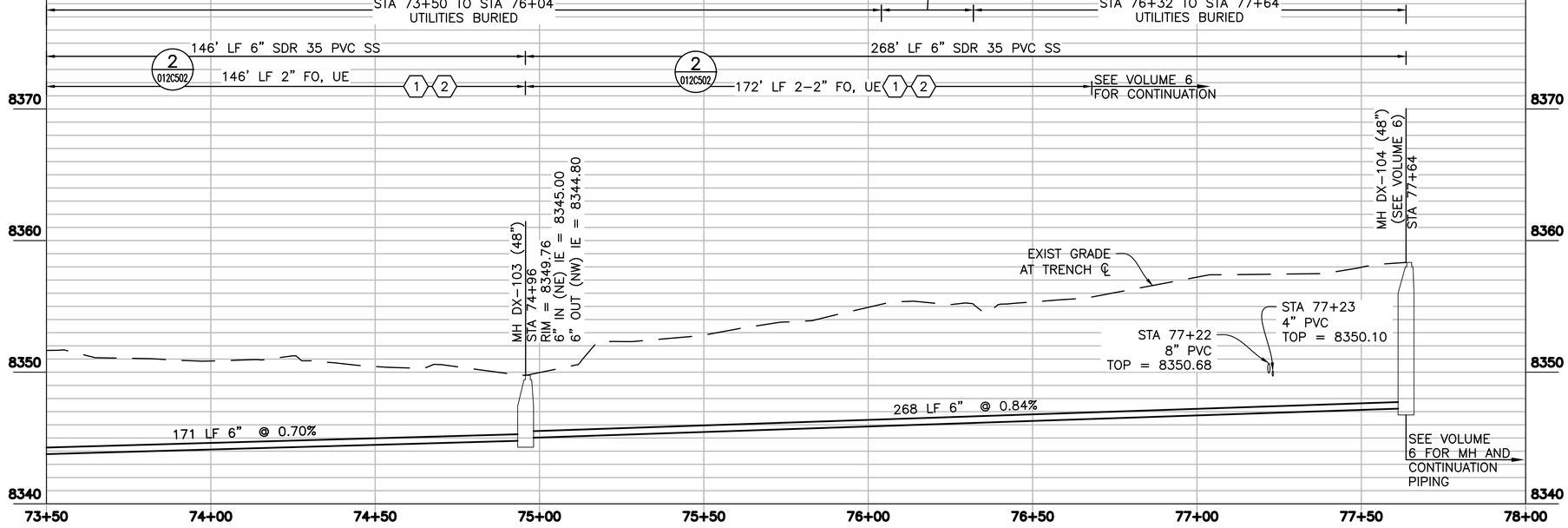
GENERAL NOTES:

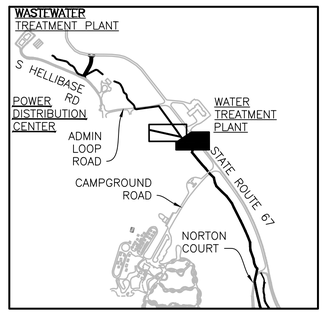
A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. BEND UE AND FO WITH A MINIMUM RADIUS OF 20 FT.

C. SEE VOLUME 0, SHEETS 002C101 THRU 002C102 FOR POTHOLE (PH) INFORMATION AND TABLE.

D. CONTRACTOR SHALL MAINTAIN A MINIMUM OF ONE LANE OF ACCESS PER SPECIFICATION OF 01 50 00.





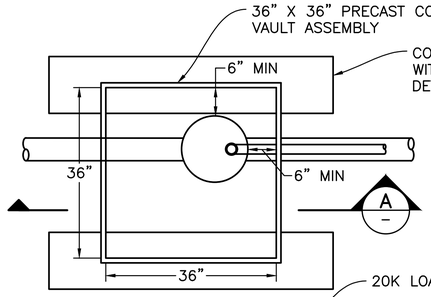
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

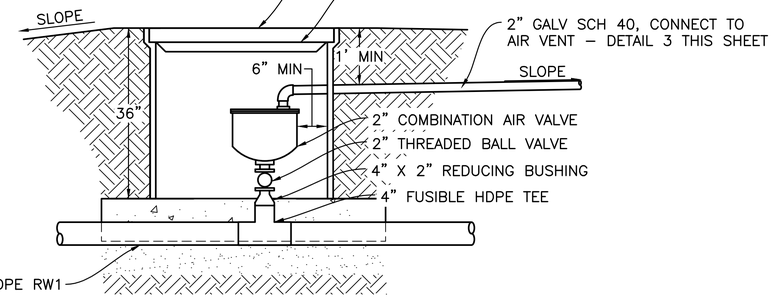
Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 8-31-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	SUB SHEET NO. 012 C157	TITLE OF SHEET UTILITIES MH D-103 TO WTP UTILITES PLAN AND PROFILE STA 73+25 TO 78+00 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 71 OF 87

8/18/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C156.dwg



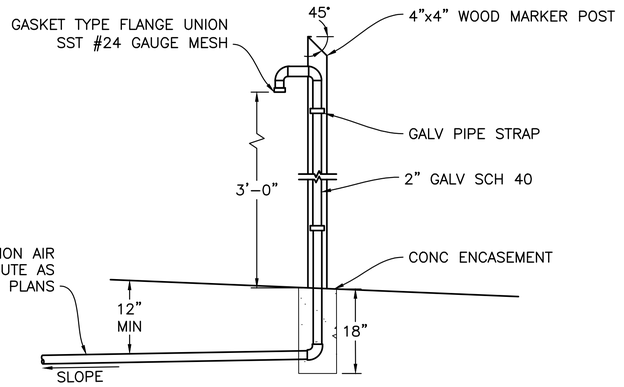
PLAN



SECTION A

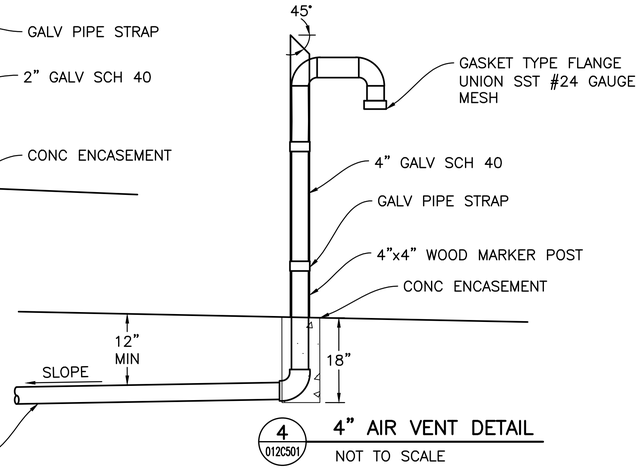
1 012C501 **COMBINATION VALVE VAULT DETAIL**

NOT TO SCALE



3 012C501 **AIR VENT DETAIL**

NOT TO SCALE



4 012C501 **4" AIR VENT DETAIL**

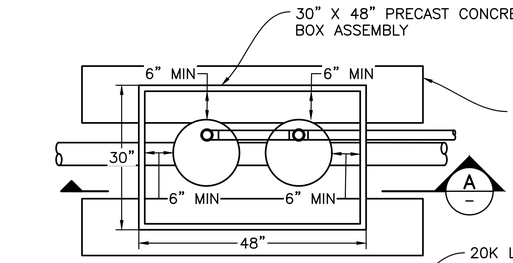
NOT TO SCALE

COMBINATION AIR VALVE VAULT SCHEDULE
RW1 LINE

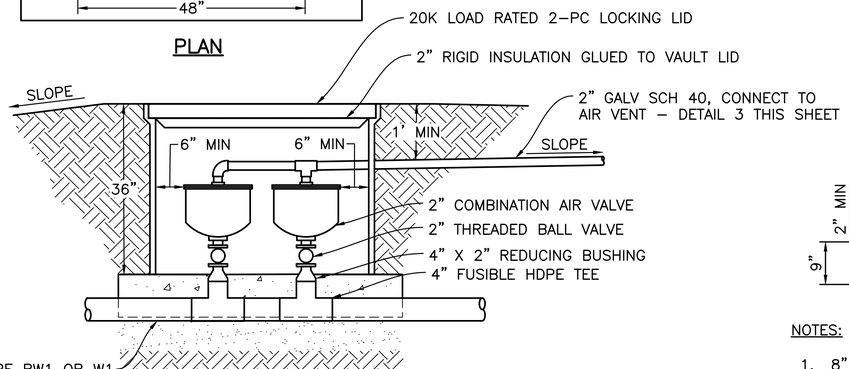
VALVE STATION	NUMBER VALVES AT STATION	SIZE
STA 1010+15	2	4"
STA 111+43	1	2"
STA 119+85	1	2"
STA 127+40	2	2"
STA 136+71	2	2"
STA 142+27	1	2"
STA 149+32	1	2"

COMBINATION AIR VALVE VAULT SCHEDULE
W1 LINE

VALVE STATION	NUMBER VALVES AT STATION	SIZE
STA 1000+15	2	4"



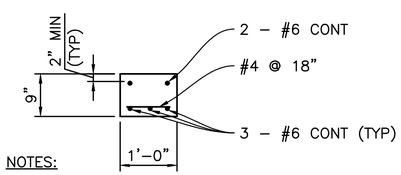
PLAN



SECTION A

2 012C501 **DUAL COMBINATION VALVE VAULT DETAIL**

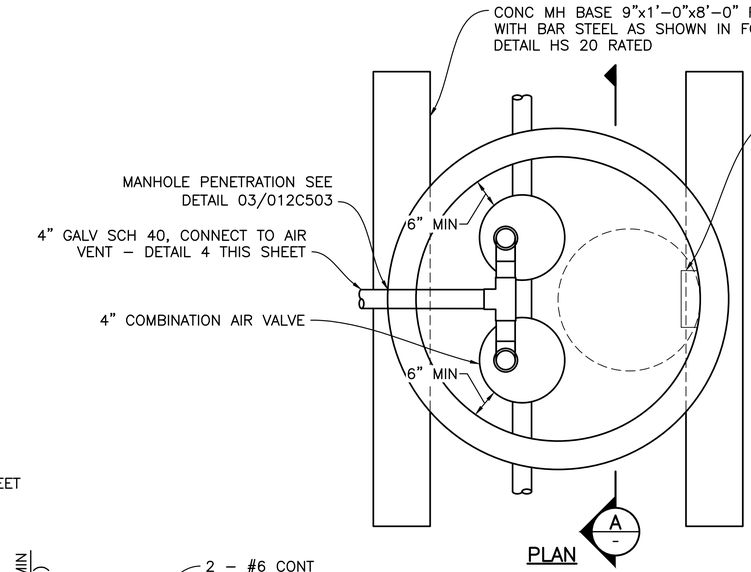
NOT TO SCALE



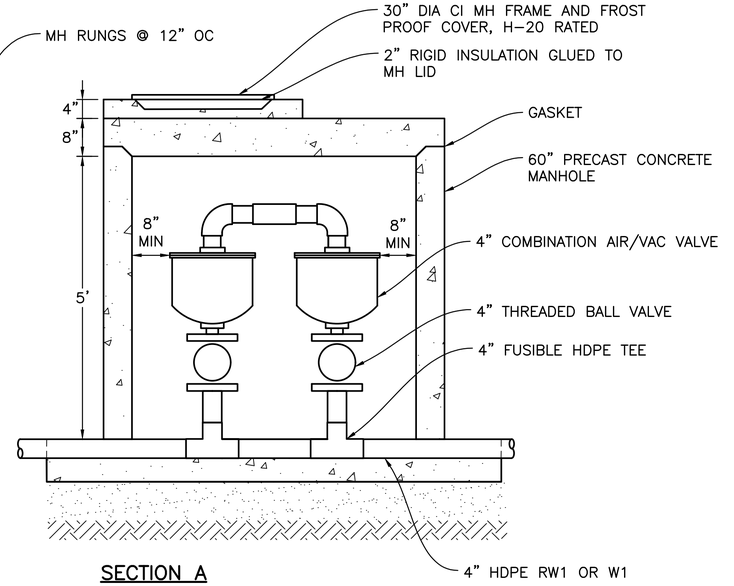
NOTES:

1. 8" LAYER OF GRANULAR BEDDING UNDER FOOTING AND EXTENDING TO TOP OF FOOTING BEAMS.
2. UNDISTURBED EARTH UNDERNEATH GRANULAR BEDDING.

FOOTER DETAIL



PLAN



SECTION A

5 012C501 **4" COMBINATION AIR VALVE VAULT DETAIL**

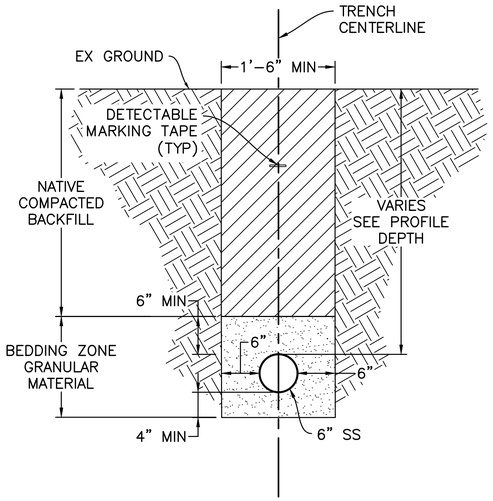
NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Signed ARIZONA, U.S.A.

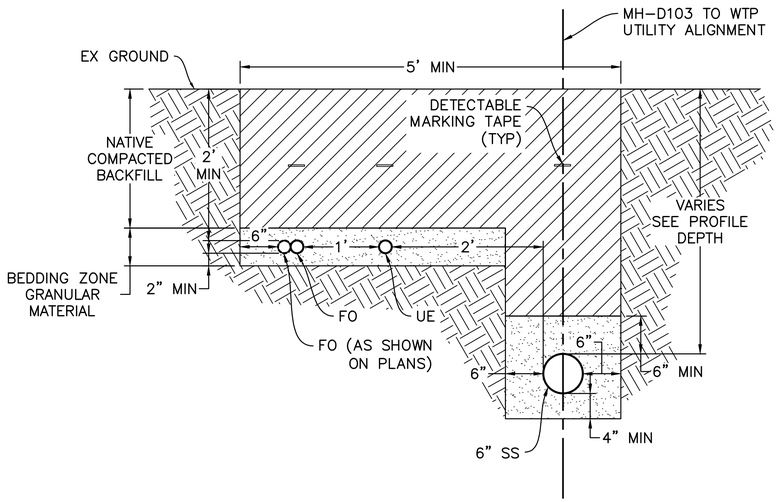
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL DRAWN: J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C501	TITLE OF SHEET UTILITIES DETAILS CARV VAULTS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 72 OF 87

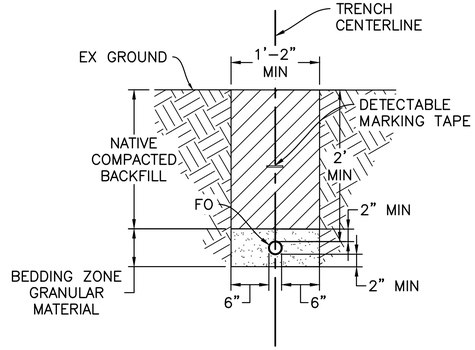
4/27/2026 Race, Sara C:\Users\grace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C501.dwg



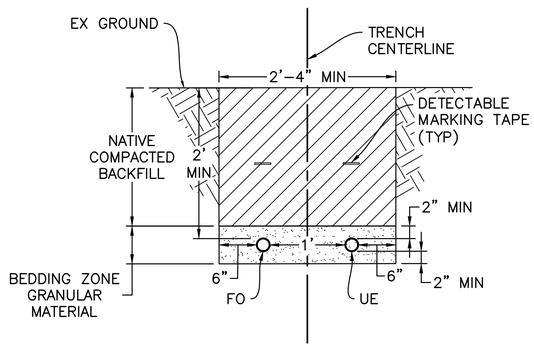
1 012C502 **TRENCH – SANITARY SEWER**
SCALE: 1" = 1'



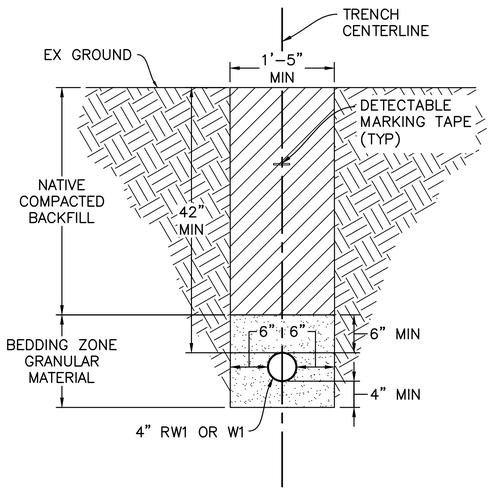
2 012C502 **SHARED TRENCH – SANITARY SEWER**
SCALE: 1" = 1'



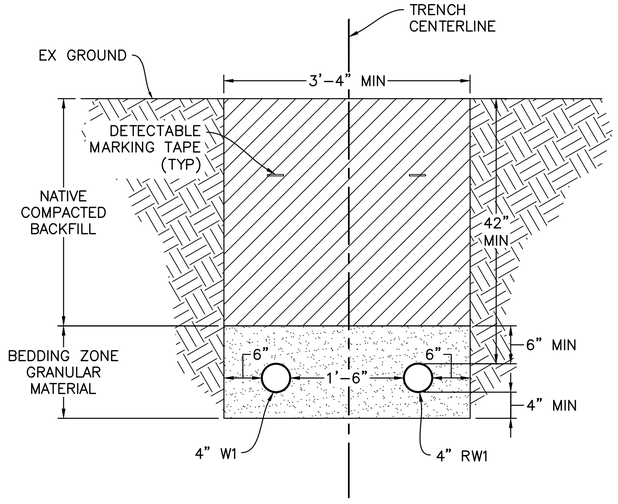
3 012C502 **TRENCH – FIBER**
SCALE: 1" = 1'



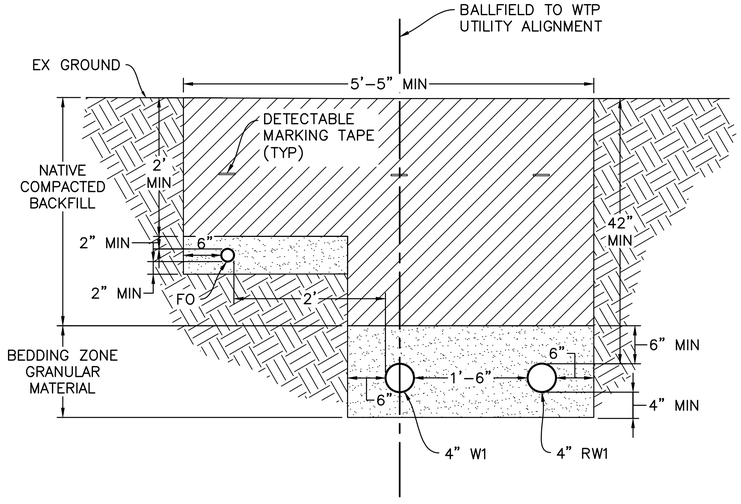
4 012C502 **SHARED TRENCH – FIBER AND ELECTRIC**
SCALE: 1" = 1'



5 012C502 **TRENCH – WATER**
SCALE: 1" = 1'



6 012C502 **SHARED TRENCH – PARALLEL WATER**
SCALE: 1" = 1'



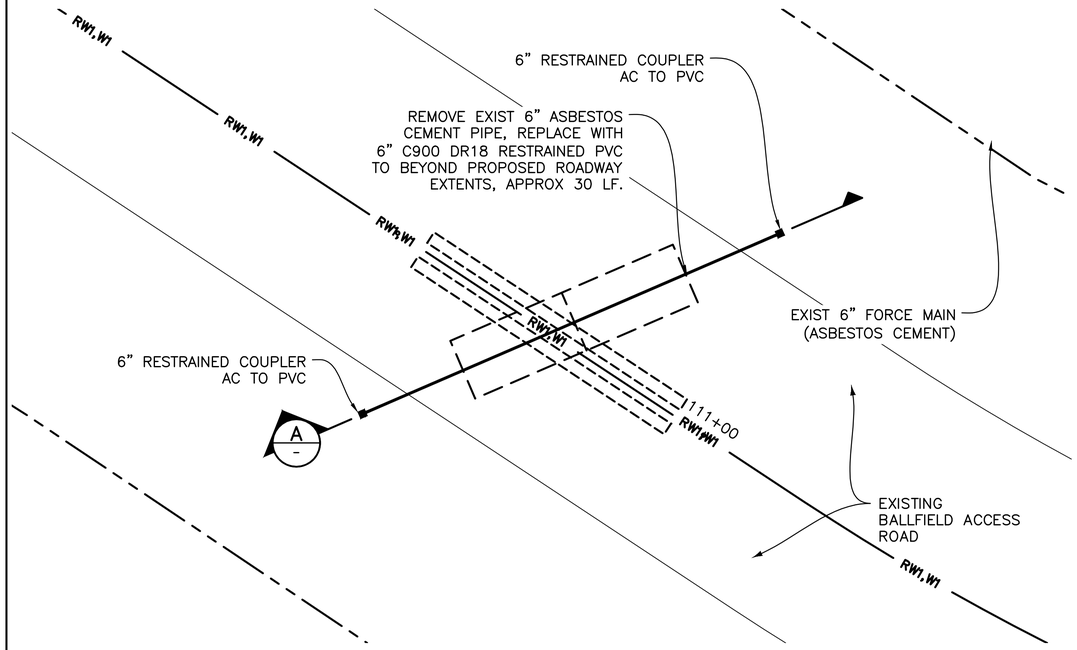
7 012C502 **SHARED TRENCH – WATER AND UTILITIES**
SCALE: 1" = 1'

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

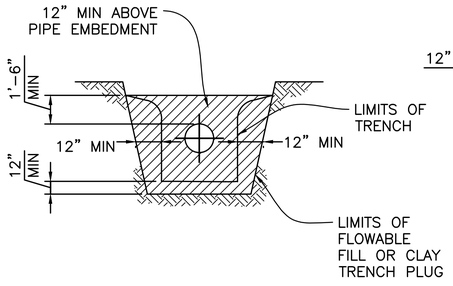
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: A. DALL CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C502	TITLE OF SHEET UTILITIES CIVIL DETAILS TRENCHING NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 73 OF 87

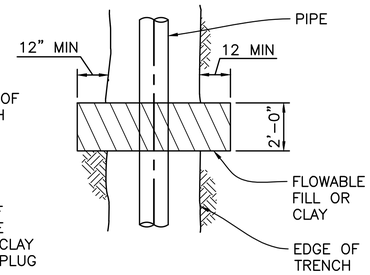
4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C501.dwg



PLAN VIEW



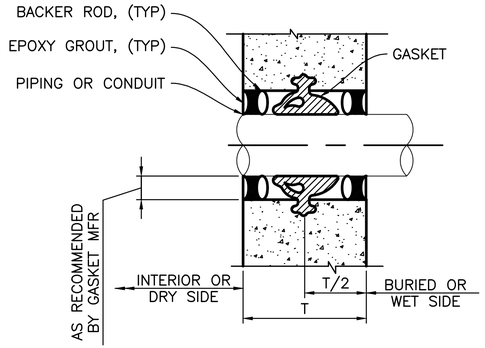
SECTION



PLAN

NOTES:
1. TRENCH PLUG MATERIAL SHALL EXTEND 12" MIN INTO UNDISTURBED SOIL ON EACH SIDE AND ON BOTTOM OF TRENCH.
2. MATERIAL USED FOR TRENCH PLUGS SHALL BE FLOWABLE FILL, CONCRETE, OR CLAY, SEE SPECIFICATIONS.

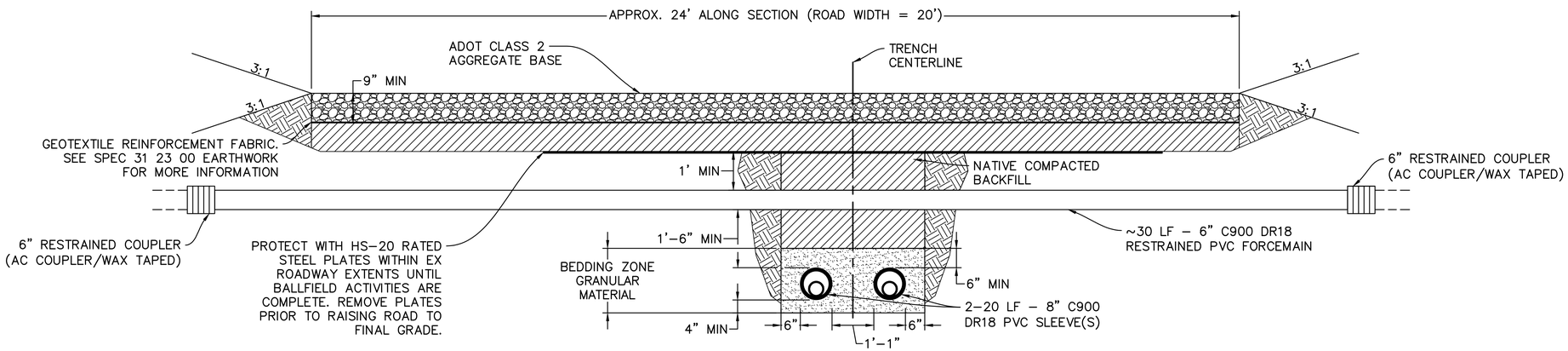
2 / 012C503 TYPICAL TRENCH PLUG
NOT TO SCALE



3 / 012C503 PENETRATION DETAIL – MH SECTION
NOT TO SCALE

NOTES:

1. SLEEVES AND FORCEMAIN REPLACEMENT TO BE COMPLETED PRIOR TO BALLFIELD CONSTRUCTION ACTIVITIES. PROTECT IN PLACE NEW FORCEMAIN WITH HS–20 RATED STEEL PLATES UNTIL BALLFIELD CONSTRUCTION COMPLETE. REMOVE PLATES, AND BRING ROAD TO FINAL GRADE AS INDICATED IN PLAN AND PROFILE DRAWINGS.
2. CONTRACTOR TO CONFIRM EXISTING AC PIPE DIMENSIONS PRIOR TO PROCURING RESTRAINING COUPLERS. RECORD DRAWINGS INDICATE 6", POTHOLE OBSERVATION NOTES 4".
3. AC (ASBESTOS CEMENT) PIPE CUTTING, REMOVAL, AND DISPOSAL TO ADHERE TO ALL APPLICABLE ENVIRONMENTAL, HEALTH, AND SAFETY REGULATIONS.
4. COUPLERS TO BE WAX TAPE WRAPPED PRIOR TO BACKFILL.



SECTION A

1 / 012C503 STA 110+38 CROSSING DETAIL
NOT TO SCALE

Registered Professional Engineer (Civil) CERTIFICATE No. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD J. GREEN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 012 C503	TITLE OF SHEET UTILITIES DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 74 OF 87

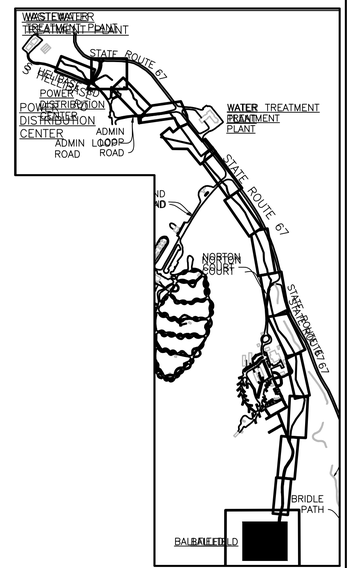
4/17/2026 Race, Sara C:\Users\srace\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\CIVIL\012-C501.dwg

PLAN
1"=20'

SEEDING AREA SPECIES MIX PROVIDED BY PARK.

SEEDING QUANTITIES

1. DISTURBANCE AREA: 182,843 SF (4.2 AC)



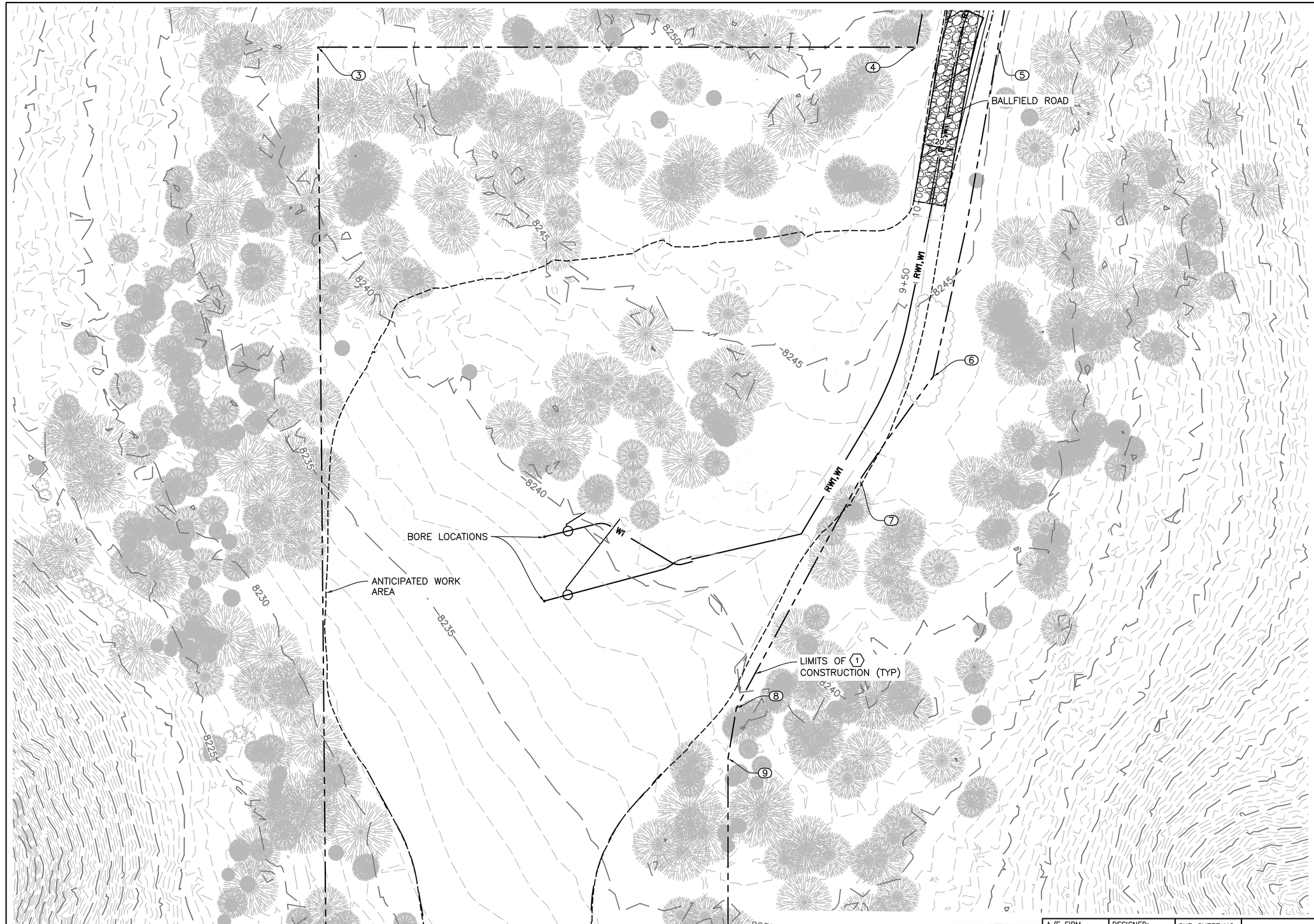
KEYMAP

30 0 30 60
SCALE OF FEET
1" = 30'

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
Date Signed 5/1/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD T. MCCALLAN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 013 L101	TITLE OF SHEET LANDSCAPE LANDSCAPE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 75 OF 87

4/17/2026 Dall, Abby C:\Users\adall\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\LANDSCAPE\013-L101.dwg



GENERAL NOTES:

A. EXISTING CONDITIONS ARE SHOWN ON THIS SHEET FOR SITE REFERENCE. BORING CONTRACTOR SHALL DETERMINE NECESSARY IMPROVEMENTS BASED ON SPECIFIC EQUIPMENT REQUIREMENTS AND CONSTRUCTION METHODOLOGY.

B. PRIOR TO SITE IMPROVEMENTS FOR BORE CONSTRUCTION, CONTRACTOR SHALL SUBMIT EARTHWORK PLAN, SEQUENCING, SITE IMPROVEMENTS, ETC. TO CONTRACTING OFFICER'S REPRESENTATIVE. WORK SHALL NOT COMMENCE UNTIL APPROVALS ARE GIVEN BY THE CONTRACTING OFFICER.

KEY NOTES: (X)

1. AVAILABLE CONSTRUCTION LIMITS ARE SHOWN IN THE LIMITS OF CONSTRUCTION TABLE.

LIMITS OF CONSTRUCTION		
POINT NO.	NORTHING	EASTING
1	13145678.29	1328830.41
2	13145677.75	1328580.41
3	13146248.92	1328575.26
4	13146248.92	1328947.54
5	13146248.76	1328998.28
6	13146043.95	1328958.02
7	13145979.43	1328912.19
8	13145837.61	1328835.55
9	13145806.35	1328830.41

30 0 30 60 1" = 30'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
Date Signed
ARIZONA, U.S.A.

A/E FIRM	
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	
DESIGNED:	G. YOUNG
CADD	P. OXBORROW
TECH. REVIEW:	GOUGH/LUSTER
DATE:	MAY 2026
SUB SHEET NO.	013 C101
TITLE OF SHEET	SITEWORK OVERALL BALLFIELD GRADING PLAN EXISTING SITE CONDITION
	NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	76 OF 87

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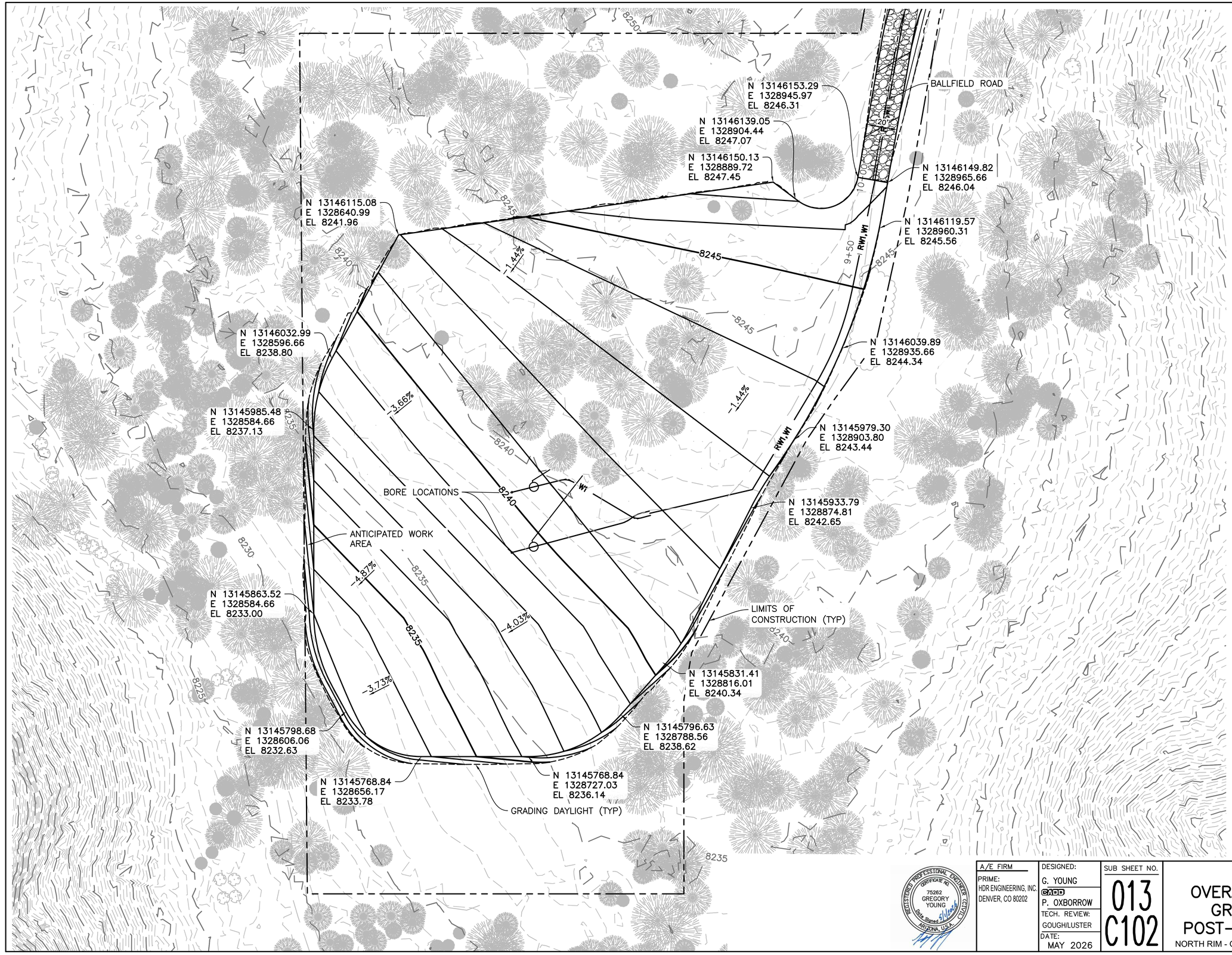


GENERAL NOTES:

A. QUANTITIES PROVIDED BELOW ARE FOR THE BALLFIELD POST–CONSTRUCTION EARTHWORK, RESULTING FROM EXCAVATION ACTIVITIES FROM VOLUMES 1, 6 , 7 , & 8. QUANTITIES DO NOT TAKE INTO ACCOUNT FACTORS SUCH AS MATERIAL SHRINK AND SWELL. THE BALLFIELD SITE CAN ACCOMMODATE A TOTAL OF 2,450 +/– CY OF FILL, WHICH IS MORE MATERIAL THAN ANTICIPATED TO BE GENERATED FROM CONSTRUCTION ACTIVITIES.

A.A. CUT = 4450 +/– CY
A.B. FILL = 2500 +/– CY
A.C. NET CUT = 1950 +/– CY (EXCESS CUT GENERATED FROM CONSTRUCTION ACTIVITIES ON THE NORTH RIM SHALL BE PLACED AT THE BALLFIELD SITE. QUANTITY NOT TO EXCEED 2450 CY.)

B. EXCESS CUT MATERIAL FROM CONSTRUCTION ACTIVITIES ON THE NORTH RIM SHALL BE HAULED TO THE BALLFIELD, GRADED AND COMPACTED IN THE GENERAL AREA SHOW ON THIS SHEET. HEIGHT OF FILL SHALL NOT EXCEED 18" AT THE HIGHEST POINT.

C. EXCESS CUT MATERIAL FROM CONSTRUCTION ACTIVITIES THAT WILL INCREASE THE OVERALL HEIGHT OF FILL, OVER 18", WILL BE THE CONTRACTORS RESPONSIBILITY TO DISPOSE OF OUTSIDE THE PARK BOUNDARIES.

D. UNSUITABLE CUT MATERIAL WILL BE THE CONTRACTOR'S RESPONSIBILITY TO DISPOSE OF OUTSIDE THE PARK AT AN APPROPRIATE DISPOSAL SITE, FOLLOWING ALL STATE AND LOCAL REGULATIONS.

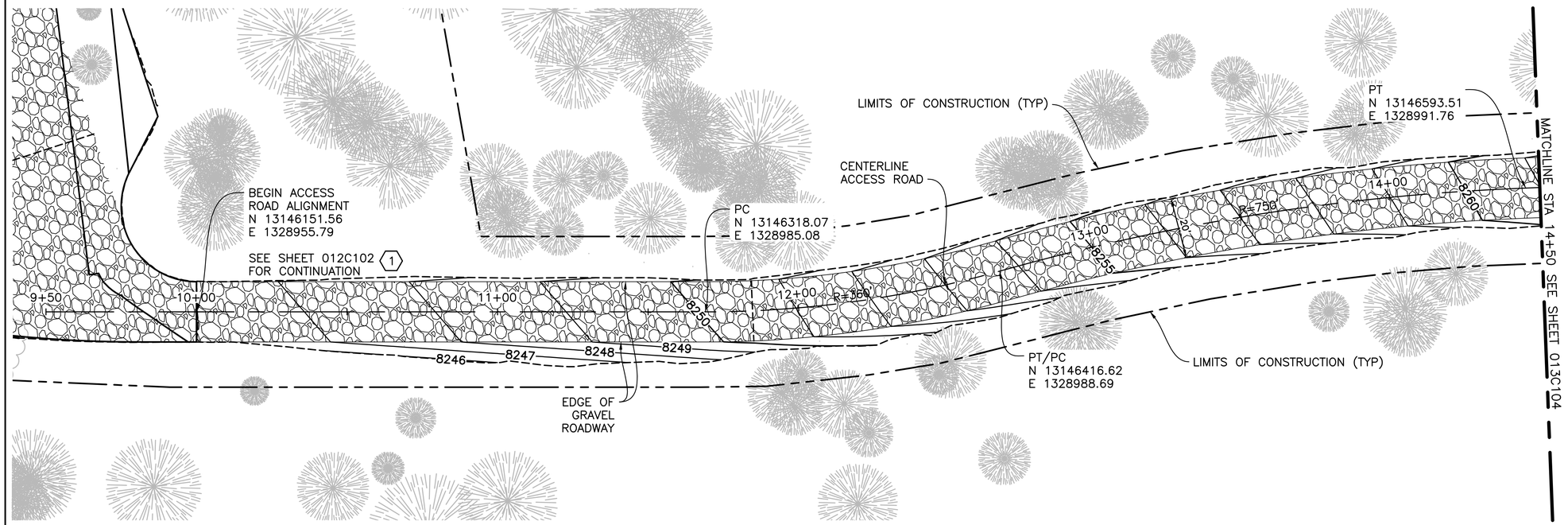
E. GRADING SHOWN ON THIS SHEET IS THE MAXIMUM QUANTITY OF EARTHEN FILL (EXCESS CUT GENERATED FROM CONSTRUCTION ACTIVITIES ON THE NORTH RIM) THAT CAN BE PLACED AT THE BALLFIELD SITE. IF THE QUANTITY OF EXCESS CUT DOES NOT GENERATE ENOUGH FILL MATERIAL TO GRADE BALLFIELD SITE PER THIS SHEET, CONTRACTOR SHALL GRADE THE ACTUAL QUANTITY OF FILL MATERIAL CONTINUOUSLY THROUGHOUT THE SITE CREATING A UNIFORM SURFACE. MINIMUM PIPE BURY DEPTH MUST BE MAINTAINED, BASED ON THE EXISTING PRE–CONSTRUCTION GRADE, PER DETAILS ON SHEET 012C502. CONTRACTOR SHALL NOT BE REQUIRED TO PROVIDE ADDITIONAL FILL MATERIAL FROM SOURCES OUTSIDE THE PARK TO MATCH GRADING ON THIS SHEET.

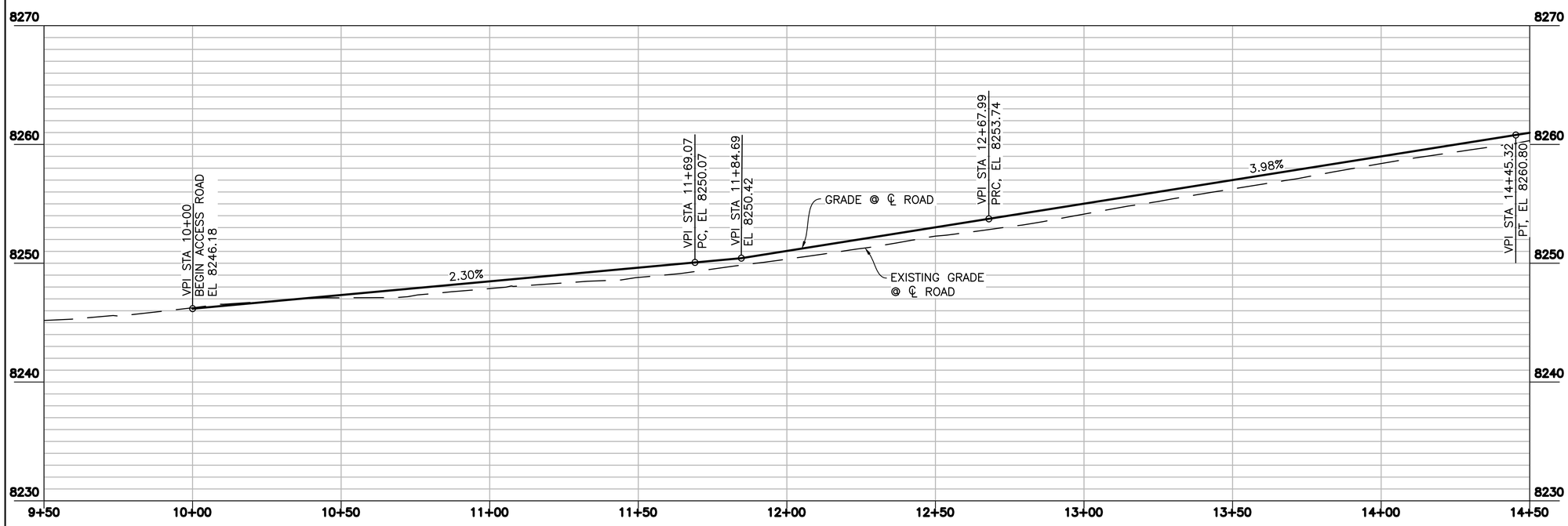
30 0 30 60
SCALE OF FEET
1" = 30'

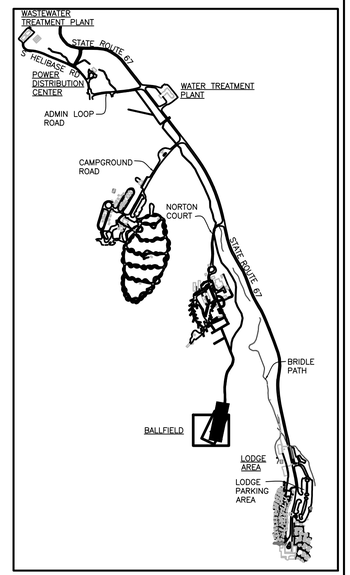
REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75262
GREGORY YOUNG
State of ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: P. OXBORROW TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 013 C102	TITLE OF SHEET SITEWORK OVERALL BALLFIELD GRADING PLAN POST–CONSTRUCTION NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 77 OF 87

4/16/2026 Flansburg, Jade C:\Users\jflansburg\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\013-C101.dwg







KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

GENERAL NOTES:

A. SEE SHEETS 011X101–011X107 FOR TREE REMOVAL LOCATIONS.

B. SEE DETAIL 1 ON SHEET 01C301 FOR GRAVEL ACCESS ROADWAY SECTION.

C. SEE 012C103 TO 012C108 FOR GESC SHEETS.

D. SEE 012C131 TO 012C137 FOR UTILITIES PLAN AND PROFILE.

E. CONTRACTOR TO VERIFY LOCATION OF ALL UTILITIES PRIOR TO CONSTRUCTION.

KEY NOTES: (X)

1. SEE BALLFIELD OVERALL GRADING SHEET FOR MORE INFORMATION ON SPOILS HANDLING.

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 75282 GREGORY YOUNG Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD T. MCCALLAN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 013 C103	TITLE OF SHEET BALLFIELD ROAD PLAN AND PROFILE STA 9+50 TO 14+50 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 78 OF 87

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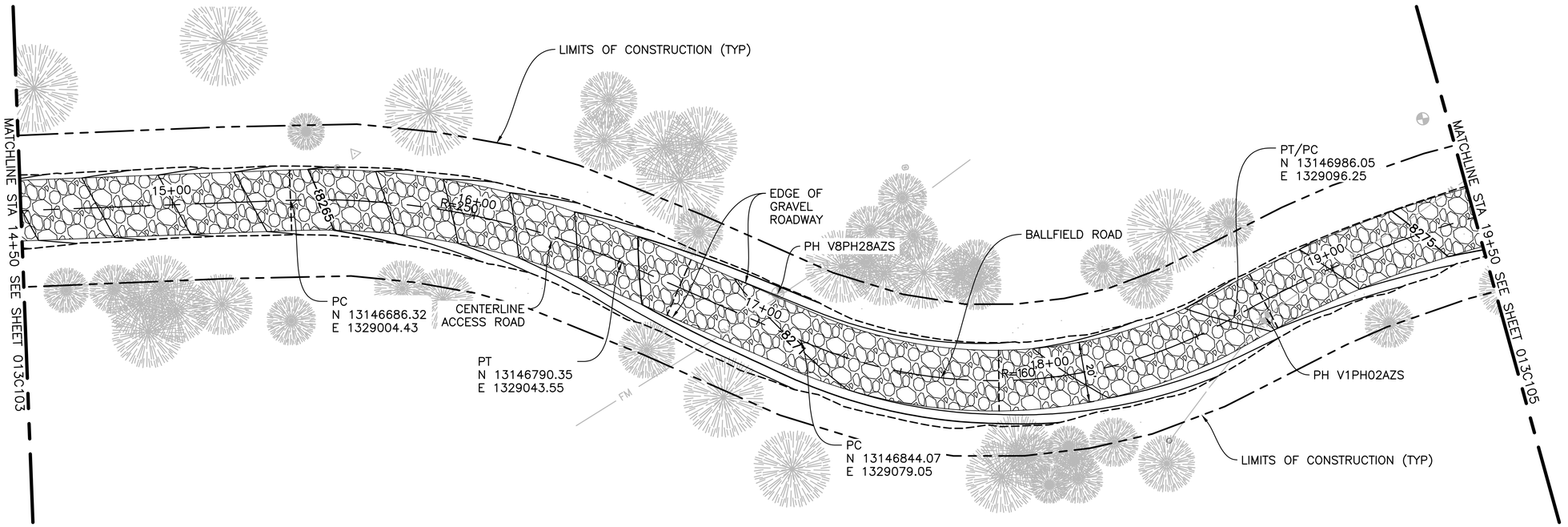


GENERAL NOTES:

A. SEE SHEETS 011X101–011X107 FOR TREE REMOVAL LOCATIONS.

B. SEE DETAIL 1 ON SHEET 01C301 FOR GRAVEL ACCESS ROADWAY SECTION.

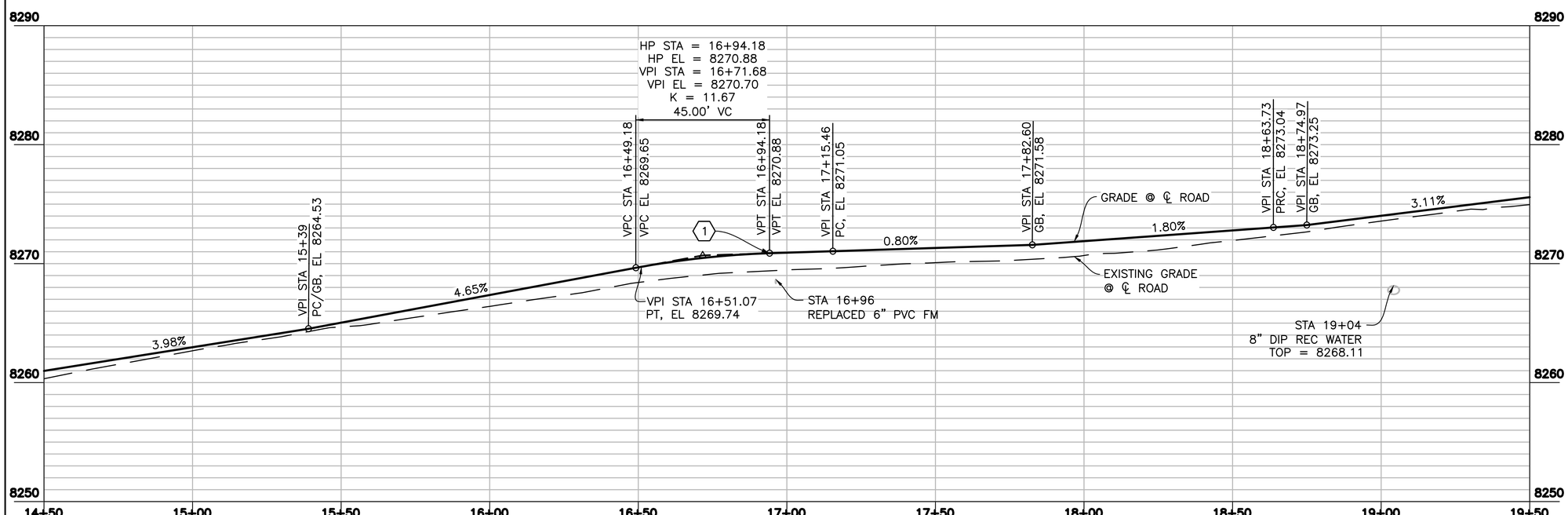
C. SEE 012C103 TO 012C108 FOR GESC SHEETS.

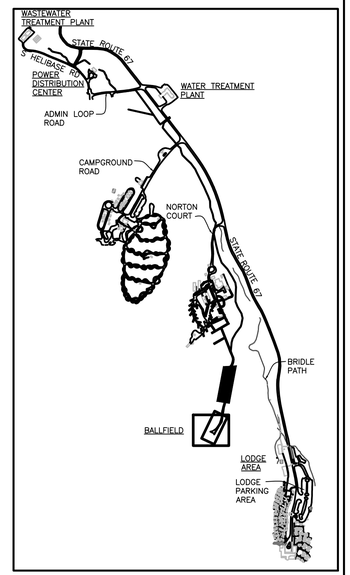
D. SEE 012C131 TO 012C137 FOR UTILITIES PLAN AND PROFILE.

E. CONTRACTOR TO VERIFY LOCATION OF ALL UTILITIES PRIOR TO CONSTRUCTION.

KEY NOTES: (X)

1. FINAL ROAD GRADE TO BE A MINIMUM OF 2 FT ABOVE THE CROWN OF REPLACED 8" PVC FORCEMAIN.





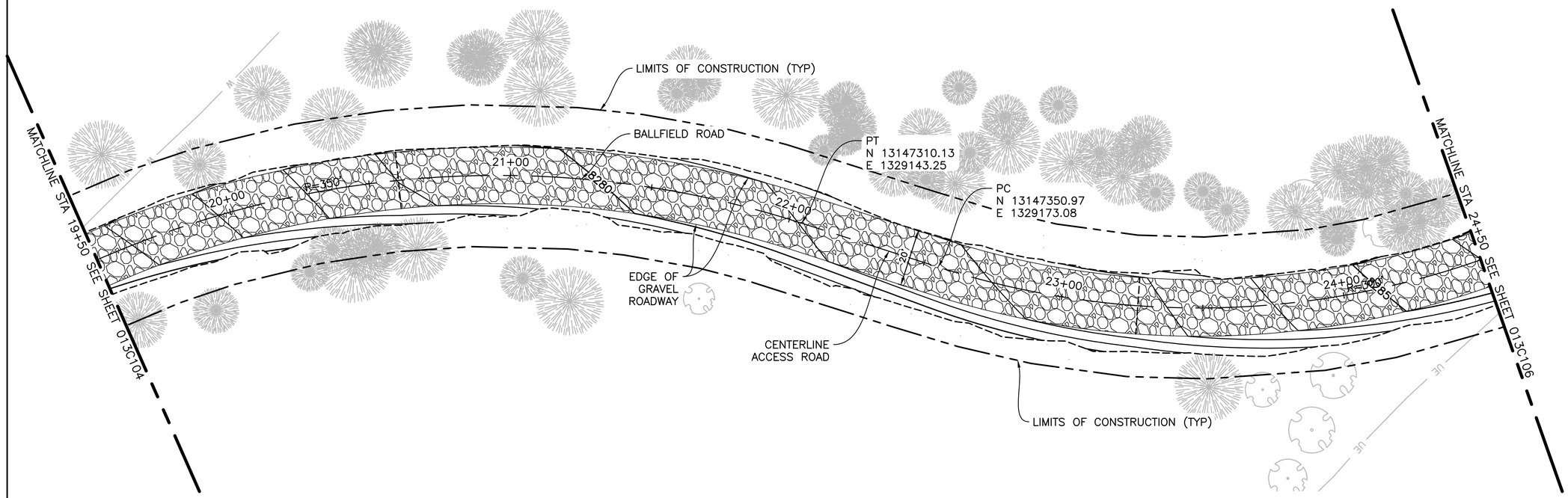
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 75282 GREGORY YOUNG Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: T. MCCALLAN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 013 C104	TITLE OF SHEET **BALLFIELD ROAD** PLAN AND PROFILE STA 14+50 TO 19+50 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 79 OF 87

4/17/2026 Dall, Abby C:\Users\adall\DC\ACCDocs\HDR\103505545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\013-C103.dwg



GENERAL NOTES:

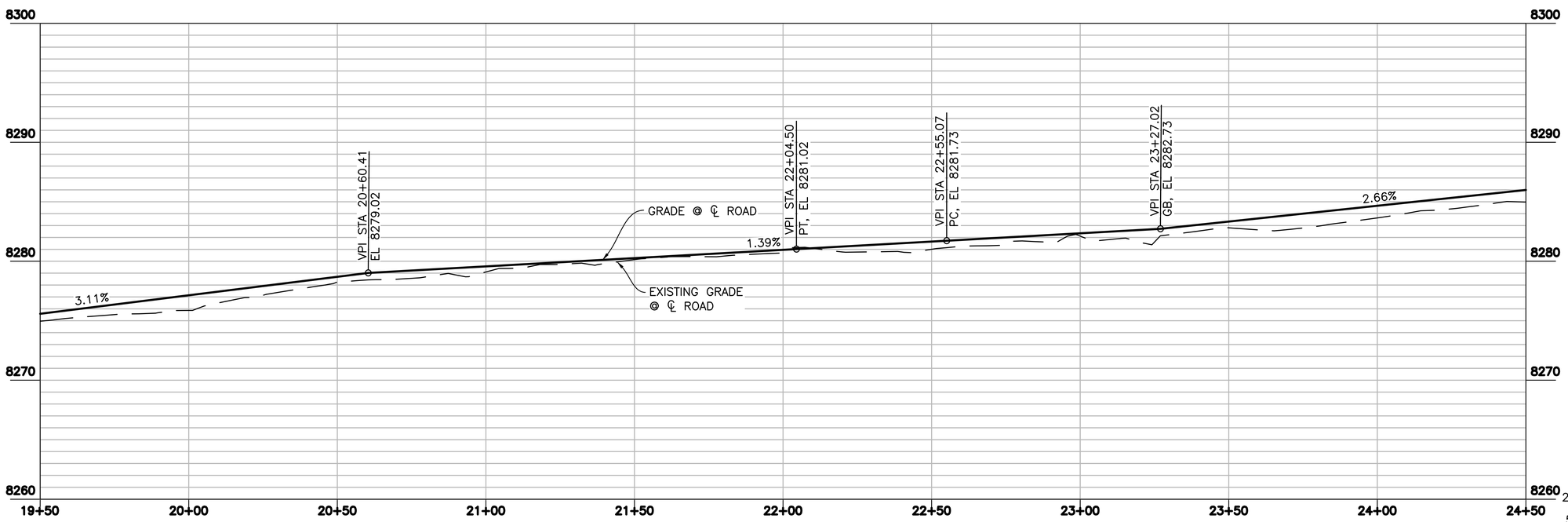
A. SEE SHEETS 011X101–011X107 FOR TREE REMOVAL LOCATIONS.

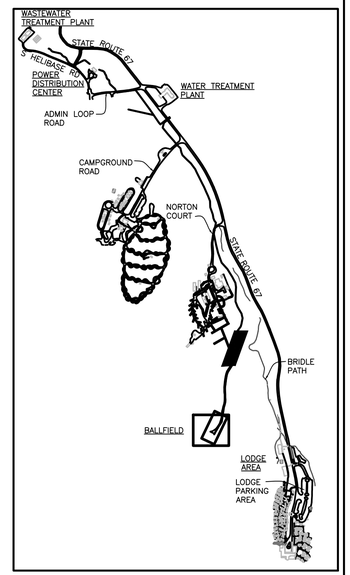
B. SEE DETAIL 1 ON SHEET 01C301 FOR GRAVEL ACCESS ROADWAY SECTION.

C. SEE 012C103 TO 012C108 FOR GESC SHEETS.

D. SEE 012C131 TO 012C137 FOR UTILITIES PLAN AND PROFILE.

E. CONTRACTOR TO VERIFY LOCATION OF ALL UTILITIES PRIOR TO CONSTRUCTION.





KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 75282 GREGORY YOUNG Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD T. MCCALLAN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 013 C105	TITLE OF SHEET BALLFIELD ROAD PLAN AND PROFILE STA 19+50 TO 24+50 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 80 OF 87

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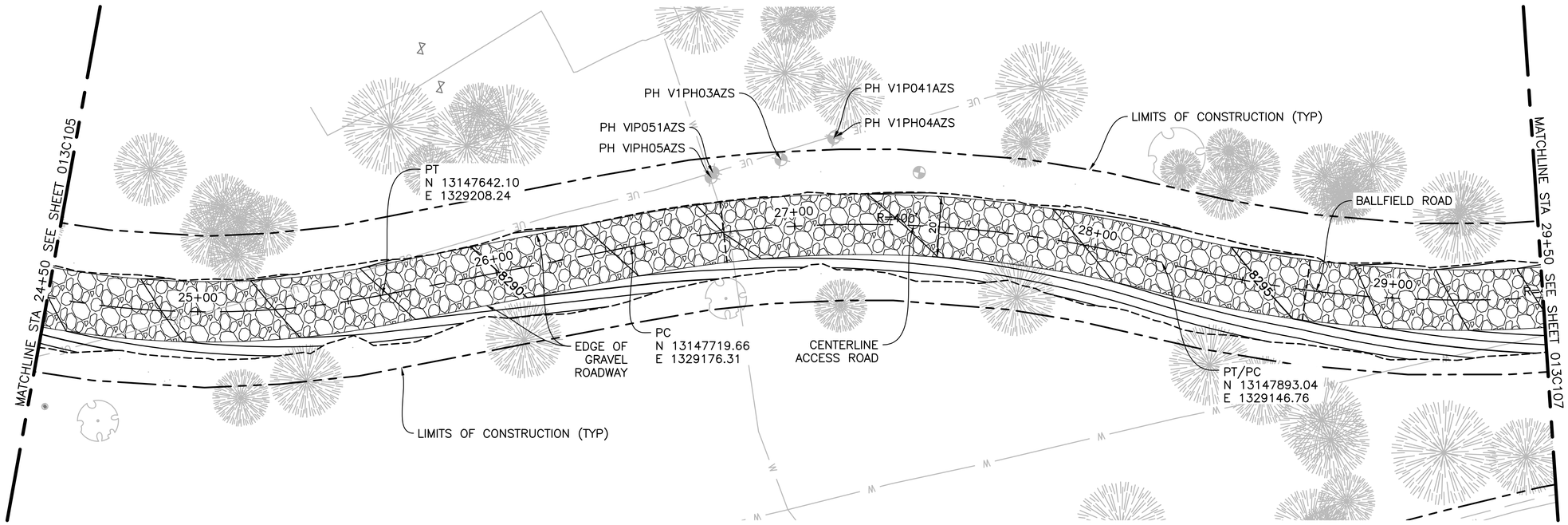


GENERAL NOTES:

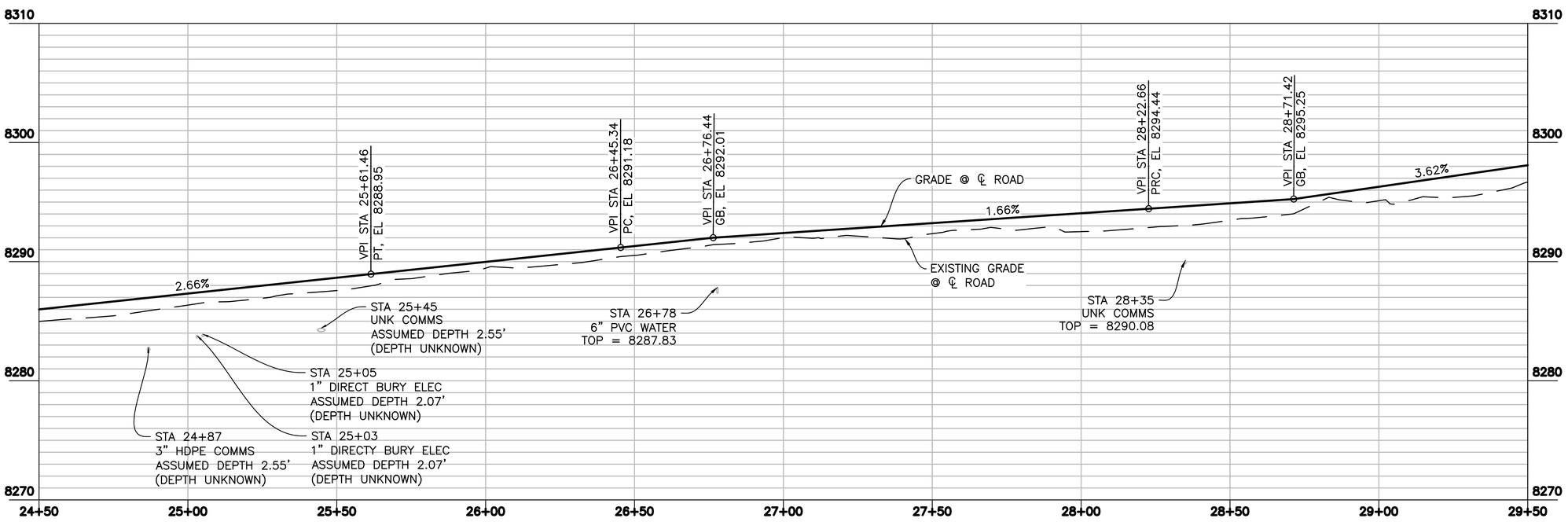
A. SEE SHEETS 011X101–011X107 FOR TREE REMOVAL LOCATIONS.

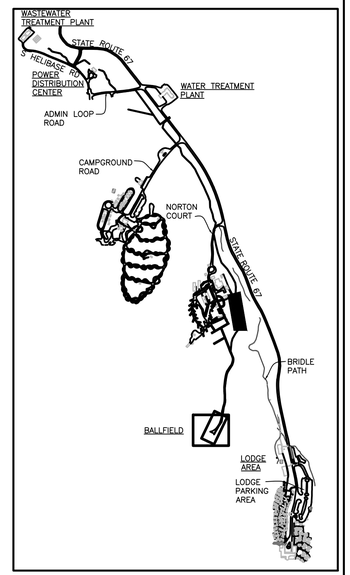
B. SEE DETAIL 1 ON SHEET 01C301 FOR GRAVEL ACCESS ROADWAY SECTION.

C. SEE 012C103 TO 012C108 FOR GESC SHEETS.

D. SEE 012C131 TO 012C137 FOR UTILITIES PLAN AND PROFILE.

E. CONTRACTOR TO VERIFY LOCATION OF ALL UTILITIES PRIOR TO CONSTRUCTION.





KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75282
GREGORY YOUNG
Date Signed
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: T. MCCALLAN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 013 C106	TITLE OF SHEET **BALLFIELD ROAD** PLAN AND PROFILE STA 24+50 TO 29+50 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET **81** OF 87

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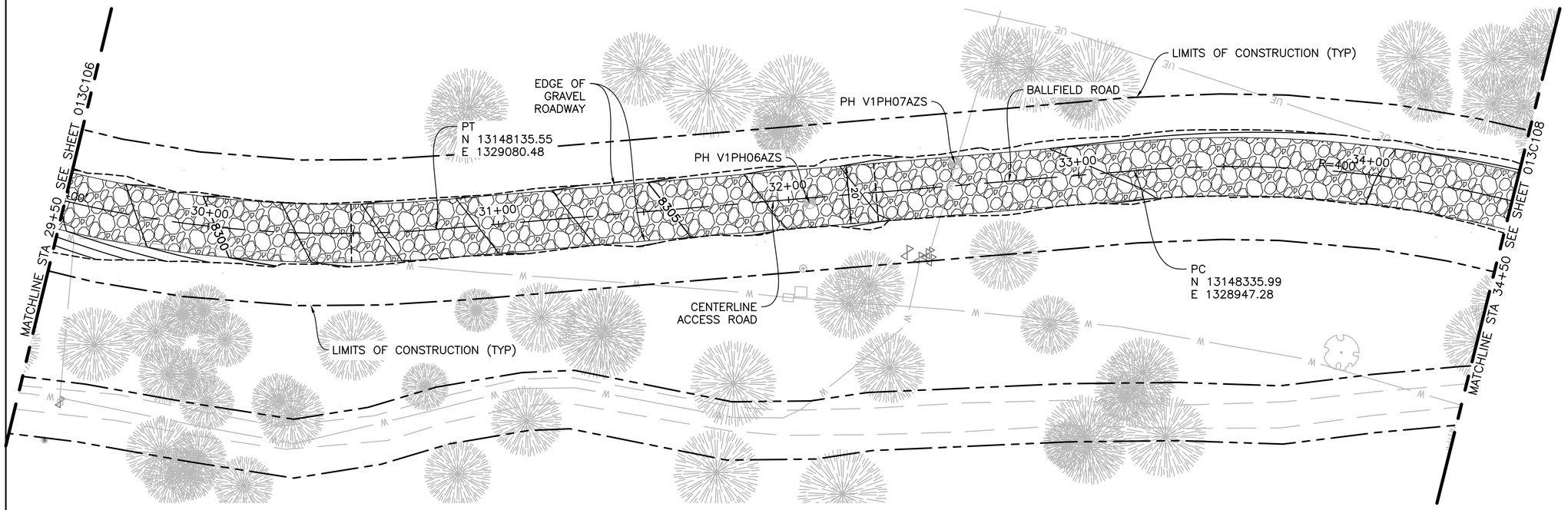


GENERAL NOTES:

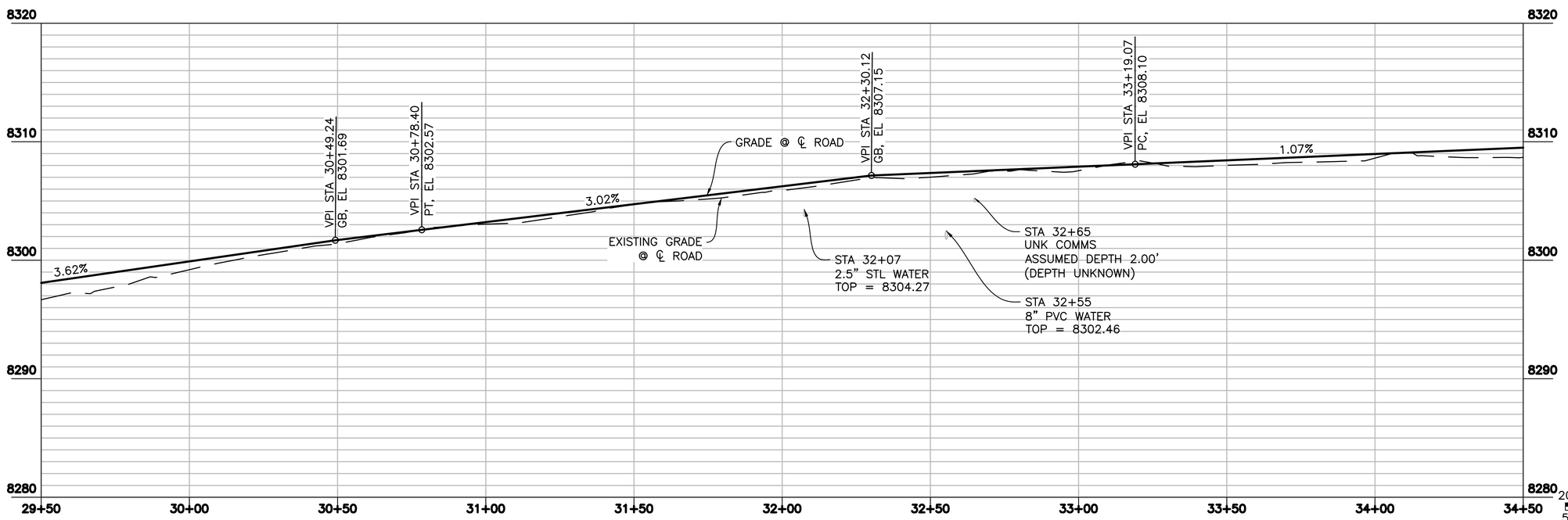
A. SEE SHEETS 011X101–011X107 FOR TREE REMOVAL LOCATIONS.

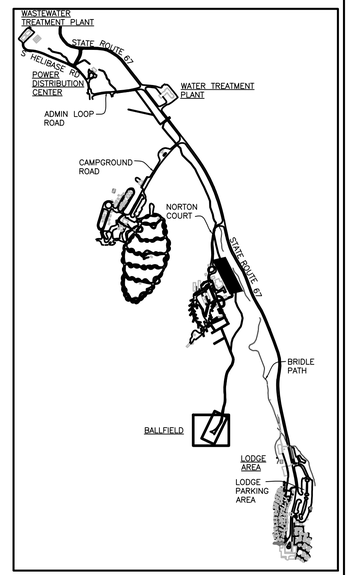
B. SEE DETAIL 1 ON SHEET 01C301 FOR GRAVEL ACCESS ROADWAY SECTION.

C. SEE 012C103 TO 012C108 FOR GESC SHEETS.

D. SEE 012C131 TO 012C137 FOR UTILITIES PLAN AND PROFILE.

E. CONTRACTOR TO VERIFY LOCATION OF ALL UTILITIES PRIOR TO CONSTRUCTION.





KEYMAP

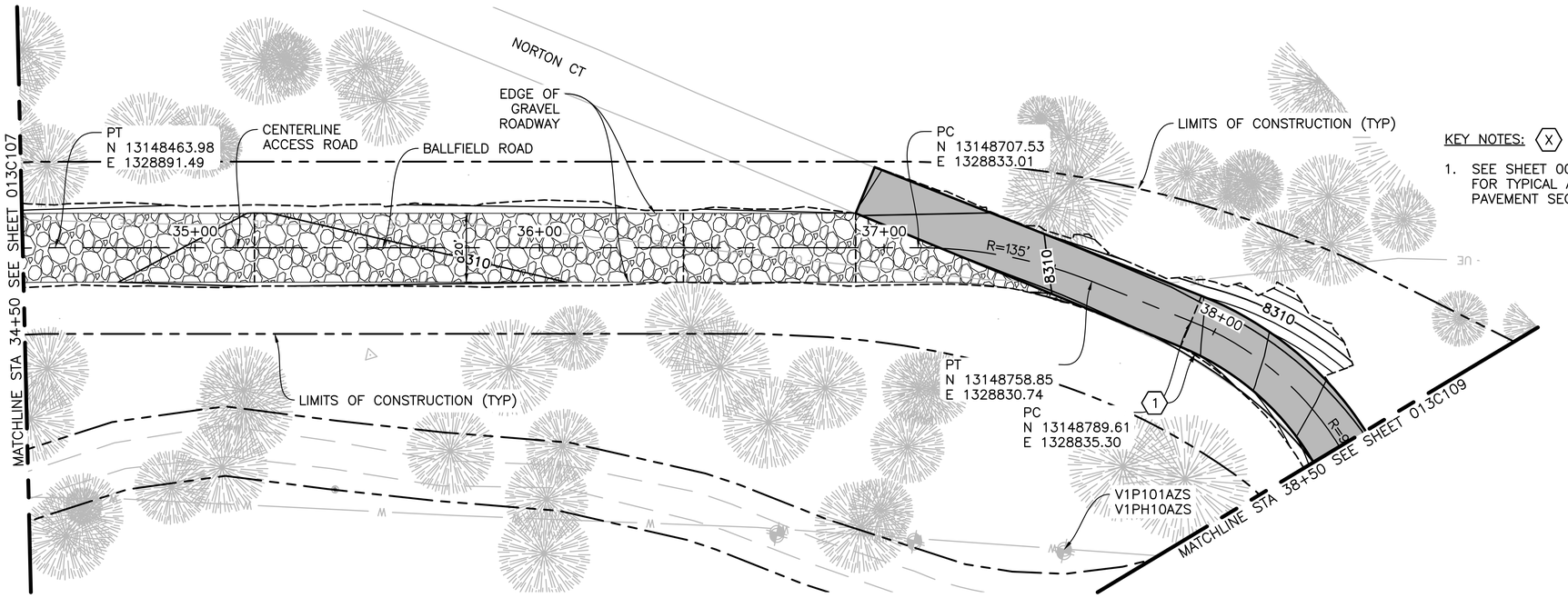
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 75282 GREGORY YOUNG Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: T. MCCALLAN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 013 C107	TITLE OF SHEET **BALLFIELD ROAD** PLAN AND PROFILE STA 29+50 TO 34+50 NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113
	182752
PMIS/PKG NO.	318719
SHEET	**82** OF 87

4/17/2026 Dall, Abby C:\Users\adall\DC\ACCDocs\HDR\103505545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\013-C103.dwg



KEY NOTES: (X)

1. SEE SHEET 002C505 DETAIL 1 FOR TYPICAL ASPHALT PAVEMENT SECTION.

GENERAL NOTES:

A. SEE SHEETS 011X101–011X107 FOR TREE REMOVAL LOCATIONS.

B. SEE DETAIL 1 ON SHEET 01C301 FOR GRAVEL ACCESS ROADWAY SECTION.

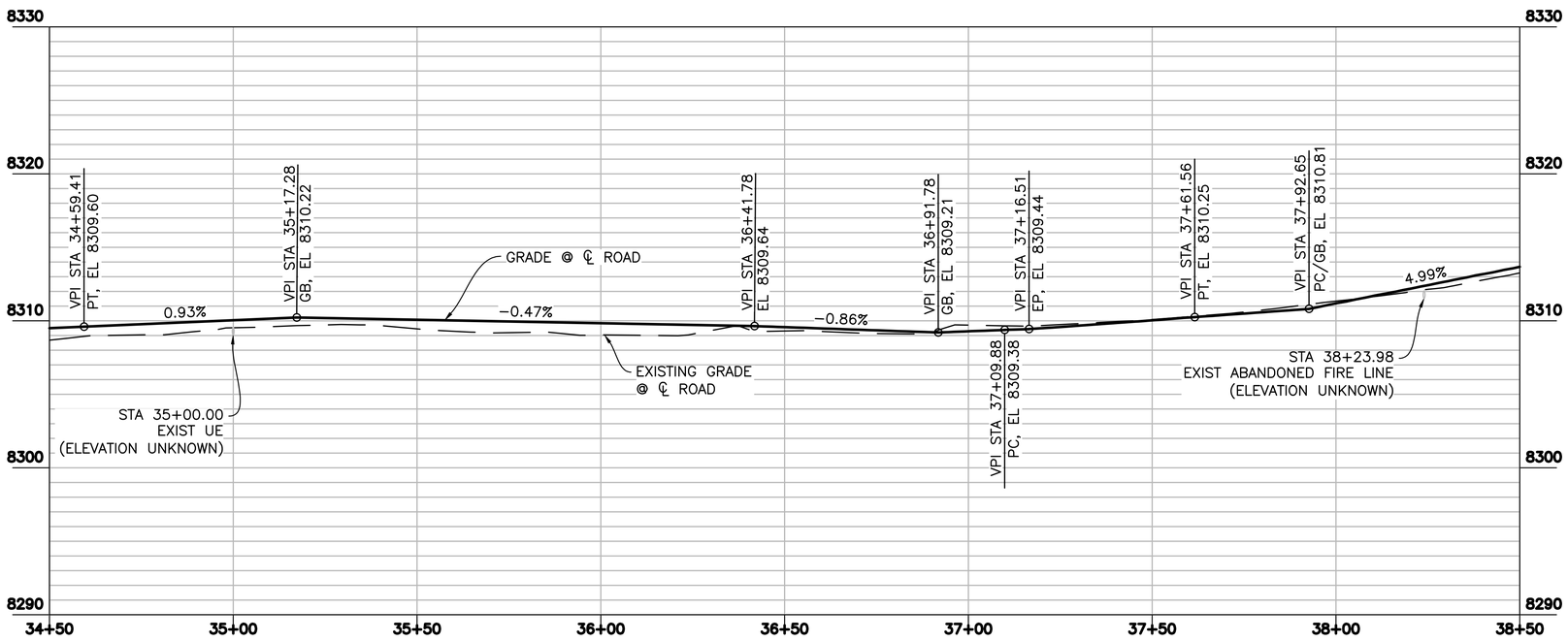
C. SEE 012C103 TO 012C108 FOR GESC SHEETS.

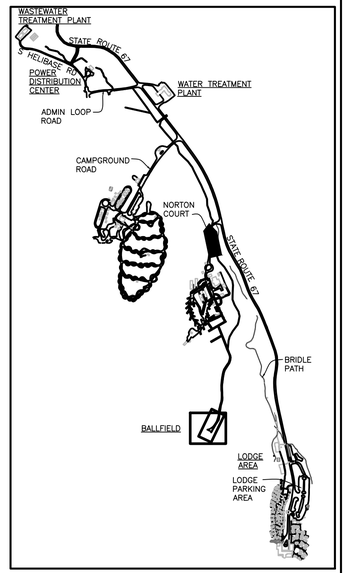
D. SEE 012C131 TO 012C137 FOR UTILITIES PLAN AND PROFILE.

E. CONTRACTOR TO VERIFY LOCATION OF ALL UTILITIES PRIOR TO CONSTRUCTION.

F. QUANTITIES PROVIDED BELOW ARE FOR THE NORTON COURT ENTRANCE POST–CONSTRUCTION EARTHWORK. QUANTITIES DO NOT TAKE INTO ACCOUNT FACTORS SUCH AS MATERIAL SHRINK AND SWELL.

F.A. CUT = 35.6 +/– CY
F.B. FILL = 48.0 +/– CY
F.C. NET FILL = 12.4 +/– CY





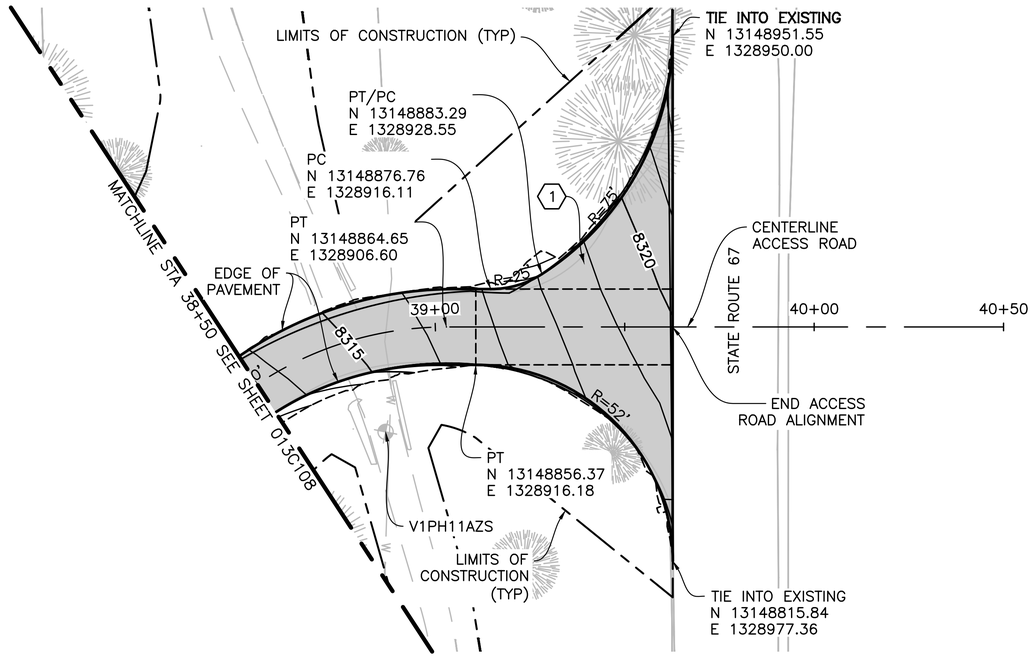
KEYMAP

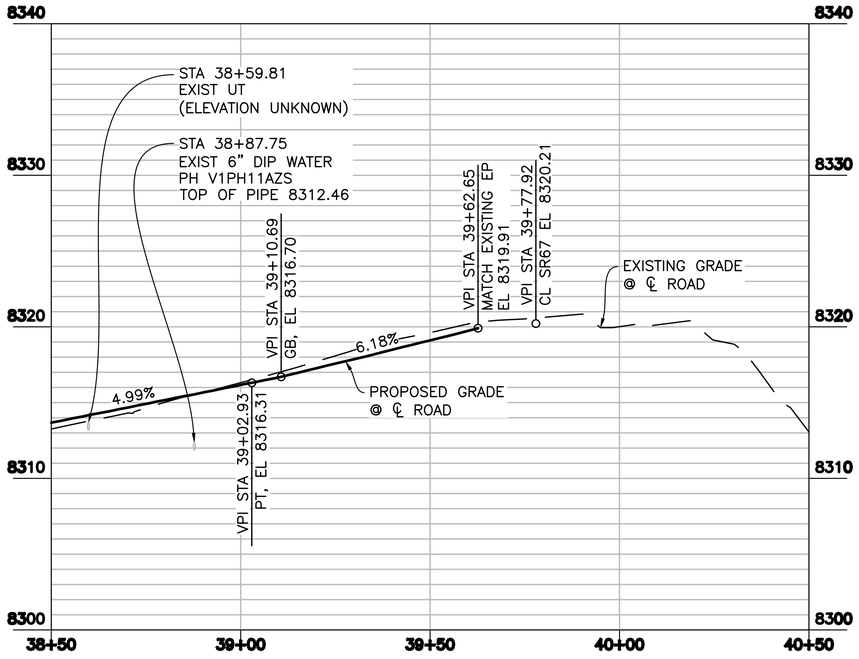
20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75282
GREGORY YOUNG
Date Signed
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: T. MCCALLAN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 013 C108	TITLE OF SHEET BALLFIELD ROAD PLAN AND PROFILE STA 34+50 TO 38+50 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 83 OF 87

4/17/2026 Dall, Abby C:\Users\adall\DC\ACCDocs\HDR\103505545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\013-C103.dwg





KEY NOTES: (X)

1. SEE SHEET 002C505 DETAIL 1 FOR TYPICAL ASPHALT PAVEMENT SECTION.

GENERAL NOTES:

A. SEE SHEETS 011X101–011X107 FOR TREE REMOVAL LOCATIONS.

B. SEE DETAIL 1 ON SHEET 01C301 FOR GRAVEL ACCESS ROADWAY SECTION.

C. SEE 012C103 TO 012C108 FOR GESC SHEETS.

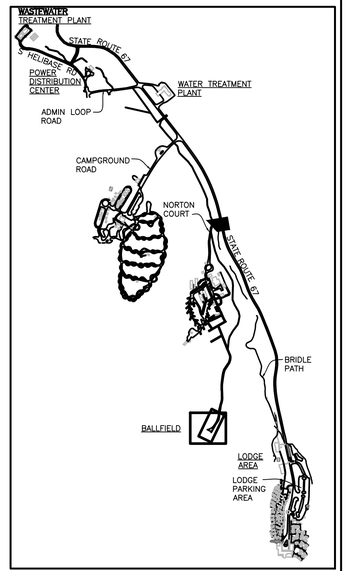
D. SEE 012C131 TO 012C137 FOR UTILITIES PLAN AND PROFILE.

E. CONTRACTOR TO VERIFY LOCATION OF ALL UTILITIES PRIOR TO CONSTRUCTION.

F. QUANTITIES PROVIDED BELOW ARE FOR THE NORTON COURT ENTRANCE POST–CONSTRUCTION EARTHWORK. QUANTITIES DO NOT TAKE INTO ACCOUNT FACTORS SUCH AS MATERIAL SHRINK AND SWELL.

F.A. CUT = 35.6 +/– CY
F.B. FILL = 48.0 +/– CY
F.C. NET FILL = 12.4 +/– CY

G. CONTRACTOR SHALL MAINTAIN A MINIMUM OF ONE LANE OF ACCESS PER SPECIFICATION 01 50 00.



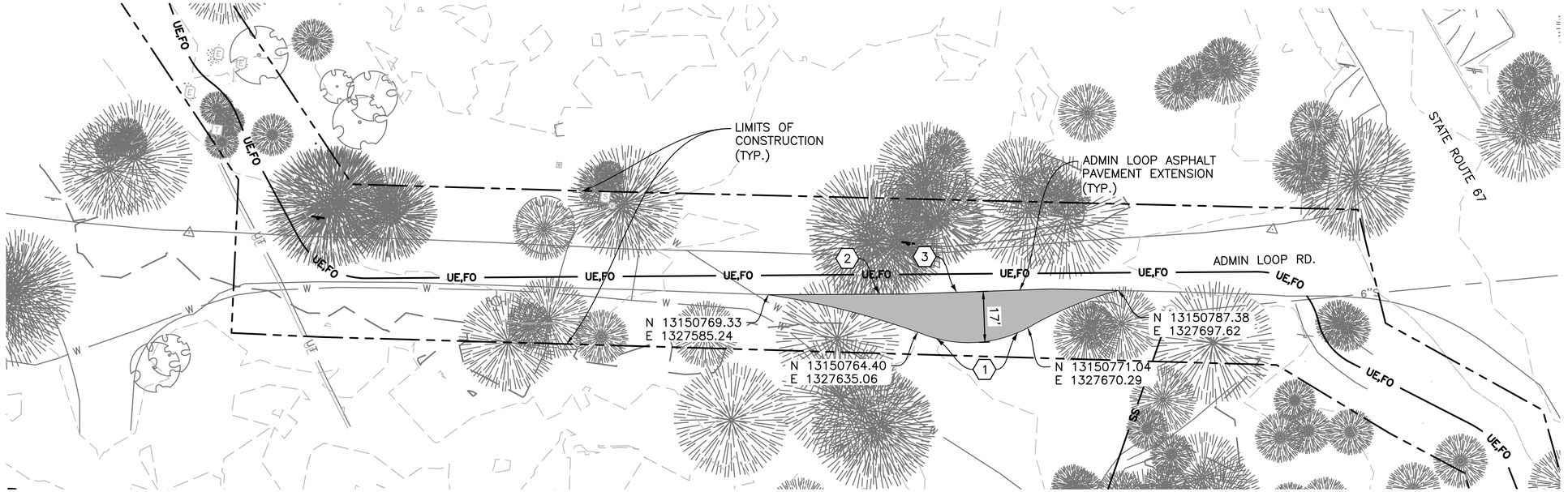
KEYMAP

20 0 20 40 HORIZ. 1" = 20'
5 0 5 10 VERT. 1" = 5'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75262 GREGORY YOUNG STATE OF ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: T. MCCALLAN TECH. REVIEW: GOUGH/LUSTER DATE: JUNE 2026	SUB SHEET NO. 013 C109	TITLE OF SHEET **BALLFIELD ROAD** PLAN AND PROFILE STA 38+50 TO 40+50 NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET **84** OF 87

8/18/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\013-C103.dwg



PLAN
1"=20'

GENERAL NOTES:

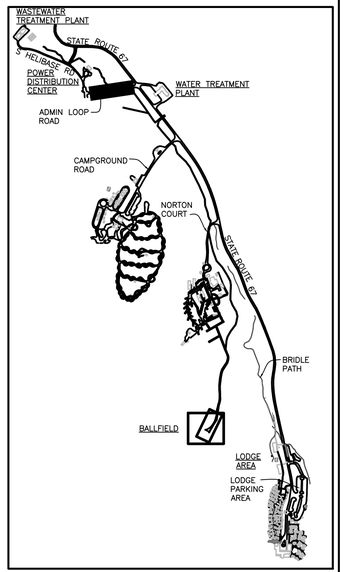
A. SEE SHEET 011–X115 FOR TREE REMOVAL LOCATIONS.

B. SEE SHEET 012–C116 FOR EROSION CONTROL MEASURES.

C. SEE 012–C145 FOR PIPE ROUTING PLAN AND PROFILE.

KEY NOTES: (X)

1. CONTRACTOR TO TAPER EDGE OF PAVEMENT EXTENSION TO AVOID TRUNKS OF NEAREST TREES AND PROVIDE MIN OF 17 FT ADDITIONAL ROADWAY WIDTH FOR VEHICLE PASSAGE.
2. ASPHALT QUANTITY = 108 SY (APPROX.)
3. SEE SHEET 002C505 DETAIL 1 FOR TYPICAL ASPHALT PAVEMENT SECTION.



KEYMAP

20 0 20 40 1" = 20'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 75262 GREGORY YOUNG Date Signed 5/1/2026 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG DRAWN: T. MCCALLAN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 013 C110	TITLE OF SHEET ADMIN LOOP ASPHALT PAVING PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 85 OF 87

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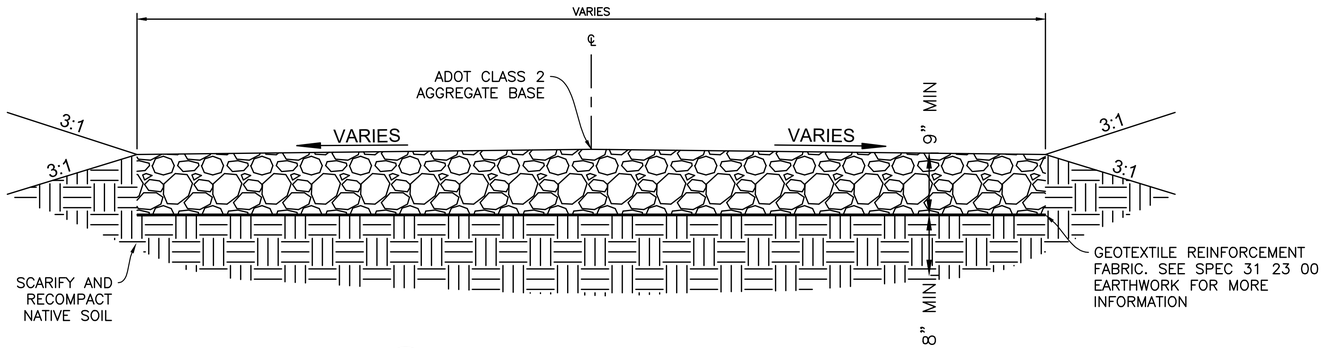

1 / 013C301 GRAVEL ACCESS ROADWAY SECTION

NOT TO SCALE

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REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO.
75262
GREGORY
YOUNG
Date Signed 5/1/2026
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: G. YOUNG CADD T. MCCALLAN TECH. REVIEW: GOUGH/LUSTER DATE: MAY 2026	SUB SHEET NO. 013 C301	TITLE OF SHEET **BALLFIELD ROAD** SECTIONS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET **87** OF 87